ORACLE METHOD[SM]

APPLICATION IMPLEMENTATION

PROCESS AND TASK REFERENCE

Applications Global Service Line
Volume 1
Release 3.0.0
August, 1999
Part Number A75150-01

ORACLE

Application Implementation Method (AIM) Process and Task Reference, Volume 1
Release 3.0.0
Part No. A75150-01

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In memory of a great friend and colleague, Josee, who would not let us forget the silent constituents — the end users — who must live and work in the environments and systems we construct and implement.

Preface

The Application Implementation Method *Process and Task Reference* provides detailed descriptions of the tasks and deliverables of the Application Implementation Method (AIM). AIM defines a set of organized and flexible process steps that guide a project team through the Application Implementation process.

The material in this reference includes a description of AIM tasks and deliverables. In addition, guidelines on prerequisites; approach and techniques; roles and responsibilities; deliverable components; component descriptions; audience, distribution, and usage; completion criteria; and deliverable template information are provided. This reference is provided to assist in expediting and standardizing all of the tasks executed and deliverables produced in an Application Implementation project.

This reference, and the Application Implementation Method itself, are part of Oracle Method℠ — Oracle's integrated approach to technical and functional delivery.

Audience

The Application Implementation Method *Process and Task Reference* is written for implementers, developers, team leaders, project team members and project leaders. Implementers, project team members, and team leaders use this reference in detail to execute tasks and create deliverables in AIM. Team leaders and project leaders use this reference as an overview to better understand the nature of tasks and deliverables in order to better manage the execution of tasks and creation and completion of deliverables.

How the Reference Is Organized

This reference consists of an introduction followed by a chapter on each of the processes of the Application Implementation Method. Due to the size of this reference, it is divided into three volumes. Volume 1 contains chapters 1 through 4; volume 2 contains chapters 5 through 8; and volume 3 contains chapters 9 through 11 and the appendices.

Introduction: The introduction presents a brief overview of processes and phases. It also provides conventions used within the reference.

Process Chapters: Each process chapter consists of two main sections: an overview, and detailed task guidelines. The overview section provides the following for each process:

- **Process Flow** — depicts the flow of tasks for this process
- **Approach** — describes the approach for this process
- **Tasks and Deliverables** — lists the tasks executed and the deliverables produced. This table also provides the responsible role and the type of task. Task type definitions can be found in the Glossary at the end of the Application Implementation Method Handbook. Task types are:

  - SI — singly instantiated task (standard task)

  - MI — multiply instantiated task

  - MO — multiply occurring task

- IT — iterated task

- O — ongoing task

- **Objectives** — describes the objectives of the process
- **Deliverables** — lists and defines the deliverables for the process
- **Key Responsibilities** — lists and defines the key roles for the process
- **Critical Success Factors** — lists the success factors for the process
- **References and Publications** — lists any references and publications for the process

The task detail section provides the following for each task:

- **Optional Criteria** — where applicable, criteria specifying under what conditions the task, or some of its task steps, should be executed
- **Prerequisites** — lists the prerequisites for the task and their source
- **Task Steps** — lists the steps of the task
- **Approach and Techniques** — describes the approach and suggested techniques for the task
- **Linkage to Other Tasks** — identifies which downstream tasks use the deliverable produced as input
- **Role Contribution** — provides the role contributions for the task
- **Deliverable Guidelines** — lists the guidelines and deliverable components for the task and describes each section of the deliverable document
- **Tools** — lists the template and tools to use when creating the deliverable for each task

Appendix A: This appendix provides an index which cross-references the AIM tasks in version 2.0 and version 3.0.

Appendix B: Appendix B provides a description of roles used in the Application Implementation Method.

How to Use this Reference

The *Application Implementation Method Process and Task Reference* provides the detailed task and deliverable information that makes up the Application Implementation Method. Each task produces a deliverable and these deliverables are described in this reference. Each task has an identifier (task ID) which makes it easy to locate. The task ID corresponds directly to the deliverable ID of the task that creates the deliverable.

All users of this reference should read the Introduction to better understand tasks, deliverables, and processes.

Oracle recommends that users of all of the AIM handbooks, and the Application Implementation Method itself, take advantage of AIM training courses provided by Oracle University. In addition to the Application Implementation Method handbooks and training, Oracle Consulting or one of Oracle's Implementation Services Partners can also provide experienced AIM consultants, automated work management tools customized for AIM, and Application Implementation Method deliverable templates.

Conventions Used in this Manual

We use several notational conventions to make this reference easy to read.

Capitalization

Names of tasks, deliverables, process and phases are always give title capitals. For example, Design Audit Facilities task, System Data Model deliverable, Technical Architecture process and Build phase.

Abbreviations and Acronyms

Occasionally, it is necessary to use abbreviations and acronyms when adequate space is not available. The Glossary lists meanings of all acronyms and abbreviations.

UPPERCASE TEXT

Uppercase text is used to call attention to command keywords, object names, filenames, and so on.

Italicized Text

Italicized text indicates the definition of a term or the title of a manual.

Bold Text

Bold text is designed to attract special attention to important information.

Attention

We sometimes highlight especially important information or considerations to save you time or simplify the task at hand. We mark such information with an attention graphic, as follows:

Attention: Since project team training occurs simultaneously with this task, some recommendations (or decisions) from training may be implemented in the mapping environment. In this case, these training inputs become predecessors to this task.

For More Information

Throughout the reference we alert you to additional information you may want to review. This may be a task section, appendix, manual reference, or web site. We highlight these references with an easy-to-notice graphic. Here is an example:

Reference: For more information about content for the Solution Design presentation, review the Critical Success Factors, page 3-7.

Web Site: You can find further information on Oracle's Home Web Page http://www.oracle.com/

Suggestions

We provide you with helpful suggestions throughout the reference to help you get the most out of the method. We highlight these suggestions with an illuminated light bulb. Here is an example of a suggestion:

Suggestion: Verify your backup and recovery plan with your hardware and software vendors.

Warning

Considerations that can have a serious impact on your project are highlighted by a warning graphic. Read each warning message and determine if it applies to your project. Here is an example:

Warning: Any time you insert data directly into Oracle Application tables, you run the risk of corrupting the database. Oracle strongly discourages inserting data directly into Oracle tables that are not designed as an Open Interface.

Optional Criteria

Where applicable, optional criteria specifying under what conditions the task, or some of its task steps should be executed is highlighted by a delta symbol graphic. Here is an example:

Δ If your project includes *process change*, you should perform this task.

Related Publications

Books in the Application Implementation Method suite include:

- AIM *Method Handbook*
- AIM *Process and Task Reference*, volume 1 (this book)
- AIM *Process and Task Reference*, volume 2
- AIM *Process and Task Reference*, volume 3
- AIM *FastForward Add-In Handbook*

You may also refer to the following Project Management Method (PJM) suite of reference books:

- PJM *Method Handbook*
- PJM *Process and Task Reference*

Obtaining Additional AIM Advantage Licenses

Each key member of your project team should have a licensed copy of AIM Advantage installed on their workstation. Key members are those individuals who will be leading areas of the implementation project and generating key deliverables.

Oracle provides AIM Advantage licenses to select Oracle Applications Implementation Partners and Oracle Consulting for their project staff. Customers should also equip their key project team personnel with licensed copies of AIM Advantage. This facilitates improved understanding of the implementation process by all team members, as well as improved overall cross-project team productivity.

Additional copies of AIM Advantage may be obtained from the Oracle Direct Marketing or telesales group in your country, or you can contact your local Oracle sales representative.

EMM Advantage and Oracle Application Upgrades

EasiPath Migration Method (EMM Advantage), Oracle's packaged methodology for application upgrades, is complementary to the Applications life-cycle of installation and subsequent upgrades. Produced by Oracle Corporation, EMM Advantage is available to help you structure and manage your Oracle Applications upgrade project.

EMM Advantage includes:

- EasiPath Migration Method (EMM) — a proven, structured approach used successfully worldwide by Oracle consultants
- Project Management Method (PJM) — a standardized Oracle approach to project management

The EMM Advantage toolkit, in combination with your skills, experience, and business knowledge, helps promote a higher-quality migration and lead you to business results faster. It is available from the Oracle Direct Marketing or telesales group in your country, or you can contact your local Oracle sales representative.

Your Comments are Welcome

Oracle Corporation values and appreciates your comments as an Oracle AIM user and reader of the reference. As we write, revise, and evaluate our documentation, your comments are the most valuable input we receive. If you would like to contact us regarding comments on this or other Oracle Method manuals, please use the following address:

email: aiminfo.us@us.oracle.com

Contents

Introduction

Processes, tasks, and deliverables are the basis of Oracle's Application Implementation Method. They are the building blocks from which project managers construct Application Implementation projects. The AIM *Process and Task Reference* provides the details of every process that plays a part in the Application Implementation Method and every task included in each process. As an introduction, this section provides a brief overview of the concepts of *process*, *task*, and *deliverable*, and describes the information in this reference that is given for each.

What is a Task?

A **task** is a unit of work that results in the output of a single deliverable. Tasks are the most elementary unit of work that one would put into a project plan — they provide the basis of the work breakdown structure. A task produces a single, measurable deliverable and is usually assigned to be the responsibility of a single team member (although many others may play contribution, review, and approval roles). Project progress is usually measured by the successful completion of tasks.

Task IDs and Names

Each task in AIM has a unique ID. That ID is composed of the alphabetical ID of the process in which the task plays a part (more on this later), followed by a three digit sequence number, that usually increments by 10. The sequence number generally reflects the order in which tasks are to be completed. For example, you can be reasonably sure that task BP.020 will be completed well in advance of BP.090 within the Business Process Architecture (BP) process.

Note that the task ID does not reflect the strict order or phase in which the task occurs within a project. A project manager may combine tasks and processes in different ways to meet the needs of different development approaches. Therefore tasks may have different sequences, relative to each other, in different types of Application Implementation projects.

Each task also has a unique name. A task name always consists of a verb (such as *create...*, *determine...*, *design...*), followed by an object. In most cases the object is the name of the deliverable that the task produces. You will find that the text always refers to task names with title case letters, for example, Develop Current Process Model (BP.040).

Task Information

Each task in AIM has a task guideline. If you know a task's ID, it is easy to find the guideline for that task in the AIM *Process and Task Reference*. Locate the process chapter by the first part of the ID, and then locate the task within the chapter by using the numerical sequence number part of the task ID.

If you only know the name of a task, you can use Appendix A to find the ID. Appendix A contains an alphabetical listing of tasks by task name. It also contains an alphabetical listing of tasks by deliverable name.

Optional Criteria

Task are divided into two types — Core and Optional. A core task must occur on every implementation. An example of a core task is Setup Applications (PM.050). An optional task is performed only if the project requirements mandate its use. An example of an optional task is Design Database Extensions (MD.060), which only needs to occur on projects where application extensions that drive database changes will be developed.

Many of the tasks in the AIM *Process and Task Reference* have criteria that define when the task or some of the task steps should be executed. The optional criteria, where applicable, is located just below the task description. In the case of optional task steps, the delta symbol (Δ) will appear to the left of the task step ID.

Prerequisites

Each task assumes that certain things (such as information, programs, and hardware platforms) have been previously produced or compiled, and are available for use. In most cases, these **prerequisites** of the task are specific deliverables of previous tasks. In some cases, they are expected from the client business organization.

Each task guideline lists that task's prerequisites. Under each prerequisite you will find an indication of which components or specific information within the prerequisite is used in the task. The text also indicates how you will use that component or information when carrying out the task.

Task Steps

Many tasks may be broken down into smaller units of work called **task steps**. In some cases, a task guideline may indicate a suggested set of task steps to follow. Many times, the team member responsible for the task (the "owner" of the task) will want to specify the task steps. The task owner will want to base those steps upon techniques that are appropriate to the overall development approach and the tools and resources that are available to the project. Any set of task steps that reaches the deliverable is acceptable as long as it includes adequate quality assurance steps.

From time to time, the reader will see a task step that has a delta symbol to the right of the task step ID. This indicates that the task step may be optional. The reader should consult the task optional criteria located just below the task description for advice regarding when a particular task step may be optional. If no delta symbol is present, then the task is assumed to be recommended as mandatory.

Role Contribution

In addition to the task owner, many other project team members may spend time on a task. Each of these team members will be fulfilling a particular **role**. Their responsibilities may be to contribute information, analyze, document, set up, review, administer, or control. The effort they spend on the task may be significant or nominal.

Each task guideline provides a suggested list of roles that may play a part in completing the task. Next to each role is an indication of the percentage of the total task effort that that role may contribute. These are suggestions that can be used in planning — actual role contributions will depend on the task steps that the task owner assigns.

What is a Deliverable?

AIM is a deliverable-based method. This means that each task produces one or more deliverables, or output, whose quality you can measure. Deliverables can have many formats, such as documents, schedules, program code, or test results.

Each deliverable in AIM is **recognizable** (it has a specific name and ID) and **measurable**. Each deliverable has the same ID as its corresponding task. Each deliverable also has a unique name, which the text always refers to using title case letters. An example is the Current Process Model (BP.040). If you know the name of a deliverable, you can find it's ID, and the name of it's corresponding task, by using Appendix A.

An AIM deliverable can be further broken down into smaller units called **deliverable components**. For example, the deliverable Current Process Model (BP.040) contains the following deliverable components:

- Enterprise Process Model
- Core Process Models
- Process Flow Diagrams
- Process Step Catalog
- Event Catalog
- Performance Measures

A **deliverable definition** is a specification of the content, organization, and usage of the product from an AIM task. This reference attempts to specify a standard set of deliverable definitions for the Application Implementation Method.

Dependencies between Application Implementation Method tasks are based on AIM deliverables. Each task requires specific deliverables from prior tasks before it can commence. A single task may impact many subsequent tasks in the current and future phases, based on the deliverable it produces.

The AIM *Process and Task Reference* lists the prerequisite deliverables for each task in the task guidelines, and also indicates this information on the process flow diagram at the start of each chapter. The AIM *Method Handbook* does the same in the diagram at the start of each phase chapter. This makes it easy for a project team member or manager to verify which deliverables must be collected as input to a particular task, before that task can be started.

Project Deliverables

You identify your deliverables using the project workplan. Each task in the project workplan should produce a single unique deliverable. As you tailor the tasks in your workplan, you need to tailor your deliverables as well.

When you begin preparing the project workplan using AIM routes (work breakdown structures), each Application Implementation Method task initially refers to the name of its corresponding deliverable. As you create or revise the tasks in your workplan, make sure that your deliverable names are unique and meaningful. For example, if you create a separate instance of an AIM task for multiple project teams, you would append a qualifier, such as the team name, to the deliverable name for each new task as well.

Project tasks and dependencies can also be tailored based on the prior availability of project deliverables. If a deliverable is already available prior to a project or phase, the task that normally produces it can be reduced to a "review" task. In some cases, it can be eliminated.

Deliverable Review

The production of deliverables is a way to measure the progress of a project. You verify successful completion of a deliverable by performing a quality review. The quality review should be based on the quality criteria specified for the AIM deliverable definition in this reference. You can also establish alternate or additional criteria, such as those required by the business management. Whatever the case, make sure that the completion criteria for each deliverable are clearly understood by the entire project staff.

Deliverable Documentation

Project deliverables can take many formats. Paper and electronic formats are the most common, but other formats include computer hardware and software (for example, System Test Environment), and even human beings (for example, Skilled Users). In many cases, you will want to produce not only the project deliverable itself, but also a record or representation of that deliverable that may be easier to review, record, and signoff. For example, in addition to producing the actual Skilled Users deliverable, document the learning events that were actually attended by each student, including an indication of which users were not prepared as expected.

You should keep in mind that in many cases, the document only **represents** the project deliverable, or only documents the parts or aspects of the deliverable that are most relevant to communicate. Much more information can often be required to actually meet the quality criteria of the deliverable. In some cases you may not need to produce a document at all. The production of the document alone should not be the goal.

Deliverable Control

You should determine the level of control of each project deliverable as either **controlled** or **uncontrolled**. You control a deliverable and its corresponding documentation in order to protect its integrity and manage its approval and revision.

As a rule, every key deliverable of the project should be controlled. You control the content of the deliverable using configuration control procedures, to restrict access and changes to the deliverable to only those authorized. You also track each version of the deliverable over time and reconstruct any previous version as needed.

You control documentation for a deliverable using document control procedures, to define how documents are prepared, approved and distributed. A controlled document is assigned a version number and its date and distribution list is clearly indicated. You may also want to number each copy of the document with a copy number. As authorized changes are made to the contents of the document, new versions are periodically created and sent out to the original distribution list (at a minimum). You also include a log of changes within each version. If you had numbered copies, you may also want to request that superseded copies be returned.

A deliverable may be uncontrolled because it will not be updated or will be shortly replaced by a controlled deliverable. Changes to an uncontrolled deliverable would not go through the project's configuration control process. However, you should still review the deliverable for quality and retain a copy of the deliverable in the project's repository.

In the case of uncontrolled documents, such as meeting minutes or memos, document control requirements can be reduced. You will not need to associate a version number with an uncontrolled document. You may not even need to formally distribute it, although a copy should always be retained in the project library.

For more information on configuration control and document control, see the *Project Management Method Process and Task Reference*.

What is a Process?

Major project objectives are achieved by processes. A **process** is a series of **tasks** that results in one or more critical project deliverables. The tasks in a process are usually highly dependent upon one another, and often span much of the project. For example, the Data Conversion process begins early in the development life cycle by defining the scope of the conversion project. This is followed by designing and building the programs and tools for conversion. After testing, the data is converted and available for production.

Figure I-1 shows the processes that are a part of AIM and their relative durations.

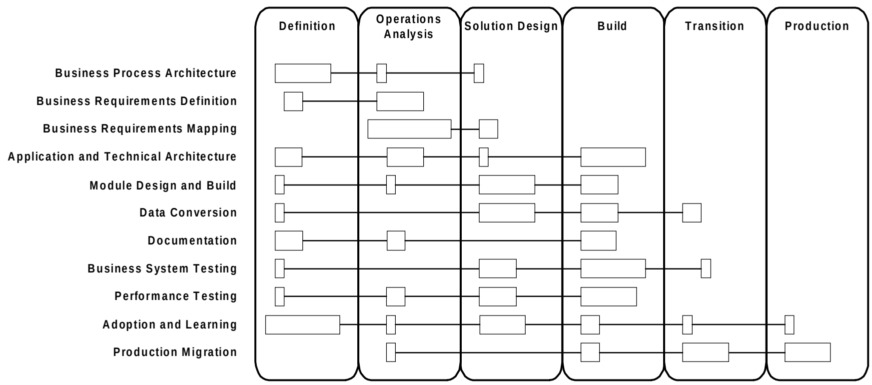


Figure I-1 AIM Context

Process Guidelines

Each chapter of the AIM Process and Task Reference is devoted to a single process. The first part of each chapter gives guidelines on the process as a whole. It shows the relationships of tasks within the process, lists the critical deliverables of the process, and provides guidance on the skills needed to execute the process.

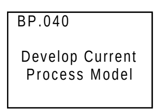
The process guidelines do not indicate exactly where each task falls in the project task plan, since this may vary by the development approach chosen. For more information on choosing and structuring a development approach using Application Implementation Method processes, see the AIM *Method Handbook*.

Process Flow Diagrams

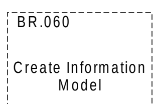
Each process is represented by a process flow diagram. This diagram shows the tasks that are part of the process, and the dependencies between those tasks. It also shows the key deliverables of the process, and the roles that are responsible for each task. The "Approach" section that follows explains the reason behind the organization of the tasks in the process flow diagram.

One thing to keep in mind is that a process flow diagram may indicate that two tasks are strictly dependent upon one another (task "B" may not begin until task "A" has completed) when, in fact, the two tasks will most likely overlap in real life. An example is in the Documentation process. The task Produce Documentation Prototypes and Templates (DO.50) is a predecessor task to Publish User Reference Manual (DO.060). Ideally, all problems would be analyzed before making any changes to the application so that changes to the modules could be made efficiently. However, this is rarely the case in the course of a project, where schedule demands usually require that application changes be made as soon as possible.

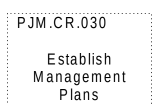
The following describes the symbols that are used in the process flow diagrams.



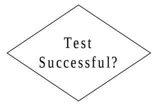
Core Task. This shows a core task that is contained in AIM. The text provides the AIM ID and the task name.



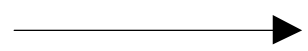
Optional Task. This shows an optional task that is contained in AIM. The text provides the AIM ID and the task name.



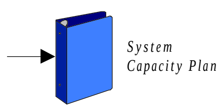
External Task. This symbol depicts a task that should be performed, but is not contained in the same process.



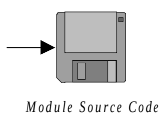
Decision. A decision indicates that there are two or more possible branches to the process flow, depending upon the outcome of the stated question. A decision symbol does not indicate another task -- all work that is done in order to indicate a particular outcome is included in previous tasks.



Process Flow. An arrow between two tasks signifies that the task at the end of the arrow generally should not start until the previous task has been completed. Example: you should not start the task *Map Business Requirements (BR.030)* until you have finished the task, *Gather Business Requirements (RD.050)*. In some cases, it may be desirable to overlap such tasks in an actual project plan.

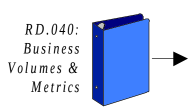


Key Document Deliverable. This represents an important textual output of the process. It includes the name of the deliverable.



Key Software Deliverable. This represents an important software product of the process.

Other symbols are used for key deliverables, as appropriate.



Required Prerequisite. This represents a key input deliverable for a task. The name of the deliverable and the ID of the task that produces the deliverable are given. Different symbols may be used to represent the medium of the deliverable.

System Architect

Agent Role. Each diagram is divided horizontally into sections or *agent channels*. Each agent channel is labeled with a role. All tasks within that section are the responsibility of that project role.

☞ **Attention:** Required Prerequisites and Deliverables do not correspond to the agent channel in which they are drawn.

CHAPTER

1

Business Process Architecture (BP)

This chapter describes the Business Process Architecture process.

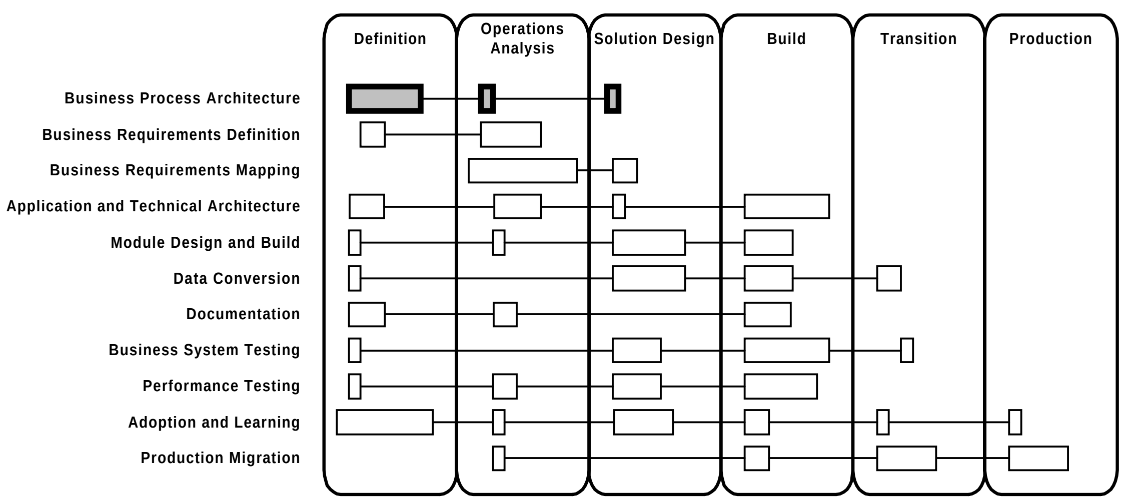


Figure 1-1 Business Process Architecture Context

Process Flow

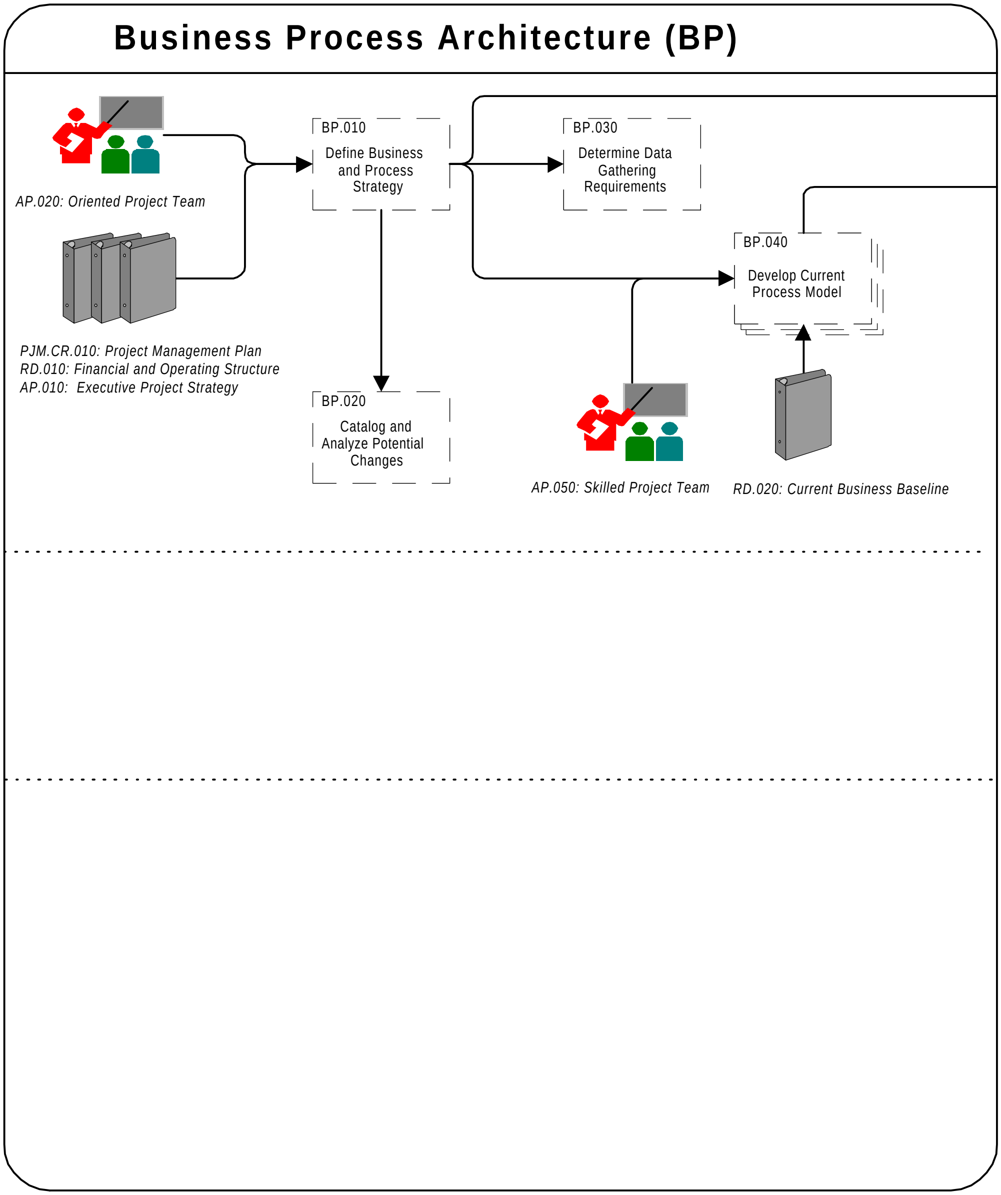


Figure 1-2 Business Process Architecture Process Flow Diagram

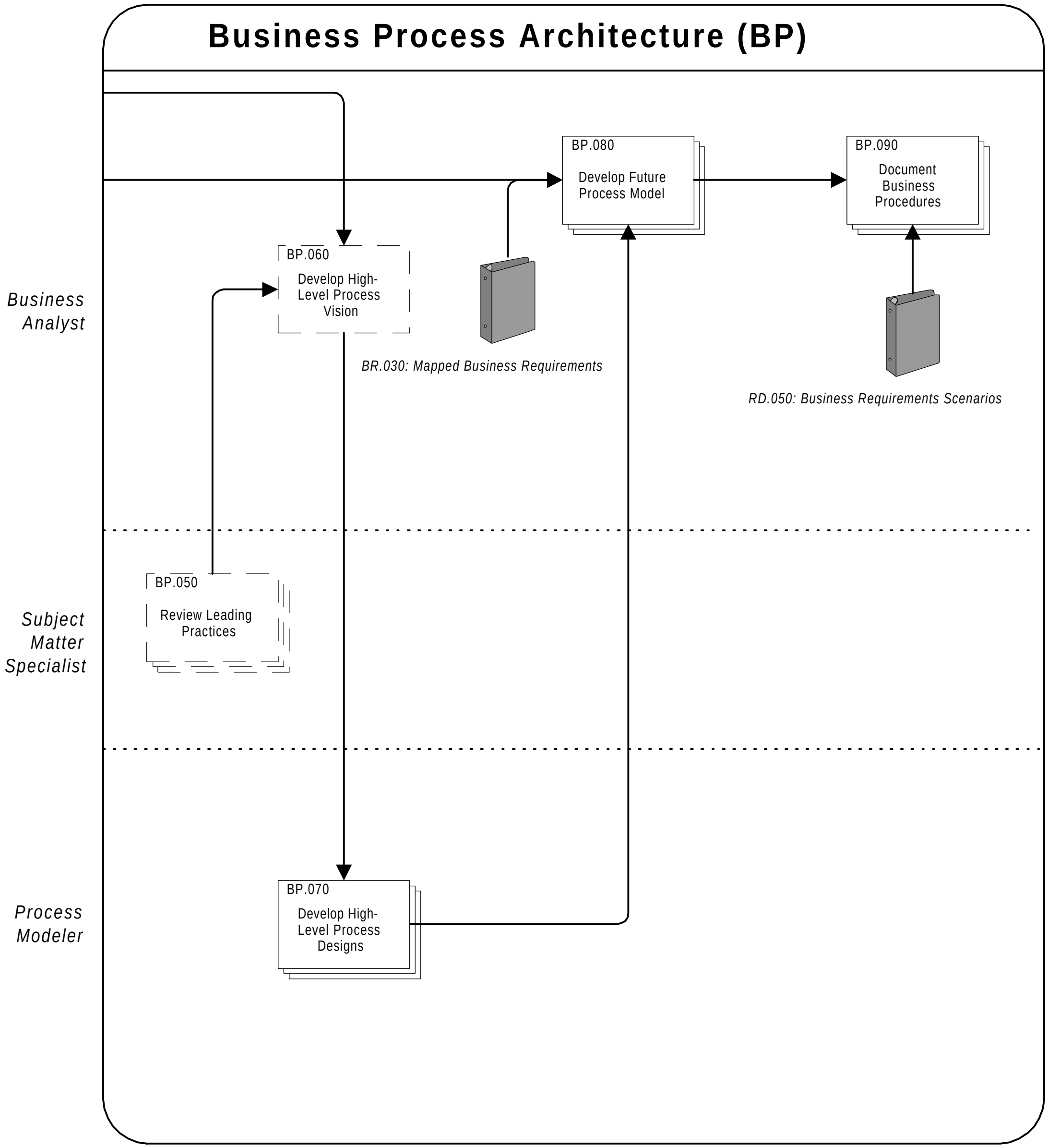


Figure 1-2 Business Process Architecture Process Flow Diagram (cont.)

Approach

The objective of Business Process Architecture is to provide the framework for combining change in business processes with implementation of software applications. In this process, standard templates are used to map the application's functions and features to standard business processes. The project team can examine the processes appropriate for the application and then make business-focused decisions either to change the current processes to suit the application or to customize the application.

Business Process Architecture is relevant whenever the organization's existing business processes may need to change as a result of implementing the applications. It focuses on high-level business processes and operations generally applicable to the organization.

This process begins with a high-level assessment of the organization and its processes. The assessment includes a definition of the organization's business processes that will be analyzed and improved upon during the course of the implementation project, along with a determination of current process performance data. The current process performance data is used to provide a baseline against which process improvements can be measured. The process also consolidates the organization's existing business and uses it to align the objectives of the process improvement activities to be carried out later.

Business Process Architecture makes use of a number of work sessions to assist the organization in creating a vision of the future. In the work sessions, you review leading practices relevant to the project and use them to create targets and stretch goals for the organization's improved processes. The work reviewed creates the business vision itself, which provides a general, high-level statement of how the new processes will operate. A separate task translates this vision into a high-level design, showing the steps in the process that will support the vision.

During Business Process Architecture, a Change Catalog (BP.030) is established and then maintained as a common reference for all potential changes to the applications, as well as changes to the organization's processes. This catalog, along with the high-level design, is used to conduct a gap analysis in the Business Requirements Mapping (BR) process. To perform a gap analysis, you compare the envisioned process design with the processes supported by the applications. The differences (gaps) revealed by this analysis need to be resolved by producing an approach that balances changes in the application against changes in processes and organization.

Business Process Architecture results in:

- a vision of the future for the processes that will be impacted by the implementation of the application in the target environment
- a modified process design reflecting the vision of the future and a cost-justified solution for the organization
- current and future process models
- a list of potential changes to the application needed to achieve this solution

Solution sets and process models for Oracle ® Applications, such as Oracle Business Models (OBM), directly support Business Process Architecture, as they represent the business processes supported by Oracle Applications. They provide a consistent framework for modeling enterprise processes. They also provide a consistent approach for determining the potential changes to the applications needed to support alternative processes.

Oracle Tutor™ is a tool, complementary to OBM, that picks up from level 3 of the OBM framework to define detail procedures in levels 4 and 5. Oracle Tutor provides a format for documentation of procedures and a repository of procedures and courseware to support Oracle Applications.

Business Process Definitions

Process modeling views processes as systems that are triggered by events that receive inputs and produce outputs. Process modeling shows the steps that occur within a process to transform the inputs into the process outputs. Process models can also represent flows of information and material between process steps. By showing which departments or employee roles are responsible for the various steps within each process, process modeling places processes in their organizational context.

Process

From the perspective of process modeling, a process is defined by the following information:

- the business event that triggers the process
- the inputs and outputs
- all the operational steps required to produce the output
- the sequential relationship between the process steps
- the business decisions that are part of the event response
- the flow of information and material between process steps

Process Flow Diagram

The process flow diagram presents process information visually. The result is a visual map of the finished work and the information that is needed during the execution of the business process. People who understand how the work is actually performed can review the visual models, identify inaccuracies, and make corrections. In addition, they can more easily identify problem areas and visualize opportunities for process improvement and process redesign.

Business Processes

A business process is a combination of operational steps and management controls that, taken together, produce a product or deliver a service. Central to this definition is the fact that a business process produces an output of some kind. The product or service may be something received by an external customer (such as filling a customer order). Other business processes produce outputs that are not visible to the external customer, but are essential to the effectiveness of the organization (for example, hiring an employee).

Business processes occur as preplanned responses to business events. An event is an occurrence inside or outside of the business environment that causes an event response to be executed. Event responses follow an established, repeatable set of steps that make up the complete response to the business event. The combination of an event and its event responses forms a process. The entire set of business processes in a high-level business function area makes up a planned response system.

Events

All business processes are planned responses to business events. An event is simply an occurrence that causes a response to be executed. Some examples of events that trigger processes are shown in the table below.

Event	**Response**
Customer Places Order	Fulfill Customer Order
Position Opens	Hire Employee
Request for Proposal Arrives	Prepare Proposal
Fiscal Year Ends	Prepare Tax Return

Table 1-1 Event and Response Examples

Types of Events

Events may be external to the organization *(customer places order)*, internal to the organization *(position opens)*, or temporal *(year ends)*. A temporal event is one that initiates at a predetermined point in time.

Just as processes may be high- or low-level, events may also be high- or low-level, although in process modeling events are not formally decomposed. A high-level event is one that represents several similar events. Consider the example of the event c*ustomer requests service*, causing the process response p*rovide customer service*. This event may actually represent several types of service request events. A manufacturing company, for example, may have many types of customer service events:

- The customer requests change of billing address.
- The customer orders inventory items.
- The customer orders warranty contracts.
- The customer requests credit approval.

The event *customer requests service* is the most general case of these different types of events. An example of a low-level event is the arrival of a specific customer order.

Event Mechanisms

Event mechanisms identify the various ways in which an event is recognized. For example, a time sheet may arrive by electronic mail, surface mail, or fax. A customer may place an order by phone or through the mail. The ultimate deliverable — the fulfilled order — is the same, regardless of the mechanism used, but details of the process execution may be different. If the customer places an order by phone, a clerk may enter the order in the system while the caller is on the phone. When a customer sends an order by mail the process steps may be:

1. Stamp the order with the date received.
2. Check the order for errors.
3. Enter the order into the system, if there are no errors.
4. Return the order to the customer with a request to clarify the order information, if there is an error.

Steps 1, 2, and 4 are not performed when the order is received by phone.

If there is more than one mechanism by which an event can be recognized, and the event response is significantly different depending on the mechanism, it is customary to model the differences. Consider each event-mechanism combination different events and produce a process flow diagram for each. Using the above example, there are two different events:

- Customer Phones in Order
- Customer Mails in Order

Develop a process flow diagram for each event.

If the responses to an event recognized through more than one mechanism are similar, use one diagram, but show each mechanism as a separate event. If the variation in process steps is small, use decision diamonds and conditional branching to model the alternate responses.

Agents

An agent is the party responsible for a process step. An agent may be an organizational unit, a functional unit, an employee role, or a party external to the organization (such as a customer or supplier).

As with processes and events, agents can represent any level in an organization, from a business unit down to a specific job description. Some examples of agents that might be involved in the process of making a sale are:

- sales
- account executive
- area vice president
- regional sales manager
- administrative assistant
- sales accounting

These examples show the different levels at which agents can be identified. *Sales* is a business unit, and *account executive* is a job role. *Area vice president*, *regional manager*, and *administrative assistant* identify specific roles. *Sales accounting* represents a functional area. Assuming that *sales accounting* is a separate system, the last agent example could represent both a functional area and a business system; process flows to and from this agent's process steps could represent system interfaces.

The generic definition of agent as *any responsible party* gives the business analyst a high degree of modeling freedom. In some situations, the level of detail represented in the specification of an agent may be driven by the process for which the agent is responsible. For example, the manufacturing agent identifies the functional area responsible for building a product in an order fulfillment cycle. Viewing the process *build product* in more detail, it is possible to divide *manufacturing* into *shop floor manager*, *assembler*, *inspector*, *packager*, and so on.

These examples illustrate the broad use of the concept of agent. Its value is in representing who (in the generic sense) is performing the work of a process. From the perspective of application implementations, agents represent job functions that will be carried out on the applications.

Suggestion: If you specify agents at the job role level, they can be used to develop custom responsibilities in the application.

Tasks and Deliverables

The tasks and deliverables of this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
BP.010	Define Business and Process Strategy	Business and Process Strategy	Project includes *process change*	SI
BP.020	Catalog and Analyze Potential Changes	Change Catalog	Project includes *process change*	SI
BP.030	Determine Data Gathering Requirements	Data Gathering Requirements	Project includes *process change*	SI
BP.040	Develop Current Process Model	Current Process Model	Project includes a *need to model current business processes and practices*	MI
BP.050	Review Leading Practices	Leading Practices Review	Project includes *process change*	MI
BP.060	Develop High-Level Process Vision	High-Level Process Vision	Project includes *process change*	SI
BP.070	Develop High-Level Process Designs	High-Level Process Designs	Always	MI
BP.080	Develop Future Process Model	Future Process Model	Always	MI
BP.090	Document Business Procedures	Business Procedure Documentation	Always	MI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 1-2 Business Process Architecture Tasks and Deliverables

Objectives

The objectives of Business Process Architecture are:

- Identify the core processes of the organization and the processes that are within the scope of the project.
- Create a baseline of existing business process performance and goals (for measuring future performance).
- Document current process models.
- Create or consolidate a strategy for improving the business and refining the processes.
- Review leading practices and models of best use of the application for the industry and other industries with relevant processes.
- Produce a vision and high-level designs of how processes would operate after implementation of the application.
- Define detailed future process models.

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
Business and Process Strategy	A business and process strategy, and a preliminary business case.
Change Catalog	A catalog and initial analysis of the costs, benefits, and risks of changing the application or processes.
Data Gathering Requirements	Current relevant process data collected and a list of process data to be collected during the remainder of the project.
Current Process Model	Process flow diagrams of the current business processes and functions.

Deliverable	Description
Leading Practices Review	A review of leading practices for relevant organization processes and best uses of relevant application features.
High-Level Process Vision	A general, high-level statement of how the new processes will operate to meet the organization’s objectives. Identification of the major influences on the organization’s future and the opportunities that will be opened by implementing the applications. Documents challenges and opportunities to be addressed by the vision.
High-Level Process Designs	A generic process design, relevant process descriptions, and generic technology design.
Future Process Model	Process flow diagrams of the events and business processes that the applications and the associated functions of the business area will support.
Business Procedure Documentation	Documentation of the procedures supporting the business processes.

Table 1-3 Business Process Architecture Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Application Specialist	Provide a knowledge of the Oracle Application. Help determine feasible changes to the application and the organization's operations. Highlight the limitations and constraints of current and envisioned releases.
Business Analyst	Provide strategy expertise. Lead the project team through leading practices review, work sessions, and visioning. Identify and assess changes to the processes.
Business Line Manager	Provide expertise about the operation of the current business process. Provide knowledge of the intentions and objectives of the business. Identify and group core processes into process clusters.
Client Executive	Define business and process strategy and prepare project delegation to the steering committee and project team. Participate in work sessions.
Client Project Manager	Determine which clusters of processes are appropriate for change. Determine the mission, goals, and objectives of the organization.
Client Staff Member	Provide information about the organization's current processes and business events. Participate in defining future business events.

Role	Responsibility
Key User	Contribute knowledge of the organization’s strategy, goals, and processes. Determine and evaluate the core capabilities of the organization.
Organizational Development Specialist	Evaluate the effects of change on the organization and processes.
Process Modeler	Develop the process design and gain agreement with the organization representative. Document the deliverable components for generic process design and the written description of how the process will operate.
Project Sponsor	Lead and facilitate alignment behind project vision and momentum.
Subject Matter Specialist	Provide specialist expertise and knowledge relevant to the organization’s industry in business processes and practices.
System Architect	Provide system functionality expertise. Responsible for highlighting the limitations and constraints of current and envisioned releases.
Tool Specialist	Provide knowledge and guidance regarding specific tool functionality, for example, procedural documentation software, such as Oracle Tutor.
User	Provide acceptance to the overall concept and framework of expected solutions to be implemented.

Table 1-4 Business Process Architecture Key Responsibilities

Critical Success Factors

The critical success factors of Business Process Architecture are as follows:

People

- effective and continuing sponsorship at a high-level in the organization
- involvement by representative samples of stakeholders in the project
- full-time involvement of a dedicated resources with relevant skills and working knowledge of business process change

Process

- effective process scoping to determine which processes are within the scope of the project and the elements of those processes
- understanding of the organization’s business strategy and the objectives of each relevant business process
- use of *solution sets* and process models for Oracle Applications, such as Oracle Business Models (OBM)
- clear understanding of the difference between process and functions or business units
- measurability of process performance and determination of key performance indicators (KPIs) for business processes
- avoidance of too much effort spent on detailed descriptions of current processes (“analysis paralysis”)

Technology

- availability of expertise in relevant application features and how the features are used to support business processes
- awareness of the best uses of Oracle technologies, through employment of tools such as Oracle Business Models (OBM)
- extension of the application, where necessary (when there is a measurable gain to the organization)

References and Publications

Reference: Oracle Business Models, Release 2.

Reference: Gary Born, *Process Management to Quality Improvement*, John Wiley & Sons, 1994. ISBN: 0471942839.

BP.010 - Define Business and Process Strategy (Optional)

In this task, you work closely with executive management on a strategy for improving the business and refining its business processes.

∆ If your project includes *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **Business and Process Strategy**. It identifies the organization's mission and objectives for improving the business and refining its business processes.

Prerequisites

You need the following input for this task:

❑ **Project Management Plan [*initial*] (PJM.CR.010)**

The initial Project Management Plan defines the scope, objectives, and approach for the project, that can impact the business scope.

❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure portrays organization structure information that may be important.

❑ **Executive Project Strategy (AP.010)**

The Executive Project Strategy describes facilitation the general direction of the project. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

❑ **Oriented Project Team (AP.020)**

The Oriented Project Team has been exposed to the project direction and is now able to contribute to strategy development.

Optional

You may need the following input for this task:

- ❑ **Enterprise and Business Process Reengineering Strategy Studies** (if available)

Enterprise and business process reengineering (BPR) strategy studies may indicate the need for a complete overhaul of current processes to be supported by the new applications. The output of a BPR study may provide most of the content for the Future Process Model (BP.090).

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Create a high-level model of the enterprise, identify mission, strengths and gaps, and major processes.	High-Level Enterprise Model
2.	Define relevant process groups for process change.	Process Clusters, Sequence, and Dependency
3.	Evaluate existing business performance and produce a baseline for process improvement. Assess potential changes.	Initial Change Assessment
4.	Identify existing process improvement projects.	Process Improvement Projects
5.	Define a high-level business strategy.	Business Strategy
6.	Define a high-level process strategy.	Process Strategy

No.	Task Step	Deliverable Component
7.	Summarize the business and process strategy	Introduction
8.	Define a preliminary business case.	Preliminary Business Case
9.	Review the Business and Process Strategy with executive management.	
10.	Secure acceptance of the Business and Process Strategy.	Acceptance Certificate (PJM.CR.080)

Table 1-5 Task Steps for Define Business and Process Strategy

Approach and Techniques

The Business and Process Strategy is carried out in work sessions with executives. Identify the primary mission of the organization and define how the primary mission is supported by the organization's goals, objectives and core capabilities. Assess the strengths and gaps in the core capabilities of the organization and identify the major business processes supporting the organization's innovation and delivery capabilities.

In addition, focus on existing business performance and goals. Create a baseline that will be refined in Business Volumes and Metrics (RD.040) against which future improvements, such as the Effectiveness Assessment (AP.180), can be measured. Use the baseline and the assessment of what will be changed to create the Business and Process Strategy. Finally, create the preliminary business case.

Additional analysis and background research may be required for preparation of the business case.

Mission, Goals, and Objectives

The organizational mission and the goals and objectives are developed or updated at one or more work sessions. The mission, goals, and objectives identify the strategy of the organization that provides the context for process change.

Core Capabilities Assessment

Core capabilities represent the response of an organization to the environment. They represent the organization performance against goals and objectives. These capabilities are the competencies that allow the organization to survive and flourish. Capabilities may be divided into three categories:

- innovation
- delivery
- infrastructure

Innovation

These capabilities help the organization to understand its internal needs and develop something of value to meet those needs.

Delivery

These capabilities help the enterprise to actually deliver the value and to subsequently generate cash back to the organization. The value delivered is defined as a product or service.

Infrastructure

These capabilities represent the basic skills of the organization. These skills allow the organization to grow, learn, and adapt to the changing demands of the environment. These skills should be applied to the everyday business processes that are performed, to exhibit the innovation and delivery capabilities.

The following is a sample core capabilities assessment:

Core Capabilities Assessment
Innovation—develop and improve new electronic products quickly and at low cost, for the automotive electronics market. **World Class**
Supply highly qualified engineering skills to address current and new product and market opportunities. **Good**
Manufacture products within cost, cycle time, and quality targets, meeting or exceeding expectations and requirements. **Excellent**
Manage efficiently a supply chain of local and foreign suppliers. **Average**
Infrastructure Project Management—manage costs, time, and resources to achieve project goals. **Average**
Organizational Learning—adopt and adapt leading practices and processes based on the experience of the organization. **Poor**

Figure 1-3 Core Capabilities Assessment

Core Processes and Process Change Projects

As part of this task, core business processes that add value to the organization's performance are identified.

These processes are then grouped together into clusters to determine which will provide suitable combinations for a process change project. As a rule, processes should be clustered when they are strongly interconnected and when changes to one of these processes have a significant impact on the others in the cluster.

Defining a series of process change projects requires consideration of many factors. The enterprise process model is a roadmap to this effort. However, it does not provide sufficient information to select process change projects and phases.

Projects selected must be reviewed for feasibility. If the scope of the selected projects is too narrow, the project will be overly restricted and unable to effect real and beneficial change. Too large a scope may create a situation where the organization wants to reinvent the business which may not be justified.

It is easy to become caught up in a mechanical approach to developing process clusters, forgetting that the primary objective is to define scope boundaries for potential project change projects. Clusters should pass the common sense test: *Does it make sense to address these areas together? Are you likely to be successful with a project like this?*

Initial Change Assessment

A combination of workshops and analysis sessions is used to determine the initial assessment of changes to the application and the working practices. The workshops are used to develop very high-level visions of the process changes, while the analysis sessions investigate the implications of the change, in terms of costs, business performance, and working practices.

☞ **Attention:** Make sure that the impact of process change on each organizational system (work, people, organizational structure, organizational culture, and physical layout) is considered.

Sessions with Executives

Executive's schedules should be coordinated across processes, such as Executive Project Strategy (AP.010), to avoid holding too many sessions. Consolidated sessions help align strategies and create acceptance from the executive stakeholders in the organization.

Linkage to Other Tasks

The Business and Process Strategy is an input to the following tasks:

- BP.020 - Catalog and Analyze Potential Changes
- BP.030 - Determine Data Gathering Requirements
- BP.040 - Develop Current Process Model
- BP.050 - Review Leading Practices
- BP.060 - Develop High-Level Process Vision
- BP.070 - Develop High-Level Process Designs
- AP.030 - Develop Project Team Learning Plan
- AP.060 - Develop Business Unit Managers' Readiness Plan

- AP.080 - Develop and Execute Communication Campaign
- AP.090 - Develop Managers' Readiness Plan

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	80
Organizational Development Specialist	20
Client Executive	*
Business Line Manager	*

Table 1-6 Role Contribution for Define Business and Process Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the Business and Process Strategy to determine the extent of changes to the organization's business processes. Address the whole organization, rather than an individual department or function. The enterprise model component should be built first, as it contains the core business processes and shows their interactions.

Separate work sessions are used to investigate and develop the components of this deliverable. *Process workshops* are used to investigate the high-level steps of the core business processes. The initial change assessments are only intended as a preliminary indication of the potential changes and will be modified significantly as the project progresses.

This deliverable should address the following:

- high-level model of the enterprise process clusters, focusing on relevant business processes for process improvement
- key industry dynamics
- business objectives
- key opportunities
- analysis of the current situation
- key business issues
- current initiatives mapping opportunities to application mix
- financial justification for application mix

Deliverable Components

The Business and Process Strategy consists of the following components:

- Introduction
- High-Level Enterprise Model
- Process Clusters, Sequence, and Dependency
- Initial Change Assessment
- Process Improvement Projects
- Business Strategy
- Process Strategy
- Preliminary Business Case

Introduction

This component summarizes the business and process strategy.

High-Level Enterprise Model

This component provides a high-level model of the entire organization that illustrates:

- existing organization and roles
- mission statement

- goals and objectives
- business drivers
- core capabilities and assessment
- critical success factors
- key performance indicators and their current values

The High-Level Enterprise Model contains a process flow diagram that illustrates the high level flows between core processes and identifies the process owners.

Process Clusters, Sequence, and Dependency

This component groups related tasks into process clusters after the key business process flows are described in the Enterprise Model. These process clusters should contain closely connected activities, linked by process flows. Each process cluster should be capable of being improved or changed, largely independent of the other parts of the enterprise model.

Initial Change Assessment

This component documents the findings made during work and analysis sessions based on the high-level vision of the change implications (in terms of changes on cost, business performance, and working practices).

Process Improvement Projects

This component identifies the following elements for each business process improvement project:

- scope (described in terms of process clusters)
- sequence and dependency (relative to other reengineering projects and organization change initiatives)
- expected duration of phase
- chief objective for the project phase
- potential team constituency

Business Strategy

This component documents the actions for strengthening the organization's competitive position, satisfying customers, and achieving performance targets. It helps build the strategy for outlining the recommended business change programs that best address the identified opportunities, and identifies the business processes to concentrate on (for example, those that contribute most to achieving the organization's vision).

This component summarizes the business strategy (if it has already been prepared prior to this task).

Process Strategy

This component focuses on the organization's strategy for improving its business processes and aligning them with the applications to be implemented. This component probably includes an evaluation of existing processes and the directions that the organization intends to take in improving them. It may also include key process measures and goals for their improvement.

Preliminary Business Case

This component uses the preliminary business case as a high-level justification for the strategy. It includes an initial cost-benefit analysis to justify the proposed investment.

It helps to assess return on investment and impact on the organization. It uses the business drivers from the Initial Change Assessment to justify the recommended strategy.

Tools

Deliverable Template

Use the Business and Process Strategy template to create the deliverable for this task.

BP.020 - Catalog and Analyze Potential Changes (Optional)

In this task, you compare the costs and benefits involved in making changes to the business processes and roles of the organization, as well as to the application.

To provide a sound basis for making these comparisons and reaching decisions, the project establishes a Change Catalog, which documents these potential changes.

Δ If your project includes *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **Change Catalog**. It is a catalog and initial analysis of the costs, benefits, and risks of changing the application or processes.

Prerequisites

You need the following input for this task:

- ❑ **Business and Process Strategy (BP.010)**

The Business and Process Strategy includes key programs that will require change and need to be taken into account in the Change Catalog. It also includes information, relevant to the organization's priorities, that will be needed in evaluating the costs and risks of change. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review initial assessment from the Business and Process Strategy (BP.010)	
2.	Identify potential changes to applications, the organization and processes.	
3.	Create an assessment of the likely costs and benefits of the changes.	Introduction
4.	Create an assessment of the risks involved in making changes.	Introduction
5.	Update the Change Catalog, throughout the Definition and Operations Analysis phases, with new potential changes and risks, as they are revealed.	

Table 1-7 Task Steps for Catalog and Analyze Potential Changes

Approach and Techniques

The Change Catalog is established in this task; it will be maintained until all relevant decisions have been taken on the appropriate mix of changes for the organization's needs. Initially, the Change Catalog is prepared with an initial assessment of likely changes (mainly gathered during Define Business and Processes Strategy - BP.010). This task will be activated repeatedly throughout the project, as new, potential changes are uncovered and need to be assessed.

Entries in the Change Catalog should be specific and relate to areas of change within the scope of the engagement. For example, it would normally be appropriate to include any modification to the application. It is also within scope to include changes to the business processes and related skills, as long as they were viewed as within scope during Define Business and Processes Strategy (BP.010). However, changes to the organization's career structure and remuneration is done in Align Human Performance Support Systems (AP.110).

Linkage to Other Tasks

The Change Catalog is an input to the following tasks:

- BR.010 - Analyze High-Level Gaps
- AP.100 - Identify Business Process Impact on Organization
- AP.180 - Conduct Effectiveness Assessment

Role Contribution

The percentage of total task time required for each role follows:

Role	**%**
Business Analyst	60
Application Specialist	40

Table 1-8 Role Contribution for Catalog and Analyze Potential Changes

Deliverable Guidelines

Use the Change Catalog to group related changes together as options. Later, in Analyze High-Level Gaps (BR.010), some further consolidation of changes into options may occur, but will be simplified if the Change Catalog is already grouped into options.

Where possible, quantify the potential changes to enable a cost-justified assessment in Analyze High-Level Gaps (BR.010).

This deliverable should address the following:

- the cost of making the change (including related or incidental costs)
- the benefit of making the change
- the risk of making the change, including the impact of the change and the likelihood that the risk will materialize

In providing this quantitative information about the changes, remember the approach is based on a cost-justified assessment of all potential changes resulting from the implementation. These changes impact far more than possible modifications or extensions to the application, and may include:

- organization business processes and operations
- organization roles and infrastructure
- on-going maintenance of the applications
- competitive position of the organization, in terms of production efficiency and speed of response

Information about all of the costs, benefits, and risks of a change will later enable the project team to make sound decisions about the best balance of changes for this particular organization and its future needs.

Deliverable Components

The Change Catalog consists of the following component:

- Introduction

Introduction

This component is a single table listing potential changes and their associated costs, benefits, and risks.

Tools

Deliverable Template

Use the Change Catalog template to create the deliverable for this task.

BP.030 - Determine Data Gathering Requirements (Optional)

In this task, you determine the data that needs to be collected during Business Process Architecture.

Δ If your project includes *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **Data Gathering Requirements**. It documents current relevant process data, as well as the process data to be collected during the remainder of the project.

Prerequisites

You need the following input for this task:

❑ **Business and Process Strategy (BP.010)**

The Business and Process Strategy includes an initial assessment of what will be changed. It also identifies process clusters and performance data on existing business processes. If Define Business and Processes Strategy was not performed, this deliverable will not exist (See the task description for BP.010 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

❑ **Existing Process Data (Internal)**

A description of relevant data already gathered by the organization, and existing data gathering activities, may assist the project.

❑ **Tender Documents**

The organization's statement of requirements and the proposal documents, including interview notes during preparation of the bid, may provide valuable input to this task.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify data already collected and data collection activities that may be useful for the project.	Data Collection Requirements
2.	Analyze and summarize performance of current business processes and operations.	Process Performance
3.	Determine additional data that needs to be collected and plan for its collection.	Data Collection Requirements

Table 1-9 Task Steps for Determine Data Gathering Requirements

Approach and Techniques

The Data Gathering Requirements should be carried out quickly. A quick assessment of relevant data is normally all that is required. The purpose of this task is to determine what data will need to be collected for the Business Process Architecture, so that the effort involved can be properly planned. The actual collection of data or data analysis is inappropriate at this stage in the project.

Business and Processes Strategy (BP.010) may provide most of the input required for this task.

Data on current business process and system performance is collected after defining the metrics that are appropriate for measuring performance for the organization's business area. After defining the measurements, client staff members may be asked to collect this data, if it has not already been recorded.

Linkage to Other Tasks

The Data Gathering Requirements are an input to the following tasks:

- RD.040 - Gather Business Volumes and Metrics
- BR.030 - Map Business Requirements

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	100
Client Staff Member	*

Table 1-10 Role Contribution for Determine Data Gathering Requirements

* Participation level not estimated.

Deliverable Guidelines

Use the Data Gathering Requirements to identify current data collection activities and existing business processes and system performance data. Data listed in the system performance component will usually be collected as part of Gather Business Volumes and Metrics (RD.030).

This deliverable should address the following:

- data collected
- existing data gathering activities
- data to be collected

Deliverable Components

The Data Gathering Requirements consist of the following components:

- Data Collection Requirements
- Process Performance

Data Collection Requirements

This component documents the discussions with key staff about data collection activities that may be relevant to the project. Current data collection may include earlier studies and projects that were carried out by external consultants. After evaluating this material, the data and data collection activities that will assist the project team members in their work should be documented. In particular, data that will be useful in understanding and evaluating business processes should be included. The existing data and data gathering activities specified should be reviewed before determining whether the data still needs to be collected.

Process Performance

This component makes use of the process clusters identified in Business and Processes Strategy (BP.010). Performance data should relate to business processes whenever possible, since these will be the basis for improvements enabled by implementation of the application. It includes existing business performance measures, as well as measurements that your team will take.

Relevant existing measurements should be documented. However, if the relevant measurements have not been collected, teams should be organized to collect the measurements data in Gather Business Volumes and Metrics (RD.030).

☞ **Attention:** Oracle Business Models, Release 2 contains solution sets that identify standard business process measures for core business processes.

Tools

Deliverable Template

Use the Data Gathering Requirements template to create the deliverable for this task.

BP.040 - Develop Current Process Model (Optional)

In this task, you examine the current business processes and practices to identify how the existing business system meets current business requirements.

∆ If your project needs to *model current business processes and practices,* you should perform this task.

Deliverable

The deliverable for this task is the **Current Process Model**. It includes process flow diagrams of the current business processes and functions.

Prerequisites

You need the following input for this task:

❑ **Current Business Baseline (RD.020)**

The project team uses structured process questionnaires as another process analysis technique to collect business and current system information during a business baseline interview for a given process.

❑ **Skilled Project Team (AP.050)**

The Skilled Project Team has learned modeling and mapping techniques.

❑ **Existing Reference Material**

The project team uses existing reference material to gain an understanding of the existing practices, processes, and systems that support the business objectives.

Optional

You may need the following input for this task:

- ❑ **Business and Process Strategy (BP.010)**

When *process change* is applicable, use the high-level enterprise model to determine the core processes to be modeled in this task. If Define Business and Processes strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Define the scope of current business analysis.	
2.	Review the Current Business Baseline (RD.020).	
3.	Review any current business materials that may enhance team understanding of the current business processes.	
4.	Develop, schedule, and confirm the process definition sessions.	
5.	Produce a high-level current process view of the entire organization.	Enterprise Process Model
6.	Identify and describe the events to which the current business processes respond.	Event Catalog

No.	Task Step	Deliverable Component
7.	Conduct workshops to build each existing core process model — one diagram per core business process.	Core Process Models
8.	List existing business performance measures.	Performance Measures
9.	Conduct business process analysis meetings to further define relevant existing business processes.	
10.	Construct process flow diagrams for each relevant existing business process.	Process Flow Diagrams
11.	Identify any issues from current business analysis.	
12.	Review the Current Process Model with users and management.	
13.	Secure acceptance of the Current Process Model.	Acceptance Certificate (PJM.CR.080)

Table 1-11 Task Steps for Develop Current Process Model

Approach and Techniques

Prepare the Current Process Model based on information collected during Conduct Current Business Baseline (RD.020). Benefits of this exercise include the opportunity to recognize potential process improvements by capitalizing on the new application functionality and the ability to establish a current business benchmark against which to gauge the ultimate success of the new business model.

Process Flow Diagramming

Process flow diagramming supports the analysis of current business processes by:

- clearly communicating the current business processes
- allowing analysts to document the current business in a structured way
- facilitating business requirements mapping to the applications

Develop process flows for the current business system, not the current set of applications (commonly called "the system") used to run the business. The current business "System" (big "S") represents how a business runs, the formal and informal systems, processes, and procedures that are in place. The set of applications or "system" (little "s") are merely the tools/formal mechanism where data are transformed into useful bits used for making business decisions and providing information.

Current business process models may be created in several ways. If the organization has existing process models they developed internally, you may be able to use them. Review them for suitability without modification, or possible further development. If no process models exist, consider starting with generic business flows and tailoring them to your business, or build them from scratch. The important thing to remember is that they represent the way the business processes are running today. Form is less important than substance.

Process Modeling Using Oracle Business Models

Use solution sets and process models for Oracle Applications, such as Oracle Business Models (OBM), as a focal point to stimulate discussion of the organization's current business processes. Modify them as required to reflect the organization's view of their current business and their operations.

The models modified from those in OBM will show the differences between the organization's current processes and the standard processes that Oracle Applications support.

There are two recommended approaches to modifying Oracle Business Models to model current organization processes. When working with client staff members in work sessions, you should produce hard copies of the relevant models, enlarged if possible. Then, working with these printed models, the team can modify them to reflect the actual processes. In general, it is recommended that the new process flows and process steps are clearly marked with red, and that deleted (or moved) flows and steps are clearly marked with blue or green notations.

The second approach works directly on the models in OBM, in either their Visio or Oracle Designer format. In this situation, an experienced process modeler extracts information about the organization's current processes and translates them directly into modifications to the standard models in OBM.

It is easy to combine these two approaches. Following a workshop that produces modified paper versions of the models, the process modeler uses these paper versions as a basis for updating the software versions of the models, as described above.

Review and Signoff

As each process team completes its current business baseline, you should review it with the project team, area users, management, and any cross-process integration teams.

Linkage to Other Tasks

The Current Process Model is an input to the following tasks:

- BP.080 - Develop Future Process Model
- RD.030 - Establish Process and Mapping Summary
- RD.050 - Gather Business Requirements
- RD.070 - Identify Business Availability Requirements
- DO.030 - Prepare Glossary
- DO.070 - Publish User Guide
- AP.060 - Develop Business Unit Managers' Readiness Plan
- AP.100 - Identify Business Process Impact on Organization

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	100
Business Line Manager	*
Project Sponsor	*
User	*

Table 1-12 Role Contribution for Develop Current Process Model

* Participation level not estimated.

Deliverable Guidelines

Use the Current Process Model to reflect a common understanding of your existing business processes. The objective is to gain an understanding of the integrated current business requirements.

This deliverable is an important input for developing the Future Process Model (BP.080), which in turn becomes the basis for mapping process scenarios to the application.

This deliverable should address the following:

- business events to which the organization responds
- high-level model of enterprise processes
- detailed process flow diagrams of current business processes
- detailed listing of process steps performance measures

Deliverable Components

The Current Process Model consists of the following components:

- Enterprise Process Model
- Core Process Models
- Process Flow Diagrams
- Process Step Catalog
- Event Catalog
- Performance Measures

Enterprise Process Model

This component prepares a high-level context diagram showing the major business process areas or core processes and the information flows between them. A core process is a process essential to the running of the organization and provides added value to the output produced (such as procurement or order fulfillment).

If you have carried out the Define Business and Process Strategy (BP.010) task, then you have already produced a high-level enterprise model for the organization's current processes. The process flow diagram from that model can be reused for this component.

Core Process Models

This component constructs a process flow model showing the activities in the process for each core process, as identified in the enterprise model. Each core process should be at a lower level of detail, generally corresponding to levels 2 and 3 of the Oracle Business Models.

Process Flow Diagrams

This component uses the application-supported process flows provided in OBM (or similar process modeling tools for Oracle Applications) as a starting point for creating the current process flow diagrams. These flows document processes directly supported by the Oracle Applications, but only show the process steps performed on the applications. These process steps can be modified by adding the process steps required by the organization to support its unique processes. In many cases, one application process flow will be the basis for several business process models. For example, when events and responses differ, a planning flow may need to vary by type of product family or subprocess.

Process Step Catalog

This component lists the process steps that make up the process. Each step should have a brief, descriptive title and represent an elementary business function. The process step catalog should also record the agent responsible for the execution of each step, and classify each step by desired degree of automation: manual, computer assisted, or fully automated. If desired, you can use procedure documentation software such as Oracle Tutor to assist with this component.

Event Catalog

This component lists all of the events the business responds to; each event has a name, a brief description, frequency, and conditions under which it occurs. These may be external events (a customer order), internal events (the completion of production), or temporal events (end of month).

Performance Measures

This component lists the performance measures used to measure current process performance for each core process.

Tools

Deliverable Template

Use the Current Process Model template to create the deliverable for this task.

Oracle Tutor

Development of the Process Step Catalog component, can be done with the help of Oracle Tutor, which includes a publishing tool and a library of generic procedures.

BP.050 - Review Leading Practices (Optional)

In this task, you review which leading practices are relevant and appropriate for the organization.

Δ If your project includes *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **Leading Practices Review**. It is a review of leading practices and best uses of the application relevant to the organization's processes.

Prerequisites

You need the following input for this task:

- ❑ **Business and Process Strategy (BP.010)**

The Business and Process Strategy defines the process areas relevant to the project. It also includes information relevant to the organization's priorities that is needed to evaluate leading practices. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review solution sets and process models for Oracle Applications relevant to the organization's business strategy.	
2.	Review existing leading practice material after literature searches and other research.	Introduction
3.	Determine whether additional research is required and collect missing leading practice information.	
4.	Hold a workshop to review the results and to determine relevance to the project.	Leading Practices Conclusion

Table 1-13 Task Steps for Review Leading Practices

Approach and Techniques

The Leading Practices Review provides examples of what is possible within the application. Leading practices are reflected in solution sets and process models for Oracle Applications, such as Oracle Business Models (OBM), and contain the following:

- solution sets, representing leading practices
- process models, representing best use of Oracle Applications for a complete range of processes for the majority of organizations

Using the models of leading practice and best use of Oracle Applications as a guide, the organization determines the processes and practices most appropriate to the goals in implementing new applications.

In almost all cases, a work session format is appropriate for this task. This allows process and other experts from the business to work with an application specialist and business analysts to determine which leading practices are relevant and appropriate to the application and its implementation.

Linkage to Other Tasks

The Leading Practices Review is an input to the following task:

- BP.060 - Develop High-Level Process Vision

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	40
Subject Matter Specialist	40
Application Specialist	20
Business Line Manager	*

Table 1-14 Role Contribution for Review Leading Practices

* Participation level not estimated.

Deliverable Guidelines

Use the Leading Practices Review as an input to determine the visioning of the organization's business processes.

This deliverable should address the following:

- relevant solution sets or process models reflecting the leading practice or best use of Oracle Applications
- relevant leading practice process designs from other sources
- key design features and performance characteristics of solutions
- leverage points that have enabled other organizations to align culture and beliefs with business objectives to produce successful change

The illustration below provides an example of a leading practice description for Accounts Payable.

Accounts Payable - Leading Practice Descriptions **EDI / Financial EDI** Establish Electronic Data Interchange (EDI) linkages with suppliers for order placement, invoicing, and payment. Utilizing EDI linkages reduces the paperwork involved in processing vendor payment requests and enhances the level of invoice details kept in the system. Combined with Financial EDI, which enables companies to exchange payment and payment-related remittance information and online approval of invoices, EDI significantly reduces the time and workload involved in the payment cycle, resulting in lower transaction cost. It also achieves the organization's requirement for accurate and timely information for cash outflow analysis, budget and project analysis, and inventory value. **EFT/Wire Transfers** Electronic Fund Transfers (EFT) and wire transfers automate the funds exchange portion of the payment cycle. They are suitable for making payments that do not require remittance information. EFT and wire transfers also minimize the number of checks generated, thus reducing the amount of check handling. This leads to increased timeliness in the overall payment cycle. **Electronic Filing of Travel and Entertainment (T&E) Reports** Instituting electronic filing of expense reports expedites the expense reporting process. Benefits will be realized in combining the preparation of expense reports and the recording of expenses and their reimbursements into one step, thus reducing the work load of the Accounts Payable Expense Processing group. **Corporate Cards for Travel and Entertainment (T&E) Charges** Using corporate credit cards to pay for travel-related expenses minimizes the need to issue cash advances, resulting in a reduction of processing volume. Travel data is also available from most credit card companies that allow companies to perform analysis that may lead to various organization initiatives. These initiatives may result in a reduction of travel expenses.

Figure 1–4 Leading Practice Example

Deliverable Components

The Leading Practices Review consists of the following components:

- Introduction
- Leading Practices Conclusion

Introduction

This component documents steps in the leading practice review and identifies which leading practices have been reviewed.

Leading Practices Conclusion

This component documents decisions, made during workshops, regarding which relevant leading practices the organization wants to adopt.

Tools

Deliverable Template

Use the Leading Practice Review template to create the deliverable for this task.

BP.060 - Develop High-Level Process Vision (Optional)

In this task, you provide a general, high-level statement for each business process describing how the new processes will operate. The statement determines any relevant principles that will regulate the operation of the new processes.

Δ If your project includes *process change*, you should perform this task.

Deliverable

The deliverable for this task is the **High-Level Process Vision**. It is a general, high-level statement of how the new processes will operate to meet the organization's objectives. The High-Level Process Vision identifies the major influences on the organization's future and the opportunities that will be opened by implementing the applications, and also documents the challenges to be addressed by the vision.

Prerequisites

You need the following input for this task:

- ❑ **Business and Process Strategy (BP.010)**

The Business and Process Strategy provide a starting point for constructing a vision of the future. If Define Business and Process Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

- ❑ **Leading Practices Review (BP.050)**

The Leading Practices Review provides insight into the way the "best of class" organizations carry out their business, and can often provide a model for the organization. If Review Leading Practices was not performed, this deliverable will not exist. (See the task description for BP.050 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Conduct work sessions to identify high-level opportunities and challenges.	SWOT Analysis
2.	Develop final high-level opportunities and challenges list.	SWOT Analysis
3.	Develop, finalize, and document process vision.	Process Vision
4.	Secure acceptance of the High-Level Process Vision.	Acceptance Certificate (PJM.CR.080)
5.	Summarize the results of this visioning process.	Introduction

Table 1-15 Task Steps for Develop High-Level Process Vision

Approach and Techniques

The High-Level Process Vision is determined through a short series of high-level work sessions with senior staff. As part of visioning, the organization and project team work together to identify the major influences on the organization's future and opportunities. Generally, the sequence is to identify opportunities and challenges, followed by development of the process vision. Develop the informal process vision during the work sessions. Later formalize, document, and agree on the process vision with the organization's project sponsors.

At the end of this task, the project has a clear definition of where the organization intends to be in the future, and how the use of the selected application will fit into the new organization.

Linkage to Other Tasks

The High-Level Vision is an input to the following tasks:

- BP.070 - Develop High-Level Process Designs
- BP.080 - Develop Future Process Model
- AP.070 - Develop Project Readiness Roadmap
- AP.180 - Conduct Effectiveness Assessment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	70
Subject Matter Specialist	30
Business Line Manager	*
Key User	*

Table 1-16 Role Contribution for Develop High-Level Process Vision

* Participation level not estimated.

Deliverable Guidelines

Use the High-Level Process Vision to define a high-level statement of how the new processes will operate.

This deliverable should address the following:

- SWOT analysis (strengths, weaknesses, opportunities, and threats)
- key challenges (classified by process)
- major opportunities for process change
- written statements summarizing the vision of the future for each process

- high-level process maps to accompany the vision statements
- performance measures and stretch goals identified to test the vision when it becomes a reality

An example of key challenges and major opportunities is shown below.

Accounts Payable — Summary of Key Challenges and Major Opportunities

The review of the *As Is* activity flows for Accounts Payable resulted in the identification of several key challenges and major opportunities that are summarized here and which will later be resolved or addressed in the *To-Be* environment.

- Capture an adequate level of invoice detail to do proper analysis.
- Reduce paper and manual effort by utilizing EDI and EFT.
- Process treasury wire transfers through the consolidated system.
- Improve the invoice approval process by using online approval.
- Automate the netting of employee advances against expense reimbursements.
- Streamline the filing of employee expenses.
- Ease the preparation of 1099 forms and filing.
- Clean up the vendor file.
- Eliminate reconciliation errors due to manual checks.
- Automate the collection of credits across divisions.
- Automate the preparation of dunning letters.
- Enhance the up-front duplicate payment checking capability.
- Automate setting beginning check numbers and due dates to produce checks.

Figure 1–5 Key Challenges and Major Opportunities

An example of part of a process vision, along with proposed process measures and future goals follows:

Accounts Payable—Process Vision			
Current Process	**New Process**	**Process Measures**	**Targets**
Currently 95% of the input into the Accounts Payable system is manual data entry from paper invoices. Coding and approval of a typical invoice are accomplished via manual routing and processing which takes approximately 4 - 18 days. Exception handling may add another 2 - 6 days.	Set up Electronic Data Interchange (EDI) relationships to receive invoices electronically from most vendors. This will reduce the amount of manual data entry. Working with Harbinger EDI will allow even small vendors access to EDI technology.	Percentage of suppliers covered electronically Accuracy of invoice information Cycle time (standard) Cycle time (exceptions)	Set up EDI relationships to receive invoices electronically with more than 90 percent of vendors Fewer than 5 percent of invoices rejected because of incorrect or insufficient information Reduce standard cycle time to 4 hours Reduce median exception cycle time to 2 working days
	EDI transmissions should include all identification codes needed for internal processing. When used, blanket purchase orders or procurement cards will help purchasers provide the coding to the vendor at the time of purchase.	EDI transmissions without full identification codes (and thus requiring manual handling)	More than 99.5 percent of transmissions will have full identification codes
	Recurring EDI invoices will have coding and approval set up once. Only invoices out of tolerance will need to be handled before payment. Examples of this type include utility or telephone invoices.	Setup time Percentage of invoices out of tolerance and needing to be handled before payment	Median setup time under 5 minutes Fewer than 5percent of invoices out of tolerance
	Nonrecurring EDI invoices will be imported directly into the Accounts Payable system and then be coded (if necessary) and approved online, eliminating the need for manual routing.	Percentage of nonrecurring EDI invoices routed manually	Fewer than 5percent of invoices routed manually

Figure 1–6 Process Vision Statement

Deliverable Components

The High-Level Process Vision consists of the following components:

- Introduction
- SWOT Analysis
- Process Vision

Introduction

This component contains relevant summary information, such as the organization’s strategy, performance measures, current process model, and leading practices.

SWOT Analysis

This component contains the strengths, weaknesses, opportunities, and threats (SWOT) analysis that can be carried out during a short work session or assigned to a team for work outside the work session. This analysis is used as a starting point to identify major challenges and opportunities. A series of interviews is carried out to establish an initial list. After analysis and classification by the facilitation team, use a workshop to discuss the issues in depth and develop a final list.

Process Vision

This component supports the work sessions that focus on the process vision of the organization as it will exist in the future after the selected applications are implemented as part of the daily business. Consider the following:

- product or service offerings
- core competencies and skills sets
- competitive positioning
- growth rates
- innovations required
- enablers
- major cost savings
- major improvements in market share and / or margins

Tools

Deliverable Template

Use the High-Level Process Vision template to create the deliverable for this task.

BP.070 - Develop High-Level Process Designs (Core)

In this task, you produce high-level designs documenting how the new organization will operate after the applications are implemented.

Deliverable

The deliverable for this task is the **High-Level Process Designs**. It includes a generic process design, relevant process descriptions, and generic technology design.

Prerequisites

You need the following input for this task:

- ❑ **Business and Process Strategy (BP.010)**

The Business and Process Strategy includes an initial assessment of what will be changed. It also identifies process clusters, and performance data on existing business processes. If Define Business and Processes Strategy was not performed, this deliverable will not exist. (See the task description for BP.010 for more information on when this task should be performed.)

- ❑ **High-Level Process Vision (BP.060)**

If your project includes *process change*, you should derive the initial design from the High-Level Process Vision. The vision describes the objectives and goals of the new processes, but generally does not describe how the processes are to be carried out, which is the purpose of this task. If the Develop High-Level Process Vision was not performed, this deliverable will not exist. (See the task description for BP.060 for more information on when this task should be performed.)

- ❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides a clear understanding of the current organization structure.

❑ **Executive Project Strategy (AP.010)**

The Executive Project Strategy describes the general direction of the project. If Define Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for more information on when this task should be performed.)

❑ **Return on Investment (ROI) Analysis**

A return on investment analysis, such as Oracle's Solution Value Assessment (SVA), can document the business case and the return on investment the executive management team can realistically expect.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
Δ 1.	Review the High-Level Process Vision - BP.060 (if your project includes process change).	
2.	Analyze Oracle Business Models (OBM) process models or equivalent, to determine changes required to represent the High-Level Process Designs.	
3.	Produce High-Level Process Designs at the highest level of detail.	Generic Process Designs
4.	Produce High-Level Process Designs at the lower levels of detail.	Generic Process Designs

Table 1-17 Task Steps for Develop High-Level Process Designs

Approach and Techniques

The High-Level Process Designs take into account the costs, benefits, and risks of changing the application and its processes.

The High-Level Process Designs are done *generically* across the whole of the organization. Designs for variants, such as business units or departments, are produced later in Develop Future Process Model (BP.080).

Generally, a separate high-level design is produced for each major business process in the project. Subsequently, the design is revised iteratively as a result of Analyze High-Level Gaps (BR.010). The generic process model explains in high-level terms how work will flow in the future and who will be responsible for completing the activities. It is composed of the following components:

- **Event Catalog** — Lists all events to which the business responds; each event has a name, a brief description, frequency, and the conditions under which it occurs. There may be new events introduced by the new applications.
- **Process Descriptions** — Provides an introductory, textual description of the process that is executed in response to each event, together with an identification of the process' main inputs and outputs.
- **Process Step Catalog** — Lists the process steps that make up the process. Each step should have a brief, descriptive title. An objective of this task is to confirm that the analysis is sufficiently detailed to provide a basis for more detailed design later in the project. The agent responsible for the execution of each step is also recorded. Each step should also be classified by degree of automation: manual, computer assisted, or fully automated.
- **Process Flow Diagram** — Provides a description of the high-level design in graphical form; normally a modification of a process model from Oracle Business Models or equivalent.

Process descriptions are high-level descriptions of the operations that will be carried out using the generic processes described above. They focus on the movement of information through the organization.

Linkage to Other Tasks

The High-Level Process Designs are an input to the following tasks:

- BP.080 - Develop Future Process Model
- BR.010 - Analyze High-Level Gaps
- AP.080 - Develop and Execute Communication Campaign

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Process Modeler	60
System Architect	40
Business Line Manager	*

Table 1-18 Role Contribution for Develop High-Level Process Designs

* Participation level not estimated.

Deliverable Guidelines

Use the High-Level Process Designs to illustrate how the new organization will operate after the application are implemented.

This deliverable should address the following:

- generic process designs
- written description of how the new processes will operate

Deliverable Components

The High-Level Process Designs consist of the following component:

- Generic Process Designs

Generic Process Designs

This component contains the generic process designs. If the Oracle Business Models (OBM) are being used as a basis for creation of the design, produce the process designs by modifying the models. Since the design is only intended to show high-level activities and interactions, you use levels 1 - 3 of OBM.

Tools

Deliverable Template

Use the High-Level Process Designs template to create the deliverable for this task.

BP.080 - Develop Future Process Model (Core)

In this task, you define the future business model in the form of integrated process flows built on the business processes supported by the new applications.

Δ If your project includes *process change,* you should perform the additional task steps indicated below.

Deliverable

The deliverable for this task is the **Future Process Model**. It includes process flow diagrams of the events and business processes that the applications and the associated functions of the business area will support.

Prerequisites

You need the following input for this task:

- ❑ **Current Process Model (BP.040)**

The team can use material from the Current Process Model to gain an understanding of the processes and systems that support the current business. It may identify processes that offer opportunities for process improvement in cycle time, cost, and quality. It may also show major processes that are candidates for process redesign. If Develop Current Process Model was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

- ❑ **High-Level Process Vision (BP.060)**

When a process change approach is followed, a visioning task will be carried out before this task. This is usually done in the form of work sessions. The output from these work sessions is recorded as the vision of the future, which was converted into the High-Level Process Designs (BP.070) of the previous task and will be converted into the Future Process Model in this task. If Develop High-Level Process Vision was not performed, this deliverable will not exist. (See the task description for BP.060 for more information on when this task should be performed.)

- ❑ **High-Level Process Designs (BP.070)**

The High-Level Process Designs are used to illustrate how the high-level vision of business process will be put into practice through a series of process steps and flows.

- ❑ **Business Requirements Scenarios (RD.050)**

The Business Requirements Scenarios provide a formal statement of the detailed business requirements for the business processes.

- ❑ **Skilled Project Team (AP.050)**

The project team members must understand application concepts and capabilities in order to develop process models based on leading practices. This prerequisite helps them understand what process steps are supplied by application-specific functions.

Optional

You may need the following input for this task:

- ❑ **Business Process Reengineering (BPR) Studies**

Business Process Reengineering Studies are only appropriate if process change is relevant, and earlier business process change tasks were not carried out — most significantly, Define Business and Process Strategy (BP.010), Catalog and Analyze Potential Changes (BP.020), and Develop High-Level Process Vision (BP.060). Process reengineering studies may recommend the complete overhaul of current processes to be supported by the new applications. The output of a BPR study may, in fact, provide most of the content of the Future Process Model.

- ❑ **Future Business Strategy**

This prerequisite includes other internal business documents that describe the future business processes. For example, the original request for quote and management business strategy documents on the manufacturing, distribution, financial, and administrative processes may be useful during this task.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review any documented future business requirements.	
2.	Identify and describe the events to which the business responds.	Event Catalog
3.	List each process and write a summary description of each, identifying the event to which it responds and its main inputs and outputs.	Process Listing and Process Descriptions
4.	Construct the top level of the hierarchy from information provided by interviews with senior management and Current Business Baseline (RD.020) information.	Function Hierarchy
5.	Use the generic high-level process models created in Develop High-Level Process Designs (BP.070) as a starting point for the more detailed process models produced by this task. Identify the steps that make up each process, their sequence, any conditions that determine alternative execution paths, and the agent responsible for each step; validate that each step maps to an elementary business function.	

No.	Task Step	Deliverable Component
6.	Identify the steps that make up each process, their sequence, any conditions that determine alternative execution paths, and the agent responsible for each step; validate that each step maps to an elementary business function.	Process Step Catalogs
7.	Construct process flow diagrams for processes with more than two steps or with conditional steps showing the sequence of process steps and the flows between them. Show conditional steps where appropriate.	Process Flow Diagrams
Δ 8.	Break down the detailed Future Process Model into a list of the steps carried out by all participants in the process (if your project includes process change).	To-Be Activities
Δ 9.	Translate the to-be process design into the functions of the Oracle Application, and determine high-level changes (if your project includes process change).	
10.	Construct the intermediate and lower levels from application reference material and other required business functions.	Function Hierarchy

No.	Task Step	Deliverable Component
11.	Review the Future Process Model with users and management.	
12.	Secure approval of project and business line management.	Acceptance Certificate (PJM.CR.080)

Table 1-19 Task Steps for Develop Future Process Model

Approach and Techniques

The Future Process Model identifies the complete set of events to which the business function responds in order to meet its objectives, describes the future processes that the business performs in order to respond to each of the events, and identifies each of the steps in those processes.

Definition of Process Modeling of Future Business Processes

Process modeling of the future business processes results in a common and comprehensive picture of the project's scope of relevant processes and, if desired, expected metrics and costs.

Processes are Driven by Business Objectives

Process modeling needs to capture those processes that are the essence of the mission statement for the business. Often the design and execution of the business processes are the mechanism by which a competitive advantage or parity is achieved. In fact, it is possible that the mission statement embodies the reason for selecting this application system in the first place.

Advantage of Process Modeling

The advantage of process modeling is that business processes are real to users in a way that entities and functions are not. Users are the ones who actually perform business processes. Such processes represent the work of the organization and show how information is used. The outputs they produce are tangible and essential to the organization's mission. In fact, one of the primary means for measuring the success of a process model is that it is intelligible to the users who perform the process. By linking business requirements to modeled processes, the quality of defined requirements will be better than informal wish lists or nonintegrated requirements listed by functional area.

Business Events Identification

Use individual interviews or working sessions with the focus groups to identify business events. Ask staff members to define all the events to which they must respond in the future. Client staff members should identify and list each event, and describe each one in business (not system) terms. Each event should be identified as one of the following:

- external event (initiated outside the business area)
- temporal event (initiated at a point in time)
- internal event (initiated inside the business area)

Ask questions regarding the fulfillment of customer (internal or external) requests to build a complete list of events. Then capture high-level descriptive information about the event, such as its frequency and response requirements.

Finally, work with the business staff to produce a summary description of the process that responds to the event and to identify its main inputs and outputs. The following examples of events from a customer information system for a manufacturing concern show the event identifier and event description:

- OE-10 Customer requests change of billing address
- OE-23 Customer orders inventory item
- OE-17 Customer orders warranty contracts
- OE-30 Customer requests credit approval

Assign ownership of each event to a functional area. You can base ownership on who performs the majority of process steps in the process or on the functional area that initiates the business process responding to the event. Recognize that events create responses by processes that may be self-contained within one functional area or which may cross functional boundaries.

It may be difficult to identify events or to distinguish an event from an event mechanism. For example, receiving internal information is not necessarily an event, but is simply the natural flow of information between process steps in a process. If you are unsure whether something is an event, determine if it would still be an event if the roles of sender and receiver were performed by the same person.

Process and Function Levels

It is important to understand the concept of levels, and the differentiation between high-level and low-level processes, when modeling business processes. These terms appear throughout the following discussion.

In defining business functions, the level relates directly to the diagrammatic representation of functions in a hierarchy. A function hierarchy is a grouping of business functions into one or more hierarchical levels. Typically, the highest level corresponds to the organization, the middle level corresponds to the grouping of application functions, and the lowest level corresponds to elementary business functions. Functions near the top of the hierarchy are referred to as high-level functions, while functions near the bottom are called low-level functions. Sometimes the term *summary level* is used synonymously with high level.

Because a functional hierarchy is a classification scheme, the concept of level carries the connotation of how broadly or narrowly defined a function is. Broadly defined functions exist at the top of the hierarchy and include many different (although related) activities within them. Human Resources, for example, may include the functions *benefits*, *compensation*, *employee relations*, *hiring*, and so on.

High-level functions allow you to refer to an extensive group of related activities in a summary way, such as Human Resources. Thus, the concept of level also implies the amount of detail that is *visible on the function hierarchy*. The function Human Resources does not give you a direct view of all of the details involved in this function. To access this detail you must navigate down the functional hierarchy. The further down the hierarchy, the more narrowly the function is defined, and the more detail is included in the definition. Elementary business functions represent the most narrowly defined functions in the functional hierarchy and include the most detail in their definition.

The key concepts in process modeling — business process, process step, event, and agent — all have a *leveling* dimension similar to functions in functional decomposition. A critical difference, however, is that in process modeling you are not attempting decomposition. In process modeling the analyst is free to choose the level at which each object (process, process step, event, and agent) is defined. This freedom extends to the use of objects at any level of detail in the same process flow diagram.

These concepts are summarized in the figure below:

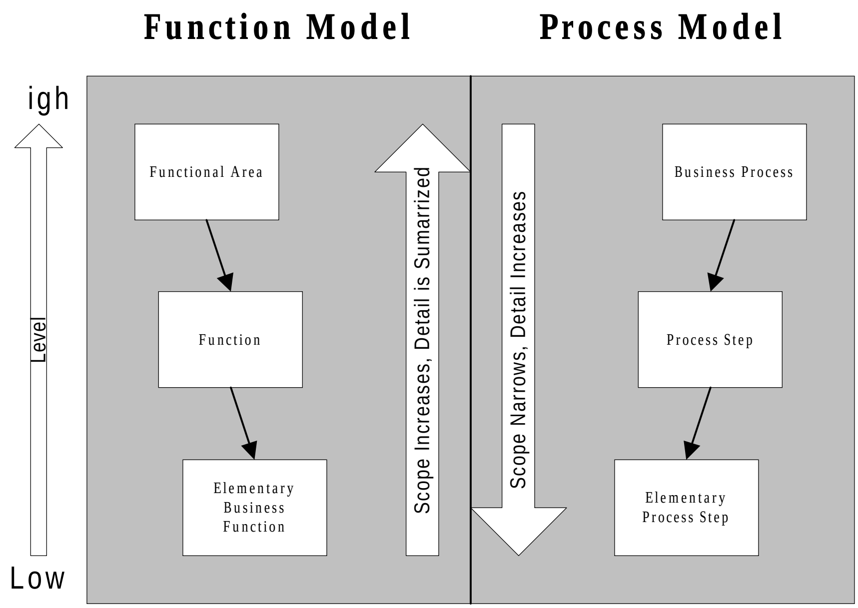


Figure 1–7 Concept of Level in Process and Function Modeling

From the perspective of the applications, the functional model usually has only three conceptual levels: the first level, or top, is an organizational one; the second is an application-related function; the third is the elementary business function.

Functions and Processes

Functions are defined as something an organization must do to achieve its objectives. This definition does not carry with it the same constraints as the definition for process — a planned response to a business event consisting of a sequence of operational steps that produces a product or delivers a service. Functions are identified and defined without the criteria that are inherent in the definition of processes.

At the highest level in a functional decomposition, functions are groupings of related activities — *sales, marketing, manufacturing,* *distribution*, and so on. High-level functions include processes, but from a modeling perspective, you should not attempt to achieve a one-to-one correspondence in a functional hierarchy between functions and processes at the top levels. In fact, a very important advantage that process modeling brings to the table is a cross-functional perspective. Process modeling allows you to understand business activity as it crosses functional boundaries. It accomplishes this by highlighting interactions between process steps from different higher-level functions.

When processes are fully specified, the process steps are defined at the elementary business function level. At this level there is *necessarily* a one-to-one correspondence between the elementary business functions in the function model and the elementary process steps in the process model. As a quality assurance step, you should make sure that each elementary process step equates to an elementary business function in the function model and that each elementary business function, either appears in the process model, or is a single-step process in itself.

The process model designates the flow of all work through the business area. In order to do this, it should indicate all elementary process steps that form part of multi-step processes. The function hierarchy, on the other hand, should include those functions, that are supported by the application system, manual steps, as well as single steps and maintenance steps.

The table below designates types of elementary process steps and elementary business functions and the models in which they are included.

EPS/EBF Type	Process Model	Function Model
system-assisted	**include**	**include**
manual	**include**	**include**
single-step / standalone	exclude	**include**
maintenance	exclude	**include**
external	**include**	exclude

Table 1-20 Elementary Business Functions and Process Steps by Model

Business Processes Specification

The goal of this step is to specify the full business response to each event, identifying each step in the process, the sequence of all steps, and the conditional flow.

Perform this step by conducting workshops or interviews with personnel who are representative of those actually responsible for executing the process.

Working on a suitable medium such as a whiteboard, the process modelers and the client staff members who perform the process should create a process flow diagram to represent the response. This diagram should depict the events, the steps and their sequence, and all related decisions made during the process.

The process steps that make up the process flow will be built around the function hierarchy that is supported by the applications. Insert additional steps into the hierarchy and the process flows to supplement application-supported process steps as required by the unique responses in the organization.

The function performed by each step should be at the level of elementary business function and will match the elementary functions shown on the function hierarchy. For each step, get members of the organization's staff to specify who will be responsible and to identify the inputs and outputs.

Use these functions to provide input to the identification of entities and attributes. In addition, determine the degree of automation required for each step and specify whether it is to be performed manually, with computer assistance, or entirely automated through the system.

Process Modeling Using Application Functions

Construct future processes by combining system-assisted process steps and manual steps. The diagram may further divide the system-assisted steps into user-executed process steps and those done automatically by the applications. An example of a user-driven step is the creation of a work order; an example of a system step is the execution of a concurrent job.

As you construct processes, application consultants and analysts identify process steps that can be supported by the target applications. The application process steps for a given process usually reside within the same applications (intra-application process steps), but process steps can also reside within other applications.

Client staff members add manual steps to the system-assisted process steps as necessary.

Linkage with Current Process Model

It is important to recognize that there may be significant differences between the depth, robustness, and content of current process modeling across different projects, depending on each situation. One organization may choose not to create current process models , another may choose to do only a very high-level version, while a third may implement the rigorous process modeling method described, both for its current and its future process model.

The most important point to recognize is that future processes must cover the minimum business requirements that current processes cover, to supply the information required to achieve the business objectives and, possibly, to yield business improvements in cost, time, and quality. Construct the future processes to respond to the events that become approved business drivers. They will consist of process steps, which may be manual or automated. In the Future Process Model evaluate each process step conducted by the current business system to confirm that it has been addressed, either by a manual or an application step, or that it is no longer required.

Δ Process Change

When *process change* is applicable, create the Future Process Model in two stages. Initially, create the Future Process Model at a high level, corresponding to OBM levels 1 to 3. This corresponds to a generic process model, applicable to the whole organization. After that model is accepted, additional detail is added to the model, corresponding to OBM levels 4 and 5 that are contained in Oracle Tutor. This corresponds to variations to the generic process model needed to cater to the needs of individual business units.

An additional deliverable component, the description of the to-be activities, is produced, following production of the detailed Future Process Model. This component breaks down the detailed Future Process Model into a list of the activities carried out by all participants in the process. It corresponds to the activity descriptions provided by the Oracle Tutor (level 4 - 5) element of the OBM.

Linkage with Business Requirements Scenarios

Gather Business Requirements (RD.050) will feed the future process model in several iterations. First, the Business Requirements Scenarios (BRS) are used to create the initial process model. Next, the initial process model feeds into the creation of more detailed Business Requirements Scenarios. Finally, the Business Requirement Scenarios feed into the creation of the final Future Process Model.

A Business Requirements Scenario (BRS) is a formal statement of the detailed requirements for a business process, the source of those requirements, and how those requirements will be satisfied (for example, by the application, manual process steps, a workaround, or other application systems). In a sense, it is a recipe for a process step — specifically how to execute it. Since the application-supported process steps map directly to application functions, the BRS then becomes a major input to the task of Map Business Requirements (BR.030).

Full Event and Process Coverage

Evaluate the entire catalog of events to confirm that associated processes fully capture the entire system’s response, that all processes are comprehensively portrayed, and that there are no disconnects when one process stops and another continues. One way to address the last point is through an integration team that has this evaluation as one of its responsibilities.

Process Analysis with Business Process Teams

Process analysis is a logical technique used to drive the task of defining the future business model. The recommended approach is to hold a series of working sessions in which the teams of analysts and business staff work together to develop a business process model for the processes that represent the scope of the implementation. During this time, these teams perform the following functions:

- Participate in a series of working sessions to construct the future processes.
- Identify the process steps for each process.
- Capture the process flow using flowcharting software.
- Hold feedback and review sessions with appropriate business area management.

- Iteratively refine the model throughout process design activities.

The primary technique for obtaining and validating the information in the model is through these interview, working, feedback, and review sessions.

Rapid Implementation Techniques

Implementations projects can derive the future process model faster by following these guidelines:

- Use an application references model such as Oracle Business Models as a target.
- Confirm that process is suitable.
- Create future process designs on an exemption basis.
- Focus on high-impact areas.
- Minimize the time spent on optimizing business processes.

Changes in Process Recognition

Since the solution has yet to be defined, it is too early to get complete commitment from the users. Therefore, you should obtain an agreement from users on overall direction. Users should agree with the overall concept and framework of expected solutions. The flows should also represent some obvious benefits to users, such as a reduction in material or information handling. Such a reduction implies reduced cycle times and less potential for communication errors. In the Future Process Model, highlight the flows that provide more direct approaches than those in the current process.

Differences Between the Current and Future Models

Some project teams may want to see the differences between the current and future models quantified. If this is the case, it will be helpful to capture these changes in Mapped Business Requirements (BR.030), although it is also possible to summarize them in the mapping scenario listing in the Process and Mapping Summary (RD.030).

Business Procedure Documentation

As you are starting to produce the various catalogs of events, process steps and so on, start capturing the information in your Business Procedure Documentation (BP.090).

Staffing

Depending on the number and complexity of the business processes, it may be necessary to divide the task workload among multiple teams. However, make every effort to minimize the number of teams and business analysts involved in order to reduce communication overhead and improve efficiency. Ideally, these teams should have one or more business analysts to work with the appropriate business representatives. Assign each team to a set of closely related processes, from which to develop a model. When deploying multiple teams, require that they coordinate their models. Under no circumstances should teams work in isolation, or they may produce inconsistent models, subsequently requiring considerable effort to integrate.

When managing the construction of future business models, move as rapidly as possible through this task. Because there will be many resources engaged in this task, watch for and avoid the decision paralysis that can lead to scope and time creep. To meet the scheduled completion dates, the project manager and team members must see that decisions are made, issues resolved, and consistent progress achieved. Do not hesitate to elevate delays, increase the urgency, and demand date accountability from the process modeling teams.

Linkage to Other Tasks

The Future Process Model is an input to the following tasks:

- BP.090 - Document Business Procedures
- RD.030 - Establish Process and Mapping Summary
- RD.040 - Gather Business Volumes and Metrics
- RD.060 - Determine Audit and Control Requirements
- RD.070 - Identify Business Availability Requirements
- BR.020 - Prepare Mapping Environment
- BR.030 - Map Business Requirements

- BR.060 - Create Information Model
- BR.090 - Confirm Integrated Business Solutions
- AP.090 - Develop Managers' Readiness Plan
- AP.100 - Identify Business Process Impact on Organization

Role Contribution

Core Task Steps

The percentage of total task time required for each role follows:

Role	%
Business Analyst	40
Process Modeler	30
Application Specialist	20
Tool Specialist	10
Business Line Manager	*
Project Sponsor	*
User	*

Table 1-21 Role Contribution for Core Task Steps of Develop Future Process Model

* Participation level not estimated.

Optional Task Steps

If your project is addressing the optional task steps for this task, you need to allocate additional time and the following resources:

Role	%
Business Analyst	40
Process Modeler	30
Tool Specialist	10
Application Specialist	20
Business Line Manager	*
Project Sponsor	*
User	*

Table 1-22 Role Contribution for Optional Task Steps of Develop Future Process Model

Deliverable Guidelines

Use the Future Process Model to define a model of the new business system (after conducting current business baseline activities).

This deliverable should address the following:

- what events the business responds to
- what processes make up the response
- what process steps make up each process
- how business decisions are supported by the new system
- how processes will be streamlined
- how desired process changes can be accommodated
- how resources will interact in the new system

- what necessary control elements (approval steps) should be established
- how process and performance will be communicated to management for improvement

Deliverable Components

The Future Process Model consists of the following components:

- Event Catalog
- Process Listing and Process Descriptions
- Function Hierarchy
- Process Step Catalogs
- High-Level Designs
- Process Flow Diagrams
- To-Be Activities

Event Catalog

This component lists all events to which the business responds; each event has a name, a brief description, frequency, and conditions under which it occurs. These may be external events (a customer order), internal events (the completion of production), or temporal events (end of month). You can bring many events forward from the Current Process Model (BP.040), but the new applications may also introduce new events.

Process Listing and Process Descriptions

This component provides a textual description of the process that is executed in response to each event, along with an identification of the main inputs and outputs of the process.

Function Hierarchy

This component contains the following information:

- **Function** — a brief description of the function, or its short name
- **Function Owner** — the agent with overall responsibility for the function

- **Date** — date form is being initiated
- **Function Number** — the code that uniquely identifies each business function
- **Function Description** — a more complete description of the function

Process Step Catalogs

This component lists the process steps that make up the process. Each step should have a brief, descriptive title and represent an elementary business function. You also record the agent responsible for the execution of each step, and classify each step by desired degree of automation: manual, computer assisted, or fully automated.

High-Level Designs

This component provides a diagram of the high-level design for the future processes. This should be taken from the High-Level Process Designs (BP.070). If a vision was produced in Develop High-Level Process Vision (BP.060), then this design will show how the vision will be carried out through the organizations new processes.

Process Flow Diagrams

This component provides a diagram for any process with conditional steps or more than two steps.

You can use the application-supported process flows provided in Oracle Business Models as a starting point for creating the future process flow diagrams. These flows document processes directly supported by the Oracle Applications, but only show the process steps performed on the applications. You can then augment these process steps by adding the process steps required by the organization to support their unique processes. In many cases, one application process flow will be the basis for several business process models. For example, in Oracle Business Models, the Oracle Purchasing application is part of three processes, order fulfillment, procurement, and materials management.

To-Be Activities

This component provides details of the activities to be carried out as part of the process flow. If there are variants of a generic process, then corresponding variants of activities must also be described for each process flow.

Tools

Deliverable Template

Use the Future Process Model template to create the deliverable for this task.

Oracle Designer

The Oracle Designer process modeler can also be used as a tool to construct process models.

BP.090 - Document Business Procedures (Core)

In this task, you document the procedures supporting the business processes enabled by the implementation of new applications.

Documentation of procedures helps provide consistency of quality work and drives the learning content for job incumbents.

Deliverable

The deliverable for this task is the **Business Procedure Documentation**. It documents the procedures supporting the business processes.

Prerequisites

You need the following input for this task:

❑ **Future Process Model (BP.080)**

The Future Process Model describes the complete set of events the business function responds to in order to meet its objectives and identifies each of the steps in those processes.

❑ **Mapped Business Requirements (BR.030)**

For scenarios with workarounds or proposed custom extensions, business requirements mapping forms are necessary to specify the job-level details for the business process.

Optional

You may need the following input for this task:

❑ **Documentation Standards and Procedures (DO.020)**

The Documentation Standards and Procedures determine the *look and feel* of the project documentation and the procedures the project team uses to develop documentation. If Define Documentation Standards and Procedures was not performed, this deliverable will not exist. (See the task description for DO.020 for more information on when this task should be performed.)

❑ **Documentation Environment (DO.040)**

The Documentation Environment details all hardware, software, and utilities needed to support documentation development. If Prepare Documentation Environment was not performed, this deliverable will not exist. (See the task description for DO.040 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Describe and identify the business process.	Procedure (Scope)
2.	List the system forms and reports accessed during this procedure.	Procedure (System References)
3.	List the policies that limit or affect how the business process can or will function.	Procedure (Policy)
4.	List the user roles who participate in the procedure.	Procedure (Responsibility)
5.	List the user roles who should receive a copy of this procedure documentation.	Procedure (Distribution)
6.	Indicate who maintains this procedure.	Procedure (Ownership)
7.	Indicate the business events that trigger this procedure	Procedure (Activity Preface)

No.	Task Step	Deliverable Component
8.	Document the procedural steps for each process task/step.	Procedure (Tasks)
9.	Repeat the task steps for each Business Procedure Documentation.	
10.	Secure acceptance of the Business Procedures Documentation.	

Table 1-23 Task Steps for Document Business Procedures

Approach and Techniques

Business Procedure Documentation provides a restatement of the information contained in business process designs in a narrative form.

Definition of Business Procedure Documentation

Business Procedure Documentation is a job-level description of a business process design. Business Procedure Documentation defines how the work is done and accomplishes the following:

- confirms the process design and how systems/files/tools/forms are to be used within each process step
- creates prerequisite material and lays the foundation for the User Guide (DO.070), role-based user learning (AP.170), and user certification or other types of readiness testing (if required)

Write Business Procedure Documentation for each business process. In general, there should be a one-to-one correspondence between Business Requirements Scenarios (RD.050) and the documentation.

Business Procedure Documentation is like a topical essay. It describes a business solution by explaining how the new process supports the business requirements. Users, managers, and developers use these essays to establish the framework for designing new procedures, and developing custom modules.

☞ **Attention:** When writing the Business Procedure Documentation, clearly state who is the principle audience. Write topical essays for a general business audience. Refrain from making this document technical. Discuss technical details in detailed design documents.

Topical essays describe the mechanics of the process flow, detailing how inputs are converted into outputs within the boundary of the process.

From Business Requirements Scenarios to Business Procedure Documentation

Business Procedure Documentation represents a decomposition of a Business Requirements Scenario (RD.050). Each business process step on a BRS can be related to a series of procedures, application screens, reports, and inquiries. The lowest level of detail for a BRS is the process step, which is at an elementary business function (EBF) level. However, each process step represents a work activity — and for each activity, there is usually a set of procedural steps that describe how to accomplish that step. Each activity is:

- usually a set of tasks or operations (often including a system transaction) logically grouped together
- composed of manual, clerical, or online (system-assisted) types of work
- executed to achieve file/records update, notification, decision-support, and so on
- performed by a person or piece of equipment (or combination)

Two examples of an activity are:

- A receiving clerk inspects goods, updates a manual receiving log, and enters a receiving transaction into the system.
- A quality control inspector moves discrepant material to a holding area, hangs a tag on the material, and records a transfer transaction into the system.

An important quality objective is to verify process designs at the operator job level. Business Procedure Documentation is a step in that direction. All Business Procedure Documentation development work is reusable downstream during the creation of learning materials and the User Guide (DO.070).

Comparison with Other Deliverables

Whereas a BRS (RD.050) is an instrument for answering the question, *What are the business requirements of the process and its steps?*, Business Procedure Documentation answers the question, *How will the process job steps be performed?*

The User Guide (DO.070) helps make sure that people can demonstrate mastery of the process steps and use of systems / files / tools / forms, and so on. Business Procedure Documentation provides key input into such activities.

How does Business Procedure Documentation differ from a BRS?

- Business Procedure Documentation is the basis for *how* to perform the role steps (of the process), whereas BRSs are for identifying *what* the business requirements are.
- Business Procedure Documentation typically identifies expected step duration where feasible.
- Business Procedure Documentation is a set of directions for performing a role, and therefore begins to build understanding at the user level for confirming that the process design proposed is the correct and best one.
- Business Procedure Documentation uses work language, not codes, to communicate with the user.

How does the Business Procedure Documentation differ from the User Guide (DO.070)?

- The User Guide (DO.070) has field-level references (and refers to sample forms, reports, and so on).
- Business Procedure Documentation has zone-level references (for example, Header / Line-Item) and does not refer to field level.
- The User Guide (DO.070) generally includes a section on how to correct errors.

Writing Teams Organization

Creating Business Procedure Documentation requires two primary skills:

- strong business writing abilities
- understanding of the business and systems concepts

Business understanding is important, since creation of Business Procedure Documentation includes a level of quality assurance — verifying that each procedure makes sense and fits with the characteristics and roles of anticipated users.

When possible, writing teams should be organized around the same grouping as mapping teams. In this way, people who write Business Procedure Documentation are involved during detailed business process design in their respective areas.

Approaches to Process Design

In order to engineer a business process so that people and equipment can reliably perform the steps time after time, a great deal of effort must go into specifying the characteristics of each step. Delineating each step also improves quality, because expectations are set regarding the policies and tools that drive execution.

Prepare Business Procedure Documentation in the same way you would prepare specifications for a manufacturing process, providing very detailed instructions for each step along the way.

The input-process-output technique is a generic modeling tool that was designed for framing complete process specifications by identifying inputs, rules, process step descriptions, tools, and outputs. The BRS (RD.050) provides much of the material necessary for employing such a technique in developing procedures:

- process step descriptions in the identification, requirements and sources, and solutions components
- process inputs in the data profile component
- rules in the data profile component
- tools in the support profile and test specification components

- process outputs in the process analysis component

Many techniques can be used to develop sound Business Procedure Documentation. Possibly the most important technique is the use of a physical walkthrough to verify each procedure. This is not prototyping, where the goal is to prove that a design satisfies business requirements, or solution testing, where the goal is to confirm that various elements of the solution will work together to create the output. Instead, this walkthrough technique focuses more on the ability of the resources involved to perform the work consistently — and having access to the appropriate tools/support systems (like reports, scanners, and so on) as well as knowledge of driving policies.

A policy is defined as a principle, plan, or course of action. Policies are the rules that strengthen a business process and make it workable and acceptable.

Warning: If a Business Requirements Scenario (BRS) must be revised as a result of creating Business Procedure Documentation (for instance, a task turns out to be less feasible than planned), then that BRS must be tested again — possibly resulting in project delay.

Linkage to Process Modeling

System process models can be developed in conjunction with, or at least as a facilitator of, Business Procedure Documentation.

The system process model has two objectives:

- to define the requirements for the system technical architecture
- to define the requirements for the user interface

System process models enable users to visualize how the process will be executed with the new system. This goal can be accomplished in several ways and with varying levels of sophistication.

Typically, system process models are process flow diagrams that are derived from the business process model and annotated with details about how the processes will be automated. Process steps are identified as manual or automated and as batch or online. The type of user interface, frequency of execution, sites where the process is executed, inputs, interfaces to other systems, and current systems support, if any, are also identified.

Linkage to Other Activities

The Business Procedure Documentation forms a strong link with these types of activities:

- mapping
- documentation
- adoption and learning
- business systems testing
- business solutions testing
- user certification and readiness testing

Mapping produces a technically feasible solution that is reasonable in terms of process flows and steps. However, certain assumptions regarding underlying procedural steps are not made until they can be documented in detail. Business Procedure Documentation is the bridge between BRSs (RD.050) and the User Guide (DO.070). User-learning materials should draw heavily from content found in the Business Procedure Documentation.

Because Business Procedure Documentation describes business processes in terms of how work will be performed, they provide valuable input toward proving that business process steps are feasible and can be supported by the resources that must ultimately perform them. Business Procedure Documentation helps narrow the scope of Business System Testing (TE) activities.

Although business solutions are generally approved before Business Procedure Documentation writing begins, solutions drive the procedures and the procedure may modify the way solutions are presented or described. Additionally, the Develop System Test Script (TE.040) and Develop Systems Integration Test Script (TE.050) tasks provide data profiles that describe business conditions required for the application system to function properly in support of the business.

User certification and readiness testing draws heavily from the content found in the Business Procedure Documentation. Such testing is important for quality. Each business process must meet the following performance criteria:

- perform consistently under the same trained resource, using the same tools and achieving the same measurable results
- produce consistent response in terms of acceptable process elapsed (or cycle) time
- meet cost target
- meet customer expectations (usually as required by the next process)

Impact of Business Procedure Documentation

Typical Questions and Answers Regarding Business Procedure Documentation

Question: *Can we implement without Business Procedure Documentation?*
Answer: No. This document is necessary for proving the procedures that support business processes and for developing user training material.

Question: *Can we put off creating these documents until late in the project?*
Answer: While Business Procedure Documentation work could be combined with creation of User Guides (DO.070) at a later stage of the project, this leaves little time for getting the "ergonomics" of job design right, so it is advisable to create this material as early as possible.

Question: *Can Business Procedure Documentation be developed before custom development is complete?*
Answer: Yes. Procedures are living documents.

Question: *What is the biggest benefit of writing Business Procedure Documentation?*
Answer: While demonstrating user-level task understanding is very important, a more important reason is to begin creating process step job designs that are efficient, user friendly, and reliable.

Review and Signoff

As each Business Procedure is completed, have it reviewed by:

- the project sponsor
- management
- representatives from process design and mapping teams
- formal integration teams

Linkage to Other Tasks

The Business Procedure Documentation is an input to the following tasks:

- MD.050 - Create Application Extensions Functional Design
- DO.070 - Publish User Guide
- TE.030 - Develop Link Test Script
- AP.110 - Align Human Performance Support Systems
- AP.130 - Conduct User Learning Needs Analysis
- AP.180 - Conduct Effectiveness Assessment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	80
Tool Specialist	20
Business Line Manager	*

Table 1-24 Role Contribution for Document Business Procedures

* Participation level not estimated.

Deliverable Guidelines

Use the Business Procedure Documentation template to document the procedures supporting the business processes enabled by the implementation of new applications.

This deliverable should address the following:

- **Identification** — describes the business process being documented
- **Inputs** — lists primary inputs into the business process
- **Controls** — describes rules, policies, and other constraints
- **Mechanisms** — identifies support tools required during execution of the business process
- **Procedure** — describes the detailed procedural steps for each process step or task

Deliverable Components

The Business Procedure Documentation template consists of the following components:

- Procedure

Procedure

This component contains the following:

- Scope
- System References
- Policy
- Responsibility
- Distribution
- Ownership
- Activity Preface
- Tasks

Acceptance Criteria

The Business Procedure Documentation should contain heading information such as:

- process number
- process description

In order to pass a quality review, the following conditions must be met:

- Each BRS (RD.050) must be covered by one business procedure, but one business procedure may cover multiple, related BRSs.
- When multiple BRSs are covered by one business procedure, they must have the same general flow with the same basic responsibilities and differ only in minor logic.
- Business Procedure Documentation should mention *how* all logs/forms/tools are to be used within each process step.
- Business Procedure Documentation describes the sequence of job activities for a process from request/trigger to fulfillment/completion.
- Job activities should be defined to include all details pertinent to user training, except reference to individual fields on manual/computerized forms.
- The procedure component should be written as an instruction, each sentence beginning with an action verb (use the *imperative mood* as you would in giving directions).
- A user responsibility should appear first in each numbered paragraph of the Business Procedure Documentation (either in bold or with a trailing colon).
- Avoid acronyms in the Business Procedure Documentation; use simple, clear language.

Tools

Deliverable Template

Use the Business Procedure Documentation template to create the deliverable for this task. Prepare a Business Procedure Documentation for each Business Requirements Scenario created during Gather Business Requirements (RD.050).

Oracle Tutor

Documentation of procedures can be done with the help of Oracle Tutor, which includes a publishing tool and a library of generic procedures. Use of this tool brings numerous advantages to the preparation of Business Procedure Documentation:

- provides a consistent format
- saves time because of the library of generic procedures that only need tailoring
- allows for central distribution and maintenance
- provides access to Oracle courseware on the application

CHAPTER

2

Business Requirements Definition (RD)

This chapter describes the Business Requirements Definition process.

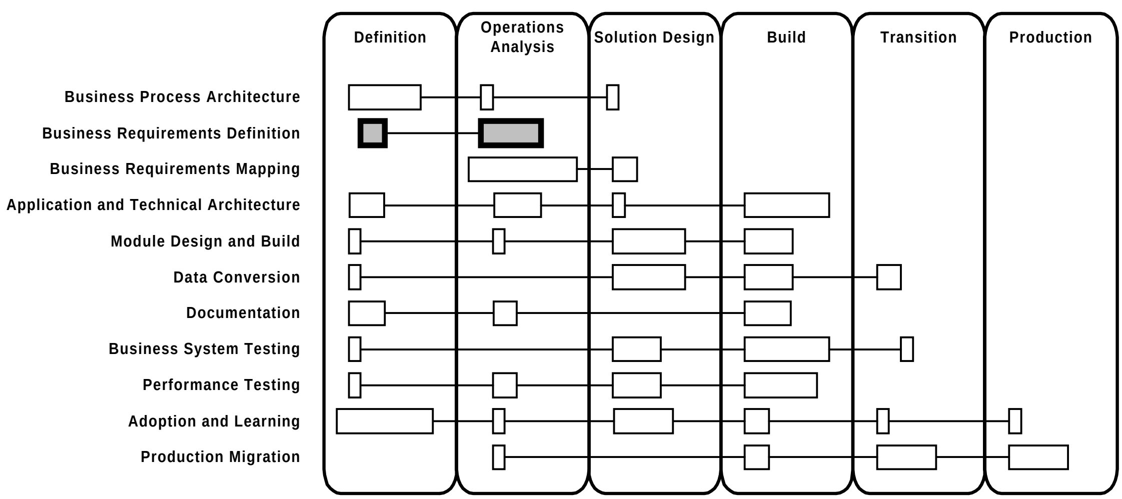


Figure 2-1 Business Requirements Definition Context

Process Flow

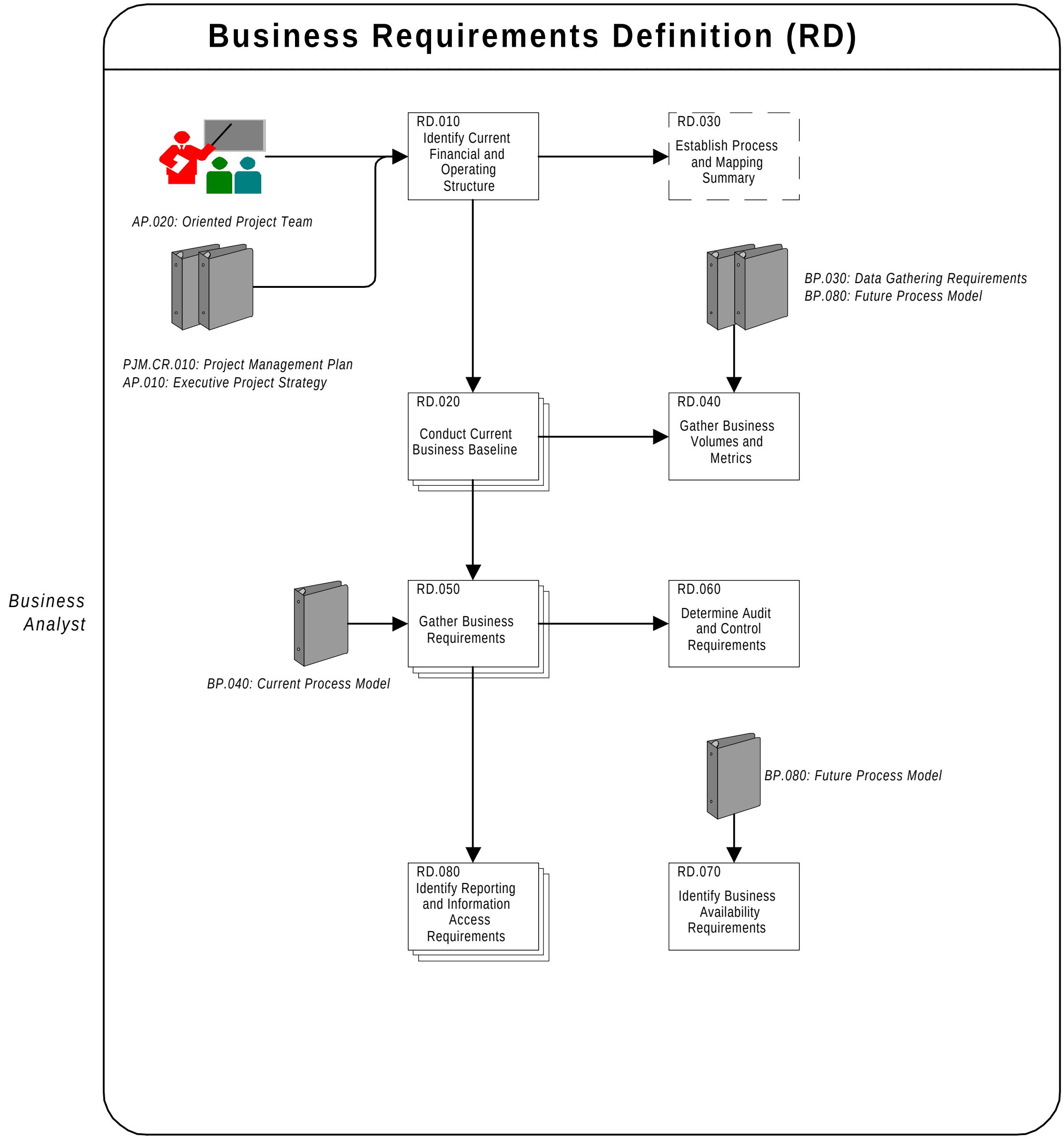


Figure 2-2 Business Requirements Definition Process Flow Diagram

Approach

The objective of Business Requirements Definition is to define the business requirements of the new applications system. These requirements are used as input to the Future Process Model (BP.080).

The Business Requirements Scenarios (RD.050) list the necessary steps for executing a business function within a business process; some of the steps may be application-specific, and some may be manual. To confirm that business needs are met, you will later map these requirements against application functions and features in Business Requirements Mapping (BR).

During Business Requirements Definition, you:

- Examine current processes and practices to understand and document the main business factors that currently benefit the business.
- Gather business transaction and data volumes from the future business model to help assess the system's ability to support current and future business volume.
- Carefully document audit and control requirements to satisfy financial and quality policies.
- Identify the business operating requirements that the technical architecture will need to support.
- Analyze and identify the reporting requirements for the business.

Tasks and Deliverables

The tasks and deliverables for this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
RD.010	Identify Current Financial and Operating Structure	Current Financial and Operating Structure	Always	SI
RD.020	Conduct Current Business Baseline	Current Business Baseline	Always	MI
RD.030	Establish Process and Mapping Summary	Process and Mapping Summary	Project is of *medium or high complexity*, or impacts a *large user population*	SI
RD.040	Gather Business Volumes and Metrics	Business Volumes and Metrics	Always	SI
RD.050	Gather Business Requirements	Business Requirements Scenarios	Always	MI
RD.060	Determine Audit and Control Requirements	Audit and Control Requirements	Always	SI
RD.070	Identify Business Availability Requirements	Business Availability Requirements	Always	SI
RD.080	Identify Reporting and Information Access Requirements	Master Report Tracking List	Always	MI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 2-1 Business Requirements Definition Tasks and Deliverables

Objectives

The objectives of Business Requirements Definition are:

- Understand the business and business area requirements for the Applications implementation project.
- Understand and document the organizational and financial structure of the business.
- Quantify the transaction and data volumes required by those processes.
- Develop a complete set of business requirements scenarios that can be used to map business requirements to application functionality.
- Define the audit and control requirements for the business information generated.
- Document the reporting requirements of the organization.

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
Current Financial and Operating Structure	A depiction of the current financial structure and the operating organizations of the entity.
Current Business Baseline	An analysis of the current business processes and functions.
Process and Mapping Summary	A summary of the key processes, business requirements, and mapping decisions.
Business Volumes and Metrics	An inventory of key transaction, data volumes, and measurements by business functional area.

Deliverable	Description
Business Requirements Scenarios	A set of formal statements of the detailed business requirements for each business process, the source of these requirements, how these requirements will be satisfied (either by the application, manual process steps, workarounds, or by other applications), and what prototyping steps must be taken to prove the designs.
Audit and Control Requirements	A statement of security, monitoring, and review of requirements for business transactions.
Business Availability Requirements	A statement of what the business needs are, in terms of operating hours, system uptime, response, and turnaround time.
Master Report Tracking List	A consolidated master list of reporting requirements.

Table 2-2 Business Requirements Definition Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Application Specialist	Provide answers to functionality questions. Support and provide interpretation for tools, templates, and methods.

Role	Responsibility
Business Analyst	Conduct interviews and working sessions. Identify business requirements, and perform analysis to resolve issues.
Business Line Manager	Participate in interviews and working sessions, and provide access to existing systems.
Configuration Management Specialist	Manage deliverables, assist in process integration, update deliverable status, and manage project data and documentation.
Project Manager	Conduct project planning and assign tasks, establish roles, brief project staff, manage team, manage changes, manage resolution of issues, maintain quality, and obtain approval.
Project Sponsor	Confirm availability and commitment of users, facilitate resolution of major issues, review and approve deliverables.
Technical Analyst	Review backup and recovery needs for new applications and technical architecture.
User	Participate in interviews and working sessions, and communicate current business requirements and processes.

Table 2-3 Business Requirements Definition Key Responsibilities

Critical Success Factors

The critical success factors of Business Requirements Definition are as follows:

- dedicated resources for conducting analysis
- project team understanding of application functionality and the leading industry practices
- a committed project sponsor who maintains a consistent and high level of team commitment
- active involvement and support of management
- active involvement and support of knowledgeable business area specialists
- full access to information about relevant business areas, their processes, data generation, and use
- involvement of skilled analysts to work with management, project, and area teams to obtain information required for requirements definition

References and Publications

Reference: Hickman, Linda and Longman, Cliff. *CASE*Method Business Interviewing*. Addison-Wesley, 1994. ISBN: 0-201-59372-6.

Reference: Millsap, Cary V., *Designing Your System to Meet Your Requirements,* OAUG Conference Proceedings, Fall 95.

RD.010 - Identify Current Financial and Operating Structure (Core)

In this task, you develop a complete picture of the organization's current financial and operating structure.

Deliverable

The deliverable for this task is the **Current Financial and Operating Structure**. It depicts the current financial structure and the operating organizations of all of the business entities within the enterprise.

Prerequisites

- ❑ **Project Management Plan [*initial*] (PJM.CR.010)**

The initial Project Management Plan defines the scope and roles of the project.

- ❑ **Executive Project Strategy (AP.010)**

The Executive Project Strategy describes the executive perspectives and direction for the project, goals, and how they relate to the organization's strategy. If Establish Executive Project Strategy was not performed, this deliverable will not exist. (See the task description for AP.010 for information on when this task should be performed.)

- ❑ **Oriented Project Team (AP.020)**

The Oriented Project Team represents the current team members who have a thorough knowledge of the project vision, direction, and strategies, and their individual mandates.

Optional

You may need the following input for this task:

- ❑ **Current Business Reports**

Current business reports can be useful when trying to understand the Current Financial and Operating Structure or determining the chart of accounts.

- ❑ **Contractual and Business Agreement Documentation**

Collective and ancillary documents such as vendor agreements, contract/labor agreements, system requirements documentation, board level directives and communications, and requests for proposal. These documents assist in the identification of the business drivers for the project and provide details on the scope of the implementation, the modules purchased, the agreements, and so on.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Interview organization management to obtain a clear understanding of current and proposed entity structure.	Introduction
2.	Develop a chart showing the current organization structure.	Organization Legal Entity Structure Chart
3.	Develop a business organization overview and listing.	Business Organization Listing and Overview
4.	Define and chart the financial operating environment.	Financial Operating Environment

No.	Task Step	Deliverable Component
5.	Define the financial and management reporting environment.	Financial and Management Reporting Environment
6.	Review the Current Financial and Operating Structure with project management and secure approval.	
7.	Review the Current Financial and Operating Structure with appropriate organization management and secure acceptance for the deliverable.	

Table 2-4 Task Steps for Identify Current Financial and Operating Structure

Approach and Techniques

The Current Financial and Operating Structure portrays the legal and operating entity structures, business organizations, financial operating environments, revenue and cost centers, and financial consolidation path. It may also catalog other operating entities.

Organization Information

Information about current and proposed organization structures may have been discussed during the applications acquisition cycle, and may have been one of the deciding factors in the selection of the application package. Be sure to gather this information, if available.

Start the internal organization analysis by interviewing the highest ranking financial official possible, since that position will likely have the most knowledge of the financial and operating structure of the organization.

An organization’s operating structure drives the business and has a strong influence on the setup and use of the applications; however, the financial statements will reflect the operating structure to allow profitability, balance sheet and cash flow reporting, and analysis against that structure. Reporting and analysis begins by capturing and valuing operational transactions that occur at the specific event level. These events occur at a specific site that must be properly defined and ultimately set up at the correct organizational level and with the appropriate organizational attributes.

Types of Organization Levels

In your analysis, look for the following types of organization structures:

- **Legal Entities** — an organization considered as one legal or fiscal entity, based on regulatory or reporting requirements
- **Consolidation Entities** — these may be of two varieties: a single set of books with a division segment in the general ledger for the consolidation, or two (or more) sets of books consolidated into one parent set of books
- **Set of Books** — a financial operating environment combining a particular chart of accounts, fiscal calendar, and functional currency
- **Operating Units** — an operational business unit that uses a common order entry, receivables, payables, or purchasing function
- **Reporting Entities** — entities that may have different management reporting requirements, depending on classification as revenue, cost, profit, or investment center
- **Organizations** — a physical or logical site used in manufacturing or distribution that holds inventory or other assets

Financial and Management Reporting Environment

The reporting environment is divided into two categories:

- **Financial Reporting** — addresses all the reporting requirements of the legal entity
- **Management Reporting** — based on reporting for the following types of relationships with the parent entity: revenue centers, cost centers, profit centers, and investment centers; management reporting also encompasses operational reporting

Project Impact of Current Financial and Operating Structure Analysis

Once you create a clear picture of the organization structure, you may want to reexamine the project plan for the application implementation, especially in the case of a financial implementation. It may make more sense to implement sequentially by organization, rather than simultaneously across the entire enterprise. This may also impact the Application Deployment Plan (TA.130), which occurs later in the project.

Linkage to Other Tasks

The Current Financial and Operating Structure is an input to the following tasks:

- BP.010 - Define Business and Process Strategy
- BP.070 - Develop High-Level Process Designs
- RD.020 - Conduct Current Business Baseline
- RD.030 - Establish Process and Mapping Summary
- RD.060 - Determine Audit and Control Requirements
- RD.080 - Identify Reporting and Information Access Requirements
- BR.060 - Create Information Model
- BR.110 - Design Security Profiles
- TA.030 - Develop Preliminary Conceptual Architecture
- TA.040 - Define Application Architecture
- DO.030 - Prepare Glossary
- AP.060 - Develop Business Unit Managers' Readiness Plan

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	100
Business Line Manager	*

Table 2-5 Role Contribution for Identify Current Financial and Operating Structure

* Participation level not estimated.

Deliverable Guidelines

Use the Current Financial and Operating Structure to capture all the organizational structure information in one place. The Current Financial and Operating Structure provides a place for compiling all the relevant attributes of the organization to be addressed in setting up and running the applications.

This deliverable should address the following:

- organization information
- reporting environment

Deliverable Components

The Current Financial and Operating Structure consists of the following components:

- Introduction
- Organization Legal Entity Structure Chart
- Business Organization Listing and Overview
- Financial Operating Environment
- Financial and Management Reporting Environment

Introduction

This component introduces the business, states its business purposes, and documents the way in which the present organization supports the business purpose. This information is often found in the initial Project Management Plan (PJM.CR.010).

Organization Legal Entity Structure Chart

This component displays the structure of the organization in chart format, showing the highest level of consolidation at the top. The organization legal entity structure chart reflects the operating or geographical entities rolled up into the consolidated level.

In addition, it shows the specific business units at which the operating events or business transactions occur beneath the operating or geographical entities. The specific business units represent those locations where manufacturing, inventory, order entry, purchasing, and other similar business processes and functions are executed.

Business Organization Listing and Overview

This component documents an overview for each of the business units and explains the nature of the facilities, the events and processes engaged in, the type of products handled, and a preliminary statement of business volumes and staffing levels.

Financial Operating Environment

This component documents the financial structures in more detail, according to the following elements:

- chart of accounts
- functional currencies
- accounting calendars
- legal entities

All of the elements above are relevant for the organization when it is performing a particular function on its own behalf or on behalf of another organization.

Financial and Management Reporting Environment

This component categorizes the enterprise's locations by type of financial, supply chain, sales chain, and management reporting requirements based on its type of reporting structure.

Tools

Deliverable Template

Use the Current Financial and Operating Structure template to create the deliverable for this task.

RD.020 - Conduct Current Business Baseline (Core)

In this task, you examine current processes and practices to understand and document the main activities that keep the organization operating today.

Deliverable

The deliverable for this task is **Current Business Baseline**. It provides an analysis of the current business processes, activities, and functions.

Prerequisites

You need the following input for this task:

- ❑ **Project Management Plan [*initial complete*] (PJM.CR.010)**

The team uses the Project Management Plan to further clarify the scope of the project.

- ❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides a clear understanding of the current and proposed organization structure.

- ❑ **Existing Reference Material**

The team uses existing reference material to gain an understanding of the existing practices, processes, and systems that support the organization.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Schedule, confirm, and prepare for process definition sessions by business area.	
2.	Identify the core business processes (level 1 if you are using Oracle Business Models) and write a summary description of each process.	Process Listing and Process Descriptions
3.	Conduct interviews using the questionnaires and other sources of information to clarify questions you have identified.	Process Questionnaires
4.	Gather any other current business materials that may enhance team understanding and documenting of current business process requirements.	Current Business Process Documentation
5.	Identify any issues regarding the current business analysis.	
6.	Review the Current Business Baseline with users and business area management.	

No.	Task Step	Deliverable Component
7.	Secure acceptance of the Current Business Baseline from business area management.	
8.	Secure acceptance of the Century Date compliance approach described.	

Table 2-6 Task Steps for Conduct Current Business Baseline

Approach and Techniques

Current business baselining is the process of analyzing current business activities, processes, functions, and practices to determine current business operations and requirements. Those requirements represent the organization's responses to external and internal events that drive the business today.

Benefits of Baselining

Conduct baselining to develop a common understanding within the project team and across the organization of what cross-functional processes are in place to support the achievement of business objectives in the current environment. Awareness of current business requirements and unique processes today will educate team members and prepare them for the construction of future business processes. Because many current environments may not have been developed in an integrated fashion, and people have entered the organization at different times during the evolution of the current business systems, many people and departments may not fully understand the processes and requirements of other departments. An added benefit of baselining is the resulting cross-functional knowledge gained by team members. This information is invaluable during subsequent mapping of future business processes and requirements to new application functionality.

Baselining highlights the added value of the current processes. Key processes differentiate the organization from the competition, potentially creating a competitive edge. It is very important that the team understands and protects these special features and functions and can differentiate between a *must have* and an *always have done it this way* requirement.

Minimum Requirements for Baselining

Management may choose to minimize the investment in reviewing the current business — particularly when it is expected that the new application system or new business processes will be dramatically different. This may be appropriate when a business process reengineering project has preceded or is proceeding as part of the application implementation.

Warning: The minimum review requirements are to understand the integrated nature of current business processes and the business information needs of each process within and across high-level business functions, as well as across organizations.

Century Date Compliance

In the past, two-character date coding was an acceptable convention due to perceived costs associated with the additional disk and memory storage requirements of full four-character date encoding. As the year 2000 approached, it became evident that a full four-character coding scheme was more appropriate.

In the context of the Application Implementation Method (AIM), the convention Century Date or C/Date support rather than Year2000 or Y2K support is used. Coding for any future Century Date is now considered the modern business and technical convention.

Every applications implementation team needs to consider the impact of the century date on their implementation project. As part of the implementation effort, all applications, processes, functions, customizations, legacy data conversions, and custom interfaces need to be reviewed for Century Date compliance.

As you review all current systems that will remain in use after the implementation of Oracle Applications for Century Date compliance, make sure that all interfaces are reviewed as well.

Baselining Teams Organization

Approach baselining by organizing project members and supporting resources into process teams within each high-level business function. Appoint business analysts who have experience in the business function and a good cross-functional grasp of current business processes, such as:

- Human Resource Management
- Financial Management
- Sales and Market Management
- Procurement
- Order Fulfillment
- Planning and Forecasting
- Product Development
- Enterprise Management
- Product Development
- Materials Management
- Quality Management
- Post-Sales Service
- Operations

The critical objective is to understand the business requirements associated with the processes (for example, key data, transaction volumes, critical timing of the data, deciphering short cut codes, and so on). Optimally, the workshop leaders will be well-versed in facilitation techniques.

The project manager should consider assembling business analysts for a general, introductory kickoff meeting to review the baselining strategy and to verify that the analysts understand the deliverable and the task completion commitments to the project. During this session, the schedule for each business process team to complete their questionnaires and business requirements definitions should be confirmed.

All process teams should conduct their current business baselining in the same way, using the same methods (for example, questionnaires, identify exceptions, note input sources, and output medium) — the deliverables should be consistent in content and quality.

All relevant, high-level business functions to be covered by a process should be represented on the process teams and key users from those functions should participate in process requirements definition sessions. They will have the best insight into how the process works today, may be vocal on what does not work, and can help to recognize process improvement opportunities.

Suggestion: It may be helpful to present an example completed deliverable to act as a guide for the process teams. This approach often supports consistent results.

In order to work effectively, all process requirements definition sessions should be accompanied by an agenda that includes the following:

- session location and duration
- role assignments
- the business process where requirements are to be defined and its process boundary
- activities and their sequence
- required materials (for example, prerequisite deliverables) and assignments for ensuring their presence
- expected outputs and the criteria for successful closure of the sessions

Process Performance

Benefits of this task include the following:

- the opportunity to recognize potential and additional process improvements to capitalize on the new application functionality
- the ability to establish a current business benchmark (for instance, turn around time for current purchase orders, number of invoices with errors, types of errors, bottlenecks, current system workarounds) against which to gauge the ultimate success of the new business model

Business Process Questionnaires

Use structured process questionnaires to collect business and current system information during a business baseline interview for a given process. These questionnaires can be modified to help make sure that the team interviews net the following factors:

- business events — triggers for action (for example, receive invoice)
- location, nature, and sequence of transactions — data added
- magnitude and frequency of transactions
- performance metrics core processes or critical transactions
- key factors for success
- key processes and process cost drivers
- representative families or products and transactions
- opportunities, constraints, risks, and issues
- underlying structures of static data organization
- bottlenecks to the flow of information and material
- the particular value of current business processes
- data gathering methods that drive technology requirements
- current system configuration options

Always customize the questions for your audience and their particular needs and business conditions. Because these questions target current legacy systems, avoid references to particular features of the applications being implemented.

Suggestion: Get assistance from users when creating the questionnaires. This helps develop ownership in key people who represent interviewees.

Use local business terminology to facilitate development of content and to avoid misunderstanding words, phrases, and concepts during the interview. It is sometimes helpful to review the terms definitions and document them to assist cross-functional users in accepting a department's / process' jargon. For instance: *COB* can mean *Close of Business* in one area and *Cost of Benefit* in another.

Before using a questionnaire, think about how to organize the information (into families and categories) and reuse the questions to cover representative products and processes for each group. For instance, the process baseline for finished goods shipments may be completely different from that of the shipment of spare component parts. In this case, if you ask questions about only one class of material, you may miss critical information about how the business works.

Although baseline information concerns the current business system, you can usually reuse much of the information during future business process modeling. For instance, current business process interview sessions should uncover a list of current events that drive business processes. The events identified should be very close to the future events.

Review and Acceptance

As each process team completes its current business baseline, it should be reviewed with the project team, area users and management, and any cross-process integration teams. Formal acceptance of the deliverable should be by the business area manager.

Multi-Phase, Multi-Site Implementation

If the implementation is multi-phase or multi-site, the task of developing the Current Business Baseline may be more complex. It is helpful in a multi-phase implementation effort to complete the entire Current Business Baseline, especially at the higher levels during the first phase. A completed Current Business Baseline, at a high level, gives the project focus and improves decision making and assists in defining the limits of the scope of the project. Additional detail or levels can be documented when the appropriate phase is started.

With multi-site implementations, the differences in the processes can be annotated as belonging to the specific site where the process differs. Occasionally, process modeling team members learn a *better* way in these sessions and business process improvements are affected more quickly. When this happens, determine if the *better* way maps adequately to the Oracle Applications and any custom designs.

Linkage to Other Tasks

The Current Business Baseline is an input to the following tasks:

- BP.040 - Develop Current Process Model
- RD.040 - Gather Business Volumes and Metrics
- RD.050 - Gather Business Requirements
- RD.060 - Determine Audit and Control Requirements
- BR.020 - Prepare Mapping Environment
- BR.040 - Map Business Data
- TA.030 - Develop Preliminary Conceptual Architecture
- CV.010 - Define Data Conversion Requirements and Strategy
- DO.030 - Prepare Glossary
- AP.060 - Develop Business Unit Managers' Readiness Plan

The Current Business Baseline sets the stage for the more formal process modeling of the current business requirements that will occur in Develop Current Process Model (BP.040) and future business requirements that will occur in Develop Future Process Model (BP.080). As a result of baselining activities, teams develop an appreciation for cross-functional processes and the information requirements that span those processes. They will be ready to take on the task of developing an integrated set of process models that addresses the unique needs of their business.

Gather Business Volumes and Metrics (RD.040) begins with an identification of current business levels of volume, driven by current process steps. As you define the future business processes, you will translate current volumes into projected future levels, which you will then use to estimate information storage needs.

The System Availability Strategy (TA.050) captures the technical information about legacy systems that support current business processes.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	100
Business Line Manager	*
Project Sponsor	*
User	*

Table 2-7 Role Contribution for Conduct Current Business Baseline

* Participation level not estimated.

Deliverable Guidelines

Use the Current Business Baseline to reflect a common understanding of your existing business processes, activities and functions. The objective is to gain an understanding of the integrated current business requirements.

Details gathered from interviews with key members of the staff are included within the components of this deliverable. This deliverable is an important input for developing the Current Process Model (BP.040) and the Future Process Model (BP.080), which in turn become the basis for mapping process scenarios to applications.

This deliverable should address the following:

- process research
- process descriptions

Deliverable Components

The Current Business Baseline templates consist of the following components:

- Process Listing and Process Descriptions
- Current Business Process Documentation
- Process Questionnaires

Process Listing and Process Descriptions

This component provides an introductory, textual description of the processes that are executed in response to current business events, together with an identification of each process's main inputs and outputs.

Current Business Process Documentation

This component describes the following:

- identifies critical requirements that differentiate the organization from the competition; identifies what is working well
- lists benchmark activities
- identifies current bottlenecks and their perceived impact on business
- identifies critical exception processing requirements
- documents insights discovered during the working sessions including candidates for improvements

Process Questionnaires

This component provides content and structure for the interviewing activity during process research.

Tools

Deliverable Template

Use the Current Business Baseline template to create the Process Listing, Process Descriptions, and the Current Business Process Documentation components for this task.

Use the Process Questionnaire template to create the Process Questionnaire component of this task.

Century Date Acceptance

To document the review and approval of Current Business Baseline for Century Date compliance considerations, prepare a separate Acceptance Certificate (PJM.CR.080) replacing the standard acceptance language with the following:

The above deliverable has been reviewed by <Company Long Name> and fully meets the Century Date compliance objectives expressed by <Company Short Name> and <Client Project Manager> and passes the acceptance criteria established for Century Date support specified by <Company Long Name>.

After obtaining acceptance and appropriate signatures, this Century Date Acceptance Certificate should be filed with the deliverable in the project library.

RD.030 - Establish Process and Mapping Summary (Optional)

In this task, you create the repository for key project findings and decisions that occur during the process and functional requirements gathering activities.

∆ If your project is of *medium or high complexity* or impacts a *large user population*, you should perform this task.

Deliverable

The deliverable for this task is the **Process and Mapping Summary**. It provides a single source of information on key mapping decisions for project members and executive management as well as resources joining the project.

Prerequisites

You need the following input for this task:

❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure portrays an organization structure that may be important for application and technical architecture.

Optional

You may need the following input for this task:

❑ **Current Process Model (BP.040)**

The Current Process Model lists events and processes that the new business system may continue to support. In addition, it may identify processes that require redesign or reengineering. If Develop Current Process Model was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

❑ **Future Process Model (BP.080)**

The function hierarchy assists in assessing the completeness of the Future Process Model.

❑ **Configured Mapping Environment (BR.020)**

During the requirements definition and mapping cycle, you will identify and prototype scenarios and create a summary listing. Although the Configured Mapping Environment is not identified as a dependency within the base project workplan template, the Process Mapping and Summary can be used as a mechanism to track the status of the mapping activities for the project.

❑ **Original Justification for Business Systems Investment**

This document explains the business objectives for investing in new applications and describes items, objects, processes, events, and financial payoffs. It may be important to include in the Process and Mapping Summary.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review and capture high-level management policies that might affect application usage and architecture configuration.	Key Business Requirements
2.	Capture key mapping decisions.	Key Mapping Decisions
3.	Identify the key issues for each of the core business processes.	(Updated) Process and Mapping Tracking Summary
4.	Capture future metrics and update summary.	Updated Metrics Summary

No.	Task Step	Deliverable Component
5.	Review the Process and Mapping Summary with the project manager and organization management as part of regular reporting.	

Table 2-8 Task Steps for Establish Process and Mapping Summary

Approach and Techniques

Definition of the Process and Mapping Summary

The Process and Mapping Summary is a facilitating mechanism for team members to familiarize themselves with any key project developments, discoveries, and decisions. It is also a point of reference for team members external to the Business Requirements Definition (RD) and Business Requirements Mapping (BR) processes. It is valuable for the organization, because it provides an audit trail of reasons behind key decisions. The project manager may use it to get status information quickly (particularly during the early part of the project) on the progress of baselining, process definition, mapping, custom module build, and testing.

Original System Justification Review

Revisit the original system justification to assess the key requirements, assumptions, and financial and operating objectives made in the original investment proposal. These objectives will become components of analysis to be used in the post-implementation audit after project completion. The Process and Mapping Summary can be the starting point and repository for capturing much of this key project information.

Current and Future Business Processes, Events, and Metrics

The original investment decision is justified on the basis of some or all of the following:

- process, quality, or cost improvements
- additional sales revenue

- ability to develop new products and markets
- customer responsiveness and satisfaction
- state-of-the-art information management
- state-of-the-art technical architecture
- employee satisfaction initiatives

There may also be key management directives, mandated as strategic business initiatives, that may have overarching impact on the application and technical architecture. Be sure to clearly document and adhere to these directives as the implementation unfolds. For example, the future business requirement to have a chart of accounts that consolidate in a specific way, may have significant impact on technical considerations or user procedures for the applications' use.

As the implementation progresses, the analysis to support the implementation will formalize or generate significant information about supported events, processes, resources, costs, and other metrics; many of these can be captured en route, so that the pieces gradually come together to enable efficient audit to determine whether original business objectives were met.

Linkage to Other Tasks

The Process and Mapping Summary is an input to the following tasks:

- TA.040 - Define Application Architecture
- TA.070 - Revise Conceptual Architecture
- TA.080 - Define Application Security Architecture
- TA.130 - Define Application Deployment Plan

This task supplies Application and Technical Architecture (TA) with information about major business operating decisions and strategies that will need to be supported.

- PT.020 - Identify Performance Test Scenarios
- PT.030 - Identify Performance Test Transaction Models

Although the Process and Mapping Summary is created in Business Requirements Definition (RD), you will update it with information gathered later. As other key implementation findings and decisions are made, append them individually onto the Process and Mapping Summary, based on the specifics of each implementation.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	75
Project Manager	25
Business Line Manager	*

Table 2-9 Role Contribution for Establish Process and Mapping Summary

* Participation level not estimated.

Deliverable Guidelines

Use the Process and Mapping Summary to create the repository for key decisions that occur during the implementation.

This deliverable should address the following:

- key business decisions, process definitions, and organizational structures that support the application implementation in achieving the original business justification
- the intended application setup and usage to support those business processes across organizations, which may impact the application and technical architecture
- status information for the project manager on the deliverables being produced during the following tasks:
  - Develop Current Process Model (BP.040)
  - Develop Future Process Model (BP.080))
  - Gather Business Requirements (RD.050)

– Map Business Requirements (BR.030)
– Test Business Solutions (BR.080)
– Create Application Extensions Functional Design (MD.050)
– Create Application Extensions Technical Design (MD.070)
– Create Application Extension Modules (MD.110)
– Perform Unit Test (TE.070)
– Perform Link Test (TE.080)

Deliverable Components

The Process and Mapping Summary consists of the following components:

- Key Business Requirements
- Key Mapping Decisions
- Process and Mapping Tracking Summary
- Updated Metrics Summary

Key Business Requirements

This component records any key business policies, strategies, or enterprise-level requirements.

Key Mapping Decisions

This component records any key mapping decisions made during the project.

Process and Mapping Tracking Summary

This component records completion percentage or status for the implementation progress on business processes by organization and by function.

Updated Metrics Summary

This component consolidates any metrics that were gathered during Develop Current Process Model (BP.040) and Map Business Requirements (BR.030). After gaining sufficient experience with the new production system, this summary is updated with metrics captured from the execution of newly designed business processes.

Tools

Deliverable Template

Use the Process and Mapping Summary template to create the deliverable for this task.

RD.040 - Gather Business Volumes and Metrics (Core)

In this task, you research the current business volumes and estimate the future projected operational processing volumes, transaction patterns, and data storage volume requirements.

This information provides the basis for determining the size and processing capacity of the new production system.

Deliverable

The deliverable for this task is the **Business Volumes and Metrics**. It provides an inventory of key transaction, data volumes, and measurements by business functional area.

Prerequisites

You need the following input for this task:

- ❑ **Data Gathering Requirements (BP.030)**

Use the Data Gathering Requirements to define the metrics that are appropriate for measuring performance for the organization's business area. If Determine Data Gathering Requirements was not performed, this deliverable will not exist. (See the task description for BP.030 for more information on when this task should be performed.)

- ❑ **Future Process Model (BP.080)**

The Future Process Model provides the process context for the steps that drive the business volumes. The Function Hierarchy, also included in this deliverable, documents all the elementary process steps that will have transaction, data, and storage volumes associated with them.

Optional

You may need the following input for this task:

- ❑ **Current Business Baseline (RD.020)**

If the current and future business process volumes will be similar, the Current Business Baseline is used as a high-level check for estimating future volume.

- ❑ **Return on Investment (ROI) Analysis**

A return on investment analysis, such as Oracle's Solution Value Assessment (SVA), includes information on expected business metrics that needs to be incorporated into the Business Volumes and Metrics.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the existing documentation.	
2.	Gather business process measurements.	Business Process Measures
3.	Extract business volumes from the current business and the Business Requirements Scenarios (RD.050).	
4.	Summarize the business transaction volume statistics by functional area.	Business Transaction Volumes
5.	Gather total data volume requirements.	Total Data Volume Requirements
6.	Determine the critical processing periods window.	Critical Processing Periods

No.	Task Step	Deliverable Component
7.	Gather system user counts by functional area.	System User Counts
8.	Review volume requirements with area managers.	

Table 2-10 Task Steps for Gather Business Volumes and Metrics

Approach and Techniques

The Business Volumes and Metrics documents the data volumes and processing frequency of the transactions on the new production system. To begin this task, examine all major business processes that transact moderate to high volumes of data (for example, customer orders, purchase orders, purchase requisitions, manufacturing orders, manufacturing receipts, invoices paid, and journal entries). Concentrating on the resource-intensive areas allows you to assess when the new system will inherit the same volume and performance challenges. If you are working on a small, single-site implementation, it may seem that there are no performance risks. Do not minimize the importance of this task. The number of possible configurations can lead to performance problems, even for smaller implementations.

Capture of Business Metrics Consolidation

When you interview users, managers, and analysts on current business processes, include questions that draw out measurable information regarding operations. It may be preferable to give worksheets to the business area contacts for gathering the detailed quantitative information offline.

Another approach that can be beneficial and more accurate is to arrange for operations to monitor system activity such as reports, processes, and transaction volumes for selected time periods, and gather the detailed information in this way. Be careful when using system counts of database activity; depending on the type of legacy system, they can greatly overestimate the actual business volume.

Multi-Phase, Multi-Site Implementation

If the implementation is multi-phase or multi-site, the task of gathering business volumes and metrics may be more complex.

Using the multi-phase approach, business volumes can be intelligently estimated for the next phases by following the guidelines in defining the current business baseline. At each phase it is necessary to review the initial findings and alter them as new data are found.

Multi-site implementations require a close coordination of the individual site teams. Rotating site work sessions or forming a central project team that includes critical representatives from each site is essential. By having a central project team, you can help make sure that issues and exceptions are addressed in a more timely manner.

Linkage to Other Tasks

The Business Volumes and Metrics are an input to the following tasks:

- BR.010 - Analyze High-Level Gaps
- TA.070 - Revise Conceptual Architecture
- TA.090 - Define Application and Database Server Architecture
- TA.110 - Define System Capacity Plan
- TA.140 - Assess Performance Risks
- CV.030 - Prepare Conversion Environment
- PT.020 - Identify Performance Test Scenarios
- PT.060 - Design Performance Test Data
- AP.110 - Align Human Performance Support Systems
- AP.180 - Conduct Effectiveness Assessment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	100
User	*
Business Line Manager	*

Table 2-11 Role Contribution for Gather Business Volumes and Metrics

* Participation level not estimated.

Deliverable Guidelines

Use the Business Volumes and Metrics to determine the sizing and performance needs of the production system.

This deliverable should address the following:

- critical volume and frequency processing requirements (such as completing overnight job stream by morning)
- online inquiries
- online transactions
- batch processing
- batch reporting
- average and peak volumes
- average number of users and peak period users
- transactions per day within a critical processing period (such as quarter close)
- total number of periods of data to be kept online
- percentage change due to growth
- total data volume requirements

Deliverable Components

The Business Volumes and Metrics consist of the following components:

- Business Process Measures
- Business Transaction Volumes
- Total Data Volume Requirements
- Critical Processing Periods
- System User Counts

Business Process Measures

This component defines metrics for each business area that are appropriate to measure for the organization. After measurements are defined, users will be asked to collect this data (if it has not already been recorded).

Business Transaction Volumes

This component lists the key transaction processes (online entry, batch creation, and reports) of each business function and further subdivides the functions by business object. The other dimension of the worksheet matrix captures the current detailed transaction volumes by month and projects future volumes by using the additional columns.

Total Data Volume Requirements

This component documents the volume changes in the key business objects for each business function and the total number of periods of data likely to be present in the production system at maximum data population (just before the next archive or purge of the production data).

Critical Processing Periods

This component lists the key transaction volumes for the business function by precise day of the critical processing period, relative to the average day's processing activity. In the case of financial-based periods, the critical processing window would often be around month or quarter end, with the main processing day (day 0) being the period end.

System User Counts

This component profiles the number of current users by function or user group and estimates the projected growth over future periods.

Tools

Deliverable Template

Use the Business Volumes and Metrics template to create the deliverable for this task.

RD.050 - Gather Business Requirements (Core)

In this task, you define detailed business requirements and perform an initial assessment of application fit to these requirements.

Deliverable

The deliverable for this task is the **Business Requirements Scenarios (BRS)**. The Business Requirements Scenarios provide a formal statement of the detailed business requirements for the business processes. At least one BRS is created for each business process and links requirements to process steps.

Prerequisites

You need the following input for this task:

- [ ] **Current Process Model (BP.040)**

Current information for relevant processes is useful in determining minimum business requirements based on functionality that supports the current business. Identification of transactions (their volumes, costs, and metrics) is important because these are the bases against which future business processes are measured. If Develop Current Process Model was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

- [ ] **Current Business Baseline (RD.020)**

Current business requirements and existing procedures for current processes may yield future business requirements information.

Optional

You may need the following input for this task:

- ❑ **Business Requirements Scenarios (Sales Cycle)**

For core processes, it is possible that preliminary BRSs were created during the sales cycle in response to the request for proposal (RFP). Use of such material strengthens the implementation by forming a solid link with the reasons for acquiring the application package. This can be a key facilitator of rapid implementation.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Train all assigned team members in use of the methods and tools for BRS development, and in the boundary and characteristics of the target business process.	
2.	Check the prerequisite deliverables and get a control number assigned for the new BRS.	
3.	Assign preliminary research topics to team members and confirm that they complete all research before the first design session begins.	
4.	Revise the Future Process Model (BP.080) based on research findings.	

No.	Task Step	Deliverable Component
5.	Construct a process identification for the target future business process. If it exists, use the preliminary BRS collected during initial project planning.	Business Process Identification
6.	For each process step, document business requirements and indicate the source of those requirements.	Requirements and Sources
7.	Assess initial fit of application functions to business requirements at the elementary business function level. Make references to application documentation or navigation and indicate where known gaps exist.	
8.	Publish the BRS, distribute it to reviewers, get feedback, and make adjustments.	
9.	Secure acceptance of the Business Requirements Scenarios.	

Table 2-12 Task Steps for Gather Business Requirements

Approach and Techniques

Definition of a Business Requirements Scenario

When you create a narrative description of your process model, you are describing business requirements so that they are intelligible to business people and systems users. The rule is: You must express all business requirements in the context of process models and business requirements scenarios. In other words, you must be able to trace all detailed business requirements to business processes.

A BRS is a formal statement of the detailed business requirements for a business process, the source of these requirements, how these requirements will be satisfied (either by the application, manual process steps, workarounds, or by other applications), and what prototyping steps must be taken to prove the designs.

Since BRS development sessions are design sessions, you can expect additions and corrections to be made to initial process models and function models during the course of design. Although it is possible to create a BRS at the same time that its process model is being developed, it is actually better to start with a graphical representation of the business process before going into descriptive detail since people tend to respond better to pictures than to words. Using visualization techniques during the early stages of any design is crucial to understanding and common agreement.

Design Teams Organization

Break up each business area into manageable groups of business processes, with a team assigned to each group of processes based on the team's background, business expertise, and demonstrated effectiveness as a team. In this way, BRS development can proceed in parallel — reducing the elapsed time to complete all BRS deliverables. All expected functions to be covered (identified as elementary business functions in the process models) should be represented. Each team must have a thorough understanding of the events that drive their processes and process boundaries. If process models have already undergone a rigorous quality review (especially for interprocess integration), this risk will be minimized.

Key users should participate in design sessions. It is best to use storyboard and other similar techniques to make each process and its steps realistic and visual. As decisions are made and agreements reached, capture them in the BRS so that the final product reflects the proposed business process design.

It is very important that the library management role be properly executed, especially in regard to the following areas:

- updating a function model that is shared by several teams
- providing advice regarding process integration

- providing deliverable status
- providing configuration management (to help maintain proper use of tools and cataloging of deliverables)

In order to work effectively, all design sessions should be accompanied by an agenda that includes:

- session location and duration
- role assignments
- the business process to be designed and its process boundary
- activities and sequence
- the inputs required (for example, prerequisite deliverables)
- the expected outputs and criteria for successful closure of the design session

Requirements Determination

The objective of BRS development is to compile a comprehensive list of requirements against which to map Oracle Application features (Map Business Requirements - BR.030). You can develop detailed business requirements and process flows concurrently, but the process flow diagrams created during process modeling provide the best input. As a starting point, use vendor evaluation lists or the request for proposal (RFP) to begin summarizing the future requirements for the new system. Review this list with the project team and verify that there is a common understanding of the requirements between the team and (Oracle) applications specialists. Pay special attention to features that were rated unsatisfactory during the vendor evaluation process.

Establish a strong linkage with the reasons for the selection of the new application system. The idea is to focus on the new system and de-emphasize the old system as much as possible. This is a key technique for rapid application implementation.

If preliminary BRS material was not created during the presales cycle, review the original justification for business systems investment document referred to in Establish Process and Mapping Summary (RD.030) to understand the business objectives that supported the investment.

Attention: Documents may exist that describe key business requirements, architectural assumptions, or policy statements — all of which could influence process design. These documents should be referenced in the BRS. It is especially important to transition properly from the RFP wish lists that are often created by functional organization, into process-driven, detailed business requirements. Pay particular attention to the role that the new technology will play in driving business requirements—it can even cause revisions to project scope (Project Management Plan - PJM.CR.010) if not understood early enough during the life of the project.

Watch for requirements that do not match the business processes identified in Develop Current Process Model (BP.040) or Develop Future Process Model (BP.080). Team members may list features of their old system as requirements, even if they no longer need those features.

New requirements are likely to emerge as the analysis progresses. Confirm that the new requirements fit within the scope of the project before adding them to the business requirements list. Set aside time to finalize these requirements.

Attention: Avoid describing requirements based exclusively on legacy system functionality. Such requirements may reflect a current system constraint or workaround. In many cases these shortcomings will no longer apply.

Process Research

Assign team members preliminary process research just after the team kickoff and before intensive design sessions begin for each business process. The research goals are to test feasibility of the process models and gather facts.

Since BRS development sessions are process design sessions, there is much at stake. Final designs will dictate how the business operates. Therefore, team members must be informed about:

- process characteristics and idiosyncrasies
- transaction volumes
- exception processing — both real and perceived (for example, legacy workarounds)

Follow the *Rule of 3-2-1*: Roughly 3 hours of research are required for 2 hours of process design and 1 hour of formal entry (capturing the BRS using a template or another tool). Perform research at the job site (shop floor, user office, and so on). If this task is performed in an isolated environment like a conference room, the product will not be as complete or accurate.

Process research also helps you to establish the minimum business requirements that must be met to keep the business running. If the Current Business Baseline (RD.020) and the Future Process Model (BP.080) map closely to this baseline and the same team is employed in both baselining and BRS development, process research may be a minimal activity.

Preliminary Application Fit

A key principle of rapid application implementation techniques is the concurrent assessment of application fit while formal generation of detailed business requirements is in process. In fact, the new application functionality should drive these requirements to the extent possible. The preliminary application fit will simplify the implementation because it encourages team members and participating users to adapt more quickly. The rule is: Do not design a new system from scratch after you have just purchased one.

As requirements are being formulated into a BRS for a business process, gaps between the application and some set of requirements may appear. Use the solution component of the BRS template to document any potentially feasible workarounds with a status of *Pending Active*. This will serve as a placeholder for further research and investigation during later detailed mapping activities (Map Business Requirements - BR.030). Remember that the primary mission is identification of necessary business process steps and their associated requirements. Too much emphasis on finding gaps can be counterproductive to a good design.

Linkage with Process Modeling

Process modeling is a structured approach used to identify, define, and document the activity performed by a business to produce a specific business deliverable. A BRS is a formal statement of the detailed business requirements for a business process. Creation of a BRS will cause the refinement of its associated process model.

Revisions to the Functional Hierarchy may also be necessary. For instance, it may be very helpful to establish this rule regarding the Functional Hierarchy: Define the first level above the elementary business functions in the hierarchy as application-specific functions (for example, *order promising, receiving*, and so on). Functions drawn from the organization and structure of the application will facilitate better process development.

Define each process step on a BRS at an elementary business function level. This provides a quality check against the Functional Hierarchy, and helps make sure that BRSs are developed at an appropriate and consistent level.

If during the course of process design, questions arise regarding the boundaries for a particular process, suggest the following approach to help with boundary identification:

- a process extends from the first activity by the group that receives the initial request (triggering event) to the final activity of the last group that fulfills the request and thus closes the loop with the requester
- ask "how FAR (<u>F</u>ulfill <u>A</u> <u>R</u>equest) should we go with this process?"
- note that business processes often end with the deliverable being completed at one step and placed in storage, followed by a delay before it is used or acted on by some new business process

Inter-Team Integration

Make sure that processes do not overlap and that process steps are not left hanging and unclaimed by any team. These mistakes occur when process models have not been quality assured for consistency and completeness, or when teams assume certain tasks are the responsibility of other teams.

The best preventive approach is to have an integration team that monitors process designs when they are in process, and looks for inter-team integration problems.

A helpful technique is to cross-reference all events with process outputs. This is useful because most process outputs also serve as events that trigger other processes. In this way you can identify overlapping and missing processes or process steps early, thereby avoiding rework.

When an agent is an automated functional area outside the business area or outside the domain of a design team, that could indicate that there are system interfaces that must be considered when making assumptions about business requirements.

Review and Signoff

As you complete a BRS for each process, you should review it with the project sponsor, management, representative business users, and any formal integration teams. In this way you can get progressive confirmation of your process designs and business requirements.

Linkage to Other Tasks

The Business Requirements Scenarios are an input to the following tasks:

- BP.080 - Develop Future Process Model
- RD.060 - Determine Audit and Control Requirements
- RD.080 - Identify Reporting and Information Access Requirements
- BR.020 - Prepare Mapping Environment
- BR.030 - Map Business Requirements
- BR.040 - Map Business Data
- BR.050 - Conduct Integration Fit Analysis
- BR.060 - Create Information Model
- MD.050 - Create Application Extensions Functional Design
- TE.020 - Develop Unit Test Script
- TE.100 - Prepare Key Users for Testing

A preliminary business requirements scenario may have been created during the RFP response or demo aspect of the presales cycle. Validate this information to confirm that it is still current; it may prove to be valuable.

Within the Business Requirements Mapping (BR) process, and particularly during the Solution Design phase, you must refine BRSs through detailed mapping. Mapping includes the following activities:

- proving application fit through demonstration
- identifying gaps
- bridging the gaps

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	60
Application Specialist	30
Configuration Management Specialist	10
User	*

Table 2-13 Role Contribution for Gather Business Requirements

* Participation level not estimated.

Deliverable Guidelines

Use the Business Requirements Scenarios to organize detailed business requirements by business process and process owner. The BRS provides a linkage that spans business requirement decisions made during the RFP activity of the presales cycle, generation of detailed business requirements, and requirements mapping (during Solution Design).

This deliverable should address the following:

- detailed business requirements
- business processes and process owner

Deliverable Components

The Business Requirements Scenarios consist of the following components:

- Business Process Identification
- Requirements and Sources

Business Process Identification

This component describes the business process and its detailed process steps in a recipe format with references to *events, business functions, and agents*

Requirements and Sources

This component traces the need for each process step to its source.

Three other components of the Business Requirements Scenarios will be completed in the Business Requirements Mapping (BR) process during Map Business Requirements (BR.030):

- **Process Analysis** — facilitates quantitative analysis and measurement of each process step
- **Solution** — documents how the requirement is satisfied by either the application or some other solution or workaround
- **Mapping Scenario** — supports solution prototyping

Four other components of the Business Requirements Scenarios will be completed in the Business Requirements Mapping (BR) process during Test Business Solutions (BR.080):

- **Mapping Scenario Identification** — describes the unique path through the process flow diagram for a business process; a response is a set of actions that together create some unique output based on inputs and decisions
- **Data Profiles** — describes the initial business conditions necessary to test the application system; to design a process that will exhibit consistent results, properly anticipate controllable variables

- **Test Specifications** — defines the test script execution for a scenario
- **Support Profiles** — identifies the support tools required during the execution of a test

Form and Function

A BRS should provide the following information:

- identification of detailed process steps
- association of business requirements with process steps
- references to the source of requirements
- ability to support process analysis
- references to solutions: either application, manual, or other (generally this portion of the deliverable is completed during mapping within Solution Design)
- ability to support multiple prototyping mapping scenarios from each business process

The BRS should contain heading information such as:

- process control number
- priority
- owner
- process or scenario description
- triggering events

Organize detailed information by process step. Each process step should be in a recipe format, with the requirement description written in the imperative ("do this," "move that," "enter this," and so on).

The BRS serves as a test script and an operational procedure, so complete it to that level of detail. Each BRS is generally driven by one event, and has a one-to-one correspondence with a process model.

Create a line on a BRS for each process step, and also create a line whenever one of these conditions occurs:

- a transaction is entered into the system

- ownership of a task is different from that of the prior task
- a zone changes during transaction entry
- unique output is produced
- additional inputs are added or brought into the task

☞ **Attention:** Define major exception flows in a separate BRS (for example, when there are two long parallel lists of tasks on the same flow resulting in separate deliverables or outputs).

Acceptance Criteria

In order to pass a quality review, a BRS needs to meet the following criteria:

- The BRS must be developed using the approach discussed in this task.
- A qualified peer who signs off as the first reviewer must review the BRS. This signoff indicates that the BRS is accurate and consistent; for best results, the review should include the process owner as well as the customer of the process (either the requester, or the owner of the next process down the line within the business area or both).
- Major exceptions to a process must reference separate BRSs.
- The total number of tasks on a BRS should not exceed 20 in order to be considered concise, accurate, and consistent.

At the process-step level there should be a one-to-one correspondence with an elementary business function (EBF). The *rule of thumb* is that a sub-step within the EBF must be completed once started, or else rolled back, and never left in a partial state of completion (otherwise record accuracy problems may result).

Tools

Deliverable Template

Use the Business Requirements Scenarios template to create the deliverable for this task.

RD.060 - Determine Audit and Control Requirements (Core)

In this task, you identify the high level requirements and policies that affect business and system security, control, and procedures.

Deliverable

The deliverable for this task is the **Audit and Control Requirements**. These requirements help the organization plan the future procedures for maintaining and controlling the new system.

Prerequisites

You need the following input for this task:

- ❑ **Current Financial and Operating Structure (RD.010)**

Consider Audit and Control Requirements for each organization.

- ❑ **Current Business Baseline (RD.020)**

System security specifications and existing operations procedures for current processes may yield future Audit and Control Requirements information.

- ❑ **Business Requirements Scenarios (RD.050)**

Specific scenarios may have Audit and Control Requirements.

- ❑ **Existing Reference Material**

Policies and procedures may exist that contain Audit and Control Requirements that need to be perpetuated.

Optional

You may need the following input for this task:

- ❑ **Future Process Model (BP.080)**

Business policies provide a good source of Audit and Control Requirements. Use of the Function Hierarchy component, developed during this task, will point you toward relevant functions where policies may reside.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review current security, manual operations procedures, and future Business Requirements Scenarios (RD.050).	
2.	Evaluate audit specifications for division of responsibility in finance and operations.	
3.	Create a list of security requirements for the organization, operating system, application, and database to support future business processes.	Audit and Control Requirements Questionnaires
4.	Create a list of user security requirements.	User Security Requirements

No.	Task Step	Deliverable Component
5.	Review the audit and controls with business area management.	
6.	Secure acceptance of the Audit and Control Requirements.	

Table 2-14 Task Steps for Determine Audit and Control Requirements

Approach and Techniques

Audit and Control Objectives

The overall objective of Audit and Control Requirements is to consider audits and controls that will reduce or minimize the risk of those transactions being executed that place organization assets or information in jeopardy. Such transactions, if executed, should be detectable and their recurrence prevented.

It is helpful to think about both application and general controls and risks. The following tables may help frame the thinking process:

Application Risks	Application Controls
Unauthorized application access	Logical access controls
Incorrect data entry	Input controls
Rejected items resolution	Processing controls
Incorrect processing / reporting	Output controls

Table 2-15 Application Risks and Controls

General Risks	General Controls
Unauthorized system access	System access controls
Unauthorized program changes	Program change controls
Inadequate information systems operations	Organization controls
Business interruption	Disaster recovery controls

Table 2-16 General Risks and Controls

From an auditing standpoint, these are the questions you are attempting to answer:

- What was changed?
- Who changed it?
- When did they change it?
- Why did they change it?
- Who logged in?
- Why did they log in?
- What kind of monitoring takes place of transaction processing?

From the standpoint of financial controls, consider the following questions:

- What steps need to be taken to facilitate external auditors' review of procedures and controls?
- What controls are in place to prevent insider trading, fraud, non-malicious errors, and so on?
- What approval hierarchies need to be in place?
- What are the electronic commerce and web-based systems security requirements?
- What business transactions are permissible in a web-based system?

- What policies will prevent unauthorized intranet access?
- What kind of special security is required for financial and confidential information systems (such as General Ledger, Accounts Receivable, Human Resources, Electronic Funds Transfer, Electronic Data Integration, and Data Warehouse)?

Key User Interviews

Interview the financial auditors, database administrators, and system administrators for specific security and period-close acceptance criteria. Interview managers for direct report security requirements.

Financial Perspective

Schedule and interview financial auditors for security and any period-close needs. Financial auditors often have minimum information that they require when examining operations and financial data. Financial and legal policies may influence the amount of history that the organization maintains.

In many large organizations, the business requires built-in process audit checks. For example, the process should not permit an accounts payable clerk to issue checks to himself. Purchasing requesters should not be authorized to execute purchase order approval on their own requisitions. Collect these types of audit and control requirements to confirm that the design can support it.

System Administrator Perspective

As part of daily operations, database and system administrators schedule and maintain backup and recovery procedures. In case of system failure, these administrators will retrieve and restore specific data for a minimum period to continue running the business. Verify that the data retention period can adequately support the business operations.

Additionally, interview managers for control needs based on individual areas of responsibility. When defining control requirements, keep in mind that job functions may change based on the final design of business solutions. Gather information that will provide insight on data that should be segregated based on business function, controlled because of sensitive information, or prohibited due to required authorization.

Other system administration topics that fall under audit and control are:

- new accounts and passwords
- transaction monitoring
- log and output file purge tracking

Linkage to Other Tasks

The Audit and Control Requirements are an input to the following tasks:

- BR.030 - Map Business Requirements
- BR.060 - Create Information Model
- BR.110- Design Security Profiles

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	100
Business Line Manager	*
User	*

Table 2-17 Role Contribution for Determine Audit and Control Requirements

* Participation level not estimated.

Deliverable Guidelines

Use the Audit and Control Requirements to plan the future procedures for maintaining and controlling the new system. To capture audit and control requirements, define the types of fundamental audits and control you need to run your business. Remember to include functional and manual audits and controls, as well as automated background controls. By categorizing these types, you can further describe the needs of each area.

This deliverable should address the following:

- Who or what is dictating control?
- Why do certain data, functions, or other elements require control?
- Why do certain data, functions, or other elements require audit?
- Who or what will perform these audits or controls?
- Is timing or sequencing critical to the requirement?
- What is the level and type of control (field, form, process, host, and so on)?
- What impact does audit and controls create for normal business operations?

Deliverable Components

The Audit and Control Requirements consist of the following components:

- Audit and Control Requirements Questionnaires
- User Security Requirements

Audit and Control Requirements Questionnaires

This component provides a set of self-assessment questions to:

- facilitate compliance with generally accepted application audit and control objectives
- serve as a checklist of issues for consideration during application setup
- assist the functional leads or application owners in defining audit and control requirements

The seeded questions should be tailored to your enterprise by adding additional ones or deleting those that do not apply. The specific questionnaire categories follow:

- Application Owner and User Security Requirements
- Separation of Duties

- Sensitive Programs
- Logical Access Controls
- Vital Records
- Change Control
- Transaction Origination Controls
- Communication Controls
- Processing Controls
- Output Controls
- Documentation

Application Owner and User Security Requirements

The application owner is generally responsible for accomplishing the following for each application:

- identifying sensitive programs and establishing controls
- classifying data
- making sure controls are in force
- establishing adequacy of security testing

Separation of Duties

The duties of the developers, computer operations personnel, and user groups are reviewed to verify that problems do not exist with the separation of duties. In other words, no one individual should control activities within a process in a way that permits errors of omission, fraud, or theft.

Sensitive Programs

Sensitive programs are application programs whose improper use could cause serious financial loss that normal application controls cannot prevent.

When considering application security, the names of sensitive programs, identification and protection of application data files, and the arrangements for archiving and recovering vital records should be examined. The proper identification of sensitive programs is critical to an effective control procedure. The sensitive program control concept is valuable only when a relatively small number of programs are identified. Once identified, controls must be in place to prevent unauthorized modification or use.

Logical Access Controls

While a site procedure generally controls sensitive programs, application data has a formal access control mechanism. Through this mechanism, the application owner can authorize and control who has access to data (by user ID) and how the data can be accessed (read, update, and alter).

Vital Records

Vital records are essential to the continuing operation of the application system. You must identify and safeguard all such records.

Change Control

A change control system should be in place to evaluate, justify, and control changes to programs or procedures, including those for backlog management, library management, and system testing.

Transaction Origination Controls

Accuracy and completeness of information should be confirmed before it enters the application system. Owner and user responsibilities include activities up to the point of converting data to machine readable format.

Communication Controls

The integrity of data should be confirmed as it passes through the communication lines from the input device to the reception devices. This covers hardware controls, message transmission, message accountability, and reception controls.

Processing Controls

Controls for entry of data into the computer application system should be applied to help maintain accuracy and completeness of data during processing.

Output Controls

Output controls help maintain the integrity of the output data from conclusion of computer processing to delivery to the user.

Documentation

The quality and completeness of existing program documentation determines the degree to which programs could be efficiently reconstructed if they were destroyed.

User Security Requirements

The process flow, transaction, and department that relate to a given audit or control should be identified. In addition, the level of control (field, form, and process) should be identified as well.

Tools

Deliverable Template

Use the Audit and Control Requirements template to create the deliverable for this task.

RD.070 - Identify Business Availability Requirements (Core)

In this task, you identify the business operating requirements that the operating functions and information systems functions agree are necessary to support the business processes.

This task also helps to summarize contingency situations the organization must guard against and documents the approach to managing delay, failure, or other unexpected results that could imperil daily business operations.

Deliverable

The deliverable for this task is the **Business Availability Requirements**. It provides a statement of what the business needs are, in terms of operating hours, system uptime, response, and turnaround time.

Prerequisites

You need the following input for this task:

- ❑ **Future Process Model (BP.080)**

The Future Process Model describes the processes that the business is required to perform in response to the events that drive it. The Future Process Model determines the requirements for business availability and implies systems availability to support those requirements.

Optional

You may need the following input for this task:

- ❑ **Current Process Model (BP.040)**

The Current Process Model provides an understanding of the existing systems that the new system will replace. The Current Process Model may contain the current contingency procedures and can be used as a checkpoint to verify that business availability requirements are not overlooked. If Develop Current Process Model was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

❑ **Current Business Contingency Plan and Procedures**

There may be an existing contingency plan that defines how to respond to business interruptions. If so, the Current Business Contingency Plan may serve as a guide to start the redefinition of contingency plans for the new software and technical architecture.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify the critical window of time in which interrupted business processes must be reestablished and normal operations must resume.	Business Availability Requirements
2.	Identify contingency situations and causes of business interruptions.	
3.	Describe the method for supporting contingency situations.	Business Contingency Requirements
4.	Secure acceptance of the Business Availability Requirements.	

Table 2-18 Task Steps for Identify Business Availability Requirements

Approach and Techniques

What constitutes business availability is often regarded as the sole responsibility of the information systems organization. Because the information systems organization is responsible for the technical behavior of the business systems and the technology determines how fault-tolerant the systems are, the assumption may be made that the information systems organization controls the business availability. This conventional reasoning does not give adequate representation to the business operations and user communities who may have a different view of what availability is necessary in order to perform critical business functions and provide customer satisfaction.

This task is designed to facilitate a discussion between the operational and information systems parts of the business in order to arrive at a consensus on what is an acceptable percentage system uptime and the contingency measures to implement during a system outage. The system architect, database administrator, and system administrator should represent the information systems organization. Key business analysts and lead users should represent the business operations and user communities in the discussion. This task should have the following results:

- a detailed description of the business availability requirements, taking into account any variations in criticality of different applications, application modules, organizations, and so on
- a binding service-level agreement that states the minimum systems-level availability
- a discussion of the contingency tools and procedures (if any) that will be used to provide availability during periods of system outage

For a technical view of a service-level agreement, see:

Reference: Millsap, Cary V., *Designing Your System to Meet Your Requirements,* OAUG Conference Proceedings, Fall 95.

Acceptable Downtime

Find out what the maximum acceptable system downtime is for daily and weekly backup procedures and the maximum acceptable recovery time in the event of a disaster. These factors will affect both the backup and recovery procedures and the type of hardware and software required to support the requirements.

Current Business Contingency Procedures

Use the current contingency procedures as a basis for defining the contingency requirements for the future business processes. Contingency requirements outline the specific situations that cause abnormal behavior or complete business process failure.

Cost, Performance, and Availability Equation

When designing your system, balance cost, performance, availability, and the means by which different system failures will be handled. The system users may favor transparent solutions to systems failure that use advanced technology, but this may also be prohibitively expensive. Consider the cost of any technical solutions realistically, including a cost-benefit assessment of whether the business really needs the extra safeguards offered by an advanced technical architecture. The term 24x7x52 is often used to describe system availability requirements that provide 99.9 percent availability. However, few businesses or individual business functions truly need this extreme level of system availability — especially if it will multiply the costs of the hardware for the new system and the soft costs of system management by a factor of two, three, or more.

Linkage to Other Tasks

The Business Availability Requirements are an input to the following tasks:

- TA.050 - Define System Availability Strategy
- PM.010 - Define Transition Strategy

This task affects the system management procedures as requirements that verify acceptable system support of ongoing operations. Those procedures are defined within the System Availability Strategy (TA.050).

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	95
Technical Analyst	5
Business Line Manager	*
User	*

Table 2-19 Role Contribution for Identify Business Availability Requirements

* Participation level not estimated.

Deliverable Guidelines

Use the Business Availability Requirements to define the operating requirements of the business process that will need to be supported by the information systems function, as well as the maximum downtime that can be tolerated with the contingency method defined.

This deliverable should address the following:

- business operating requirements
- contingency requirements

Deliverable Components

The Business Availability Requirements consist of the following components:

- Business Availability Requirements
- Business Contingency Requirements

Business Availability Requirements

This component provides the operating requirements that support the business transactions into the technical architecture process. The Business Availability Requirements catalog the transactions that have the highest volume or significance in the organization and identify the maximum allowable window for the application system to be down. They also help with preparation for abnormal business and system conditions.

The series of questions can be used to capture and formalize feedback:

- from business process owners to systems management
- from systems management to business process owners

The questions address such issues as the operational schedule, operating window, volume and frequency of transactions, and backup schedules. The Business Availability Requirements component formalizes the understanding of both groups of people in order to optimize the technical support of business activities.

Business Contingency Requirements

This component helps to determine contingency requirements by considering the root cause of situations such as organization-wide power failures, local interference, and physical limitations to accessing data. From these situations, your requirements can be addressed in the following areas:

- business systems procedures
- system downtime for normal, daily, and weekly backups
- computing and networking hardware and peripherals
- disaster recovery

The information gathered here is used in Define System Availability Strategy (TA.050).

To complement the system contingency requirements, the root cause of business failure or interruptions also needs to be assessed. In the event of system failure, it is necessary to continue with normal business. This component describes the mechanisms, subsystems, and procedures each business function needs and calculates the time needed to sustain these temporary processes. Considerations for defining contingency requirements include the following:

- paper forms
- PC tools
- related files and programs
- alternate system components

This component is used to formalize a contingency plan for each critical process in the business. Both internal and external transactions are identified. The paper forms that can be used as alternate source documents during the contingency period are listed, and other tools or systems that could be used are identified. This enables you to understand which business processes have documented procedures in place to cover the contingency.

In addition, this component describes contingency situations that need to be anticipated and prioritize them by severity. The nature of each contingency situation will help you determine the robustness of the contingency plan you require for each possible situation.

Tools

Deliverable Template

Use the Business Availability Requirements template to create the deliverable for this task.

RD.080 - Identify Reporting and Information Access Requirements (Core)

In this task, you identify the organization's future reporting requirements.

Deliverable

The deliverable for this task is **Master Report Tracking List**. It provides a consolidated master list of reporting requirements.

Prerequisites

You need the following input for this task:

- ❑ **Current Financial and Operating Structure (RD.010)**

Reporting requirements that are associated with the existing environment are identified in the financial and management reporting environment component of the Current Financial and Operating Structure.

- ❑ **Business Requirements Scenarios (RD.050)**

Reporting requirements that are linked to future process steps may be identified in the Business Requirements Scenarios.

- ❑ **Current System Reports**

The collection of Current System Reports will be the starting point for the baseline of the present system reporting output.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review current reporting materials that may enhance the team's understanding of the current state reporting environment.	
2.	Determine an approach for collecting report requirements.	
3.	Design the Master Report Tracking List.	Master Report Tracking List
4.	Update the Master Report Tracking List with information from the current reporting materials.	Master Report Tracking List
5.	Update the Master Report Tracking List with information from the BRS documents.	Master Report Tracking List
6.	Identify critical reporting issues and document them.	
7.	Forward a copy of the Master Report Tracking List to users for approval.	

Table 2-20 Task Steps for Identify Reporting and Information Access Requirements

Approach and Techniques

The Master Report Tracking List is used as the primary repository for all information collected about a report requirement. It should contain system and report name, business purpose, frequency, priority, user name, and contact information.

Report Requirements Collection Approach and Techniques

Possible methods to determine report requirements include:

- list report on the current system
- future report requirements from the Business Requirements Scenarios (RD.050)
- user survey

You may use one or a combination of these methods.

Reports on Current System

The quickest way to catalog current system reports may be to get a composite listing of all current reports from the information systems department. Analyze the reports in terms of frequency, distribution, and content.

Report Requirements from Business Requirements Scenarios

Many Business Requirements Scenarios (RD.050) may generate reports as outputs of the process. Each report generated will then link to a future business process and show functional ownership as a result.

User Survey

For large organizations where it may be difficult for the project team to determine all the critical report requirements, some form of a user survey may be necessary. If the list produced after extracting requirements from the Business Requirements Scenarios (RD.050) and researching system report files is not satisfactory, the team may need to survey the users.

☞ **Attention:** The goal of the survey is to collect business critical report requirements that may have been overlooked—not to list every report that may have been run regardless of whether or not anyone used it. Emphasize to the users that only critical business requirements are important, and that their manager must approve their list before mapping begins. Analyzing, mapping, and building reports is very expensive.

When designing a survey approach, consider the following elements:

- **Develop a Survey Rollout Plan** — Detail how you plan to distribute and collect the reporting requirements within the organization. If the organization is small, you can send a survey to each individual. However, as organizations increase in size it becomes more and more difficult to distribute, track, and summarize the survey information. In this case, the best approach is to appoint a representative user to be responsible for gathering all requirements and representing their department during analysis, construction, test, and delivery. This will reduce the number of individuals the reporting team must interact with. Make sure that the representative user understands the responsibility for their group's requirements. Also, when distributing the survey in a large organization, the best approach is to distribute it through the management chain. As managers pass the survey through their organizations, they can emphasize the importance of full participation.
- **Develop Survey Organization Chart** — This chart shows how the survey will be distributed throughout the organization. Start by separating the organization by its logical units. Identify the manager in charge of each unit, who then identifies the individuals that should represent each organization. List the manager, the representative user, and the group they represent. Use this chart to verify that all organizational areas have been covered.
- **Prepare the Survey Packet** — This packet should include a survey cover letter introducing and describing the survey process, survey instructions, and the survey itself. Examples of survey data are included in the guidelines section.

☞ **Attention:** It is extremely important, that users specify a priority for each requirement. This priority helps determine which requirements are most important, and therefore, should be considered first. The priority can be a combination priority and ranking, such as A1, A2, where A reports are needed at *go live*, and A1 is the most important *go-live* report.

Once the survey packet has been prepared, distribute the packets through management. Make sure that you emphasize the completed packet due date and that you track all returns.

Survey Elements

You should include the following elements on a survey:

- **Report Filename/System Filename** — Report names may be similar and become interchangeable. The filename is a unique name representation. Users may need help from their information systems representative to determine filenames.
- **Report Name/Report Title** (optional)
- **Business Purpose** — What business purpose does this report satisfy? Use this information to determine whether or not the report is needed in the future processes.
- **User Priority** — How important is this report to the user? Build some logic into this field so it includes both priority and ranking (for example, A1, A2, A3, B1, B2, B3, C1, C2, C3). Suggested priorities are:
  - A: highest priority, needed by *go-live* date
  - B: needed by first month-end
  - C: needed by first quarter-end
  - D: needed by year-end
  - E: nice to have
- **User Run Frequency** (optional) — How often is the report run? This is useful when setting up daily or night batch jobs.

- **User Name** — Who owns this requirement? You may have several people requesting the same report. Track each request as a new unique request. This identifies which reports are common across divisions, teams, departments, and so on. A report used by multiple divisions has a higher implied priority than a report used by one person. If the multi-divisional report is not replaced, it will have a greater negative impact. If the requirement is originated in a BRS document, enter the BRS name.
- **User Phone Extension** (optional)
- **User Email** (optional)
- **User Position** — This information is useful when determining the future reports associated to a position.
- **User Division** — The division is useful when consolidating reports.

Master Report Tracking List Development

The Master Report Tracking List is the most important document next to the completed report. From this document you are able to do the following:

- list all report requirements and their current status
- compile a current customization list
- determine the final result of each requirement
- provide users with a list mapping their old report to the new report
- calculate development costs
- track back to the original report (if necessary) to get more detail

The size of the reporting project will determine the type of tool. For small projects, you can track the information in a spreadsheet. For larger projects, a database tool is necessary. For very large projects, maintaining the Master Report Tracking List is a full-time position.

To make tracking easier, you can divide the elements of this tracking document into the following files and link them together through the report tracking number:

- survey master requirement list and Business Requirements Scenarios (RD.050)

- development
- user delivery information

Keep the Master Report Tracking List in the same directory as other project tracking documents. Because the list may actually be composed of smaller files, a separate directory may be necessary.

The following calculations show how to use the Master Report Tracking List for status reporting. If you are using a spreadsheet, this may require sorting the spreadsheet.

- total report requirements = total number of entries on the tracking list
- total unique requirements = (total number of entries on the tracking list) - duplicates [sort by assessment]
- customization list = (total builds + total modifies) [sort by assessment]
- total specs completed = sort by spec done = y
- development backlog = sort by assessment and with a subsort by development = n
- map old report to new report = sort spreadsheet by future report name
- all reports for a specific user = sort by user name

Master Report Tracking List Update

Collect all BRSs (RD.050), surveys, interviews, and so on, that contain report requirement information and transfer them to the Master Report Tracking List. It is important to catalog requirements as quickly as possible so mapping can begin. If the requirement was gathered from a BRS, it is very important to include the future process flow number in the Master Report Tracking List. Any requirement categorized as a build cannot be built without a valid future process flow number. Also, insert the BRS name as the user name. This will make it easier to identify all requirements coming from a BRS.

Master Report Tracking List Distribution to Users

It is important to forward a copy of the reporting requirements documented to each group for approval. The user community should review the resulting document, determining that:

- all requirements have been collected
- information gathered about each requirement is correct
- all priorities have been assigned and are correct

Linkage to Other Tasks

The Master Report Tracking List is an input to the following tasks:

- BR.060 - Create Information Model
- BR.070 - Conduct Reporting Fit Analysis
- BR.110 - Design Security Profiles
- TA.060 - Define Reporting and Information Access Strategy

Role Contribution

The percentage of total task time required for each role follows:

Role	**%**
Business Analyst	100
Business Line Manager	*
User	*

Table 2-21 Role Contribution for Identify Reporting and Information Access Requirements

* Participation level not estimated.

Deliverable Guidelines

Use the Master Report Tracking List reports to define typical outputs of business processes that are used as tools in support of business process operations.

This deliverable should address the following:

- report requirements
- business purpose
- frequency
- priority

Deliverable Components

The Master Report Tracking List consists of the following component:

- Master Report Tracking List

Master Report Tracking List

This component lists all of the report requirements and their current status. The Master Report Tracking List is divided into three main sections:

- General Tracking
- Analysis Activity
- Mapping

Tools

Deliverable Template

Use the Master Report Tracking List template to create the deliverable for this task.

CHAPTER

3

Business Requirements Mapping (BR)

This chapter describes the Business Requirements Mapping process.

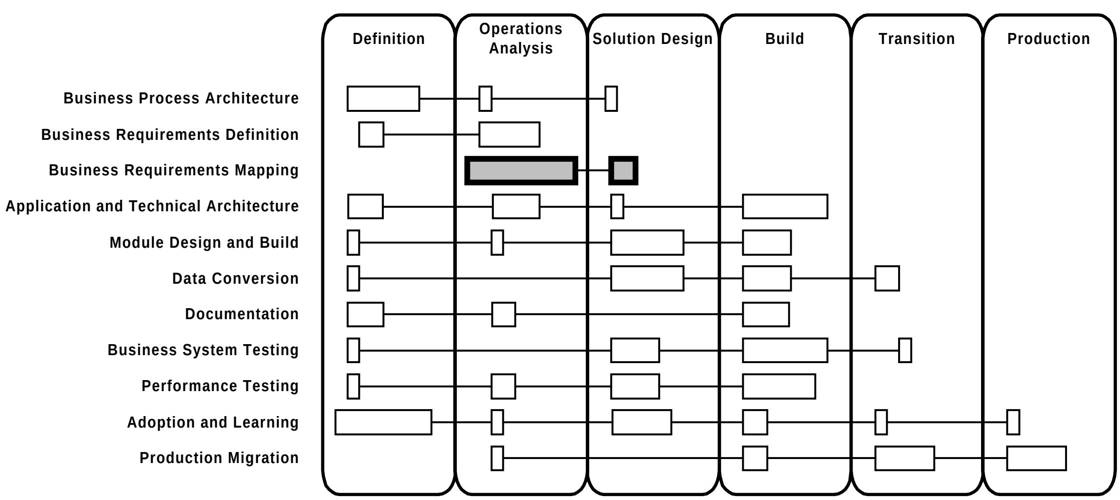


Figure 3-1 Business Requirements Mapping Context

Process Flow

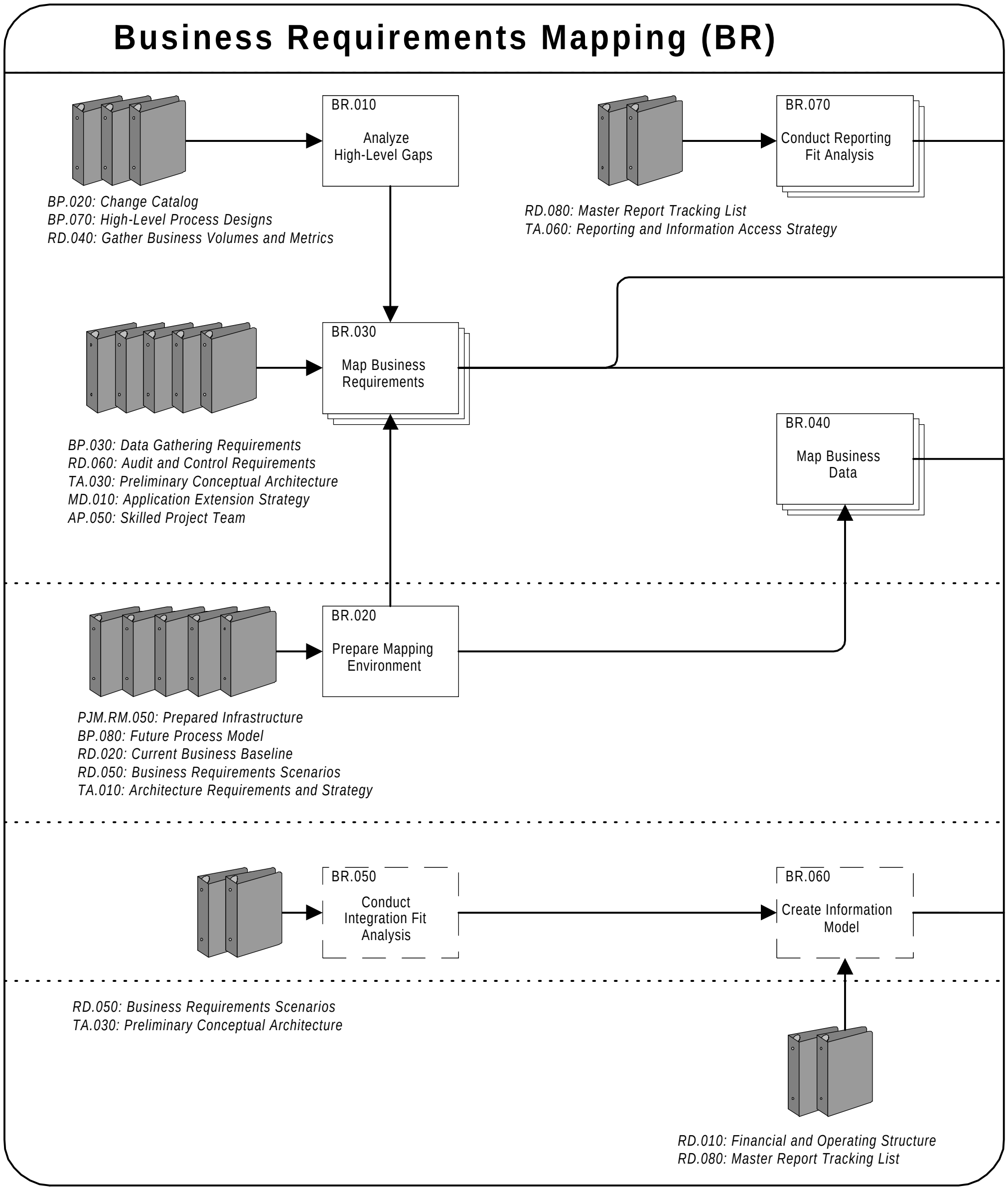


Figure 3-2 Business Requirements Mapping Process Flow Diagram

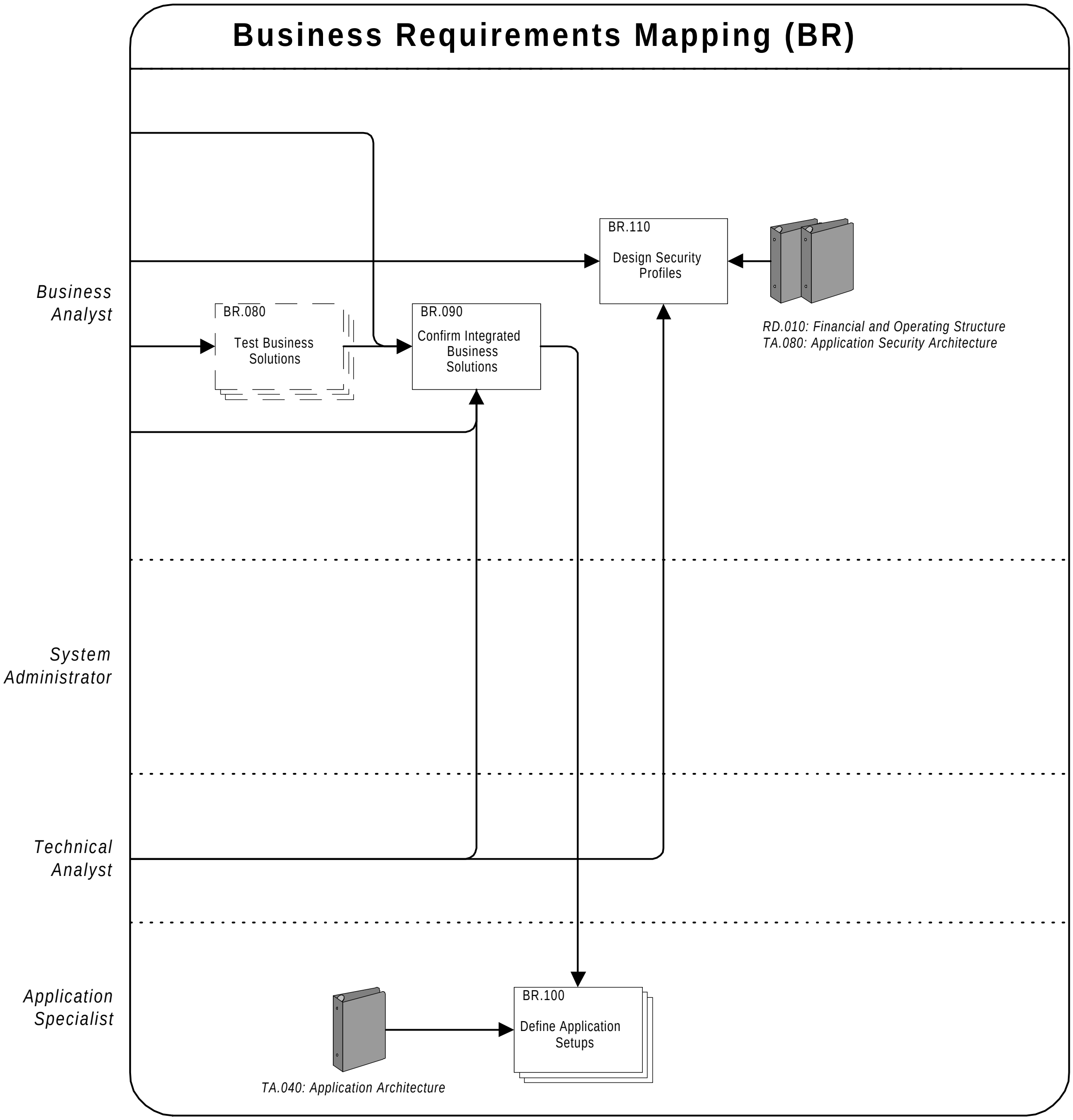


Figure 3-2 Business Requirements Mapping Process Flow Diagram (cont.)

Approach

The objective of Business Requirements Mapping is to create an acceptable and feasible solution, that is proven, and becomes well documented.

The objective of mapping is to ascertain the fitness for use of application features in satisfying detailed business requirements expressed at a business process step level — and then to make a formal disposition of any and all identified gaps. Each business process step relates to one or more application screens, reports, and inquiries. As a starting point, use the approved Business Requirements Scenarios (RD.050) for a given business process. Usually this means that the mapping team has a formal statement of requirements and their sources.

Business Requirements Mapping is closely related to Business Process Architecture (BP) and Business Requirements Definition (RD). The primary goal of the implementation process is to create a set of business processes that represent how the business will operate after implementation. Mapping helps make sure that the new application system supports these new business processes.

Business Requirements Mapping encompasses the following areas:

- mapping
- business system testing
- application setups

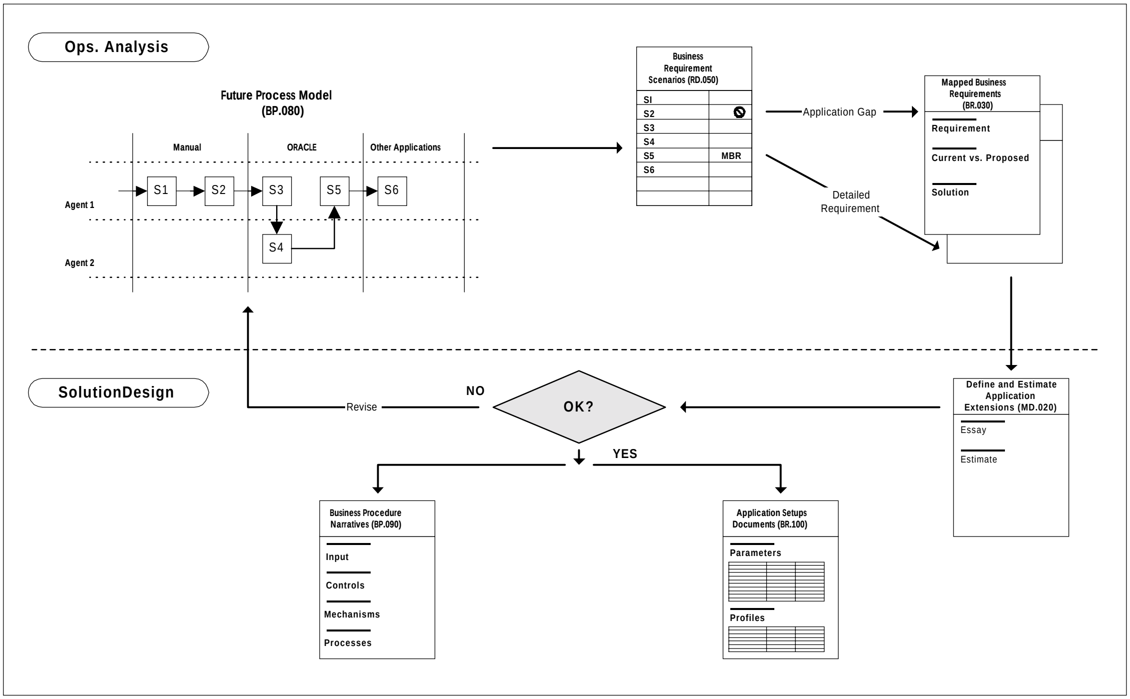


Figure 3-3 Relationship Between Phases and Deliverables

Mapping

Mapping is an iterative approach with the following objectives:

- prove business process designs through demonstration
- identify gaps in the application
- propose feasible bridges to gaps

Areas that are necessary to map include:

- business requirements attached to business process steps
- report requirements
- business data requirements

Business System Testing

Consider the business system finalized when the following have been completed:

- participation of key users in test design
- documentation of tests and results
- agreement that the application and other tools perform to process design specifications by management
- integration of process designs

Application Setups

Application setups include both static data, such as common codes and tables, and system configuration options. It is important to capture applications setups and security decisions in mapping. The approval of process designs allows use of the resulting configuration in the training and production environments.

Tasks and Deliverables

The tasks and deliverables for this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
BR.010	Analyze High-Level Gaps	High-Level Gap Analysis	Always	SI
BR.020	Prepare Mapping Environment	Configured Mapping Environment	Always	SI
BR.030	Map Business Requirements	Mapped Business Requirements	Always	MI
BR.040	Map Business Data	Mapped Business Data	Always	MI
BR.050	Conduct Integration Fit Analysis	Integration Fit Analysis	Project includes *interfaces* with external systems	SI
BR.060	Create Information Model	Information Model	Project includes *process change, interfaces* with external systems, or *complex reporting*	SI
BR.070	Conduct Reporting Fit Analysis	Master Report Tracking List	Always	MI
BR.080	Test Business Solutions	Business Mapping Test Results	Project includes *process change* or is of *medium or high complexity*.	MI
BR.090	Confirm Integrated Business Solutions	Confirmed Business Solutions	Always	SI
BR.100	Define Application Setups	Application Setup Documents	Always	MI
BR.110	Design Security Profiles	Security Profiles	Always	SI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 3-1 Business Requirements Mapping Tasks and Deliverables

Objectives

The objectives of Business Requirements Mapping are:

- Establish application fit to business requirements.
- Identify gaps and propose initial alternatives; propose feasible bridges to gaps.
- Prove designs through demonstration.
- Trace business requirements to application solutions.
- Create a model that visually depicts information flows in the business in order to specify data use within and across your organization. Define the application configuration you need to support the new system.
- Define detailed setup parameters.

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
High-Level Gap Analysis	Compares the processes as envisioned in the Develop High-Level Process Designs (BP.070) with the processes supported by Oracle Applications.
Configured Mapping Environment	New or existing application environments prepared for exclusive use during mapping activities by mapping teams.

Deliverable	Description
Mapped Business Requirements	Detailed business requirements written in business language and associated with business processes; analysis and comparison of the current system for a business requirement to the proposed system; details for the type and nature of the solution in a descriptive format.
Mapped Business Data	Verification that the underlying target application modules, business objects, and attributes will support business processes.
Integration Fit Analysis	Statement of fit and gaps for integration points between existing applications and the new application.
Information Model	A model that visually depicts information flows in the business between business organizations and applications, also depicts access to key process and organization information for reporting or security purposes.
Master Report Tracking List	A working document that supports the analysis and mapping of report requirement to future business processes and standard application reports. It contains the final disposition of every report requirement.

Deliverable	Description
Business Mapping Test Results	A plan for and record of business system testing, including: business processes involved in the test, the necessary business conditions to test the application system, definition of test script execution, support tools required during execution of the test, and a record of test actions.
Confirmed Business Solutions	Agreement by management that proposed integrated business system cover business objectives.
Application Setup Documents	Definition of detailed setup parameters that has been proven to support the system.
Security Profiles	List of role-based security specifications necessary to help maintain good controls and transaction access.

Table 3-2 Business Requirements Mapping Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Application Specialist	Provide application functionality and guidance. Support and provide interpretation for tools, templates, and methods.

Role	Responsibility
Business Analyst	Conduct mapping and working sessions. Map detailed business requirements to Oracle Applications and propose alternatives to identified gaps.
Business Line Manager	Participate in mapping and working sessions, review and approve proposed alternatives.
Client Project Manager	Promote availability and commitment of organization personnel and availability of hardware, software, and facilities. Establish commitment to the project from the appropriate users. Confirm that they review and formally approve deliverables for which they will be responsible.
Configuration Management Specialist	Provide application and reporting setup expertise. Map detailed business requirements to Oracle Applications and identify gaps.
Database Administrator	Install and configure the mapping database. Make sure all users have access to the database.
Organizational Development Specialist	Provide expertise in the human and organizational facets of change.
System Administrator	Install and configure hardware and operating system. Install and configure application code. Confirm that login IDs exist for the mapping team.
System Architect	Provide input and advice on the architectural consequences of particular gaps and alternatives.

Role	Responsibility
Technical Analyst	Propose technical alternatives to gaps. Develop data profiles.
Technical Writer	Provide writing standards, models, and guidelines. Write non-business content.
User	Participate in working sessions, review content of business requirement scenarios, and help identify gaps and alternatives.

Table 3-3 Business Requirements Mapping Key Responsibilities

Critical Success Factors

The critical success factors of Business Requirements Mapping are as follows:

- consistency of team composition across process design, mapping, narrative writing, and approval activities
- consistency of team approach across business areas or process groups
- adequate integration of processes across design and mapping teams
- project team understanding of primary and related applications and industry practices
- a committed project sponsor who will maintain a high level of commitment across the team
- active involvement and support of management and team access to business area experts
- full access to information about those business areas, their processes, and information generation and use
- clear and concise visualization regarding how information will flow across and be owned by organizations, functions, and applications
- effective management of issues, including their timely resolution
- proper sign off throughout all stages

BR.010 - Analyze High-Level Gaps (Core)

In this task, you compare the process as envisioned in the High-Level Process Designs (BP.070) with the processes supported by Oracle Applications.

The differences (gaps) revealed by this analysis need to be resolved by producing alternatives that balance change in the application against change in processes and organization.

Δ If your project includes *process change*, you should create additional deliverable components (updated change catalog, process performance differences, cost differences, differences in working practices, and staffing differences).

Deliverable

The deliverable for this task is the **High-Level Gap Analysis**. It compares the processes as envisioned in the High-Level Process Designs (BP.070) with the processes supported by Oracle Applications.

Prerequisites

You need the following input for this task:

❑ **Change Catalog (BP.020)**

This deliverable provides the context for identifying potential changes to applications and the organization and processes. If Catalog and Analyze Potential Changes was not performed, this deliverable will not exist. (See the task description for BP.020 for more information on when this task should be performed.)

❑ **High-Level Process Designs (BP.070)**

The process design makes an initial assessment of the costs, benefits, and risks of changing the application as well as the organization and its processes.

❑ **Business Volumes and Metrics (RD.040)**

When *process change* is applicable, use the business process measures component to determine the extent of improvements (stretch goals).

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
Δ 1.	Update the Change Catalog (BP.030) with the latest information about potential changes to the application or to the organization (if the project includes process change).	Updated Change Catalog
2.	Compare and evaluate the costs, benefits, and risks of the alternatives.	Gap Analysis
3.	Develop new solutions to determine an optimal set of options for the client.	Solution Options
4.	Document the proposed changes to the application and other software resulting from the selected alternatives at a high-level.	Application Modifications and Extensions
Δ 5.	Evaluate differences between current process performance and Solution Options (if the project includes *process change*).	Process Performance Differences

Δ 6.	Determine differences in operating costs between the current processes and Solution Options (if the project includes *process change*).	Operating Cost Differences
Δ 7.	Evaluate how practices would change if the Solution Options were implemented (if the project includes *process change*).	Differences in Working Practices
Δ 8.	Evaluate changes to staff numbers and costs, roles, and skills if the Solution Options were implemented (if the project includes *process change*).	Staffing Differences

Table 3-4 Task Steps for Analyze High-level Gaps

Approach and Techniques

The High-Level Gap Analysis is a large task, and deliverable components are produced in several steps.

Δ Process Change

When *process change* is applicable, we recommend that you update the Change Catalog (BP.020) to incorporate all proposed changes resulting from the latest high-level process design. In addition, produce additional deliverable components to document the gap between the organization's existing process and operations and those that would result from the solution options.

Linkage to Other Tasks

The High-Level Gap Analysis is an input to the following task:

- BR.030 - Map Business Requirements
- TA.100 - Define and Propose Architecture Subsystems

Role Contribution

The percentage of total task time required for each role follows:

Core Task Steps

Role	%
Application Specialist	50
Business Analyst	40
Organizational Development Specialist	10

Table 3-5 Role Contribution for Core Task Steps of Analyze High-Level Gaps

Optional Task Steps

If your project is addressing the optional task steps for this task, you need to allocate additional time and the following resources:

Role	%
Business Analyst	60
Application Specialist	30
Organizational Development Specialist	10

Table 3-6 Role Contribution for Optional Task Steps of Analyze High-Level Gaps

Deliverable Guidelines

Use the High-Level Gap Analysis to describe the differences (gaps) between processes as envisioned in the High-Level Process Designs (BP.070) and the processes supported by Oracle Applications.

This deliverable should address the following:

- a solution in terms of desired process and modifications to the applications
- a list of potential changes to the applications needed to achieve this requirement
- a list of changes to the *ideal* process design that was input to this task from Develop High-Level Designs (BP.070)
- an evaluation of key changes to the organization's process and operations compared with current performance and staffing

If this task results in a large number of changes to the process design, it is desirable to return to Develop High-Level Process Designs and update the process designs with the new information.

Deliverable Components

The High-Level Gap Analysis consists of the following components:

- Gap Analysis
- Solution Options
- Application Modifications and Extensions
- Updated Change Catalog
- Process Performance Differences
- Operating Cost Differences
- Differences in Working Practices
- Staffing Differences

Gap Analysis

This component documents your comparison of the envisioned process design with the standard process or processes supported by Oracle Applications.

Solution Options

This component documents the best efforts of your team to narrow gaps by providing a balance between changing the process design and modifying or extending the application.

Application Modifications and Extensions

This component documents the proposed application changes needed for the solution options at a very high level. (It is not intended to be a design specification.) Whenever possible, standard business modeling, such as Oracle Business Models (OBM), should be used to represent the processes supported by Oracle Applications.

Updated Change Catalog

This component updates the Change Catalog (BP.030) with the latest information about potential changes to the application or to the organization.

Process Performance Differences

This component documents the differences between existing performance and that expected with the solution options — the performance measures and stretch goals that were produced in the High-Level Process Vision (BP.060) should be used.

Operating Cost Differences

This component focuses on operating cost differences between the existing process and those supported by the solution options.

Differences in Working Practices

This component lists the changes that will result in how people carry out their jobs under the new business process.

Staffing Differences

This component lists the team's initial assessment of changes to staffing, including numbers, staff costs, role changes, and skills needed.

Tools

Deliverable Template

Use the High-Level Gap Analysis template to create the deliverable for this task.

BR.020 - Prepare Mapping Environment (Core)

In this task, you either install a new application, or prepare an existing application environment for detailed mapping activities.

Deliverable

The deliverable for this task is the **Configured Mapping Environment**. It is a working applications system environment that includes all of the servers, applications, and development tools required to run detailed mapping activities.

Prerequisites

You need the following input for this task:

- ❑ **Prepared Infrastructure (PJM.RM.050)**

Follow this approach to maintain consistency with resource allocations and schedules.

- ❑ **Future Process Model (BP.080)**

Process flow diagrams provide a graphical view of the employment of transactions during mapping, and therefore, mapping environment requirements.

- ❑ **Current Business Baseline (RD.020)**

An understanding of current operations is useful when setting up a mapping environment.

- ❑ **Business Requirements Scenarios (RD.050)**

Business Requirements Scenarios (BRS) help establish the scope of environment configuration and boundaries with environments used by other mapping teams.

❑ **Architecture Requirements and Strategy(TA.010)**

This Architecture Requirements and Strategy defines the application environment that can be used for mapping.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review Architecture Requirements and Strategy (TA.010) to understand the strategy for deployment of project environments in general, and the mapping environment in particular.	
2.	Update the introduction component to reflect the mapping environment hosting strategy per the Architecture Requirements and Strategy (TA.010).	Introduction
3.	Update all of the tables in the Database Tier component to reflect the configuration of all servers within the database tier.	Database Tier
4.	Update all of the tables in the Applications Tier component to reflect the configuration of all servers within the applications tier.	Applications Tier
5.	Update the Desktop Client Tier sub-components to reflect the configuration of desktop components.	Desktop Client Tier

No.	Task Step	Deliverable Component
6.	List any other software applications needed to support mapping.	Other Applications
7.	Set up the mapping environment.	
8.	Enter the data for baseline mapping purposes.	
9.	Install the conversion programs and automated conversion tools.	

Table 3-7 Task Steps for Prepare Mapping Environment

Approach and Techniques

The Configured Mapping Environment prepares an application environment for detailed mapping activities. The importance of proper environment planning for the implementation cannot be stressed enough. Not having the right environments available at the right time is one of the most common causes of substantial failure to meet milestones. Try to use an environment that accurately reflects the organization’s business to the extent known at the time. Avoid using cluttered environments like demonstration environments.

Training or Demonstration Environment

If space and time do not allow for a separate environment for mapping, you may have no other recourse than to use either the training or demonstration environment. Be aware of the limitations in using these environments for mapping. The application-level parameters and the setup in the model database may not represent the organization’s business. Eventually, these environments require refreshing. In addition, each major process group or business area may require its own environment in order to preserve the integrity of mapping and business system testing.

If possible, perform this task once per mapping team, and then refresh back to a stable starting point as needed, depending on mapping and testing progress, findings, and status.

Notes from Training Sessions

Although performance of this task is often parallel with project team training, the training may reveal the need for implementation of setup requirements in the mapping environment. Gather any notes from the training classes or meet with project team members after each applications course to identify specific decisions.

Mapping Environment Configuration

Configure the mapping environment based on decisions from the Operations Analysis phase and the Adoption and Learning process. Use Oracle's Application Implementation Wizard (AIW) as a guide during the set up for each application. Each coded question on the Current Business Baseline (RD.020) questionnaire is categorized by either *process, performance, set up, or metrics*. You can extract all setup responses and use this information as the baseline setup for mapping. You can use the Application Setup Documents (BR.100) to document your setup decisions. The goal is to prepare the environment with the basic setups that allow the project team to begin the mapping and prototyping activities quickly. It is not necessary to initially define all setup options or match the organization requirements exactly — do this after performing multiple iterations of mapping and business systems testing.

Verify that each project team has an account on the system and an applications user ID.

Suggestion: For mapping purposes, the team should not have restricted menus; all screens, reports, and programs should be accessible to each application user. Full exposure to all application features enables the organization to explore areas that would not normally be under their control.

Mapping Environment for Multiple Sites Replication

After one site has configured the mapping environment, capture or export the configuration of the application database and import the data to other site environments. For more information on duplicating environments, see the Physical Resource Plan (PJM.RM.040).

Linkage to Other Tasks

The Configured Mapping Environment is an input to the following tasks:

- RD.030 - Establish Process and Mapping Summary
- BR.030 - Map Business Requirements
- BR.040 - Map Business Data
- BR.080 - Test Business Solutions

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	60
Database Administrator	30
Business Analyst	10
Client Project Manager	*

Table 3-8 Role Contribution for Prepare Mapping Environment

* Participation level not estimated.

Deliverable Guidelines

The Configured Mapping Environment contains the minimum configuration required to support downstream mapping tasks.

This deliverable should address the following:

- all application parameters to enable transactions
- enough business data to demonstrate application features effectively
- access to definition and setup screens to appropriate users

- any links to nonstandard application or custom systems (if applicable)
- reporting and data query tools available in the mapping environment to verify correct mapping

Deliverable Components

The Configured Mapping Environment template consists of the following components:

- Introduction
- Database Tier
- Applications Tier
- Desktop Client Tier
- Other Applications

Introduction

This component describes the purpose for the Mapping Environment and the detailed configuration approach taken to implement the environment.

Database Tier

This component describes the configuration for the database, concurrent processing, and administrative servers in detail. This component also describes the configuration of the hardware platforms that these server environments execute on.

Applications Tier

This component describes the configuration for the forms and web servers in detail. This component also describes the configuration of the hardware platforms that these server environments execute on as well as the configuration of the load management and fall-over capabilities.

Desktop Client Tier

This component describes the configuration for the client web environment in detail. This component also describes the configuration of the hardware platforms that web-clients execute on.

Other Applications

This component describes the configuration of any other software applications that are required to support the project.

Tools

Deliverable Template

Use the Configured Mapping Environment template to create the deliverable for this task.

BR.030 - Map Business Requirements (Core)

In this task, you assess the fit of standard application and system features to detailed business requirements.

Deliverable

The deliverable for this task is the **Mapped Business Requirements**. These requirements are an extension of the Business Requirements Scenarios (RD.050), showing the result of the mapping activity. This task generates the Business Requirements Mapping (BRMs) Forms for complex requirements and gaps. The compilation of all of the BRM Forms represents the Mapped Business Requirements.

Prerequisites

You need the following input for this task:

- ❑ **Data Gathering Requirements (BP.030)**

When *process change* is applicable, define the metrics that are appropriate for measuring performance for the organization's business area in the Data Gathering Requirements. If Determine Data Gathering Requirements was not performed, this deliverable will not exist. (See the task description for BP.040 for more information on when this task should be performed.)

- ❑ **Future Process Model (BP.080)**

The picture of the business process will be very helpful to mapping teams in visualizing how the business will operate after implementation is complete.

- ❑ **Business Requirements Scenarios (RD.050)**

The organization of mapping activities is by business process. There will be one mapping task planned for each Business Requirements Scenarios (BRSs). The intent is to confirm that the business process and its associated detailed requirements are feasible and supportable by the application, and if not, to define a feasible alternative.

- **Audit and Control Requirements (RD.060)**

After establishing business requirements at the business process level, determine methods and procedures for controlling and auditing these processes. Map these audit requirements to the application system.

- **High-Level Gap Analysis (BR.010)**

The High-Level Gap Analysis indicates the high-level approach for resolving gaps identified earlier.

- **Configured Mapping Environment (BR.020)**

Each mapping team needs an environment in which to create initial prototypes.

- **Preliminary Conceptual Architecture (TA.030)**

Use the Preliminary Conceptual Architecture to help assess the feasibility of alternatives with respect to architectural strategies, requirements, and decisions.

- **Application Extension Strategy (MD.010)**

Consider the stated project policy and approach that constrains the formulation of alternatives that entail application extension.

- **Skilled Project Team (AP.050)**

During the mapping procedure, some online prototyping will take place. To assure the feasibility of proposed alternatives to business requirements, each mapping team must be knowledgeable in application concepts and features.

Optional

You may need the following input for this task:

- **Project Management Plan [*initial complete*] (PJM.CR.010)**

Use the defined standards and guidelines to help make sure that all teams will follow the same approach during mapping creation and review.

❑ **Application Architecture (TA.040)**

The Application Architecture provides information about the key critical setup parameters and system functions that affect the precise architecture of the application installations.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Train all assigned team members in the use of the methods and tools for mapping.	
2.	Review high-level gaps, and the approach to resolve these gaps.	
3.	Check out the prerequisite deliverables and get a block of control numbers assigned for expected new Business Requirements Mapping Forms (BRMs).	
4.	Become familiar with Business Requirements Scenarios (RD.050) for the target process in need of mapping.	

No.	Task Step	Deliverable Component
5.	Conduct mapping sessions to assess detailed application fit and create or revise alternative to business requirements. Map future business requirements to application features, programs, reports, and other standard modules.	BRS Solution
6.	Perform online prototyping and deliver a prototype demonstration.	
7.	Document major requirements.	Mapping Source
8.	Fit issues into a Requirement Essay.	Requirement Essay
9.	Perform process research; look for and document alternatives.	
10.	Identify current versus proposed process steps and assess the feasibility of proposed alternatives.	Current versus Proposed
11.	Document alternatives. Record possible alternative alternatives for application gaps.	Mapping Solution
12.	Document major operating and policy decisions.	
13.	Secure acceptance of the Mapped Business Requirements.	

Table 3-9 Task Steps for Map Business Requirements

Approach and Techniques

The Map Business Requirements task adds mapping information to the Business Requirements Scenarios (RD.050) and generates Business Requirements Mapping (BRMs) Forms for complex requirements and gaps. The combination of these documents represents the Mapped Business Requirements.

Definition of Mapping

Mapping is a set of activities that begins during the presales cycle (with preliminary Business Requirements Scenarios (RD.050), continues during the initial stages of the project (gathering of identified business requirements and application gap materials), becomes more formalized during the Operations Analysis phase (preliminary fit analysis aspect of BRS development), and concludes during Solution Design with the creation of Mapped Business Requirements.

Mapping a business process means:

- proving designs through demonstration
- identifying gaps in the application
- proposing feasible bridges to gaps

The following list includes some broader connotations of the term *mapping*:

- the basis for establishing application fit to business requirements, identifying gaps, and proposing alternatives
- the formal linkage of Future Process Models (BP.080) to application features

In this regard, mapping describes the evolution of process design for a business process. The BRS will continue to evolve and be supplemented and improved throughout all mapping tasks.

The formal mapping task, however, is very specific in that it documents key business requirements and proposed alternatives in much more detail than generally described in a BRS.

Mapping Teams Organization

Organize mapping teams around the same grouping as process design teams. This means that the same teams will work on BRS development, as well as BRS completion and BRM Form creation.

The skills required for a mapping team are similar to those for a process design team, except that mapping typically includes a heavier concentration on detailed recommend alternatives. Detailed application knowledge becomes more important during mapping.

Key users should participate in mapping sessions. It is best to perform mapping as near as possible to the actual location in which most of the process tasks take place, in order to have access to key users, and to be able to witness process execution as questions arise. Capture decisions made and agreements reached in the BRS, so that the final product reflects the proposed business process design.

It is very important that the configuration management role be properly executed, especially with regard to the following areas:

- updating BRS and BRM Forms
- providing deliverable status
- setting up configuration management (to help encourage proper use of tools and cataloging of deliverables)
- tracking events, BRSs, and BRM Forms in order to help maintain integrity (addressing all identified gaps)

Always organize mapping sessions for efficiency by publishing a thorough agenda that includes:

- session location and duration
- role assignments
- the business process to be mapped
- activities and sequence
- the inputs required (like prerequisite deliverables) and assignments for bringing them
- the expected outputs and criteria for successful closure of the mapping session

Mapping Process

It may be hard to separate mapping activities from those of business process design, especially in the following situations:

- The project is small or its target business area contains few business processes.
- The project team is pursuing rapid implementation and therefore wants to complete deliverables in parallel as much as possible.
- The process design team is intact through the mapping stage, and therefore process design and mapping form more of a continuum.

Update the BRS as a record of your mapping progress. For instance, the fields in the BRS solution deliverable component indicate whether there is a fit or a gap in functionality, and the BRS business process Identification component indicates the status of each process step.

In addition to updating the BRS, create BRM Forms as needed. This document is a proposal for changes to some process design detail (at the process step level) that needs approval or specification. It is a description of the proposal. If a BRS maps 100 percent to the application, it may not be necessary to create any BRM Forms for that business process.

☞ **Attention:** You *must* complete a BRM Form for all requirements or process steps coded as *Extension* (where there is a required Extension to the standard application).

It is also acceptable to use the BRM Form to document the purpose and reason behind a particular design. Another way to accomplish this is to put a reference to a topical essay or other standard documentation reference in the BRS (RD.050).

When mapping, keep the following step-savers in mind:

- Address critical business processes identified by the organization before seeking resolution to minor issues that crop up in business process designs and maps.
- Use standard system features, functions, and reports whenever possible.
- Use extensibility features, such as descriptive flexfield.

New business requirements could emerge during a mapping session. Verify that these new requirements fit within the scope of the project before adding them to the business requirements list. Set aside time to finalize these requirements.

☞ **Attention:** When you discover new business requirements as a result of mapping activities, you should always revisit the project workplan to confirm that downstream estimates are still valid, and update the workplan if necessary.

Make sure to identify differences between *true* business requirements and a *wish* list.

Detailed Fit Assessment

During the development of Business Requirement Scenarios (RD.050), encourage teams to perform an initial assessment of application fit to business requirements. Documenting this information will provide an important input into detailed mapping.

Detailed mapping entails looking at information requirements at a very detailed level. For instance, indication of whether the business needs the capability to *drop shipments* is captured initially on the BRS. However, you must analyze the type of information needed and its availability with the application. Use one of the following items to capture this information:

- a reference to an application feature in the BRS solution component
- a requirement document reference in the BRS requirements and sources component
- a BRM Form

You can use the Process and Mapping Summary (RD.030) as the gap list.

Prototyping

Using a prototype technique, draft an initial solution and prepare a model of the solution. Consultants can help create the alternative and set up the appropriate configuration in the mapping environment. Then the project team and key users review the alternatives, using the prototype as a guide, and interactively refine the model. These sessions also include a high degree of testing. The emphasis is on proof — proving the model, concepts, and approaches that form the core of the recommended alternatives.

Repeat this process until you arrive at an acceptable alternative. By preparing the process flow and system in advance, the project team concentrates on satisfying the business need and focuses on inputs and outputs from the process.

Suggestion: Prototyping requires a thorough understanding of integrated applications. To use this technique effectively, schedule consultants or other application-knowledgeable people full time during mapping to deliver the preliminary alternative.

It is important to involve business line managers in the mapping development process. This provides an excellent opportunity for leaders to gain practical experience configuring and testing the applications. As soon as possible, encourage the business line manager to take responsibility for driving mapping sessions, thus allowing the application specialist to facilitate and provide guidance. By presenting the chosen alternatives to the rest of the team and to users, business line managers convey support and generate confidence in the new system.

Providing one alternative may not be enough to satisfy the mapping team. Therefore, the prototyping process must be iterative regarding the design and test of alternatives. Repeat the process as needed, until the team is comfortable with the chosen business alternatives and is ready to sign acceptance certificates.

The figure below depicts the creation of a visual process model, followed by a detailed business requirements scenario. Complete a BRM Form for those task steps on a BRS with a gap (or whenever you need a detailed requirement essay). A BRM Form is like a placeholder and usually results in an Application Extension Definition and Estimates (MD.020) document during the Solution Design phase. After process refinements and decisions are made, Solution Design activities can begin, resulting in process narratives and Application Setup Documents (BR.100).

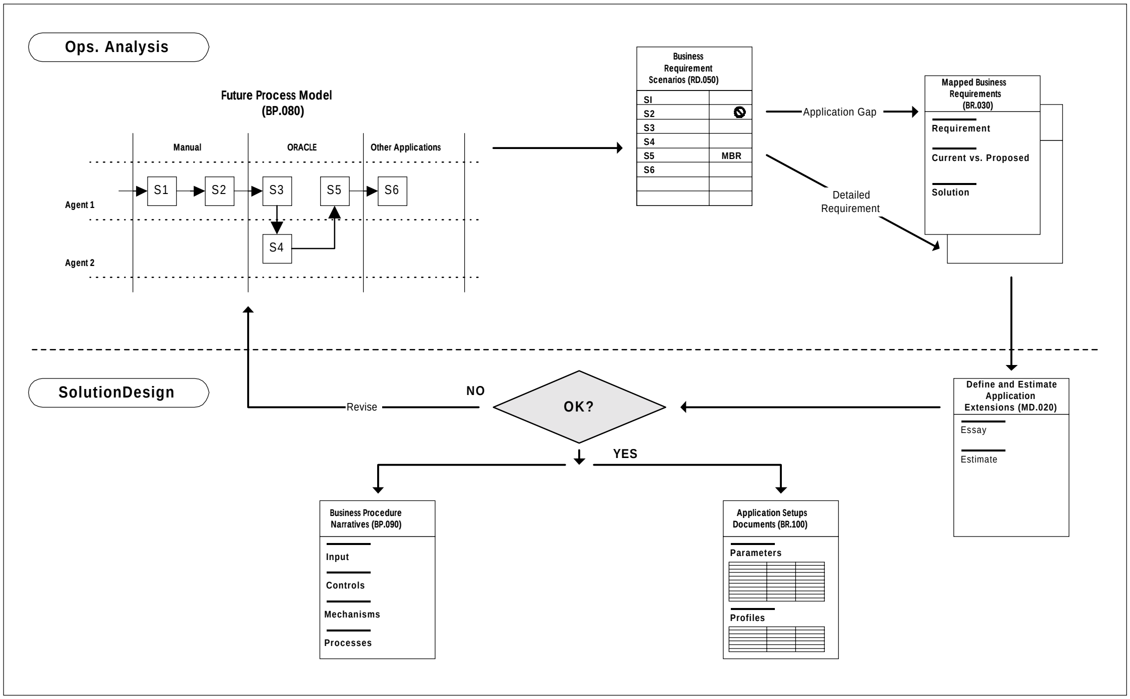


Figure 3-4 Mapping and Prototype Flow

Consider these prototyping tips:

- Quickly devise and show an essential alternative (not necessarily complete the first time).
- Focus on core processes and characteristics that drive business objectives.
- Create just the major components of a working model.
- Do not worry about cosmetics.

Process Research Cycles

After developing Business Requirements Scenarios (RD.050), process design teams should quickly begin to move into mapping activities. The research goal is to gather information on the following subjects:

- process characteristics
- ways to reduce process exceptions

- ergonomics such as ease of error correction
- how to think ahead and consider linkage of final process designs and their alternatives with user procedures and training

Just as in initial process design creation, inform team members of (1) process characteristics and idiosyncrasies, (2) transaction volumes, and (3) exception processing. The best way to approach this is through detailed interviews and observing how the business operates, in order to be in a better position to test design feasibility.

Another approach to process research is to employ a combination of workshops and key process walkthrough tests (observing how the business works). This technique requires careful management and careful picking of workshop members, but can provide instant feedback and verification of the perceptions of different groups regarding cycles, processes, and dependencies. Workshops often convey more information in a shorter time than a series of interviews, and could reduce time spent in verification and cross analyzing. Just as in the creation of Business Requirements Scenarios (RD.050), follow the "Rule of 3-2-1", roughly 3 hours of research are normally required for 2 hours of mapping and 1 hour of formal entry, and time spent toward talking with real users and reviewing real process outputs. Avoid mapping exclusively in a conference room.

Solutions to Application Gaps

There are many types of alternatives to application gaps, ranging from large subsystems to localized modifications (configuration, setup, flexfield, alert, report, and form), to simple workarounds. You still may want to create a BRM Form just for emphasis when approving a workaround built into the BRS. The revised process reflects the approach dictated by the workaround and downstream users, and reviewers of the process are able to use the reasons and support information.

Examples of large-scale subsystems are:

- custom security architecture
- reporting systems
- critical enterprise interfaces

It is best to avoid very detailed documentation in the BRM Form. Think of the BRM Form as a record of key decisions, or as a placeholder in anticipation of additional detailed design documentation.

Adequately bridging gaps is the primary purpose of the BRM Form. Consider these tips when mapping and creating BRM Forms:

- Design alternatives for the desired state of the business, rather than directly mapping to current needs.
- Always implement workarounds before designing and building a custom extension.
- When multiple alternatives are available, choose the alternative that supports organization goals or broad business areas, rather than satisfying the needs of a single department or user.

In order to obtain rapid implementation, consider these mapping tips:

- Get quick closure on gaps.
- Push hard on perceived gaps — up to 80 percent of the initial gaps identified may actually be found to be unnecessary.
- Ask the question, “What does the application do?” in order to keep the mapping session moving.
- Adjust business processes to fit Oracle Applications functionality.
- Pay special attention to prioritizing gaps in order to manage scope and budgeted resources properly.

Process and Mapping Summary

Record key decisions in the project library. This makes it easier for people who are new to the project to become familiar with important operating policy and functional decisions.

Some examples of these key decisions follow:

- use of inventory organizations
- new chart of account segment values
- multi-org structure
- inclusion of purchase orders as supply in available-to-promise rules

- use of repetitive manufacturing functionality for final assembly operations
- use of lot control for key components

Another place to look for key decisions is at the request for proposal document that drove the presales cycle (before selection of the application). This document usually contains primary requirements with its selected alternatives classified as key decisions.

Process Analysis

Generally, you should quantify in monetary terms, any alternative that requires additional cost or resources. Comparison of cost/resource for *current* versus *proposed* processes is vital to selling change to management, key business people, and systems users. It is particularly important to draw in sponsors of the change by getting them involved in the justification process itself.

The following reasons justify change:

- saving time
- reducing resource/labor or improving utilization
- increasing efficiency (and ability of process agents to take on more work due to the proposed changes)

Capture key process information for both current and proposed process steps.

Linkage to Module Design and Build (MD)

The BRM Form is a mandatory placeholder for every gap that requires a custom extension to the application. Other than for historical purposes, once created, an Application Extension Definition and Estimates (MD.020) document replaces the BRM Form, since the BRM Form is the forerunner to the Application Extension Definition and Estimates (MD.020). In one sense, the BRM Form is the event that triggers the execution of the Module Design and Build (MD) process.

Linkage to Application and Technical Architecture (TA)

The mapping process may reveal large-scale gaps that require custom extension. These gaps may affect the application and technical architecture and may require special subsystems where tasks within the architecture process own the specifications.

Review and Signoff

As you complete the BRS and BRM Forms for each process, review the documents with the project sponsor, management, representative business users, and any formal integration teams. In this way you can obtain progressive confirmation of your process designs and business requirements.

Linkage to Other Tasks

The Mapped Business Requirements are an input to the following tasks:

- BP.090 - Document Business Procedures
- BR.050 - Conduct Integration Fit Analysis
- BR.070 - Conduct Reporting Fit Analysis
- BR.080 - Test Business Solutions
- BR.090 - Confirm Integrated Business Solutions
- BR.100 - Define Application Setups
- BR.110- Design Security Profiles
- TA.040 - Define Application Architecture
- TA.100 - Define and Propose Architecture Subsystems
- MD.020 - Define and Estimate Application Extensions
- MD.050 - Create Application Extensions Functional Design
- MD.080 - Review Functional and Technical Designs
- PT.040 - Create Performance Test Scripts

Document Business Procedures (BP.090)

You may now use the BRS and BRM Forms for each business process to begin creating process narratives for the target business process. However, finalizing Business Procedures Documentation is not possible until thorough and formal testing and approval of the process design.

Test Business Solutions (BR.080)

Although the mapping technique makes extensive use of prototyping as a technique, it is not a substitute for formal testing — which is the next step, once mapping is reasonably complete for a business process.

Define Application Setups (BR.100)

After successfully mapping a business process, completing process design, and identifying and dispositioning gaps, tentatively document the application setups and configuration decisions in anticipation of confirmation during formal system testing.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	45
Application Specialist	25
Technical Analyst	20
Configuration Management Specialist	5
System Architect	5
Business Line Manager	*
User	*

Table 3-10 Role Contribution for Map Business Requirements

* Participation level not estimated.

Deliverable Guidelines

Use the Mapped Business Requirements to begin the design of business alternatives. This deliverable represents the update of one BRS (RD.050), and creation of another BRM Form (from BR.030). Refer to the Project Management Plan (PJM.CR.010) for any mapping guidelines that should be used. You should attempt to map standard features of the applications to each defined business requirement. Explore all possible workarounds before recommending custom extensions, if previously noted gaps exist.

Describe the requirements in terms of business objectives rather than application features. State any risks that might occur from an unmet requirement. This helps prioritize competing requirements.

This deliverable should address the following:

- organization of mapping alternatives
- elimination of non-value-added steps
- the best method to implement selected alternatives
- reporting and messaging needs of custom extensions
- special or additional implementation steps
- perceived shortfalls and required system enhancements

While it is ideal to use the standard application system, the reality is that each organization has unique business needs that standard features cannot always fulfill. This task investigates alternatives that use combinations of standard and custom features. Therefore, it is important to separate requirement fit, workarounds, and application extensions. This distinction helps you plan the necessary application extension design and development later.

The Mapped Business Requirements help you specify the bridging of gaps. The BRM Form closes the loop on key design issues for which an alternative must be selected.

Usage of Business Requirements Mapping Forms

Follow these guidelines regarding use of the BRM Form:

- A BRM Form is a proposal for change to some process design detail that is pending approval or even specification; it is a description of the proposal.
- Create a BRM Form for every process step coded as “C” (Custom) on the BRS (RD.050)—this will act as a mandatory placeholder.

- If there is an acceptable workaround, the BRS and process model should reflect it, and there is no need for a BRM Form (otherwise create a BRM Form).
- Use one BRM Form to support multiple gaps from a single BRS, only if there will ultimately be a single Application Extension Definition and Estimates (MD.020) document written in support of these gaps, and if there is a logical relationship.
- When listing requirements on a BRM Form, describe these requirements in terms of one or more business objectives and not application features; focus the BRM Form on *Why* you need to satisfy the requirement more so than *How* you will satisfy it.
- You may optionally use a BRM Form as a placeholder when a mapping team wishes to preserve some record of detailed design that might be useful later during detailed design documentation, even in those cases where no application gaps exist. When used for this purpose, clearly mark the BRM Form as a *design note* so it is not perceived as a gap.
- When using rapid implementation mapping techniques, it is customary to develop a BRS for the proposed process (using features from the new application package as the driver), and then have one BRM Form that reflects how the process is being improved along measurable lines. The BRM Form number can match the parent process ID, as a cross reference.

Acceptance Criteria

The BRM Form should contain the following heading information:

- process control number
- priority
- owner
- process or scenario description
- source type (gap, design note, and so on)

In order to pass a quality review:

- Develop the BRM Form using the approach listed in this task.

- The BRM Form must be reviewed by a qualified peer who signs off as the first reviewer. This signoff indicates that the BRM Form is accurate and consistent. For best results, the review should include the process owner as well as the customer of the process (either the requester or the owner of the next process down the line within the business area).

Deliverable Components

The Business Requirements Mapping Form consists of the following components:

- BRS Solution
- Mapping Source
- Requirement Essay
- Current versus Proposed
- Mapping Solution

BRS Solution

This component requires the following information:

- **Process Step Number** — a unique sequence number for each task
- **System/Module** — a reference to the application system module that satisfies this task requirement
- **Application navigation path or transaction zone** — information describing entry form used for system-assisted tasks
- **Ref Manual Page/Line** — a topical essay or other narrative in an application, user or technical reference manual
- **Tools** — reports or other mechanisms used in completing the task successfully
- **Solution Type** — used to designate workaround/extension; use “W” if there is a non-system workaround that is acceptable, although not necessarily optimal; use “E” if there is a need for extension (complete a Mapped Business Requirement form for all tasks coded with “E”)
- **BRM Ref Number** — Business Requirement Mapping Form control number; optional unless Solution Type is “E”

Mapping Source

This component facilitates the analysis and comparison between current solution to a business requirement and to a proposed solution.

The mapping source requires the following information:

- **Process** — a brief description of the process
- **Business Area** — a code designating the mapping team responsible for this process
- **Date** — date form is being initiated
- **Control Number** — a unique identifier for each business process (for example, 6-character code in format AAA.999, where AAA represents the subprocess, and 999 is a unique 3-digit code assigned sequentially from a log maintained by each team)
- **Mapping Team** — either the process modeling, design, or mapping team responsible for this business process
- **Process Owner** — the agent with overall responsibility for the process; could be the customer of the process, or the supplier (one who fulfills the request)
- **Librarian** — the person charged with maintaining this BRS deliverable and interfacing with other teams in order to maintain control and proper alignment with integration goals
- **Priority** — determined at each team's discretion; this is a good place to identify Core Processes
- **Core** — an identification of major, driver processes that affect or influence business objectives
- **Source Type** — indicate whether the BRM Form will document a gap or a design note
- **Author** — originator's name
- **Ref Doc** — reference to some other formal requirement description source document (usually such a document already exists and the BRM initiator simply wishes to avoid rekeying its contents on the BRM Form)
- **BRS Reference Number** — Business Requirements Scenarios (RD.050) control number that is the parent of this BRM Form (also enter process step number)

Requirement Essay

This component describes the business requirement in business language and in terms of business objectives, rather than application features. The following questions should be answered:

- What?
- Why?
- Who?

The reasons for the requirement, along with references to any support information (for example, policy or mission statements) should be clearly defined.

Current versus Proposed

This component documents the current process steps planned for elimination or modification, including the expected effect on cycle time, cost, and other resources. The section can be duplicated to show both current and proposed processes.

- **Practices** — narrative description of customary or usual actions
- **Policies** — principles, plans or courses of action, especially as dictated by management
- **Procedures** — narrative description of sequence of steps to be followed
- **Seq** — process step number
- **Ref/Description** — a brief description, beginning with an action verb, that captures the purpose and deliverable task; alternatively, could include a reference to another document (or just use the BRS step description)
- **Activity** — optionally may be used for activity-based analysis purposes
- **Action** — ADD / DEL / CHG status for the sequence number
- **Cost/Time/Resource** — measurable resource usage by the sequence number

Mapping Solution

This component details the type and nature of the solution in a descriptive format.

- **Workaround** — description of the proposed method for getting around an application gap to a business requirement
- **Application Enhancement** — description of the custom modification to the application whose implementation will result in satisfaction of the business requirement
- **Reengineering Opportunity** — description of simplification, elimination or enhancement of the target process
- **Solution/Design Document Reference** — if available, a document number for high-level or detailed design planned to satisfy the requirement
- **Mechanism** — resources that influence the process; people, tools, or machines affected by the BRM proposal
- **Interfaces** — description of system interface requirements necessary to satisfy the requirement
- **Solution Technique** — description and type of application feature that will satisfy the requirement (configuration, setup, flexfield, alert, report, form, and so on)

Tools

Deliverable Template

Use the Business Requirements Mapping Form template to create the supporting deliverable for this task.

The Map Business Requirements task adds mapping information to the Business Requirements Scenarios (RD.050) and generates Business Requirements Mapping Forms (BRMs) for complex requirements and gaps. The compilation of these forms represents the Mapped Business Requirements.

BR.040 - Map Business Data (Core)

In this task, you map the data elements from the legacy system to the target application modules, business objects, and attributes.

Deliverable

The deliverable for this task is the **Mapped Business Data.** This deliverable provides verification that the underlying target application modules, business objects, and attributes will support business processes.

Prerequisites

You need the following input for this task:

- ❑ **Current Business Baseline (RD.020)**

An understanding of current operations is necessary for mapping the existing source system data structures to the new target data structures and identifying conversion requirements.

- ❑ **Business Requirements Scenarios (RD.050)**

The product of business data mapping activities should be in alignment with business processes. Identify and define business processes, and while developing Business Requirements Scenarios (RD.050), analyze them for business data requirements. The intent is to confirm that the business process and its associated detailed requirements are supported by defined business objects whose sources are known. A business object is a logical business grouping of data within an application (See the next section for more information.)

- ❑ **Configured Mapping Environment (BR.020)**

A Configured Mapping Environment is needed to for mapping activities.

❑ **Existing Reference Material**

Use existing reference material to identify and gain an understanding of the existing source systems and data structures that require data conversion.

Optional

You may need the following input for this task:

❑ **High-Level Existing System Data Model**

Use the preexisting data model to identify and gain an understanding of the existing source system data structures that require conversion.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Analyze Business Requirements Scenarios (RD.070) for business objects required.	
2.	Identify the legacy source file and field / data elements that are being converted and record this information in the table provided in the business data mapping form component. Use the existing reference material and initial business data model to perform this task step.	Conversion Mapping

No.	Task Step	Deliverable Component
3.	Identify the target application, business object name, and attributes that the legacy files and fields map to and record this information in the table provided.	Conversion Mapping
4.	Record the fields that the legacy system stores but the application does not, and record the attributes which the application stores but the legacy system does not.	Conversion Mapping
5.	Secure acceptance of the Mapped Business Data.	

Table 3-11 Task Steps for Map Business Data

Approach and Techniques

The Mapped Business Data is an important deliverable, even when data conversion is not a factor since the field-by-field mapping from legacy systems to Oracle Applications is a key output. When data conversion is a factor, it is the basis for Perform Conversion Data Mapping (CV.040). The primary purpose of this task is to discover at an early point in the project life-cycle whether any business objects or attributes in the legacy system are not being stored in Oracle Applications. This task should also determine whether the application stores any required attributes that the legacy system does not currently store.

Fundamental to this task is an understanding of the term *business object*. A business object is a logical business grouping of data elements within an application. An example is a group of items in Oracle Inventory or customers in Oracle Receivables. A business object can map to one or more standard application forms and underlying tables.

In general, use the following technique when performing the business data mapping task:

- For each business object being converted, identify the application name, business object, and attributes. A business object in this context is defined as a business grouping of data that may map to one or more forms or tables. Examples of business objects are customers, vendors, invoices, items, and so on. The attributes for a business object are the adjectives or descriptive pieces of information about a business object that are either stored in the underlying tables or appear in the application forms as fields.
- Determine the required application attributes for a given business object to support future business processes. Use the application reference manuals to determine whether there are required fields in the application forms for which the legacy system does not have a data element. In these cases, the users should define default values that make sense for the organization’s business. Make sure that you map the new form fields/attributes to the legacy data and that you map the legacy data elements to the form fields in Oracle Applications.
- One way to map the legacy data elements to the application attributes is to navigate to the appropriate form and try to find a field for each legacy data element/field for a given business object. If you cannot find a place to store the legacy data element in the application forms, confirm with the users that this is a required data and it needs to be stored within the Oracle Applications. If the users require this data element, review the possibility of storing this element in an application descriptive flexfield. If many related data elements do not map to fields in the forms, then a custom form and underlying table might be required.
- Determine the source of each business object’s data elements. For example, for the business object called Inventory Items, there may be more than one source from the existing system item master. Ask to see the legacy source system file layouts and discuss the use of each of these fields in the current system with someone who understands the legacy system.

Linkage to Other Tasks

The Mapped Business Data is an input to the following tasks:

- BR.090 - Confirm Integrated Business Solutions
- MD.020 - Define and Estimate Application Extensions
- MD.060 - Design Database Extensions
- MD.070 - Create Application Extensions Technical Design
- CV.040 - Perform Conversion Data Mapping

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	50
Application Specialist	30
Technical Analyst	20
Business Line Manager	*

Table 3-12 Role Contribution for Map Business Data

* Participation level not estimated.

Deliverable Guidelines

Use the Conversion Data Mapping (CV.040) deliverable to create the Mapped Business Data deliverable.

This deliverable should address the following:

- target application modules
- business objects, and attributes

Deliverable Components

The Conversion Data Mapping (CV.040) deliverable includes the following component:

- Conversion Mapping

Conversion Mapping

Use the Conversion Mapping component of the Conversion Data Mapping (CV.040) deliverable to create this component. It is comprised of data inserted into the table columns indicated in the notes of the template. The table will later be used and completed by a conversion technical analyst during Perform Conversion Data Mapping (CV.040). One template for each standard application business object being converted to should be completed.

Tools

Deliverable Template

A deliverable template is not provided for this task. Use the Conversion Data Mapping (CV.040) template to create the supporting deliverable for this task.

BR.050 - Conduct Integration Fit Analysis (Optional)

In this task, you identify the new integration points that you require, based on your conceptual architecture and the mapping of the new applications onto the existing architecture.

> Δ If your project has, *interfaces* with external systems, you should perform this task.

Deliverable

The deliverable for this task is the **Integration Fit Analysis**. It is a statement of fit and gaps for integration points between existing applications and the new application.

Prerequisites

You need the following input for this task:

❑ **Business Requirements Scenarios (RD.050)**

Business Requirements Scenarios (BRS) represent all relevant business processes that are within the business area, including those that are supported by existing systems. Therefore, Business Requirements Scenarios are a key driver for required integration points. It is best to consider only Business Requirements Scenarios already mapped to the application. Unmapped Business Requirements Scenarios may not be complete enough to be useful. Concentrate primarily on using Business Requirements Scenarios for core business processes.

❑ **Preliminary Conceptual Architecture (TA.030)**

Use this deliverable to identify key distributed data integration points between applications, systems, and external systems.

Optional

You may need the following input for this task:

- ❑ **Current Technical Architecture Baseline (TA.020)**

Use this document to acquire technical information about the existing legacy systems, and especially the applications. Use this information to help determine integration points.

- ❑ **Mapped Business Requirements (BR.030)**

Mapped Business Requirements are detailed explanations of key requirements and/or their solutions — particularly if application gaps exist for a business process. These documents may provide information regarding interfaces identified as expectable alternatives.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the list of existing interfaces in the Current Technical Architecture Baseline (TA.020).	
2.	Perform an initial application replacement mapping.	Application Replacement Mapping
3.	Replace legacy applications in the listing of existing interfaces with the desired applications from the replacement mapping.	External Integration Point Replacement Mapping
4.	Create a diagram of the future external integration points.	External Integration Point Diagram

No.	Task Step	Deliverable Component
5.	Examine the Conceptual Architecture (TA.070) document to identify key distributed data integration points between applications, systems, and external systems.	Distributed Data (Internal) Integration Points
6.	Create a diagram of the new distributed data integration points.	Distributed Data (Internal) Integration Point Diagram
7.	Secure acceptance of the Integration Fit Analysis.	

Table 3-13 Task Steps for Conduct Integration Fit Analysis

Approach and Techniques

The Integration Fit Analysis describes how to identify and use integration points. The following diagram is an example of integration points in a business system:

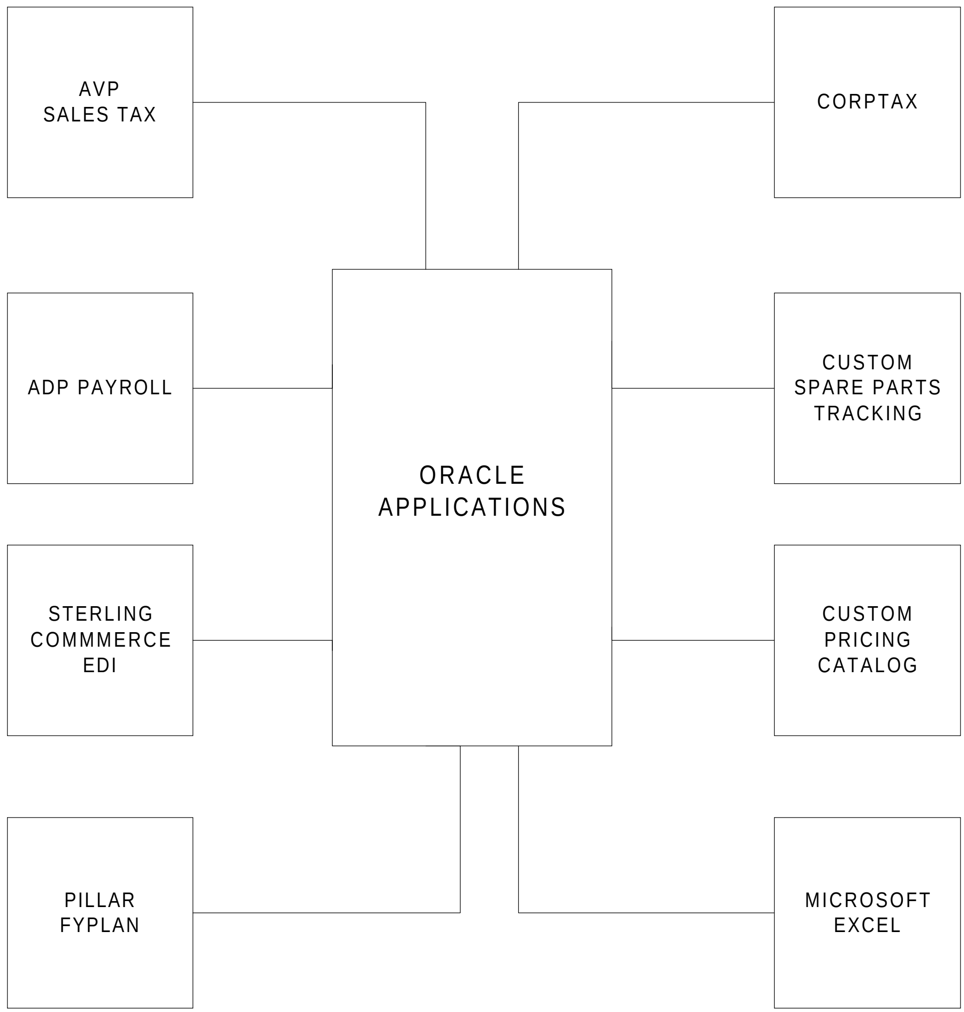


Figure 3-5 Integration Points in a Business System Example

The primary purpose is to map distributed data external integration points between standard applications and other third party applications. Once you agree on an alternative for each integration point gap, this task will result in a complete listing of all new integration points for which you must design and build interfaces. At this stage there has been no specification of interfaces; this task is merely a recognition that an integration point exists and requires attention.

External Integration Points

The most common type of integration point requiring attention in an implementation project is one that transfers data between two heterogeneous applications. External integration points are also needed between two separate installations of the same application with separate databases. These integration points are external in nature since they result in the transfer of data from application databases external to the one of interest. They usually give rise to interfaces that support the integration.

Replacement External Integration Points

The simplest way to identify the new external integration points is to perform a thorough architecture baseline analysis, then identify the changes implied and required by the replacement of certain existing legacy applications. The business requirement for each old interface should be met in some way by the new system, either as a direct mapping, or as a change in implementation or functionality.

Although the approach helps scope and validate the completeness of the mapping work, it also carries significant risks. It requires significant attention to details when you define the existing technical architecture, and it does not guarantee that new interface requirements will be noticed and documented. The technique carries the risk of re-implementing outmoded practices, rather than leveraging the capabilities of new applications, and may miss critical requirements that would only become apparent with an understanding of the role of the interface in one or more Business Requirements Scenarios (RD.050). By using a tangible mapping of old interfaces to new, you may find it easier to focus on specific implementation deliverables.

Nonstandard External Integration Points

If you are implementing a combination of custom developed applications together with Oracle Applications, also consider new integration points that you need to handle between your new, nonstandard applications and legacy applications that you are planning to retain in the new architecture. These other new integration points will be revealed by performing the replacement mapping not just for replacement by Oracle Applications, but for replacement by your entire new applications suite. Overlooking these integration points is another risk of the “cut out the old and replace with the new” method of interface development.

Integration Points from Standard Applications Distributed Data

If your implementation spans separate, standard applications databases, it is critical to identify the external interfaces that need to cross databases. This will always be the case for multi-site installations with multiple data centers. These types of interfaces can be more complex to implement, and you should not attempt to perform requirements gathering for them using this type of bottom-up analysis. However, even if you are managing the development separately, you can still record the integration points as part of this task so that you have a complete reference document for all integration points to the new applications.

Linkage to Other Tasks

The Integration Fit Analysis is an input to the following tasks:

- BR.060 - Create Information Model
- BR.090 - Confirm Integrated Business Solutions
- MD.020 - Define and Estimate Application Extensions
- TE.050 - Develop Systems Integration Test Script

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	95
Business Analyst	5
User	*

Table 3-14 Role Contribution for Conduct Integration Fit Analysis

* Participation level not estimated.

Deliverable Guidelines

Use the Integration Fit Analysis to describe the interfaces needed to integrate existing applications with the applications you are implementing.

This deliverable should address the following:

- integration points
- mapping of new applications onto existing architecture

Deliverable Components

The Integration Fit Analysis consists of the following components:

- Introduction
- Application Replacement Mapping
- External Integration Point Replacement Mapping
- External Integration Point Diagram
- Distributed Data (Internal) Integration Points
- Distributed Data (Internal) Integration Point Diagram

Introduction

This component describes the purpose and scope of the Integration Fit Analysis.

Application Replacement Mapping

This component addresses the approximate replacement mapping of the existing to new applications. The non-replaced applications, with a status of retained, should be completed if you wish to create a complete list of the application mapping.

External Integration Point Replacement Mapping

This component describes and maps the integration points of the new application system to the old system.

External Integration Point Diagram

This component provides a diagram of the future application map (and the new interfaces required).

Distributed Data(Internal) Integration Points

This component documents the new integration points that need support between the distributed Oracle Applications databases in the new architecture.

Distributed Data (Internal) Integration Point Diagram

This component diagrams the new integration points that need support between the distributed Oracle Applications databases in the new architecture.

Tools

Deliverable Template

Use the Integration Fit Analysis template to complete the deliverable for this task.

BR.060 - Create Information Model (Optional)

In this task, you create an information view of the future business by specifying the future information usage and flow requirements across business functions, business organizations, applications, and data centers.

Δ If your project includes either p*rocess change*, *interfaces* with external systems, or *complex reporting*, you should perform this task.

Deliverable

The deliverable for this task is the **Information Model**. It visually depicts information and access flows in the business.

Prerequisites

You need the following input for this task:

❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides information about the business organizations that execute the business processes.

❑ **Business Requirements Scenarios (RD.050)**

The Business Requirements Scenarios provide the process and scenario flows on which the information flow model is based. The BRS will include mapping of the business processes to application and system processes and allow the identification of processes that cross organizations, applications, or installations, and require custom extensions.

❑ **Master Report Tracking List (RD.080)**

The Master Report Tracking List is the primary repository for all information collected about report requirements.

- ❑ **Integration Fit Analysis (BR.050)**

The Integration Fit Analysis provides information about the integration points between different applications and application installations. If Conduct Integration Fit Analysis was not performed, this deliverable will not exist. (See the task description for BR.050 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

- ❑ **Future Process Model (BP.080)**

If you have modeled the future business hierarchy, it may augment the process model in summarizing the business functions that each business organization performs. If a particular business function requires specific information access, you can use this deliverable to identify the business organization that also requires the same access.

- ❑ **Audit and Control Requirements (RD.060)**

The Audit and Control Requirements may provide additional information about the general security policies, requirements and constraints. The usefulness of this deliverable will depend on precisely captured information for the audit and control requirements of the business.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Determine the information modeling approach and strategy.	
2.	Review business hierarchy and operating structures to determine major functions to include in scope.	

No.	Task Step	Deliverable Component
3.	Review Business Requirements Scenarios (RD.050) and mapping documents to determine which business processes cross major functional, organizational, or applications boundaries.	
4.	Create a table, sorted by BRS code or number, of business objects transferred across business organizations.	Information Flow Between Business Organizations
5.	Create a diagram that depicts cross-business organization information flow.	Information Flow Between Business Organizations
6.	Review Integration Fit Analysis (BR.050) and identify all integration points within the scope of the information flow model.	
7.	Create a table, by BRS, of business objects transferred between applications.	Information Flow Between Applications
8.	Review future business process and function models for secure business functions.	
9.	Review financial and operating structure.	
10.	Summarize the relevant business organizations.	Business Organizations
11.	Identify general security policies and requirements.	Security Policies and Requirements

No.	Task Step	Deliverable Component
12.	Analyze the business organization requirements for each business object type.	Organizational Information Access Analysis
13.	Analyze the inter-organization requirements for each business object type. Identify ownership and access across organizations for the business objects.	Inter-Organizational Information Access Analysis
14.	Review the information model with process and mapping teams.	
15.	Secure acceptance of the Information Model.	

Table 3-15 Task Steps for Create Information Model

Approach and Techniques

The Information Model visually depicts information and access flows in the business, resulting from and driven by the business events and business processes that form the business model. A sample Information Model can be found in the Preliminary Conceptual Architecture template (TA.040).

Select those aspects of the business that you wish to highlight in the information model, and for each aspect you determine the information and access flows between the steps of the process or scenario. Create tabular documentation of the information and access flows between process steps and present the information in graphical format for easier understanding, discussion, and communication. The selection of the aspects of the business for which you will create information models is what determines the scope and effort required for this task.

Initially, identify the information flows in the new system. This includes flows of structured (easily organized in a relational structure) business information and business documents, and unstructured data (for example text documents, video clips, and images). You should cover the following types of information flows:

- information flows between business organizations in the future business model
- information flows between applications in the future system model

Finally, describe and analyze the information access, information sharing, and information partitioning requirements in the new system. This defines control of information from a security standpoint, and access to specific information for consolidated reporting purposes. The information access requirements include both structured and unstructured data.

Business Information Flow Model

Together, the Future Process Model (BP.080) and Business Requirements Scenarios (RD.050) constitute a process view of the business, with implied information flows between the process steps, but the process-focused approach tends to submerge the actions taken on the business information. This task extracts and highlights the information view of the business processing and helps to provide a more complete picture of the business operational model in the new system. The term *information model* is preferable over the more traditional term *data flow model*, since *data* tends to imply structured data, while *information* can encapsulate structured relational data as well as nontraditional information media such as electronic text and image documents, video, and multimedia. In modern processing environments, these nontraditional media are increasingly important.

Benefits of Information Flow Modeling

In an implementation project there may be the temptation to dispense with the creation of an information flow model and use the business process scenarios as the only statement of the business operational model. This can apparently save time and resources through not having to perform these tasks, but the gains may be only perceived rather than real. The business analyst needs to understand the way information is flowing in the business. True understanding of the processes and the information flow model is necessary for other processes in the project. For example, diagramming the document and data flows for a warehouse receiving process or the checks and controls of a purchasing process is not unfamiliar to most business analysts. The issue is one of degree. The creation of an entire information flow model for all the processes and scenarios of a large business could be very time consuming and not particularly helpful in the case of a business process that mirrors the system processes of a packaged application product suite like Oracle Applications. However, you may need some amount of information flow modeling to properly understand the operational model.

The other processes that may need an information view of the business follow:

- Business Process Architecture (BP)
- Business Requirements Definition (RD)
- Application and Technical Architecture (TA)
- Module Design and Build (MD)

Application and Technical Architecture (TA) and Module Design and Build (MD) need the Information Model to properly specify the requirements for custom extensions and interfaces that require building. If the process and mapping teams do not create a formal information flow model, the technical analysts and database designers of these processes will be forced to gather the information flow model requirements in a piecemeal fashion. This is undesirable for three reasons:

- Technical individuals may try to understand and recreate the information flow requirements, often at some later time in the project. They are clearly not the best individuals to compile the information flow model.
- The information flow model requirements may be collected as individual requirements for custom extensions, thereby potentially duplicating efforts.
- A full information flow model, on the other hand, will place the information flows in the context of a full process scenario and will aid understanding of the way an architecture component or subsystem, or a custom interface, fits into the overall process.

Business Information Access Model

The term *information access model* is preferable over the more traditional *data security requirements* since *data* tends to imply structured data, while *information* can encapsulate structured relational data as well as nontraditional information media such as electronic text and image documents, video, and multimedia. In modern processing environments, these nontraditional media are increasingly important.

There are two major aspects to the business organization information access model:

- the information access requirements of a business organization
- the information access requirements of a business organization in relation to information created by other organizations

Two types of matrices are useful to summarize these requirements succinctly in analytical rather than purely descriptive form.

Organization Information Access Tables

This analysis discusses the ability of a business organization to create, change or delete information of any type. A useful device is a table of business object analyzed by organization, where each cell in the table indicates the ability of the organization to (C)reate, (R)ead, (U)pdate, or (D)elete business object information of that type. The table does not fix the medium of the business object information, although if the medium is relational data it would correspond to rows in tables for the business object. For example, a sales order might correspond to rows in order header and order lines tables. Sometimes termed a CRUD table or matrix, it is analogous to the function data usage matrices used in custom development analyses.

The following table shows which organizations need access to the system to be able to create certain business objects, those that should have only read access, and those that should have no access whatsoever. In addition, it provides an example of an organization information access table:

Business Object	Org 1	Org 2	Org 3
Chart of Accounts	CRUD	R	R
GL Entries	CRUD	CRUD	CRUD
AP Checks	CRUD	CRUD	CRUD
AP Vouchers	CRUD	CRUD	CRUD
Customer Master	R	R	R
Customer Credit Information	CRUD	CRUD	CRUD

Table 3-16 Business Objects by Organization

Inter-Organization Information Access Tables

Traditional information access analysis uses CRUD (Create, Read, Update, and Delete) tables to describe the type of access to business objects that business organizations require. When dealing with a multi-organization implementation, it is also important to identify which organizations can read, update, or delete data created by other organizations.

A technique for documenting these requirements is to use inter-organization information access tables. You create one table for each major business object (for example, sales orders) identified in the information access model as being significant from a cross-organization, security perspective. The vertical axis is a list of organizations that create an instance of the business object (for example, organizations that can create sales orders). The horizontal axis is a list of the organizations or organization types in the analysis. The contents of the cells indicate how each major organization along the horizontal axis accesses data created by organizations along the vertical axis. The symbols are RUD, since the vertical axis indicates the creator organization and the horizontal access is the consumer organization.

It is useful to note, that if only one matrix row is filled in, it signifies that only one organization creates a business object (for example, only one organization can define customers). This is a possible candidate for a data registry in a distributed data architecture. For more information, see Define Platform and Network Architecture (TA.120). The following table provides an example of an inter-organization information access table:

Chart of Accounts.	**Consumer**			
Creator	**Org 1**	**Org 2**	**Org 3**	**Org 4**
Org 1	RUD	R	R	R
Org 2				
Org 3				
Org 4				

Table 3-17 Business Objects by Creator/Consumer

Scope Definition

You need to identify the areas of the business for which you will create an information model. A suggestion for the minimum information modeling effort is to model the information flows for process scenarios under the following circumstances:

- There are custom extensions in the process.
- The process requires agents to access or manipulate information in two or more different applications (an interface between heterogeneous applications).
- The process requires agents to access or manipulate information in two or more different application installations (an interface between separate homogeneous application installations).

Avoid Inappropriate Depth or Breadth of Analysis

The type of information model required for implementing a set of packaged applications is different from the comprehensive entity and attribute analysis that takes place in a typical custom development project. You probably do not need to deal with all of the attributes of a business object until the detail design portions of the project, if your project even requires those tasks. Focus your analysis on key business requirements scenarios at the business object level or where interfaces or access may be an issue. Analyze how information flows between organizations, and which organizations can access which data.

Functional Implications

A new applications system provides a platform for simulating the physical domain that it represents in ways such as the following:

- Inventory records provide an efficient method for locating stock.
- Production process records provide in-depth source information for product costing and variances.
- Payables records provide for forecasting cash requirements.

Think of the application as a model that helps business users evaluate the potential impact of operating decisions before they are made. The key to success with any such computer-based simulation is a high level of sustainable record accuracy. To achieve high record accuracy, carefully document and visually depict the specification and ownership of information at each point of transfer.

Clearly define and confirm the transfer of information and passage of ownership across the following:

- business functions
- business organizations
- applications

Technical Implications

The precise content of the information model will vary depending on the level of detail required and the exact circumstances of the business. Typical components include the following:

- a description of data entities and key attributes required by each organization and site by business function
- reporting requirements and data flows between organizations
- data flows between applications
- timeliness requirements for shared data (this has implications for data synchronization, if distributed replication is being considered)

This information will be used to identify the specific transaction, reporting, and distributed interfaces within Application and Technical Architecture (TA) or Module Design and Build (MD). The results of this task provide the input for application and database architecture design by documenting data sharing, data flow, and initial data interface requirements. The data flows identified in the information model provide a basis for identifying required interfaces when different organizations and applications are on either end of the data flow.

The Information Model also provides the basis for identifying and designing shared registries of information that can eliminate redundant data capture and storage, thereby reducing data maintenance and improving data integrity across the enterprise. An example of a shared data registry would be a common customer master or common item master that is shared across organizations (and potentially across application or database instances).

Linkage to Other Tasks

The Information Model is an input to the following tasks:

- BR.070 - Conduct Reporting Fit Analysis
- BR.090 - Confirm Integrated Business Solutions
- BR.110 - Design Security Profiles
- TA.060 - Define Reporting and Information Access Strategy
- TA.070 - Revise Conceptual Architecture
- TA.080 - Define Application Security Architecture

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Technical Analyst	60
Business Analyst	40
Business Line Manager	*
User	*

Table 3-18 Role Contribution for Create Information Model

* Participation level not estimated.

Deliverable Guidelines

Use the Information Model to specify how information and privileges to read, manipulate, and create the information will flow across your business in response to business process scenarios.

This deliverable should address the following:

- information flow between business organizations
- information flow between applications
- how to use the information within and across your organizations and the partitioning and security requirements

Deliverable Components

The Information Model consists of the following components:

- Information Flow Between Business Organizations
- Information Flow Between Applications
- Business Organizations
- Security Policies and Requirements
- Organizational Information Access Analysis
- Inter-Organizational Information Access Analysis

Information Flow Between Business Organizations

This component documents flow details and flow diagrams.

Information Flow Between Applications

This component documents cross-application flow details, cross-application flow diagrams, cross-installation flow details and cross-installation flow diagrams.

Business Organizations

This component summarizes relevant organizations.

Security Policies and Requirements

This component documents security policies and security requirements.

Organizational Information Access Analysis

This component addresses high-level organizational data requirements and their impact on organizations.

Inter-Organizational Information Access Analysis

This component provides a statement of high-level access requirements across organizations.

Tools

Deliverable Template

Use the Information Model template to create the deliverable for this task.

BR.070 - Conduct Reporting Fit Analysis (Core)

In this task, you are required to analyze and map every reporting requirement to both a future business process and standard application report. This analysis determines the final disposition of every report requirement.

Deliverable

The deliverable for this task is the **Master Report Tracking List**. It supports the analysis and mapping of report requirements to future business processes and standard application reports. It contains the final disposition of every report requirement.

Prerequisites

You need the following input for this task:

- **Master Report Tracking List (RD.080)**

The Master Report Tracking List, created in Business Requirements Definition (RD), is updated to reflect the results of mapping activities.

- **Reporting and Information Access Strategy (TA.060)**

Planned architecture for reporting systems will influence the type of reports that will be feasible and the magnitude of the customization effort.

Optional

You may need the following input for this task:

- **Mapped Business Requirements (BR.030)**

Mapped Business Requirements specify alternatives to business requirements. Frequently, the documented alternatives will be a report.

❑ **Information Model (BR.060)**

The Information Model describes information sharing and information partitioning requirements across organizations for the key business objects and business process information. If Create Information Model was not performed, this deliverable will not exist. (See the task description for BR.060 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Map any previously unmapped report requirements to future business processes using Business Requirements Scenarios (RD.050) and Mapped Business Requirements (BR.030), and update the Master Report Tracking List (RD.080).	Updated Master Report Tracking List (RD.080)
2.	Review the reporting strategy to understand the capabilities of placing reporting systems and constraints on designs.	
3.	Decide on an approach for mapping report requirements and assign responsibilities.	
4.	Map report requirements to standard application reports.	Updated Master Report Tracking List (RD.080)
5.	Analyze reports for reduction.	Updated Master Report Tracking List (RD.080)

No.	Task Step	Deliverable Component
6.	Prioritize custom reports.	Updated Master Report Tracking List (RD.080)
7.	Secure acceptance of the Master Report Tracking List.	

Table 3-19 Task Steps for Conduct Reporting Fit Analysis

Approach and Techniques

The Master Report Tracking List documents the report mapping process.

Requirements Mapping to Future Business Processes

Map any previously unmapped report requirements to a Business Requirements Scenario (RD.050) and enter the scenario number in the future process number field of the Master Report Tracking List. Most likely any unmapped report came from a report survey, or other gathering technique. If a report does not map to a BRS, enter "No Match."

Reporting Systems Leveraging from the Application Architecture

You may be able to leverage use of special reporting systems to reduce the number of reports you need to design and build. Examples of such systems are:

- business intelligence systems
- data warehouses (operational or decision-support)
- online analytical processing (OLAP) systems
- ad hoc query systems

If the architecture work completed so far during the project has already identified the need for such systems, work with the system architect to understand how you may make use of these systems to satisfy reporting needs.

Conversely, if you identify the need for a special reporting system to address common reporting requirements, you should provide the input to the architecture process.

Requirements Mapping to Standard Application Reports

To map successfully, business analysts must have a thorough understanding of the original report requirement and the standard application reports. Otherwise, you may need to conduct a series of interviews between the user and the individual mapping the process. The following are some of the typical report mapping issues:

- **Flexfields** — Data captured in flexfields will not be part of a standard report; therefore, any report requirement using flexfield data will become a *custom extension*. Sometimes you do not know whether data will be stored in a flexfield or another application field used by a standard report. In these cases, mark the report as a match with a note to modify the report, if the data is stored in a flexfield.
- **Lack of training** —Users are often trained just before the implementation is complete. Unfortunately, mapping occurs much earlier in the project. If users are going to do their own mapping they will need the following:
  - access to a prototype environment
  - training on future processes
  - training on how to run reports

Even with training, it can be difficult for users to envision the final product because they may be too far removed from the implementation team. If there is not adequate help from the team, their mapping will be inaccurate and their reports could be useless.

Suggestion: Set up an interview process where the user and the project team member (user liaison role) do the mapping together. The user liaison must invest enough time before mapping to become familiar with all standard application reports that impact the area in question.

The organization may be such that major divisions within the organization have similar subfunctions. Purchasing might be an example. Each division or group may have a purchasing function. If the organization of the project is by division or group, the purchasing role might have a possible duplication multiple times across the project. The mapping of all purchasing reports by one user liaison will save the time and energy of the rest of the team.

Reports Analysis for Report Reduction

Some form of reduction process must take place when there are more custom reports identified for development than:

- are necessary to run the business
- can be completed in the allocated time
- are expected, potentially placing reporting development over budget

Reduction Techniques

The following are suggestions for reducing the number of report requirements. The process of analyzing and consolidating may continue until well after construction has begun, so it is important to focus on your most critical reports first, in order to allow time for the analysis to conclude.

- Eliminate reports with duplicate file names. Do not delete these requirements from the list, since they represent a valid user requirement that is necessary when preparing status documents for the user, user's department or management.
- Analyze based on function. Resort the Master Report Tracking List by function. If several reports relate to the same function, you may be able to combine the requirements from one report into another. This type of consolidation may dramatically reduce the number of requirements. The users and the team need to know which reports are being consolidated and the name of the new report.

Warning: Track consolidations carefully, especially if they cross departmental boundaries. While initially all parties may agree the consolidation looks good, changes requested by one group may not be appropriate for others. When this happens, you may need to create another new report and track it separately.

Custom Reports Prioritization

As you map reports to standard functionality, custom requirements may develop. Anything marked as a *Build* or *Modify* is a custom requirement. Sort the Master Report Tracking List by *Assessment* and make sure all custom requirements have a priority. Print this list and distribute it to the team and users, and request that they make any necessary changes to the priority. This will be an ongoing function. Priority is the basis for the drive for the entire development process and thus needs careful management. Users should always sign off on the assigned priority to avoid conflicts at later stages.

Master Report Tracking List Distribution to Users for Approval

It is important to forward a copy of the documented reporting requirements to users for approval. The user community should confirm that the following are true:

- All the reporting requirements documentation is forwarded to each group.
- The information gathered about each requirement is correct.
- All priorities have been assigned and are correct.

Linkage to Other Tasks

The Master Report Tracking List is an input to the following tasks:

- BR.090 - Confirm Integrated Business Solutions
- MD.020 - Define and Estimate Application Extensions

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	45
Technical Analyst	20
Application Specialist	20

Role	%
Configuration Management Specialist	10
System Architect	5
User	*

Table 3-20 Role Contribution for Conduct Reporting Fit Analysis

* Participation level not estimated.

Deliverable Guidelines

Use the Master Report Tracking List to update the mapping section of the Master Report Tracking List developed in Identify Reporting and Information Access Requirements (RD.080). Record the appropriate mapping information based on the report mapping activities performed.

This deliverable should address the following:

- disposition of every report requirement

Deliverable Components

The Master Report Tracking List consists of the following component:

- Updated Master Report Tracking List

Updated Master Report Tracking List

This component updates the mapping columns in the Master Report Tracking List.

Tools

Deliverable Template

Update the Master Report Tracking List, developed in Identify Reporting and Information Access Requirements (RD.080), to complete the deliverable for this task.

BR.080 - Test Business Solutions (Optional)

In this task, you perform an extended test of the business alternatives and validate integrated business processes based on the mapping decisions.

Δ If your project includes p*rocess change* or is of *medium or high complexity*, you should perform this task.

Deliverable

The deliverable for this task is the **Business Mapping Test Results.**

Prerequisites

You need the following input for this task:

- ❑ **Configured Mapping Environment (BR.020)**

Each test team needs a modeling environment in which to test individual and integrated alternatives.

- ❑ **Mapped Business Requirements (BR.030)**

Each scenario is a unique response path through the process flow diagram for a business process. Listing all responses that are possible from within a process will produce an associated list of scenarios.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the Mapped Business Requirements (BR.030).	Business Process Identification

No.	Task Step	Deliverable Component
2.	Document test scenarios for each process.	Testing Scenario List
3.	Identify inputs and outputs, data conditions, and business rules.	Data Profile
4.	Develop business testing specifications.	Test Specification
5.	Develop support requirements.	Support Profile
6.	Assign testing roles and responsibilities.	
7.	Review responsibilities and events with the project team.	
8.	Execute tests.	
9.	Summarize test results.	Test Specification
10.	Document required corrective actions.	Test Actions
11.	Update Business Requirements Scenarios (RD.050) based on testing results.	Updated Business Requirements Scenarios
12.	Update Business Requirements Mapping Forms (from BR.030) based on testing results.	Updated Business Requirements Mapping Forms
13.	Secure acceptance of the Business Mapping Test Results.	

Table 3-21 Task Steps for Test Business Solutions

Approach and Techniques

The Business Mapping Test Results document the extended test of the business alternatives and validate integrated business processes based on the mapping decisions.

Business Solution Testing

Test business alternatives in order to prove feasibility and to begin locking in acceptance and adoption by key users and the broader project team. Involving key users in test designs gains their approval in proportion to time invested.

Performance of alternative *prototyping* was performed during mapping. Business alternatives testing differs from prototyping in these ways:

- Alternatives testing is more formal than prototyping.
- Alternatives testing involves process *integration,* whereas prototyping does not.

Business alternative testing is more detailed than prototyping. In order to design and execute a well-run test you must document the following:

- test setup
- expected results
- actual results
- corrective actions in response to test results

With prototyping, you develop a quick alternative in order to sell a concept and determine the best, often rough-cut, design for a business process.

In this task you are testing business concepts rather than programs because application extensions are not yet developed. You are trying to get closure on a detailed design idea. Demonstrate well-integrated proposed business designs.

☞ **Attention:** Business alternative testing is not the same thing as module or system testing. Instead, you are verifying that a process design will satisfy business objectives using stated tools, resources, and application system components.

Testing Teams Organization

Normally testing teams organize around the same grouping of business processes as mapping teams. Testing groups will work together by high-level business function, process group, or even across processes for testing of process integration.

☞ **Attention:** A high-level business function could be a logical grouping of business processes, particularly for large projects.

The skills required for the testing team are similar to mapping, except that testing requires more discipline when it comes to recording details, whereas mapping involves more unconstrained creativity.

Request that key users join test sessions. The emphasis is on realistic assumptions, tools, and setting. The key to success is having the test simulate the new process in sufficient depth, so that everyone concentrates on the new, proposed way of doing business. Always organize testing sessions for efficiency by publishing a thorough agenda that includes:

- session location and duration
- role assignments
- business process to be mapped
- activities and sequence
- inputs required (like prerequisite deliverables) and assignments for bringing them
- props and other physical setups that help simulate realism
- expected outputs and criteria for successful closure of the testing session

Data Profiles

Data profiles describe data entities and business conditions needed to test the application system. Determine data profiles by performing a careful walkthrough of the steps of the scenario. The inputs into each scenario step will include data entities like *customer* or *master demand schedule*, data conditions (actual values, reference to some other document containing values, or even a screen shot), business rules (the policy or decision drivers that influence the process step), and type and status information (to clarify the business object). Together this profile information for a scenario step creates a business condition (also known as a *script*).

For the purposes of testing, there are three types of data profiles:

- **Preexisting Data Profile** — describes the conditions that already exist in order for a test to be properly executed
- **User Input Data Profile** — describes the data that must be entered by the user during test execution
- **Expected Output Data Profile** — describes the data conditions that are expected as the result of a successful test execution

If a scenario is named properly, its name may imply the primary business condition (for example, *schedule a non-stocked domestic finished goods item*).

The following figure illustrates the relationship between scenarios, test steps, and data profiles.

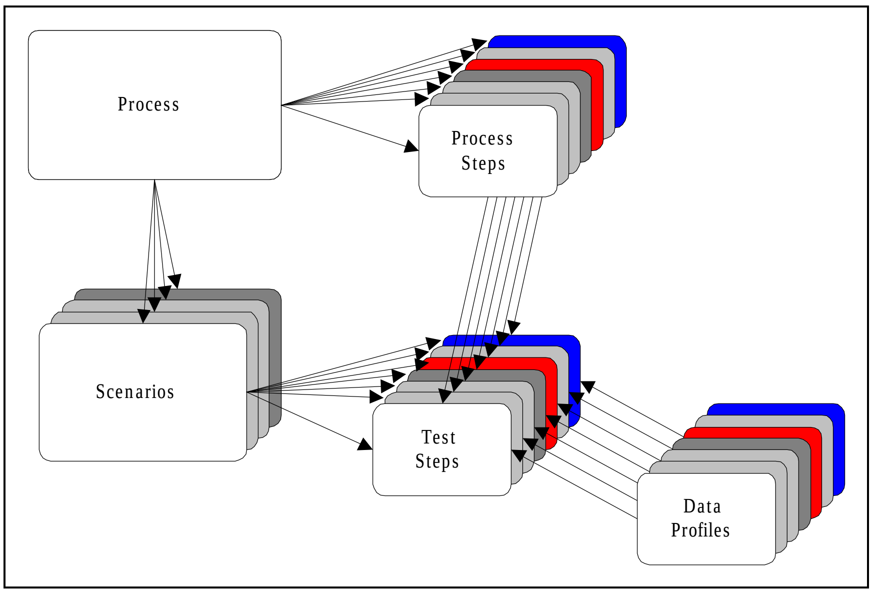


Figure 3-6 Test Step and Data Profile Relationship

Test Specifications

A test specification is the component of a test script that defines test script execution. The basis is entirely on a specific system process and consists of scenario information, process information, and a series of test steps, the order having been determined from the process. A test specification communicates the following:

- test steps, corresponding to some, if not all, system process steps, detailing the business testing requirement of a scenario
- the sequential relationship between test steps
- the data profiles involved in each test step
- the actions performed in each test step
- the expected application system responses in each test step

The figure below represents the relationship between a process, process steps, scenarios, and test steps.

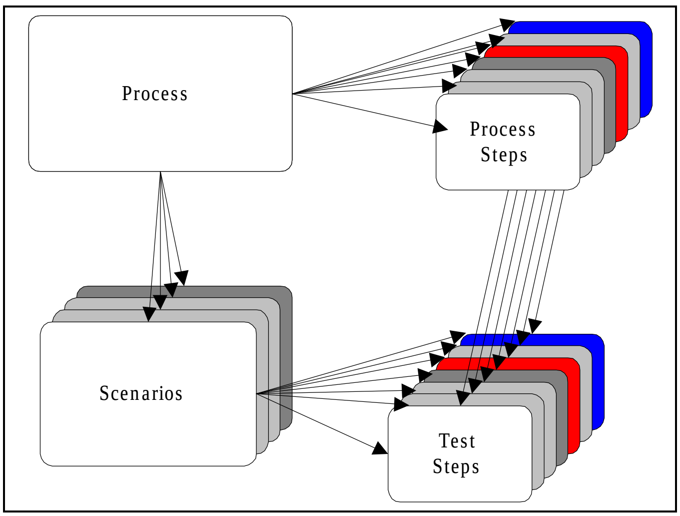


Figure 3-7 Scenario and Test Steps Relationship

☞ **Attention:** When varying initial conditions cause multiple outcomes, decide whether to remodel the process and possibly separate it into more than one process, or plan multiple scenarios from the same business process. It could be that a decision step was inadvertently omitted. It is also possible that different sets of unique initial conditions really correspond to multiple events or event mechanisms.

Test specifications also provide for recording the results of tests — both qualitative (like description of outputs) and quantitative (like measured cycle time of the actual test).

Support Profiles

The consideration of expected environmental information (such as controls and constraints) and resource capabilities (known also as mechanisms) is required for any business process step. Specifying these support requirements into a profile for each scenario accomplishes the same result as when listing assumptions in a business proposal paper. This technique alerts management regarding necessary prerequisites in order for the process step to perform satisfactorily. This is true even when a process step is system assisted, or when it is an automated step.

If support profiles are planned properly, they will provide a linkage with the Business Procedure Documentation (BP.090). Good documentation during business alternative testing saves time later and will increase the quality of deliverables.

Conference Room Pilot (CRP)

You can perform testing of business alternatives in a formal conference room pilot (CRP) or use a more informal approach as a follow-on to mapping activities. In either case, you must test *around* any identified functionality gaps because it is likely that no final alternative has been designed or built to bridge those gaps.

A CRP is a technique for evaluating a proposed alternative that simulates actual business processes in a controlled environment. It is often performed in an actual conference room so that all participants will be close together and delays will be minimized. A CRP can reveal missed problems and issues of processes tested in isolation.

A conference room pilot is usually run by business area or some other, wider grouping of business processes. This means that multiple mapping teams must be ready at the same time in order to begin. As with most testing strategies, a CRP consists of iterations, converging on a practical and feasible alternative. Once all testing is complete and confirmed, you can document final setups and selected alternatives.

Parallel Testing

A faster but riskier approach is to pursue a strategy of ongoing testing in parallel with other mapping activities. Essentially, project team members test whenever they are not participating in mapping activities. As each process area is mapped, those processes are added to an overall integrated test flow. The project team then repeats some or all of the tests to confirm that new processes integrate with the overall business flow. As testing proceeds, the team begins documenting setups and procedures (for later use in Define Application Setups - BR.100 and Document Business Procedures - BP.090 tasks) and performs them in parallel with testing.

After all processes are mapped, tested, and confirmed, documentation is completed, and the design of custom extensions (if any) can be started.

Linkage to Process Modeling and Mapping

Begin developing mapping scenarios early in the design of a business process. When possible, and for simplicity, try to design a process to include *only one* scenario. This will reduce the complexity associated with process variations.

Mapping scenarios are extensions of the BRS during the life-cycle of process design, mapping, and testing for a business process. The BRS links Business Requirements Definition (RD) and Business Requirements Mapping (BR) into a tight, holistic set of tasks that produce a final, feasible design for a business process.

Cross-Process Integration

Completion of testing faces two significant challenges:

- The tested BRS may not integrate properly with other Business Requirements Scenarios (RD.050).
- Custom extension may not yet be available.

There may be many interdependencies across Business Requirements Scenarios (RD.050). The key is to identify key internal customers who will help document and plan for testing these linkages. One way to do this is to try to match process outputs with the list of events. Outputs that have no match may mean that a process overlaps another process, or it could mean that some business process steps are not contained within any process.

Try to simulate the effect of tools that are not yet developed. List key assumptions like response time. These will be key inputs into later construction of Application Extension Definition and Estimates (MD.020) and Design Standards (MD.030). Remember, you are not testing programs in this task; instead, you are testing business concepts.

It is permissible to establish tests with unique scenario numbers that actually link scenarios across two or more business processes. This is especially useful with multiple events that drive a process source from different business processes.

Linkage with Business Systems Testing

There are many similarities between Test Business Solutions (BR.080) and Business System Testing (TE). Both involve rigorous test specification, profiling initial conditions, results collection, and corrective action. Additionally, both are concerned with cross-process integration, although Business System Testing (TE) is more formal in this regard.

While Test Business Solutions (BR.080) provides a reference point in the project for proof of concept, Business Systems Testing (TE) tests use actual converted data, developed extensions, new applications setups, modules for integration with other application components, and custom reports that could only be assumed during mapping.

Additionally, during mapping not all teams are necessarily synchronized for integrated testing since some will progress ahead of others. So the Business System Testing (TE) integration testing is more thorough.

Review and Acceptance

As each business alternative is tested, you should review documented results with the project sponsor, management, representative business users and any formal integration teams.

Linkage to Other Tasks

The Business Mapping Test Results are an input to the following task:

- BR.090 - Confirm Integrated Business Solutions

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	65
Application Specialist	30
Technical Analyst	5
Business Line Manager	*

Table 3-22 Role Contribution for Test Business Solutions

* Participation level not estimated.

Deliverable Guidelines

Use the Business Mapping Test Results as a logical extension of the work performed during process design and mapping.

This deliverable should address the following:

- **Data Profile** — describes the business conditions that are needed to test the application system
- **Test Specification** — defines test script execution
- **Support Profile** — identifies support tools required during execution of the test

In addition, test results and actions are captured.

Acceptance Criteria

The testing scenario should contain heading information, such as:

- scenario number
- scenario description

Organize detailed information by scenario step such that all testing scenario deliverable components carry information supportive of their respective step. A scenario step will be in a one-to-one correspondence with a business process step.

In order to pass a quality review:

- The testing scenario must describe all tools and user actions necessary in order to perform the test.
- Accomplish and close all documented test actions.

☞ **Attention:** As an alternative to using the detailed test actions form, you can simply document and track problems and their resolutions in an issues list.

Solution Testing Results Form

As required, the test actions component should be completed with this information:

- type of corrective action necessary
- description of the action
- responsibility
- date due

Deliverable Components

The Business Mapping Test Results consist of the following components:

- Business Process Identification
- Testing Scenario List
- Data Profile
- Test Specification
- Support Profile
- Test Actions
- Updated Business Requirements Scenarios (RD.050)
- Updated Business Requirements Mapping Forms (BR.030)

Business Process Identification

This component is used to identify the business process.

Testing Scenario List

This component documents test scenarios for each business process.

Data Profile

This component creates several data profiles per process task.

- **Scenario Step** — unique sequence number for each task as listed on the BRS for this scenario
- **Business Object** — the name of a particular data element that must be present (for example, *customer* or *master demand schedule*)
- **Data Condition** — actual values, reference to some other document containing values, or even a screen shot
- **Business Rule** — the policy or decision drivers that influence the process step
- **Type** — the entity type
- **Status** — the entity state

Test Specification

This component creates several test specification line items per process task.

- **Scenario Step** — unique sequence number for each task as listed on the BRS for this scenario
- **Test Step** — the series of procedural steps that must be taken for the scenario step in order to perform the test properly
- **Role** — step / sequence owner
- **Action or Path** — description of action to be taken by the primary resource or role during the test; either navigation, location, or directive information
- **Expected Results** — anticipated outcomes or outputs described in measurable terms
- **Actual Results** — actual outcomes or outputs described in measurable terms

- **Expected Cycle Time** — anticipated elapsed time for processing the step / sequence
- **Actual Cycle Time** — actual elapsed time consumed during processing the step / sequence
- **Status** — Active, Pending Active, Pending Obsolete, or Obsolete

Test Specification

This component documents the business testing specifications.

Support Profile

This component creates several support profile line items per process task.

- **Scenario Step** — unique sequence number for each task as listed on the BRS for this scenario
- **Test Step** — the series of procedural steps that must be taken for the scenario step in order to perform the test properly
- **Controls** — environmental information that affects what is possible: constraints or limits
- **Mechanisms** — resources that enable or facilitate the step / sequence
- **Other Input** — other tools necessary for running this step / sequence of the test
- **Status** — Active, Pending Active, Pending Obsolete, or Obsolete

Test Actions

This component document required corrective actions.

Tools

Deliverable Template

Use the Business Mapping Test Results template to create the deliverable for this task.

BR.090 - Confirm Integrated Business Solutions (Core)

In this task, you secure approval for proposed business alternatives.

Deliverable

The deliverable for this task is the **Confirmed Business Solutions**. This deliverable creates one **Confirmed Business Solution** for each set of related Business Requirements Scenario (RD.050).

Prerequisites

You need the following input for this task:

- ❑ **Future Process Model (BP.080)**

It is helpful to show the process flow diagram for each future business process as it is being approved.

- ❑ **Mapped Business Requirements (BR.030)**

Each Mapped Business Requirement represents a feasible, tested alternative to satisfying requirements for a business process.

- ❑ **Mapped Business Data (BR.040)**

Shows underlying table structures and attributes to adequately support business processes.

- ❑ **Integration Fit Analysis (BR.050)**

This deliverable lists all the integration points in the new system across applications and application installations, created from a bottom-up analysis. You should verify that these integration points are covered by Business Requirements Scenarios.

- ❑ **Information Model (BR.060)**

The specifications for the movement and access of information and data in the business by business functions, organizations, data centers and other secure user groups. This is a high-level analysis of the flow, access, security and reporting requirements in the future business model. If Create Information model was not performed, this deliverable will not exist. (See the task description for BR.060 for more information on when this task should be performed.)

- ❑ **Master Report Tracking List (BR.070)**

Acceptable business alternatives must be comprehensive in terms of stating fit and gaps for report requirements. The Master Report Tracking List (RD.080) is updated to reflect report mapping decisions during Conduct Reporting Fit Analysis (BR.070) activities.

- ❑ **Business Mapping Test Results (BR.080)**

The Business Mapping Test Results contain test results and actions as part of the alternative that must be considered when signing off on proposed process designs.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review prototype, test results, and mapping documents.	
2.	Revise business alternatives for agreed upon changes.	
3.	Prepare an Acceptance Certificate (PJM.CR.080) for integrated alternatives.	Acceptance Certificate (PJM.CR.080)
4.	Secure acceptance of the Confirmed Business Solutions.	Signed Acceptance Certificate

Table 3-23 Task Steps for Confirm Integrated Business Solutions

Approach and Techniques

The Confirmed Business Solutions present the Business Requirements Scenarios (RD.050), the Mapped Business Requirements (BR.030), and testing results to management for approval.

You may request approval for each process area as it is mapped, or you can get approval for an entire functional area (for example, manufacturing, finance, distribution, and so on). Typically, each process area must first be approved before approval of larger groups of integrated processes. Approval as a group is required for each business area in large projects composed of multiple business areas.

Confirm that "internal" customers have accepted the outputs produced during testing, as well as the assumptions surrounding the tests. Try to limit the number of cross-process group interdependencies as a way of simplifying:

- how the business works
- internal communications

In this way you can reduce the number and complexity of support systems and the processes required for their maintenance.

If alternatives are not accepted, the reasons for non-acceptance should drive a round of revisions to the affected Mapped Business Requirements (BR.030) deliverable. Another cycle of process modeling, process design, mapping, and testing will be necessary.

Linkage to Other Tasks

The Confirmed Business Solutions is an input to the following tasks:

- BR.100 - Define Application Setups
- MD.020 - Define and Estimate Application Extensions
- TE.040 - Develop System Test Script

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	50
Application Specialist	25
Technical Analyst	25
Business Line Manager	*

Table 3-24 Role Contribution for Confirm Integrated Business Solutions

* Participation level not estimated.

Deliverable Guidelines

Use the Confirmed Business Solutions to represent an acknowledgment that all relevant parties have reviewed the materials produced and agree on the proposed business alternatives. This approval task is especially important when new processes are being proposed. Dramatic changes to existing processes are likely to affect job definitions and sometimes organizational structure. The parties that approve the proposed business alternatives should address the following:

- executive management
- financial management
- operating (functional) management
- human resources
- users (responsible for processes)

The parties usually note any desired changes and agree on a course of action to implement those changes. The deliverable serves as a formal record of the parties' agreement.

This deliverable should address the following:

- deliverable being confirmed
- type and status of deliverable
- objectives of deliverable
- agreements (expressed in terms of planned future actions or policy changes)
- key decisions
- key assumptions
- exceptions or references to components requiring changes
- controls: signatures of approvers and initiators, dates, revisions, and so on
- future direction

This approval should reference the relevant list of Mapped Business Requirements (BR.030) for the proposed business alternatives being reviewed. You can use this document to clarify any issues encountered later. Confirmation is most effective when documented by a signed letter, memo, or certificate. As with all critical deliverables, verbal agreement is not as effective.

Tools

Deliverable Template

A deliverable template is not provided for this task. Use the Acceptance Certificate (PJM.CR.080) template to document acceptance of the business alternative.

BR.100 - Define Application Setups (Core)

In this task, you capture the setup decisions and implement them in the appropriate environment

Deliverable

The deliverable for this task is the **Application Setup Documents**. These documents define the detailed setup parameters that have been proven to support the selected alternative.

Prerequisites

You need the following input for this task:

- ❑ **Mapped Business Requirements (BR.030)**

Mapped Business Requirements are detailed explanations of key requirements or their alternatives. This deliverable contains intermediate decisions regarding application setups.

- ❑ **Confirmed Business Solutions (BR.090)**

Confirmation and approval of a business alternative must be obtained before implementing application setups.

- ❑ **Application Architecture (TA.040)**

If the system architect created a detailed map of the fundamental application setup parameters that affect application architecture, you must incorporate this functional architecture with the application setups.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the application configuration in the mapping environment.	
2.	Review business mapping decisions and documents.	
3.	Define the application setups intended for production.	Application Setup Documents
4.	Implement the application setups in the appropriate environments (if necessary).	
5.	Review and confirm configuration and impact of changes.	
6.	Secure acceptance of the Application Setup Documents.	

Table 3-25 Task Steps for Define Application Setups

Approach and Techniques

The Application Setup Documents record the parameters, user-defined codes, and other setups for each application. As mapping completes and design begins, the components required to complete the configuration fall into place. You can extract the setup requirements from existing documentation. Use the setup information that was recorded in the Mapped Business Requirements (BR.030) as a primary source of information. The main objective is to consolidate the configuration of all applications for centralized maintenance. With the number of separate application databases on the organization's systems, it becomes a challenge to make sure that the configurations represent the latest mapping decisions.

Warning: Only key project team members, each with specific responsibilities, should initially define the system application setups. It is important to stabilize the environment and control the changes made during the mapping process, so that any one team does not undo the settings made by another team.

It is particularly challenging to maintain setups by application module when design, mapping, and testing activities have been organized by business process. If an integration team is responsible for cross-process integrity, let them assist with maintaining application setups as well.

Application Implementation Wizard

The Application Implementation Wizard (AIW) is Oracle's workflow based setup tool that guides you through the set up steps in the appropriate sequence.

Critical Setup Parameters

Application parameters do not all carry the same importance to the business. Some are more critical in determining how the system will be operated. For instance, within the standard applications the following parameters control significant processing features and functions:

- **Set of Books** — an accounting ledger with a particular chart of accounts, functional currency and accounting calendar
- **Balancing Entity** — a division or other business unit for which you prepare a balance sheet
- **Inventory Organization** — a plant, warehouse, or other type of business unit designed to provide control and transaction functionality within one or more applications modules

Low-Volume Conversion Activities

You may use the application setup tables or spreadsheets as source documents for manual conversion activities. These spreadsheets help you record the entries needed for production. Weigh the resource time needed to enter this data against the total development time for automating the conversion.

Standard Reports

Some standard applications provide standard setup reports that capture the input data automatically. Simply run these reports and use these as source documents for later setup activities.

Screen Shots

Take online screen shots of the standard applications displaying the system and parameter selections. Be careful to capture the complete setup; many definition screens are made up of multiple screen pages.

Linkage to Other Tasks

The Application Setup Documents are an input to the following tasks:

- MD.050 - Create Application Extensions Functional Design
- MD.090 - Prepare Development Environment
- CV.040 - Perform Conversion Data Mapping
- DO.070 - Publish User Guide
- AP.160 - Prepare User Learning Environment
- PM.030 - Develop Transition and Contingency Plan
- PM.050 - Set Up Applications

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	50
Application Specialist	40
System Administrator	10

Table 3-26 Role Contribution for Define Application Setups

Deliverable Guidelines

Use the Application Setup Documents to compose profile options, application options, key and descriptive flexfields, and user-defined codes. The Application Setup Documents consolidate the setup parameters and codes of all applications for centralized maintenance as decided during business mapping. Use it to capture and communicate the final application setup decisions for implementation in the production environment.

This deliverable should address the following:

- mandatory or optional setups
- common setups across applications
- key flexfields
- descriptive flexfields
- navigation to the screen
- justification for selected options
- maintenance consolidation (if any)
- date of last update
- impact of any setup changes in the course of the implementation

Because Application Setup Documents will undergo many revisions, confirm that there are procedures in place to update and carefully control the setups. You may want to control Application Setup Documents with a configuration management tool.

The Application Setup Documents include master tables to control ownership, QA responsibility, due dates, and approval signoff for each component setup, in addition to the detail table specific to each component.

Deliverable Components

The components of the Application Setup Documents vary depending on the application that is being set up.

Tools

Deliverable Templates

Multiple Application Setup Document templates are provided. Use the deliverable template that corresponds to the module you intend to implement including common application, common system administrator, or other common setups. In addition, use the application-specific template that corresponds to the application you are setting up, to create the deliverable for this task.

BR.110 - Design Security Profiles (Core)

In this task, you gather role and function information and relate them to application security and responsibilities. As business requirements are established and mapped to application features, you also begin to define the user security necessary to support the selected alternative in a controlled environment.

Deliverable

The deliverable for this task is the **Security Profiles**. They provide a list of role-based security specifications necessary to maintain good controls and transaction access.

Prerequisites

You need the following input for this task:

❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides a list of all of the organizations within the business. You will need to identify the roles of different users within the organizations and then map these to the security features within the applications.

❑ **Audit and Control Requirements (RD.060)**

There may be requirements based on audit and control analysis that will impact Security Profiles.

❑ **Master Report Tracking List (RD.080)**

There may be requirements based on report analysis that will impact Security Profiles.

❑ **Mapped Business Requirements (BR.030)**

Mapped Business Requirements are detailed explanations of key requirements or their alternatives. This is the deliverable that contains intermediate decisions regarding application setups, including security and responsibility information you may need to refer to this deliverable.

- ❑ **Information Model (BR.060)**

The Information Model provides the organizational view of the requirements for access of information and data in the business. The organizational view of information access is the sum total of all of the individual role-based access requirements within an organization and therefore is a foundation for the individual security profiles for the roles within the organization. If Create Information Model was not performed, this deliverable will not exist. (See the task description for BR.060 for more information on when this task should be performed.)

- ❑ **Application Security Architecture (TA.080)**

The Application Security Architecture provides information about the design and usage of the applications and databases to support the organization-based security requirements. The implementation of the application security architecture may impact the way that role-based security is implemented in the new system.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify user roles across all business functions and organizations.	User Roles
2.	Identify security requirements for each user role.	User Roles
3.	Map user roles onto application security structures.	System User Roles

No.	Task Step	Deliverable Component
4.	Define application module access for each system user role.	System User Roles
5.	Secure acceptance of the Security Profiles.	

Table 3-27 Task Steps for Design Security Profiles

Approach and Techniques

The Security Profiles primary objective is to develop a security structure that naturally supports business processes. The primary technique is to map business process steps and their agents (owners) with the application-provided user responsibilities and make adjustments to the responsibilities as required.

It is important to achieve a good menu structure so that application access naturally supports the flow of information in the workplace. This will also make learning the application easier.

Linkage to Other Tasks

The Security Profiles are an input to the following task:

- PM.050 - Set Up Applications

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Business Analyst	60
Application Specialist	30
System Administrator	10

Table 3-28 Role Contribution for Design Security Profiles

Deliverable Guidelines

Use the Security Profiles to identify roles (agents from the Mapped Business Requirements - BR.030) grouped together by responsibility, so that typical business functions, along with inquiries and reports, are accessible.

Try to accomplish security objectives with the minimum number of profiles to make maintenance easier.

This deliverable should address the following:

- security structure
- roles

Deliverable Components

The Security Profiles consist of the following components:

- User Roles
- System User Roles

User Roles

This component documents user information such as role, title, and department.

System User Roles

This component documents user information such as business function, transaction, group, and audit level.

Tools

Deliverable Templates

Use the Security Profiles to create the deliverable for this task.

CHAPTER

4

Application and Technical Architecture (TA)

This chapter describes the Application and Technical Architecture process.

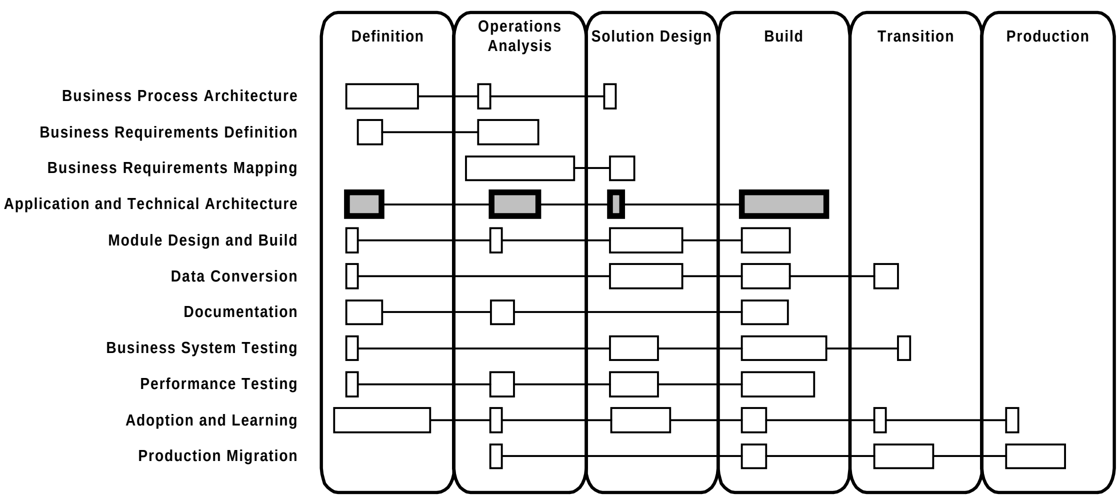


Figure 4-1 Application and Technical Architecture Context

Process Flow

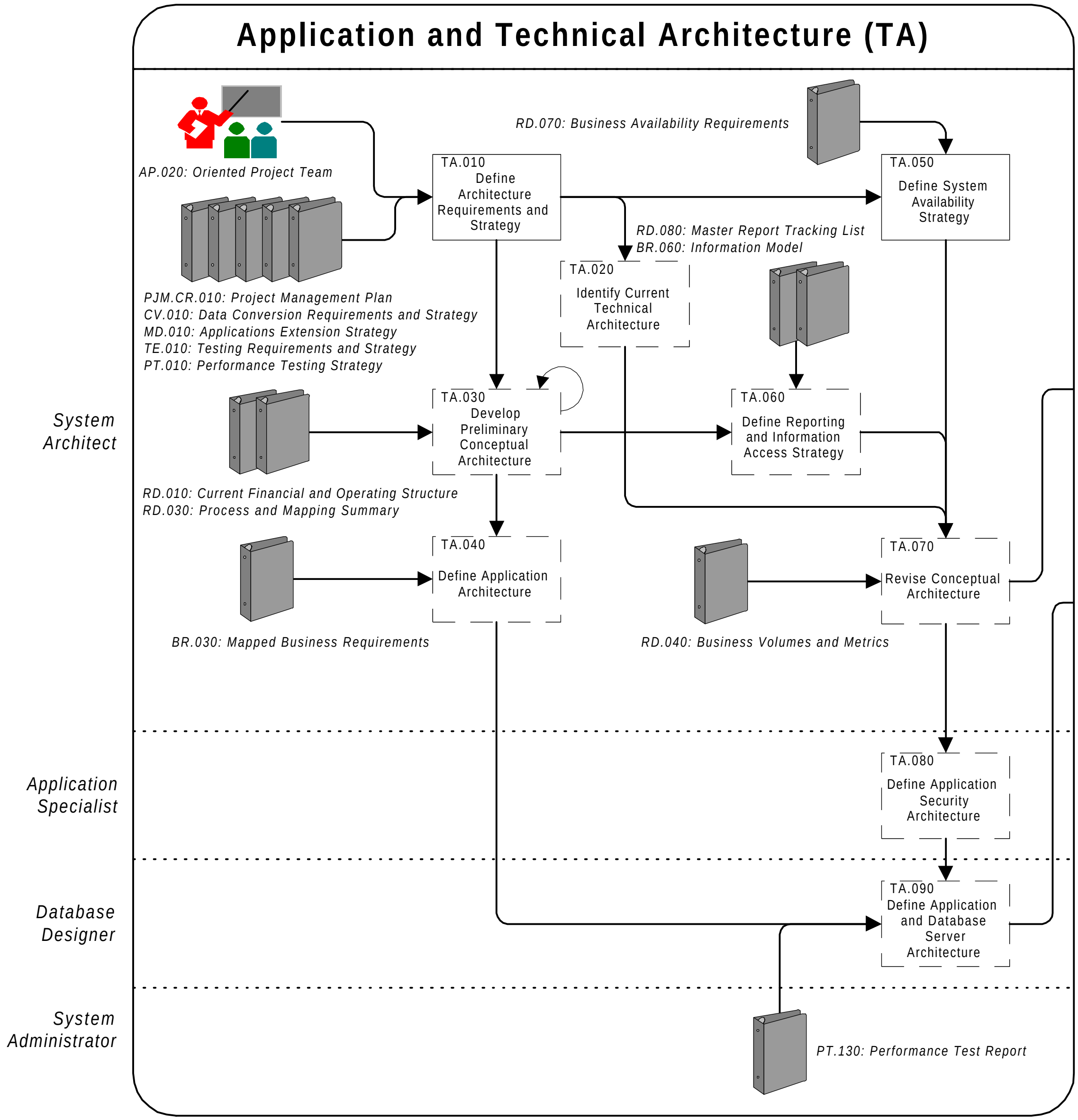


Figure 4-2 Application and Technical Architecture Process Flow Diagram

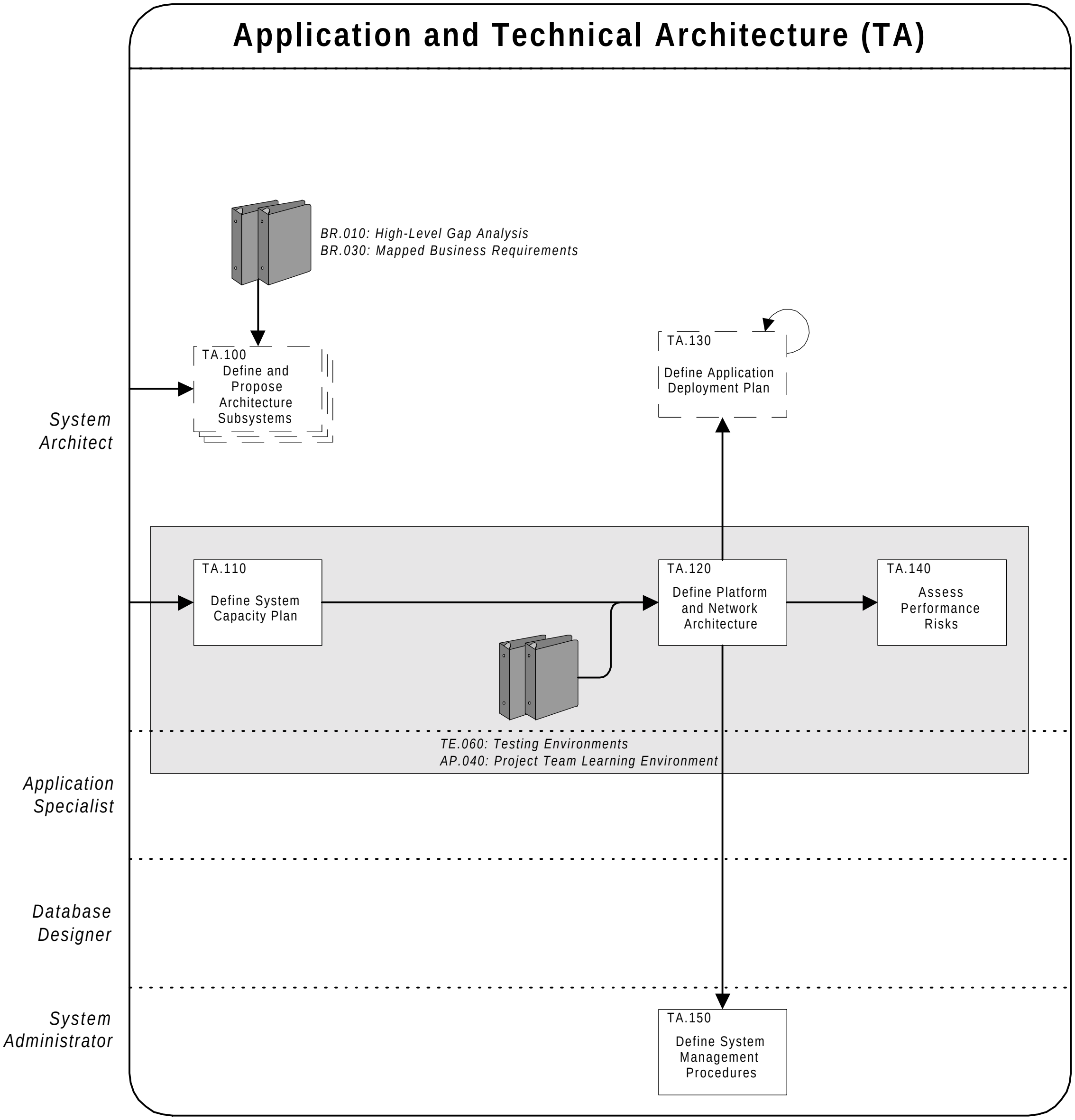


Figure 4-2 Application and Technical Architecture Process Flow Diagram (cont.)

Approach

The objective of Application and Technical Architecture is to design an information systems architecture to realize the business vision. The process takes the business and information systems requirements and develops a blueprint for deploying and configuring:

- Oracle, third-party, and custom applications
- supporting application server environments
- critical interfaces and data distribution mechanisms between applications, servers, and sites
- computing hardware, including servers and client desktop platforms
- networks and communications infrastructure

A coherent and well-designed information systems architecture is critical for the success of any implementation project. To arrive at an architecture for the newly implemented systems, consider the following complex issues:

- the best deployment strategy for the applications across one or more data centers, business organizations, and business functions
- the high-level configuration of the applications to support the financial, administration, manufacturing, customer management, supply chain, selling chain, and distribution business units
- the interface points between the applications or remote sites, including specifications, data flows, and mechanisms
- the deployment and capacity planning of the hardware and network infrastructure that will support the applications processing
- the management tools and procedures that will enable the system to continue to operate as intended

If the implementation project involves a complete replacement of legacy business systems, the architecture work needs to be comprehensive enough to define the framework for the information systems vision of the corporation, which will underpin the corporate business vision. An example of this is where you are performing a complete replacement of your Enterprise Resource Planning (ERP) systems or the implementation of a comprehensive Customer Relationship Management (CRP) system. Where the implementation is for a replacement of a portion of the business systems, the architecture of that replacement part of the system must be compatible with the overall system architecture.

Application and Technical Architecture can be divided into two major areas:

Application Architecture	Addresses the deployment and configuration of applications across data centers, servers, and hardware platforms.
Technical Architecture	Addresses the configuration and capacity planning of the hardware platform and network infrastructure and system management issues.

Application Architecture

The application architecture portion of the process includes these key areas:

- Application Deployment Map and Integration Points
- Information Security
- Reporting and Information Access Strategy
- Application Architecture (Critical Application Setup)

Application Deployment Map and Integration Points

This seemingly straightforward issue is very important for understanding and defining the applications deployment across different data centers and servers, together with the supporting databases. The information flows between the deployed applications reveal the application integration points and interfaces that may be needed.

Information Security

Most information security issues are not critical to the success of an implementation, but some may be. The modification of tables to add new columns for the provision of custom, database-level security increases implementation complexity substantially.

Reporting and Information Access Strategy

Reporting and information access strategy addresses the different types of reporting and information needs for the business, such as operational reporting strategy (data collected or transacted over the last day, week, or year), decision support strategy (usually covers data collected over a longer period of time), ad hoc reporting, business intelligence, and analytical reporting. The strategy needs to take into account physical data distribution in the business and the databases and applications within which the data is stored.

Attention: Oracle Applications have robust operational, analytical, and decision support reporting capability, but this would not cover reporting across heterogeneous systems. You may need to define operational or decision support type data warehouses, reporting, business intelligence, or other support subsystems to provide the consolidation or special reporting capability across different enterprise systems and applications.

Application Architecture (Critical Application Setup)

This area primarily addresses the configuration of the Oracle package applications (rather than in-house developed applications), where you have less control over the functional setup and must map your business to the fundamental features and functions built into the package. The choices you make about the critical application setup parameters can have a direct effect on your application and technical architecture, influencing the application deployment and data partitioning. These setups are the very few of critical elements of the overall setup activity that often cross applications, sites and business functions. These are not the specific application module setups that are addressed in the Define Application Setups (BR.100) task. These critical setups when defined, are carried forward into the Define Application Setups (BR.100) task.

Attention: In Oracle Applications, you have choices for the number of sets of books, inventory organizations, and installations of each application. Multi-organization functionality adds two further critical functional concepts: the legal entity and the operating unit. This area also covers issues in the use of key applications features that would affect application deployment. These decisions are dependent on the specific features of the application release.

Warning: Significant risk of implementation failure exists if the Multi-organization feature is not adequately configured for the organization. Careful review of the documentation, available white papers, and the engagement of a qualified Multi-organization architecture professional can reduce the risk of failure.

Technical Architecture

The technical architecture portion of the process includes the following key areas:

- Server, or Server Farm Configuration
- Web-Client / Server (Distributed) Processing Configurations (distribution of processing—number of physical tiers, N-tier configurations)
- Physical Database Design and Use of Database Features
- Platforms and Networks (Links, Routing, Bridging, and Tunneled Services)
- System Availability Strategy
- System Management Procedures
- Systems Management Tools (Network, Servers, and Client Desktops)
- Detailed Design of Distributed Data Interfaces

☞ **Attention:** The term *distributed applications* is sometimes applied to technical architectures that are web-client/server or thin client, application and database server in nature. Historically client-server was also referred to as a *distributed applications* architecture; the *processing* and user-layer interaction and validation are distributed, but the *data* may not be. The reason for the confusion is that the second qualifier, *processing* or *data*, is often omitted in common parlance, and the ambiguity is compounded by the liberal use of such unqualified terms in industry literature as well. To avoid confusion in this document and the review questions, the term *distributed* is reserved for architectures that involve multiple separate databases containing data that need to interact in some way; the more pragmatic term *web-client/server* will be used to denote distributed processing (two or more physical tiers) without *implying* a distribution of data also.

Server, or Server Farm Configuration

This area covers the design of the optimal configuration of the company-wide network of database and application servers. This also includes the review and definition of scaling and fault tolerance.

Web-Client/Server (Distributed) Processing Configurations

This area covers the various N-tier configurations for the separation of presentation, business logic, and data management processing within your applications. You have direct control over this for in-house developed applications, but packaged applications may require that you make choices regarding how you partition your processing.

You can configure Oracle Applications to run with two or three physical tiers of platforms between the user and the application database. Also, a number of clusters of middle tier servers may be employed to distribute process load. The more tiers and number of platforms used, the more complex your implementation. The technical architecture you adopt is dependent on the technology that the applications release supports, as well as the geographic nature of the organization.

The architectural options include:

- *Three tiers with web-client (web-client, application server, and database server)* — the database resides on a database server, the applications run on one or more application server platform and a browser runs on desktop client devices (for example, PCs or NCs) which act as the presentation client.
- *Two tiers with web-client (web-client, application server/database server)* — the database server and the applications server run on one platform, and a browser runs on desktop client devices (for example, PCs or NCs) which act as the presentation client.
- *Two tiers with detached mobile-client (mobile, application server, or database server)* — the database server and the applications server run on one or more platforms, and a mobile client software application runs on a PC or palm-top device, which acts as the presentation client and a detached processing client. Periodically, the detached client is attached to the network, transactions are sent to and from the database server, and refreshed local validation sets are updated to the detached client. This architecture is used for work forces, such as sales and support personnel.

☞ **Attention:** Note that the Application Server tier and the Database Server tier may be consolidated onto a single platform is some cases. This may be constrained by the nature of the operating system and hardware platform used. Define Application and Database Server Architecture (TA.090) address with this issue in detail.

Historically (pre-release 11), the architectural options across different Oracle Applications releases included:

- *One tier with Character Mode* — the database server and the application logic and presentation layer resided on a single platform. The user client connected with a basic terminal emulation application or an actual terminal.
- *Two tiers with Smart-Clients* — the database resided on a database server and 10SC clients (desktop PCs) connected using SQL*Net. The client layer performed much of the application and all of the presentation logic.

- *Two tiers with application server platform* — the database resided on a database server and the applications logic processing ran on one or more application server platforms, the tiers being connected through SQL*Net. A dumb terminal front end was used or the user client connected with a basic terminal emulation application. The applications concurrent manager optionally may have run on one of the applications server platforms. This architecture required a multi-processing application server platform, such as a UNIX-based platform. This architecture was an option for pre-10SC Character Mode Oracle Applications, but its use was more limited than 10SC, which had a PC platform for the client processing. This application is commonly referred to as *two-task*.
- *Two tiers with Oracle Applications Display Manager (OADM, thin Client-Server)* — the database and all code resided on one platform except OADM, which ran on desktop PCs and acted as the presentation client. The OADM architecture was an older client-server configuration that was designed to provide some GUI features to character mode release 9 and 10 applications. It has been be completely subsumed by release 11. OADN is no longer supported by Oracle Corporation.

Physical Database Design and Use of Database Features

This area includes the detailed layout of a database across file systems and disks and the use of the advanced database features such as multi-threaded server.

☞ **Attention:** The Oracle Database Server has several advanced features you can use to run Oracle Applications, such as multi-threaded server. Use of these features requires special knowledge of the Oracle database and appropriate tuning techniques for Oracle Applications.

Platforms and Networks (Links, Routing, Bridging, and Tunneling)

Platforms and networks constitute the technical infrastructure of the new system and must have the capacity to meet the technical requirements, performance requirements, transaction and user volumes anticipated at cutover and for some predetermined future period of time.

System Availability Strategy

This is the strategy you define for handling different kinds of planned or unplanned outages. There are many detailed techniques and tools for providing the system availability that a business needs and for guarding against loss of critical data, but there are tradeoffs to consider in selecting the mechanisms and systems to adopt.

System Management Procedures

These are the procedures and tools that you will use to keep the system functioning as it was designed to function. In a modern, open systems environment there are many complex considerations in this area.

Systems Management Tools (Network, Servers, and Client Desktops)

These are software tools that allow for monitoring and managing of the software, server hardware, and network aspects of the architecture. These tools often run on PC workstations and are able to reach out into a large wide-area network. These tools are available from Oracle and third-party providers.

Detailed Design of Distributed Data Interfaces

These are the subsystems used to pass data between databases. They must maintain the integrity of data and guarantee delivery. Use of the Oracle server asynchronous remote procedure feature is one method for building these interfaces. The building of these special interfaces is often distinguished in a project from the design of interfaces to other third-party applications, although conceptually the problem is the same in both cases. In fact, complex, tightly integrated third-party application interfaces are best treated as part of a distributed database architecture. This fact and the fact that distributed Oracle database interfaces usually affect multiple sites in an enterprise are the reasons for treating them as separate interface types.

Tasks and Deliverables

The tasks and deliverables of this process are as follows:

ID	Task Name	Deliverable Name	Required When	Type*
TA.010	Define Architecture Requirements and Strategy	Architecture Requirements and Strategy	Always	SI
TA.020	Identify Current Technical Architecture	Current Technical Architecture Baseline	Project includes *complex architecture changes* or is of *medium or high complexity*	SI
TA.030	Develop Preliminary Conceptual Architecture	Preliminary Conceptual Architecture	Project is of *medium or high complexity* or includes *complex architecture changes*	IT
TA.040	Define Application Architecture	Application Architecture	Project is of *medium or high complexity*	SI
TA.050	Define System Availability Strategy	System Availability Strategy	Always	SI
TA.060	Define Reporting and Information Access Strategy	Reporting and Information Access Strategy	Project includes *complex reporting*	SI
TA.070	Revise Conceptual Architecture	Conceptual Architecture	Project is of *medium or high complexity* or includes *complex architecture changes* <u>and</u> Develop Preliminary Conceptual Architecture (TA.030) is performed	SI
TA.080	Define Application Security Architecture	Application Security Architecture	Project is of *medium or high complexity*	SI
TA.090	Define Application and Database Server Architecture	Application and Database Server Architecture	Project is of *medium or high complexity*	SI
TA.100	Define and Propose Architecture Subsystems	Architecture Subsystems Proposal	Project is of *medium or high complexity* or includes *complex architecture changes*	MI

ID	Task Name	Deliverable Name	Required When	Type*
TA.110	Define System Capacity Plan	System Capacity Plan	Always	SI
TA.120	Define Platform and Network Architecture	Platform and Network Architecture	Always	SI
TA.130	Define Application Deployment Plan	Application Deployment Plan	Project is of *medium or high complexity*	IT
TA.140	Assess Performance Risks	Performance Risk Assessment	Always	SI
TA.150	Define System Management Procedures	System Management Procedures	Always	SI

*Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Table 4-1 Application and Technical Architecture Tasks and Deliverables

Objectives

The objectives of Application and Technical Architecture are:

- Define a systems framework to realize the business and information systems vision of the corporation, where the circumstances and scale of the legacy systems replacement project warrant it.
- Confirm that the architecture for the replacement systems is compatible with the existing information systems framework and vision, where the scale of the replacement is more limited.
- Design an architecture for the replacement systems that is compatible with the detailed business requirements.
- Design a technical architecture to support the immediate processing capacity of the new system and for some specified period of future growth.

Deliverables

The deliverables of this process are as follows:

Deliverable	Description
Architecture Requirements and Strategy	States the requirements, scope, and objectives of Application and Technical Architecture, as well as the strategy that the architectural team will take to perform the work.
Current Technical Architecture Baseline	Takes inventory of the systems, applications, hardware platforms, and networks that constitute the existing technical infrastructure for the business.
Preliminary Conceptual Architecture	Documents the conceptual architecture designs for the new system. It may contain several designs, if there is more than one possible conceptual architecture, but will also indicate the conceptual architecture model that is preferred. If there is only one possible conceptual architecture model, it will only describe one model.
Application Architecture	Addresses the design of the functional architecture for the applications. This involves identifying those key setup parameters in the applications that are relevant to the particular implementation and establishing the values for those parameters across all applications.

Deliverable	Description
System Availability Strategy	Addresses the systems and approaches that will be incorporated into the technical architecture of the new system to provide the required level of system availability.
Reporting and Information Access Strategy	This is the analysis of the reporting and information access requirements and the definition of a strategy for the new business and specifically at a high level how they will be addressed with the new architecture.
Conceptual Architecture	Is a refinement of the Preliminary Conceptual architecture developed earlier, following reviews by key members of the project team and the further gathering of information during the progressing project.
Application Security Architecture	Develops a strategy to address the applications security requirements for the project.
Application and Database Server Architecture	Provides a blueprint for the logical and physical architecture of the application and database servers. These servers are used in two of the three tiers of the architecture. This deliverable also details the configuration of these servers.
Architecture Subsystems Proposal	Documents the set of architecture subsystems that need to be incorporated into the architecture.
System Capacity Plan	States the system capacity requirements for database servers, networks, and client desktop platforms.

Deliverable	Description
Platform and Network Architecture	Describes the deployment of the key hardware platform and network components of the new system and their relationship to the application and server architecture.
Application Deployment Plan	Provides a map for the deployment of the applications to one or more functional business units as well as rollouts to hosting or data centers.
Performance Risk Assessment	Assesses the performance risks associated with the application and technical architecture.
System Management Procedures	Lists the system management procedures and tools that the operations personnel will use to manage and monitor the new system.

Table 4-2 Application and Technical Architecture Deliverables

Key Responsibilities

The following roles are required to perform the tasks within this process:

Role	Responsibility
Application Specialist	Provide input into designing the application security architecture for the new system based on the requirements generated by the Business Requirements Mapping process and the security requirements identified during the conceptual architecture design.

Role	Responsibility
Business Analyst	Provide input about the key processes, mapping decisions, and functionality that will be used in the new system; gather and communicate current and future business volumes; review proposed architecture.
Business Line Manager	Provide input in developing the reporting and information access strategy and to help define these requirements for the new system.
Client Staff Member	Provide documents and information about the existing operations procedures and policies covering applications, databases, interfaces, hardware and networks, system availability metrics, and performance.
Database Administrator	Provide input into logical database design and technical architecture; design database management and maintenance procedures; recommend database monitoring tools and techniques; plan database disk capacity requirements; assess database level performance risks.
Database Designer	Design the application and database server architecture, including custom extensions.
IS Manager	Provide input to architecture process about current and future systems and information systems strategy. Articulate the information systems policies and vision. Review key architecture deliverables.

Role	Responsibility
Network Administrator	Provide input into technical architecture planning and design, design network management and maintenance procedures; recommend network management and monitoring tools and techniques; plan network capacity requirements; assess network performance risks.
Project Manager	Conduct detailed process planning and assign tasks, establish roles, brief client and staff, manage team, manage changes, manage issues, maintain quality.
Project Sponsor	Provide review and approval authority for the development of the Preliminary Conceptual Architecture, Conceptual Architecture, and Performance Risk Assessment.
System Administrator	Provide input into technical architecture planning and design; lead design of the overall systems management and maintenance procedures; recommend systems management and monitoring tools and techniques; plan hardware capacity requirements; assess system performance risks.

Role	Responsibility
System Architect	Identify scope and strategy for the architecture process; meet with project and business managers; provide guidance during architecture design; manage dependencies to other processes; secure acceptance for architecture design work. Establish the application architecture of the new system. Meet with business analysts to identify business vision and requirements important to architecture and then translate into an application and data deployment strategy. Coordinate and lead the development of an integrated application and technical conceptual architecture. Provide input to more detailed technical design efforts, such as interfaces and custom components, to provide compatibility with the overall applications architecture. Coordinate and lead the design of technical architecture, including gathering the baseline technical information, design of the hardware platform and network deployment and system capacity planning. Gather input and advice about performance risks. Meet with the system architect to make sure the technical architecture fully supports the business and information systems vision and the application and data deployment strategy.
Technical Analyst	Provide input to the architecture process about technical designs for custom modules and functionality.

Role	Responsibility
User	Participate in interviews and provide input about the requirements for system performance and availability.

Table 4-3 Application and Technical Architecture Key Responsibilities

Critical Success Factors

The critical success factors of Application and Technical Architecture are as follows:

- proper consideration of architecture as a key determinant of the success of the implemented applications and solutions
- early initiation of an architecture process within an implementation project
- consideration of the corporate business and information systems vision in designing the architecture
- consideration of the interaction and dependencies between application architecture and technical architecture
- realistic consideration of the capabilities and the limitations of the technology to be used
- experienced application and technical architecture practitioners
- balanced input of business and technical requirements with business requirements driving the architecture
- interactions with other tasks and activities of the project that impact the final production application or technical architecture or have interim architecture requirements in support of developmental activities

A balanced input of business and technical requirements with business requirements driving the architecture is especially important. The architectural part of an implementation project is often considered to be purely technical in nature, with the risk that the business requirements are treated as subservient to the technology. A well-managed architecture project uses the business requirements and functional mapping as drivers for the optimal configuration of the applications being implemented, for the hardware and network infrastructure providing the applications processing, and for the tools and procedures needed to manage the complete system.

References and Publications

This section contains some general references; use them as a guide for the types of information sources that are available and order the most up-to-date versions when you begin the project. Specific references will are also given within task descriptions.

Reference: *Oracle Applications Documentation Library* (CD-ROM).

Reference: *Oracle Applications Installation Manual for Unix, or Linix or NT* (Operating System and Release Specific).

Reference: *Multiple Organizations in Oracle Applications,* (Release Specific).

Reference: Millsap, Cary V. *An Optimal Flexible Architecture for a Growing Oracle Database*. White Paper, Revised 1993.

Reference: Millsap, Cary V. *The System Architect's Essential Role in Open Systems Implementation.* White Paper, 1998.

Reference: Velpuri, Rama. *Oracle8 Backup and Recovery Handbook*. Oracle Press. Osborne McGraw-Hill, 1998. ISBN: 0078823897.

Web Site: Oracle's home page on the World Wide Web at: http://www.oracle.com

Web Site: The Transaction Processing Council's home page on the World Wide Web at: http://www.tpc.org

TA.010 - Define Architecture Requirements and Strategy (Core)

In this task, you identify the application and technical architecture requirements and strategy you will use for the design and development of the system you are implementing. This includes support for both the final production environment and the interim development and project support requirements.

Deliverable

The deliverable for this task is the **Architecture Requirements and Strategy**. It states the requirements, scope, and objectives of Application and Technical Architecture, as well as the strategy that the architectural team will take to perform the work.

Prerequisites

You need the following input for this task:

❑ **Project Management Plan [*initial complete*] (PJM.CR.010)**

The Project Management Plan provides a high-level discussion of the scope of the architecture work, the sites, applications and data centers/hosting facilities of the existing information systems it should address, and how the project should be organized and run. This deliverable also identifies the project systems environment high-level requirements to support Business Requirements Definition (RD), Business Requirements Mapping (BR), Module Design and Build (MD), Data Conversion (CV), Business System Testing (TE), Performance Testing (PT), and Adoption and Learning (AP).

❑ **Application Extension Strategy (MD.010)**

The Application Extension Strategy identifies technical architecture requirements in detail, to support application extension developmental activities through the life of the project. The identification of these requirements at this stage allows for the development of a consolidated technical architecture requirement for the project. Once the required hardware platforms, infrastructure, servers and network are allocated, the detailed development environment configuration is represented in the Development Environment (MD.090) deliverable. If Define Application Extension Strategy was not performed, this deliverable will not exist. (See the task description for MD.010 for more information on when this task should be performed.)

- ❑ **Data Conversion Requirements and Strategy (CV.010)**

The Data Conversion Requirements and Strategy identifies technical architecture requirements in detail to support conversion activities through the life of the project. The identification of these requirements at this stage allows for the development of a consolidated technical architecture requirement for the project. Once the required hardware platforms, infrastructure, servers and network are allocated, the detailed conversion environment configuration is represented in the Conversion Environment (CV.030) deliverable. If Define Data Conversion Requirements and Strategy was not performed, this deliverable will not exist. (See the task description for CV.010 for more information on when this task should be performed.)

- ❑ **Testing Requirements and Strategy (TE.010)**

The Testing Requirements and Strategy identifies technical architecture requirements in detail to support testing activities through the life of the project. The identification of these requirements at this stage allows for the development of a consolidated technical architecture requirement for the project. Once the required hardware platforms, infrastructure, servers and network are allocated, the detailed testing environment configuration is represented in the Testing Environments (TE.060) deliverable.

- ❑ **Performance Testing Strategy (PT.010)**

The Performance Testing Strategy identifies the architecture requirements to support the performance testing activities as well as the interrelationship between Performance Testing and Application and Technical Architecture.

- ❑ **Oriented Project Team (AP.020)**

Prior to undertaking the application and technical architecture strategy work, the project team members should have completed the project team orientation activities.

Optional

You may need the following input for this task:

- ❑ **Existing Systems Architecture Strategy or Policy Documents**

If high-level systems strategy or policy documents exist, use them to provide information about current thinking on the architecture and systems strategy.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the Project Management Plan [*initial complete*] (PJM.CR.010).	
2.	Verify the architecture description in the project-level scope, objectives, and approach contained within the Project Management Plan [*initial complete*] (PJM.CR.010).	
3.	Identify the scope of the applications and technical architecture, interfaces/integrations, data centers and hosting facilities and relationships to other projects and systems.	Scope
4.	Identify objectives and critical success factors.	Objectives

No.	Task Step	Deliverable Component
5.	Identify methods to be used for the architecture process. Review the tasks, for this process, that will be used by the architecture team.	Approach
6.	Review the deliverables that will be produced by this process.	Approach
7.	Identify constraints and assumptions that are associated with the process.	Approach
8.	Review risks to the process activities and objectives	Approach
9.	Identify procedures for scope control and issue management.	Approach
10.	Review relationships between process tasks/deliverables and other aspects of the overall project.	Approach
11.	Identify issues with existing information systems.	Information Systems Strategy
12.	Document the business and information systems vision.	Information Systems Strategy

No.	Task Step	Deliverable Component
13.	Identify the implementation roadmap, information systems policies, information systems standards, hardware and network standards, business applications and enabling technology standards, office automation and desktop standards, development standards and system management standards.	Information Systems Strategy
14.	Identify the scope and requirement for baselining of the current technical architecture.	Architecture Strategy
15.	Review and scope the requirement for capacity planning and systems management.	Architecture Strategy
16.	Identify the architecture requirements at a summary level.	Architecture Requirements
17.	Identify the architecture requirements at the detailed level.	Architecture Requirements
18.	Define the environment requirements for the testing, conversion, learning, mapping, development, performance testing, and architecture activities of the project.	Project Environment Requirements

No.	Task Step	Deliverable Component
19.	Review the draft deliverable with senior management and seek approval.	Acceptance Certificate (PJM.CR.080)
20.	Identify any material changes to project scope and associated task estimates with the project manager and update the Project Management Plan as appropriate.	Project Management Plan (PJM.CR.010)

Table 4-4 Task Steps for Define Architecture Requirements and Strategy

Approach and Techniques

The techniques that you use for capturing the information you need to complete this task may involve interviews with senior business analysts as well as the information systems and project management staff.

Information Sources

Much of the information needed for subsequent design activities may be strategic in nature, or at the very least, may require decisions by senior management or project leaders. Often the strategic information is not documented. This, combined with the fact that architecture is often a front-loaded project process, means that you may have to perform much of your information gathering in face-to-face meetings as you go along. The information systems manager or director will be an important source of decisions and information for the high-level technical and information systems requirements.

Scale of the Architecture Process

The creation of a separate statement of the Architecture Requirements and Strategy for Application and Technical Architecture may not be necessary for all types of projects.

Small- to Medium-Sized Implementations

If Application and Technical Architecture is part of a small implementation project, the scope definition, work management, and funding of the architecture may be incorporated into the management of the overall project, and this task will be greatly simplified. In such cases, you should review the existing Project Management Plan (PJM.CR.010), verify that it is an accurate statement of the architectural process, and then move to the next task. In some cases you may wish incorporate many or all of the deliverable components of the Architecture Requirements and Strategy (TA.010) into the Project Management Plan (PJM.CR.010).

Large, Complex Projects

In a large and more complex implementation project, the process may need to be semi-autonomous because of its size and complexity, and a separate architecture subproject may be needed to provide effective control. In these types of implementation projects, the Project Management Plan (PJM.CR.010) defines the high-level scope and policies, but does not provide enough detail about individual subprojects. This detail would be documented in this deliverable.

Scope of Architecture — Enterprise or Site Level

There are two broad categories of architecture projects that differ greatly in scope. As part of this task, you need to secure an understanding of the scope and agreement for what will and will not be included. If necessary, you should define the level of completion for each key area of the architecture process in which you are engaged.

Enterprise-Level Architecture

An enterprise-level architecture design will encompass all, or a major portion of, the important corporate information systems. This type of project may:

- help define strategic direction for information systems
- be subject to a program management approach to the implementation with a phased rollout strategy
- take into account multiple data centers, sites, or countries
- be designed for a mix of new applications and older retained legacy applications

- be accompanied by an upgrade or strategic change of hardware and networks

This type of architecture project covers the breadth of the enterprise and helps set strategic IS direction but may not get into the detailed design of every component of every new installation or system. Many design tasks that a site-level architecture would perform to completion are addressed only at a high level and may only set general enterprise-wide standards, policies, or procedures for the detailed design work at the site level.

Site-Level (Localized) Architecture

A site-level architecture project is narrower in scope than the enterprise-level architecture, but it goes into greater detail for the architecture components within the scope. It only deals with single installation of key applications, or a very localized set of information systems. While there is still architecture work to be done, it is organized to implement the architecture strategy and high-level architecture designs produced from the enterprise-level studies.

Task Complexity and Project Scope

The level of detail required for this task depends on the scope of the implementation project and the architecture process within it. If the architecture work is being performed for a small number of applications that need only to fit into an existing architecture or information systems strategy, you need only to document the parts of the strategy that are relevant to the limited architecture scope and proceed to the next task. At the other extreme, if this is an enterprise-level architecture for a large-scale replacement of business systems, or there is no clear architecture or information systems strategy in place already, you may need to go into some detail to document the architecture policies and expectations.

Business Vision

The vision for the new business systems is a key initial ingredient to the design of an architecture, especially where the architecture process is covering the strategic aspects of the new systems as well as the tactical design. The vision must be documented and understood early in the process. Examples of business visions are listed below:

- Users must be able to perform a global check of inventory availability across the entire corporation, regardless of the manufacturing plant source of the inventory and the distribution center location.
- The corporation will standardize all financial operations on a new single chart-of-accounts structure.
- The business must achieve differentiating or transformational business processes (for example, global available-to-promise, tight supply chain integration, or electronic receipt settlement).
- The business must streamline and reduce metrics for certain business processes (for example, process payrolls), or close the financial books in a shorter specific cycle time.

Architecture Policies and Standards

At the heart of every architecture project are directives and policies that are fundamental to the project and to the architecture design. In large projects that are replacing many of the key business applications of the company or enterprise, you may be defining these directives and policies for the first time. These will be the guiding principles for the design and configuration of the new information systems, and quite probably, for new technology as well. In projects where the implementation is more limited in scope, this may be a matter of conforming to policies and principles that the IS department has already laid down for any new information systems.

In situations where a formal information systems strategy project has preceded the implementation architecture process, a set of principles, directives, and strategies may already be in place, which you can directly input as requirements to the implementation architecture here. The directives and policies are fundamental to the strategy for new information systems in the company and should underpin any architecture that is designed for the new systems. The selection of a particular suite of applications or technology may be partly in response to the demands of the particular information systems strategy.

A common example of architecture standards is in the selection of hardware, where database servers in the new information systems are to be purchased from hardware vendor ABC, with whom the business has forged a strategic relationship.

Architecture Approach

The approach that AIM employs for architecture is robust and proven in prior implementations. There are other techniques for arriving at the performance test transactions and database, but they are all very similar to the process presented here. The advantage of the AIM approach is that it is top-down in nature and strikes a balance between the business and technical requirements, with the demands of the business driving the technology rather than vice versa. Other bottom-up approaches to the Application and Technical Architecture process can risk creating designs that lock the business into a technological solution incompatible with immediate or future business direction.

Century Date Compliance

In the past, two-character date coding was an acceptable convention due to perceived costs associated with the additional disk and memory storage requirements of full four-character date encoding. As the year 2000 approached, it became evident that a full four-character coding scheme was more appropriate.

In the context of the Application Implementation Method (AIM), the convention Century Date or C/Date support rather than Year2000 or Y2K support is used. Coding for any future Century Date is now the modern business and technical convention.

Every applications implementation team needs to consider the impact of the century date on their implementation project. As part of the implementation effort, all customizations, legacy data conversions, and custom interfaces need to be reviewed for Century Date compliance.

When designing the application and technical architecture it is essential that all dates be entered, stored, transferred and processed using the full four-digit year for compliance with Century Date standards.

Linkage to Other Tasks

The Architecture Requirements and Strategy is an input to the following tasks:

- BR.020 - Prepare Mapping Environment
- TA.020 - Identify Current Technical Architecture
- TA.030 - Develop Preliminary Conceptual Architecture
- TA.040 - Define Application Architecture
- TA.050 - Define System Availability Strategy
- TA.060 - Define Reporting and Information Access Strategy
- TA.080 - Define Application Security Architecture
- TA.090 - Define Application and Database Server Architecture
- TA.100 - Define and Propose Architecture Subsystems
- TA.110 - Define System Capacity Plan
- TA.120 - Define Platform and Network Architecture
- TA.130 - Define Application Deployment Plan
- TA.140 - Assess Performance Risks
- MD.020 - Define and Estimate Application Extensions
- MD.090 - Prepare Development Environment
- CV.030 - Prepare Conversion Environment
- TE.060 - Prepare Testing Environments
- PT.110 - Prepare Performance Test Environment
- AP.040 - Prepare Project Team Learning Environment
- AP.160 - Prepare User Learning Environment
- PM.020 - Design Production Support Infrastructure
- PM.030 - Develop Transition and Contingency Plan
- PM.140 - Propose Future Technical Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	60
Business Analyst	10
Project Manager	30
IS Manager	*

Table 4-5 Role Contribution for Define Architecture Requirements and Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the Architecture Requirements and Strategy to expand on architecture requirements defined in the Project Management Plan (PJM.CR.010). It should make reference to the main project deliverable where appropriate, and indicate those objectives and approaches that are inherited from the main project. It should not, however, be just a duplication of the Project Management Plan (PJM.CR.010).

This deliverable should address the following:

- the detailed requirements for the application and technical architectures
- scope of the architecture work
- approach to be taken within the application and technical architecture process
- risks and assumptions known at the outset of the architecture work
- specific strategy to be used to achieve the requirements
- century date compliance standards and approach

Deliverable Components

The Architecture Requirements and Strategy consists of the following components:

- Introduction
- Scope
- Objectives
- Approach
- Information Systems Strategy
- Architecture Strategy
- Architecture Requirements
- Project Environment Requirements

Introduction

This component describes Application and Technical Architecture and the purpose for it. This component also describes the downstream utility of this deliverable for other project work.

Scope

This component describes the scope of the architecture process including the relationships to other systems and systems projects, data centers and hosting facilities, existing applications and interfaces and integrations.

Architecture scope statements are made in terms of whether architecture components are in or out of scope for the project. Examples of such components that define the scope include:

- data centers
- business processes, sites, or functions
- applications
- interfaces
- technical infrastructure

In addition to discussion of the scope, this component also discusses the relationships between existing and new systems and identifies where retrofitting of existing systems is within the scope of this project.

Objectives

This component lists the high-level objectives that the business and project managers have communicated during the initial casting of the overall project. This component also discusses factors that are critical to meeting these objectives.

Since this is a strategic document, the stated objectives should not be too detailed, however, they should be specific and measurable.

Approach

This component describes the process, policies and procedures, project dependencies, and the technical background for the architecture process.

The description of process also includes a high-level discussion of the approach selected for the architecture work, the tasks and deliverables, the reasons for selection of the approach, and the benefits of the particular method adopted. Integration with other aspects of the overall project as well as approaches to manage scope are also discussed.

When the process is to use different polices, procedures, or standards from the main project in any of the key control and reporting areas, this component documents the differences in detail and explain why they differ. The following are examples of areas where the process will typically inherit standards and procedures from the main project:

- project management plan
- issue management and resolution
- change management
- reporting format
- reporting relationship to main project
- acceptance
- project policies and procedures

- process team meetings
- logistics and administrative support

Information Systems Strategy

This component documents the strategy that the organization is following in selecting and implementing new information systems. This includes:

- statement of issues with existing information systems
- business vision
- information systems vision
- implementation
- information systems policies
- information systems standards
- hardware and network standards
- business applications and enabling technology standards
- office automation and desktop standards
- development standards
- systems management standards

Architecture Strategy

This component describes the strategy for addressing key areas in the architecture project. This included the baselining approach for the current architecture, capacity planning and systems management.

These areas are highlighted and discussed because of their criticality in the architecture work, the inherent risk to the project, an innovative or unusual approach to be applied in the project, or for some other implementation-specific reason.

Architecture Requirements

This component addresses the application and technical architecture requirements.

The requirements summary summarizes the detailed requirements gathered for the business and information systems and their likely impact on the architecture and systems.

The detailed requirements detail the individual business and information systems requirements that will affect the architecture. Some of the requirements that may be included are:

- *Key Reasons For Selecting the New Application System*—the key reasons that led to the selection of the applications can be a useful indicator of the critical requirements that the systems must meet.
- *Requirement Areas*—this is a list of individual requirements affecting architecture that may include some or all of the suggested areas discussed in the approaches to this task.

Project Environment Requirements

This component describes the specific resource needs for the architecture work including software, hardware platforms, and networks.

Audience, Distribution, and Usage

Distribute and communicate the Architecture Requirements and Strategy to the following:

- project managers of the main project
- information systems manager
- architecture team members
- other process leaders who have dependent tasks with the architecture process

Quality Criteria

Use the following criteria to help verify the quality of this deliverable:

- Are the project scope and objectives clearly identified?
- Are specific critical success factors and risks documented?
- Does this document take into account the impact of dependent tasks from other processes?
- Are the architecture requirements clearly defined?

- Does the strategy discuss existing IS policies and strategies and indicate how they relate to this project?
- Is the strategy understood by those on the distribution list for this deliverable?
- Are all resource and tool requirements that could impact the architecture process stated and understood?
- Is the deliverable thorough in capturing different types of business and information systems requirements that are directly relevant to architecture?
- Have the individual requirements been stated in terms that communicate their architectural significance?

Tools

Deliverable Template

Use the Architecture Requirements and Strategy template to create the deliverable for this task.

TA.020 - Identify Current Technical Architecture (Optional)

In this task, you identify the current application and technical architecture for the existing systems. This includes identifying existing applications, interfaces, databases, networks, and hardware platforms.

> ∆ If your project includes *complex architecture changes*, or your project is of *medium or high complexity*, you should perform this task.

Deliverable

The deliverable for this task is the **Current Technical Architecture Baseline.** It is the inventory of the systems, applications, hardware platforms, and networks that constitute the existing technical infrastructure for the organization.

Prerequisites

You need the following input for this task:

- ❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work and whether or not baselining is needed (and to what extent).

Optional

You may need the following input for this task:

- ❑ **Existing System Architecture or Technical Configuration Documents**

There should be at least some system architecture or technical configuration documents in existence. The technical configuration documents may include hardware deployment, network diagrams, system interfaces, and so on. Use them to provide information about the current systems and infrastructure, but make sure they are current.

❑ **Existing System Management Procedures Documents**

If documentation about current system management procedures exists, make sure the information is current.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review overall project scope and decide the scope of the technical architecture baselining effort needed.	
2.	Review and catalog existing architecture documentation and validate its relevance to the current systems.	
3.	Identify the gaps in the current systems documentation and establish information sources.	
4.	Identify existing data centers / hosting facilities, the functions they perform, and services they provide.	Data Centers / Hosting Facilities
5.	Profile existing applications.	Applications
6.	Identify usage of applications by business units.	Applications
7.	Profile existing databases.	Applications
8.	Gather specifications of existing interfaces.	Application Interfaces

No.	Task Step	Deliverable Component
9.	Identify existing hardware, including desktop platforms, servers, and so on.	Hardware (Platforms)
10.	Gather information about existing networks, topology, and capacity.	Network Architecture
11.	Gather information about peripheral devices such as printers, modems, and others that are relevant to the scope.	Peripheral Devices
12.	Gather information about system management procedures.	System Management
13.	Gather metrics for current system availability.	System Management
14.	Identify known changes within the implementation time frame.	Expected Architecture Changes
15.	Identify opportunities for technical infrastructure improvement.	Areas for Improvement

Table 4-6 Task Steps for Identify Current Technical Architecture

Approach and Techniques

This task is essentially one of information gathering, reading, reviewing, and piecing together the components of the existing systems. If the Information Systems (IS) operations personnel have documented the existing system infrastructure, the system architect can perform the work without assistance. However, if the documentation is sparse or out of date, interviews will be necessary to gather the information.

Attention: The IS operations personnel may not be aware of the details of the functions and the processes that the existing applications provide, so you may need to talk to business analysts or users for that information.

Enterprise Multi-Data Center Projects

At the enterprise level, this task may require information gathering across multiple geographical sites. If the information gathering requires face-to-face time with local Information Systems operations personnel, you need to arrange a way to get baseline information collected at each site.

Current System Architecture Presentation

It may be useful to get the information systems manager to arrange a presentation on the current system architecture by the operations personnel. This could be a useful way to focus the Information Systems organization on the coming changes, rather than talking to individuals in a piecemeal fashion. Prepare questions on the technical standards, direction, strategy, and technical resource plan. Review the data center and site hardware, and networking requirements.

Suggestion: When documenting information systems, use diagrams wherever possible to avoid (or enhance) text descriptions and to aid visualization of system integration points. For some aspects of the system, such as network topology, a diagram may be the only reasonable way to communicate the architecture.

Baselining

A baseline of the existing system architecture is often regarded as unnecessary, but consider what the task produces before dismissing its value. Some projects may need minimal or zero baselining, but most projects require at least some understanding of the existing systems. There are several good reasons for a technical baseline:

- *Reuse of existing infrastructure, hardware or networks* — to support the redeployment or reuse the existing technical infrastructure for the new application and technical architecture

- *Replacement of existing systems* — business and technical analysts on the project need at least some elementary knowledge of the applications they are replacing, in order to be able to map business processes onto the new applications and design technical solutions

- *Integration with retained existing systems* — business and technical analysts on the project will need a knowledge of the applications and technology they are attempting to integrate

- *Integration cross-check* — confirm that all existing interfaces have either been subsumed by the new applications or have been identified as being replaced wholly or partially by new interfaces

In a project where systems integration or infrastructure reuse is intended, it can be a mistake to ignore this task and believe that project time has been saved by not doing it. In reality, individuals designing technical architecture, performing capacity planning, designing new interface mechanisms, or designing technical mapping solutions will still need to have the information before they can undertake the work assigned to their project role. There is a danger of duplicate information gathering and wasted resource time, as compared to properly planned and scoped baseline effort.

Existing Documentation

If a clear summary document already exists that discusses the current technical architecture baseline, this task can be marked as completed with little extra work other than validation that the documentation has all the details necessary. In general, however, documentation of existing systems easily falls into obsolescence over time, and some work is usually needed to restructure it and bring it up to date.

Fast Track, Limited Scope Projects

If you are working on a fast track project with complete replacement of existing systems and no integration requirements, it may be wasteful to undertake a formal baseline effort. Under these circumstances, you can drastically reduce the scope of the task, possibly to zero if you are upgrading much of your hardware and networks concurrently with the implementation.

Linkage to Other Tasks

The Current Technical Architecture Baseline is an input to the following tasks:

- BR.050 - Conduct Integration Fit Analysis
- TA.070 - Revise Conceptual Architecture
- TA.110 - Define System Capacity Plan
- TA.120 - Define Platform and Network Architecture
- TA.130 - Define Application Deployment Plan
- TA.150 - Define System Management Procedures
- PM.010 - Define Transition Strategy

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	80
Business Analyst	20
IS Manager	*
Client Staff Member	*

Table 4-7 Role Contribution for Identify Current Technical Architecture

* Participation level not estimated.

Deliverable Guidelines

Use the Current Technical Architecture Baseline as an input to subsequent tasks to describe the existing application and technical architecture in detail. This deliverable will be used for design and analysis work in the architecture process and possibly in other processes of the project. To complete this deliverable you must understand the technical direction, strategy, and technical resource plan of the business.

This deliverable should address the following:

- an inventory of the hardware and software infrastructure that is in place and in use by the organization prior to the implementation of the new systems
- an inventory of LAN/WAN networks and capabilities
- an inventory of all of the interfaces and integrations that are in place and operational, including the business objects transferred, timing and volume of interface data flows
- an inventory of the current applications and processes in place to support the systems management requirements of the current systems infrastructure
- a description of legacy applications, including the supplying vendor and the processes/functions they support
- databases supporting the applications
- known changes to the baseline architecture that will occur during the implementation time-frame
- areas in the baseline architecture where opportunities for improvement are apparent

Deliverable Components

The Current Technical Architecture Baseline consists of the following components:

- Introduction
- Data Centers/Hosting Facilities
- Applications
- Application Interfaces
- Hardware (Platforms)
- Network Architecture
- Peripheral Devices
- System Management

- Expected Architecture Changes
- Areas for Improvement

Suggestion: The company may already have documentation on these key elements. In this case, simply assemble these materials and identify how each part of the existing system was implemented. Then add a section that identifies opportunities for improvement in the current system.

Introduction

This component describes the purpose for undertaking a current architecture baseline and the contents of the deliverable.

Data Centers/Hosting Facilities

This component describes the existing data centers / hosting facilities within the scope of the architecture and the applications, systems, and user communities they support.

Applications

This component describes the current applications and databases that handle the processing in the current architecture. This discussion covers the tiered architecture, vendors and versions of the specific applications and databases, as well as their usage within particular business units.

Application Interfaces

This component describes the existing application interfaces in the current architecture. This includes the mechanisms, volumes, triggers, and frequency of these interfaces.

Hardware (Platforms)

This component describes the computing hardware in the current architecture.

Network Architecture

This component describes the high-level architecture and capacity of the networks in the current architecture. This includes topologies, bandwidth, protocols, capacity, latency, services vendors, and a high-level network diagram.

Peripheral Devices

This component describes relevant peripheral devices or device standards in the current architecture. The peripheral devices might include printers, modems, and so on.

System Management

This component documents the system management metrics, tools, and procedures that are used to manage the system architecture. It also records quantitative metrics about current system availability, where possible.

Expected Architecture Changes

This component describes the expected changes to the baseline architecture during the time frame of the implementation project.

Areas for Improvement

This component highlights specific opportunities for improvement in the systems based on the findings of the baseline analysis. Examples of these areas are:

- hardware and software limitations
- system management tools and procedures
- improvements in logistics, communications, or support

Audience, Distribution, and Usage

The Current Technical Architecture Baseline provides a summary of what is in existence currently and helps make sure that nothing is missed. The existing architecture can potentially affect significant parts of the eventual proposed architecture and the configuration and integration of the new systems. The deliverable from this task will be a useful summary of the existing infrastructure for the following processes in a project:

- Business Requirements Mapping (BR)
- Application and Technical Architecture (TA)
- Module Design and Build (MD) — especially interface modules
- Data Conversion (CV)

Distribute the Current Technical Architecture Baseline to the following:

- architecture team members
- team members for the processes listed above
- all members of the project team

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Is the deliverable comprehensive in the systems it addresses?
- Does the deliverable address all applications and systems that are within the scope of this architecture project or that will impact the project?
- Does it have the information organized in a way easy for all project technical team members to find and understand?

Tools

Deliverable Template

Use the Current Technical Architecture Baseline template to create the deliverable for this task.

Enterprise Baseline Studies

If you are performing a distributed enterprise baseline that covers several data centers, you may need to structure the template for the individual data centers. You can easily cut and paste the existing sections to create separate sections for each data center and add the data center name to section title. If you want to separate the enterprise-level aspects of the architecture (for example, shared applications or inter-data center network connections), you could also create separate sections for the enterprise level architecture baseline and then have separate deliverable sections for each data center. Within the individual data center sections, you can then document those architecture components that are local to the data center and business units that process out of that data center.

TA.030 - Develop Preliminary Conceptual Architecture (Optional)

In this task, you identify possible candidate conceptual architectures for the new information systems you are implementing, and select the preferred architecture that best fits the overall needs of the business subject to the project constraints.

This task focuses on what will be the future production environment and <u>not</u> interim environments to support the project activities.

Δ If your project is of *medium or high complexity* or includes *complex architecture changes*, you should perform this task.

Deliverable

The deliverable for this task is the **Preliminary Conceptual Architecture**. It documents the conceptual architecture designs for the new system. It may contain several designs if there is more than one possible conceptual architecture, and also indicates the conceptual architecture model that is preferred. If there is only one possible conceptual architecture model, it only describes one model.

Prerequisites

You need the following input for this task:

❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides information about the company financial hierarchy, the business organizations, and the functions the organizations perform.

❑ **Current Business Baseline (RD.020)**

The Current Business Baseline examines current process and practices of the organization and is used to help develop the Preliminary Conceptual Architecture.

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work.

Optional

You may need the following input for this task:

❑ **Physical Resource Plan (PJM.RM.040)**

The Physical Resource Plan provides information on physical resources to support design of the architecture.

❑ **Architecture Designs and Technical Documents from the Application Acquisition Process**

The designs and technical documents from the application acquisition process may provide information about the original thinking for the architecture of the new system.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather and review any materials and documents from the requirements definition and mapping processes.	
2.	Identify key business functions and organizations in the scope of the new architecture.	Business Operations
3.	Identify the data center / hosting facilities that manage the data and processing of the key business units.	Data Center / Hosting Facility Operations

No.	Task Step	Deliverable Component
4.	Identify major information and material flows for the business functions and organizations.	Business Information Model
5.	Identify critical legacy applications replaced in the new architecture.	Key Applications
6.	Identify critical legacy applications retained in the new architecture.	Key Applications
7.	Identify candidate application release versions within the project timescale and assess their base functionality and technology.	Key Applications
8.	Develop candidate application deployment strategies to support key business functions and organizations.	Conceptual Architecture
9.	Identify any specific reporting systems needed for each conceptual architecture.	Conceptual Architecture
10.	Identify possible hardware and networks that support each application deployment architecture.	Conceptual Architecture
11.	Map information flows onto each candidate application deployment.	Conceptual Architecture
12.	Identify key distributed data interfaces.	Conceptual Architecture

No.	Task Step	Deliverable Component
13.	Identify architecture subsystems needed to support each conceptual architecture.	Conceptual Architecture
14.	Perform approximate capacity analysis for each conceptual hardware and network architecture.	Conceptual Architecture
15.	Identify tradeoffs for each conceptual architecture.	Conceptual Architecture
16.	Design architecture for the project development environment.	Project Environment Architecture
17.	Perform approximate capacity analysis for the project development environment architecture.	Project Environment Architecture
18.	Review competing architecture models with senior business and project management.	
19.	Review deliverable with senior business and project management.	
20.	Secure acceptance for Preliminary Conceptual Architecture deliverable.	Acceptance Certificate (PJM.CR.080)

Table 4-8 Task Steps for Develop Preliminary Conceptual Architecture

Approach and Techniques

In this task, the system architects create one or more candidate conceptual architectures for the new system and present these to senior project and business management for review and consideration. Once a preferred candidate architecture is chosen by management, the system architects update the conceptual architecture to reflect the choice and assess its impact on the business, the information systems organization, and the implementation project.

Generally you will not have all the business and information systems requirements available to you when you start this task. The business requirements mapping may be in its early stages also, and the analysts may not have made all the key decisions at the time of starting this work. You will need to review any hard-copy information available and selectively interview business analysts and client staff as necessary.

☞ **Attention:** This task usually requires special expertise in both application and technical architecture. Verify that the individuals fulfilling these roles have the knowledge and experience to perform the task.

Conceptual Architecture

The conceptual architecture is an initial, high-level attempt at designing the application and technical architecture for the new system. In projects of a larger scope, where you are implementing an entirely new financial, distribution, or manufacturing system, the conceptual architecture should not be geared toward one particular component or system layer, but should be a slice through the applications, databases, hardware, and networks in the future system. This enables the architecture to show the relationship between the layers or components of the new system architecture. At the other extreme, in more specific or smaller scale projects (for example, where you are implementing a single business function or single Oracle Application module into an existing architecture), a formal conceptual architecture study may not be necessary.

The Role of Conceptual Architecture

The development of a conceptual architecture is an important milestone in the overall architecture design process. It is important for several reasons.

Communication of Architecture Direction

Architecture design is often viewed as a purely technical pursuit, which can be difficult for non-technical managers and business analysts to understand. The detailed designs produced in projects can be impenetrable for these types of individuals, but the fact that the conceptual architecture is high-level and abstract in nature means that the less technical members of a project team should be able to grasp the essentials of the future model. A good conceptual architecture blends business, application, functional and technical aspects into a high-level picture of the future information systems.

Competing Models — Compatibility with the Vision

Early in a project where the architecture process typically occurs, a clear vision may not yet exist of the future information systems incorporating the new applications. This is especially true if relatively few architecture directives or policies exist to dictate the future distribution of data centers, processing, or data. Possible *a priori* candidate architectures may need to be examined and assessed as models for the future business systems. The conceptual architecture design is a milestone task for identifying the candidate architectures and for selecting the working model for the future system. Alternatively, if a clear and constraining vision for the future business and information systems exists, the conceptual architecture should obviously conform to the vision. Hence this task is an important means to promote compatibility between the vision and the execution of architecture design.

Identification of Gaps and Custom Components

When designing possible conceptual architectures, issues will become apparent, such as gaps in the standard features of application products, the inability of the standard products to support specific business technology requirements, and the integration points needed in the new system. The conceptual architecture is the point where you can identify major custom architecture components or subsystems that you will have to design and build or purchase.

Assess Risk and Scope

You can use the conceptual architecture to assess risk in the new systems, thereby creating a checkpoint for architecture rescoping and replanning efforts.

Complexity of the Task

The complexity of the conceptual architecture varies with the scope of the replacement project. At one extreme, if you are replacing all financial, distribution, and manufacturing applications for a large global corporation, you may need to define the strategic architecture for the entire corporation, supporting multiple legal entities, sites, product lines, and data centers. In this situation, simply a decision about whether you are going to adopt an architecture that centralizes or distributes data and processing may be the best you can hope for at an early stage in the project. On the other hand, a single site implementation may not have the same number of architectural variables to consider, and the number of possible architectures may be small or singular.

Conceptual Architecture Development

Gather whatever information you can from existing materials and other project processes to create your designs. Unless the conceptual architecture is focusing on very specific layers in the system architecture, you need to consider application and database deployment to support critical business functions and processes; integration of heterogeneous systems and databases; and the hardware and network infrastructure to support the applications and data. To the extent that you can get business metrics at this stage, you should also perform some initial capacity planning for the database servers, desktop client platforms and networks. You perform detailed system capacity planning later. For more information see Define System Capacity Plan (TA.110).

Information Sources

Much of the information needed for designing the conceptual architecture may be strategic in nature, or at the very least, it may require decisions by senior management or project leaders. Often the strategic information is not documented. This, combined with the fact that architecture is often a front-loaded project process, means that you may have to perform much of your information gathering in face-to-face meetings as you go along. The IS manager is an important source of decisions and information for the high-level technical and IS requirements. You may have to do the same for the business requirements and obtain high-level requirements and direction in face-to-face meetings with senior business analysts and operations managers.

Communication to Senior Management

As a result of the conceptual architecture design process, there may be multiple candidate architectures, each with inherent advantages and disadvantages. If there is a specific business decision to be made about which to select as the preferred candidate architecture, you need to identify a forum or mechanism for getting management review and buy-in to one of the candidates. You can best achieve this by convening architecture review meetings and presenting a high-level view of the competing architectures to senior business and project management, including experts in particular technical or operations areas. The result of the reviews should be a consensus about the preferred candidate conceptual architecture, which will become the working model for the more detailed design tasks in the project. As the detailed design progresses, the working model may be consolidated or become untenable for the business system, in which case a revision of the conceptual architecture may be necessary.

Even in a situation where there is essentially only one candidate architecture, a review and presentation session is still valuable and may be required for management to understand the new architectural direction of the company. Management is often keen to understand how switching to open web-client/server-based computing will affect their technical and business infrastructure, and a review may be needed to explain new technology.

Architecture Subsystems

When you define the possible conceptual architectures, identify any architecture subsystems that you need to build or purchase from another vendor. Examples of such systems are:

- data registries
- message transport systems for electronic commerce or for communication between distributed databases
- data warehouses/OLAP systems
- business intelligence systems
- key enterprise-level interfaces to other applications or systems

For more information refer to Define and Propose Architecture Subsystems (TA.100).

Platform and Network Architecture

When considering the platform (hardware) and network architecture, you should be aware that although many advanced features are built into the Oracle database to support sophisticated platform configurations, you may not need those features in an architecture to support the requirements. A simpler system (for example, one with a single central database server) may be all you need to meet the basic requirements and the business needs when implementing Oracle Applications. Consider various tradeoffs in selecting possible platform configurations:

- platform capacity and performance
- scalability for future growth
- load balancing
- reliability and failure tolerance
- maintainability
- hard costs of the system components themselves
- soft costs of expertise to manage the systems

Be sure to discuss the tradeoffs in the different candidate architectures. Often the soft costs are more important than the costs of platforms themselves in the overall equation for a sophisticated architecture. The old maxim of keeping it simple is very pertinent to the development of new systems, especially with all the complexities and permutations possible in open systems.

Warning: When you propose candidate architectures using new or leading-edge technology, make sure that senior managers understand the risks and potential problems that might be incurred. Factor in the additional or special support needed for new platforms and software technology that is part of your solution.

Analysis of the Architecture Models

In order to choose between multiple competing models for the future architecture, you need to analyze the relative advantages and disadvantages of each model. Even in less complex implementations where only one feasible model for the future architecture exists, you should still perform an analysis of the architecture so expectations are properly set about what it will and will not provide. The following are general areas that you should consider during this analysis:

- realization of business vision
- realization of information systems vision
- hard and soft costs
- technical risk factors — leading-edge technology and infrastructure
- project risk factors — resources, budget, and timeline
- custom developed components — maintenance and upgrade
- rollout implications

Linkage to Other Tasks

The Preliminary Conceptual Architecture is an input to the following tasks:

- BR.030 - Map Business Requirements
- BR.050 - Conduct Integration Fit Analysis
- TA.040 - Define Application Architecture
- TA.060 - Define Reporting and Information Access Strategy
- TA.070 - Revise Conceptual Architecture

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	80
Business Analyst	20
Client Staff Member	*
IS Manager	*
Project Sponsor	*

Table 4-9 Role Contribution for Develop Preliminary Conceptual Architecture

* Participation level not estimated.

Deliverable Guidelines

Use the Preliminary Conceptual Architecture to formulate designs for the new system. It may contain several designs if there is more than one possible conceptual architecture, and also indicates the conceptual architecture model that is preferred. If there is only one possible conceptual architecture model, it only describes one model. This Preliminary Conceptual Architecture is used as the starting point for the formulation of security, reporting, application, database and application server and network architectures.

This deliverable should address the following:

- an inventory of the business operations including key operations and geographic location and scope
- the future charts of account structures, financial and operations reporting approach, consolidation approach and inter- and intra-company accounting practices

- inventory, logistics, supply-chain and manufacturing organizations and their relationships to each other
- human resources and staffing organizations
- application and business architecture

Deliverable Components

The Preliminary Conceptual Architecture consists of the following components:

- Introduction
- Business Operations
- Data Center / Hosting Facility Operations
- Business Information Model
- Key Applications
- Conceptual Architecture
- Project Environment Architecture

Introduction

This component states the purpose of the deliverable.

Business Operations

This component provides a high-level description of the business operational model that the architecture must support. It describes the business in which the company engages, the structure of the business, and the critical aspects of the business processes that the architecture must take into account. Specific discussion sections in this component are:

- Key Business Organizations and Functions
- Financial Structure
- Major Business Processes

Where possible, this deliverable should include diagrams of the geographical and organizational structure of the business.

Data Center/Hosting Facility Operations

This component describes the data centers and hosting facilities within the scope of the architecture, the particular operations that the data centers perform, the systems or applications the data centers support, and the business functions and organizations for which they perform the operations.

Business Information Model

This component describes the information flows and information access within the corporation or company.

The information flows are associated with business processes or process steps. The component does not document all flows for all processes to the lowest elementary business function level of detail, but it covers the critical business processes and functions, processes that span multiple applications or application installations, and complex processes that may require detailed custom design. A sample Information Model diagram can be found in the Preliminary Conceptual Architecture template.

The information access model part of this component describes the access to information that business units need in order to execute business process steps, but without compromising the integrity and security of the business information as a whole. Information access is concerned with:

- information (data) partitioning to provide security and maintain integrity
- information (data) aggregation, consolidation and summation to provide appropriate reporting views of the state of the business

Suggestion: If it exists and has content, you should incorporate information from the Information Model (BR.060) at this point in the project. However, it may not exist at this time, and you should try to work with the business analysts to start defining the information access requirements.

Key Applications

This component describes the major or key applications that will be part of the future architecture. The acquisition or building of new applications as well as the retention and retirement of existing applications are discussed.

Conceptual Architecture

This component describes the integrated conceptual application and technical architecture model for the business. It may be repeated more than once, if there are multiple possible conceptual architecture models to document and consider. Illustrate the architecture models diagrammatically wherever possible and considers these areas:

- application architecture and deployment
- functional architecture of the packaged applications
- key application interfaces required for this model
- technical architecture, hardware, and networks
- approximate capacity analysis
- analysis of the architecture model, advantages, and disadvantages

Project Environment Architecture

This optional component describes the architecture of the application environments needed for the implementation project. It is only needed if the architecture scope also includes architecting the implementation project infrastructure. It considers the impact of custom development on the standard package application environment.

Audience, Distribution, and Usage

Distribute the Preliminary Conceptual Architecture to the following:

- project manager
- project sponsor
- information systems manager
- architecture team members
- other process leaders who have dependent tasks with the architecture process

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Are the high-level business operations and information model requirements clearly identified?
- Are there competing conceptual architectures included, or is there a clear reason why only one is considered?
- Are all components of each conceptual architecture covered (breadth of analysis, not depth, is emphasized at this stage)?
- Have diagrams been included to aid understanding and communication?
- Does the analysis of each architecture option include advantages, disadvantages, and risks?
- If a preferred architecture has been selected from multiple choices, is it clear why this choice was preferred?

Tools

Deliverable Template

Use the Preliminary Conceptual Architecture template to create the deliverable for this task.

Multiple Conceptual Architecture Models

Repeat the conceptual architecture component to document each individual candidate architecture that may fit the requirements and the business.

TA.040 - Define Application Architecture (Optional)

In this task, you design the functional architecture for the applications. This involves identifying those key setup parameters in the applications that are relevant to your particular implementation and establishing the values for those parameters across all applications.

This task focuses on the future production environment and <u>not</u> interim environments to support the project activities.

Δ If your project is of *medium or high complexity*, you should perform this task.

Deliverable

The deliverable for this task is the **Application Architecture**. It identifies those key setup parameters in the applications that are relevant to your particular implementation and establishes the values for those parameters across all applications.

Prerequisites

You need the following input for this task:

- ❑ **Current Financial and Operating Structure (RD.010)**

The Current Financial and Operating Structure provides information about the company financial hierarchy, the business organizations, and the functions the organizations perform.

- ❑ **Process and Mapping Summary (RD.030)**

The Process and Mapping Summary provides details of the future business model, including business processes and mapping decisions made to date. If Establish Process and Mapping Summary was not performed, this deliverable will not exist. (See the task description for RD.030 for more information on when this task should be performed.)

❑ **Mapped Business Requirements (BR.030)**

The Mapped Business Requirements provide details of the mapping of the business requirements defined in requirements definition.

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements, and strategy for the architecture work.

❑ **Preliminary Conceptual Architecture (TA.030)**

The Preliminary Conceptual Architecture document provides the initial framework for the applications architecture. If Develop Preliminary Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.030 for more information on when this task should be performed.)

Optional

You may need the following input for this task:

❑ **Application Product Reference and Implementation Manuals**

The application product reference or implementation manuals should contain information about the key application setup parameters and their effect on the functional architecture and behavior of the applications.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review business requirements and mapping information to establish values of parameters already specified.	

No.	Task Step	Deliverable Component
2.	Establish a list of the sets of books needed for the implementation and map their interrelationship.	Integrated Application Business Architecture
3.	Establish a list of inventory organizations needed for the implementation and map their interrelationship.	Integrated Application Business Architecture
4.	Establish a list of human resources business groups and organizations needed for the implementation and map their interrelationship.	Integrated Application Business Architecture
5.	Create the integrated business architecture for the finance, manufacturing, distribution, and human resources functions of the business.	Integrated Application Business Architecture
6.	Map the integrated business architecture to the detailed business functions of finance, manufacturing, distribution, and human resources functions.	Integrated Application Functional Architecture
7.	Review application functional architecture with business analysts.	

Table 4-10 Task Steps for Define Application Architecture

Approach and Techniques

The goal of this task is to define the aspects of the architecture that are application specific and are broad ranging. These include organizational and organizational structural elements.

This task is related to, but distinct from, normal application setups, because it deals with setup parameters that have a profound effect on the way the application database is laid out and the basic functioning of applications. You may identify the critical parameters as part of Business Requirements Mapping (BR), but this task is where you define the relationships between the key architecture parameters and define the overall architecture for the different application installation sites or data centers in the organization as well as the cross unit and organizational features.

The Define Application Architecture task requires the identification and mapping of the key application setup parameters and the major application system functions to make sure that the overall mapping of the parameters and functions is appropriate for the business processing requirements, but is also compatible with the application architecture envisioned for the new system. The system architect gathers information about the different mapping decisions made within the mapping process, establish the key parameters and functional areas affecting architecture, and map out the high-level interrelationship.

☞ **Attention:** The system architect carries primary responsibility for this task and needs to be experienced and knowledgeable about the application architecture and functionality of the application release and application products being implemented in the project. The system architect must make decisions on how to implement the application and architecture to support the application functional architecture and the data distribution strategy developed in the conceptual architecture. These decisions require a knowledge of the fundamental technical architecture of the applications, and may also require a high-level understanding of the key business processes involved in different operations and business functions.

General Functional Architecture Considerations

The design of the applications architecture is concerned with aggregating the individual mapping decisions about the critical organizational relationships and application setup parameters into an integrated and self-consistent architecture that spans every separate application install, site, or data center in your architecture.

The implementation of a package application product is unlike that of an in-house custom designed and built application, because in the former case, the structure and processes of the business must be mapped onto the data and functional structures within the package, and there is usually very little room for altering the behavior of a package application. Even a flexible application product suite, such as Oracle Applications, has a number of critical setup parameters that profoundly affect the way applications function and the way data is organized in the application database. These parameters may affect the internal applications architecture, security, interfaces, application database configuration, and logical partitioning of the data in the database. There are usually tradeoffs to consider in determining how best to configure these parameters. The functional architecture of the applications varies from release to release of the applications, so it is important that you be aware of the functional behavior for the release that you are targeting for your implementation.

Importance of the Applications Architecture

The effort required for this task varies with the complexity of the implementation you are performing, as determined by the following factors:

- number of data centers or sites with application installations
- the business functions and products you are implementing
- the number of organizational business units and their relationships to each other
- whether you have complex interrelationships between your business units or mixed centralized / decentralized business functions
- whether your business requirements are easily met by the basic applications architecture and functionality

Regardless of the scope of your implementation, go through the exercise of mapping out your key setup parameters and their interrelationships. There are two major reasons to do so:

Independent Mapping Teams

The decisions about the critical setup parameters are often made by individual functional mapping teams or individual analysts as they proceed with the mapping of business processes. The danger is that the critical setup parameters are not assessed as a group to determine whether the overall mapping is consistent. Furthermore, the functional mapping may not have the input of a system architect, who can assess the impact of the mapping on the overall architecture and processing of the system.

Consolidation and Communication of Designs

The other key reason for performing this task is to consolidate and document the overall application architecture, as well as to communicate it to the entire project team. The application architecture design is a key piece of information for most of the separate processes in an implementation project, and should be communicated to the team so that everyone understands the framework that they should be designing for or working within.

Distributed Data Implementations

In large projects where you are implementing multiple, separate data centers, each having a separate applications database, create the application architecture for each separate data center installation of the applications, depending on what parts of the business each will support. Also remember that although you are architecting for some subset of the business supported within each data center, you may need to take account of the business application architecture elsewhere in the business. For example, if you have an interface between two separate application databases, you may need to create particular application structures to receive the data that is being transferred from the other database.

Attention: In the case of Oracle Applications, this can necessitate the creation of inventory organizations or sets of books in application installations that are not the primary sites for supporting the transactions of the particular business organization. For example, you may need to create an inventory organization in the functional architecture of one site, even though the warehouse may be directly supported out of another application data center. You may do this to manage cross-site inventory shipments and material movements. Another example is the creation of all sets of books at corporate headquarters to receive consolidation subledger transactions from other databases, even though the day-to-day operations of the remote financial entities are managed in application databases separate from corporate headquarters.

The detailed approaches and concerns for this task are closely related to the application products you are implementing, and the remainder of the discussion in this section assumes you are implementing Oracle Applications.

Critical Setup Parameters in Oracle Applications

Set of Books

A set of books is an accounting ledger with a particular chart of accounts, functional currency, and accounting calendar. It is important in Oracle Financial Applications.

Balancing Entity

A balancing entity is a division or other business unit for which you prepare a balance sheet. A fund may also have its own balance sheet. This is implemented using a balancing segment in the chart of accounts key flexfield. In non-multi-organization application architectures, the balancing segment values are used to specify legal entities within a corporation, but the balancing entity can be used more generally for any type of entity producing a balance sheet. In multi-organization architectures, an additional designation of legal entities is possible.

Inventory Organization

An inventory organization is a plant, warehouse, or other type of business unit through which you access and perform transactions within one or more Oracle Applications, such as:

- Oracle Inventory
- Oracle Purchasing
- Oracle Cost Management
- Oracle Bills of Material
- Oracle Engineering
- Oracle Work in Process
- Oracle Master Scheduling / MRP
- Oracle Capacity
- Oracle Quality

☞ **Attention:** Review the current applications documentation to determine what applications utilize the inventory organization concept.

Reference: *Oracle Applications Inventory Reference Manual* (Release Specific).

HR Business Groups and Organizations

A Human Resources (HR) business group is the largest HR organizational unit you can define to represent your company, enterprise or corporation. It contains all employee and payroll records that you enter into the Oracle HRMS system. You can define organizational hierarchies within a business group to represent reporting lines and security hierarchies.

Depending on the applications mix being implemented, HR business group information may be managed in different places.

Sets of Books Architecture

When architecting your sets of books, situations exist in which you must consider tradeoffs and where you could satisfy the business requirements with more than one approach. For example, you could fulfill the business requirements with one set of books or more than one set of books. Consider all the possible repercussions of your decision, rather than just the narrow business requirement that led to possible use of multiple sets of books.

You can create an initial architecture for the sets of books by considering just the basic parameters that define the set of books, the chart of accounts, functional currency, and calendar, and then mapping these onto the business organizations and their operating environments. Identifying the most appropriate functional currency for a business organization is not always a clear-cut decision, but generally it is the currency that is used for the financial budgeting and forecasting of the organization — that is, the currency generally used for what are regarded as ”local currency” transactions.

In foreign business organizations it is often the currency of the country where the particular office geographically resides, but this can be complicated by the flexibility in local tax laws. It is also important to separate revenue (legal) entities from the local financial office entity. For example, it is possible to have a revenue entity geographically based in the Pacific Rim that holds inventory owned by another entity and records its financial state in the currency of the owning legal entity. Having created the initial architecture, you then need to identify the ramifications of the architecture, and you may even wish to create more than one candidate architecture for review and comparison. The complexity of the sets of books architecture will increase with the implementation of the following elements:

- financial subledger products along with Oracle General Ledger
- Oracle Manufacturing products
- a corporation with multiple legal entities that consolidates within Oracle General Ledger
- multiple legal entities with different functional currencies
- application products within multiple separate data centers with separate applications databases

A key decision you need to make in the financial architecture of your implementation is whether to use multiple sets of books for multiple legal entities with the same functional currency, or to manage them within a single set of books and use the balancing chart of accounts segment to denote the different legal entities. This decision should reflect:

- security requirements in Oracle General Ledger
- handling of inter- and intra-company transactions in Oracle General Ledger and other modules
- consolidation process in Oracle General Ledger
- security requirements in associated subledger and manufacturing products
- use of accrual and cash-based accounting
- specific functionality requirements in financial subledger and manufacturing applications

These are examples of factors that influence the application architecture of implementations and are not necessarily a complete list.

Warning: The details of the sets of books architecture depend on the specific release of the applications you are implementing. Determine the specific functional capabilities of the application products and releases you are implementing prior to attempting the application functional architecture.

Reference: *Oracle Project Accounting Implementation Manual* (Release Specific).

Reference: *Oracle Manufacturing Implementation Manual* (Release Specific).

Multiple Sets of Books in Financial Subledgers

A critical limitation on the applications setup parameters configuration in the past was the inability of the financial subledger applications to support more than one set of books per installation of the applications. This has led to architectures involving multiple installations of the financial products interfacing to a single installation of General Ledger in the same physical database. With the advent of multi-organization architecture, this restriction has been lifted, and it is possible to implement multiple sets of books in a single installation of each financial application.

Attention: Unless complications exist due to an existing Oracle Applications architecture based on the old model, all new implementations that require multiple sets of books in the subledgers should be based on the multi-organization application architecture.

Warning: Oracle will no longer support multiple installation of financial subledger products beyond Release 10. Use of the multi-organization features are strongly recommended.

Inventory Organizations Architecture

As with the sets of books, consider the tradeoffs when you architect your inventory organizations and be aware that you can satisfy the business requirements with more than one approach. A common case is where you could fulfill the business requirements with one or multiple inventory organizations. Again, like the sets of books architecture, it is important to consider all the repercussions of your decision rather than the narrow business requirement.

You can create an initial architecture for the inventory organizations by mapping the existing manufacturing plants, warehouses, and distribution centers onto inventory organizations. You need to perform this mapping while also bearing in mind the requirements for the master inventory organization. You can then identify the ramifications of the initial architecture and refine it based on common manufacturing or distribution practices and business processes, data sharing or security requirements, and implications of the architecture for revenue and financial processing in the ledgers. The latter is an important consideration that you must not ignore. Although the sets of books and inventory organizations setup parameters have been discussed separately here, they are mutually interdependent and must be considered together in a full analysis of the application functional architecture.

The complexity of the inventory organization architecture increases with the implementation of the following elements:

- distribution or financial products along with Oracle Manufacturing products

- a business with multiple manufacturing / distribution units that have different business practices or manufacture different finished goods
- a business that has a tightly integrated supply chain with other external trading partners
- a corporation with multiple legal entities, some of which own inventory and generate revenue
- application products within multiple data centers with separate applications databases

A key decision in the manufacturing and distribution architecture of your implementation is whether to use multiple inventory organizations to represent the manufacturing and distribution functions of your business, or to use a single inventory organization with multiple sub-inventories. This decision should reflect the following possible features:

- security requirements for transactions generated by your manufacturing business units
- costing methods and rules for different business units
- multiple manufacturing calendars
- handling of in-transit inventory and drop shipments
- forecasting details
- planning methods
- handling of common parts across different business units (for example, different units of measure, costs, lead times, cycle count tolerance, and so on)
- consignment inventory or external supply-chain requirements

This is only a selection of the factors that influence the functional architecture of implementations and not necessarily a complete list.

Warning: The details of the inventory organization architecture are very dependent on the specific release of the applications you are implementing. You should determine the specific functional capabilities of the application products and releases you are implementing prior to attempting the application functional architecture.

Implementation of Multi-Organization Architecture

The multi-organization architecture option is a way to implement a decentralized business operational model within Oracle Applications, within a single application database, and within a single installation of every product.

The multi-organization architecture adds two more key setup parameters to the set of books, balancing entity, and inventory organization mentioned above.

Legal Entity

A legal entity is a legal tax or fiscal reporting entity. It is implemented as a classification of an organization within Oracle Applications.

Operating Unit

An operating unit is an operational business unit that performs one or more of the following business functions:

- accounts payable
- accounts receivable
- revenue accounting
- order management
- customer service
- purchasing
- receiving

Reference: *Multiple Organizations in Oracle Applications Guide* (Release Specific).

Legal Entities

The concept of the legal entity is used widely in standard reports and transactions; legal and tax reporting may also be done through the mechanism of reporting based on the balancing segment. Refer to the current release documentation for specifics.

Operating Units

The operating unit parameter provides a means to define secure business units within the Oracle Applications products mentioned above. It does not directly affect the architecture of the manufacturing application products, but has an indirect impact because the manufacturing products communicate with, and transfer data to and from the subledger and distribution products. If you are implementing Oracle financial or distribution products along with the manufacturing products, you need to make sure that the application architecture of the manufacturing part of the business is properly aligned with the financial and distribution application architecture.

The operating units are also important because of cross-business unit buying, selling, and shipping functionality in the multi-organization architecture. Different business units (that may be in different legal entities) can buy, sell, and ship product from, or on behalf of, each other. A inter-company accounting process generates the payables and receivable invoices that result from the cross-legal entity transaction. You implement this functionality at the operating unit level, and the need for this capability within the business will be an important factor in the definition of the operating units.

HR Business Groups and Organizations Architecture

If you are implementing Oracle HRMS without implementing other Oracle financial or manufacturing products, you can define your application architecture based on the HR requirements alone. Determine the reporting lines and security groupings you need to implement for your business, and then determine the business groups and the HR organizational hierarchies you will use as the framework application architecture for the rest of the data you will enter in the system.

If you are implementing human resources applications with financial applications, bear in mind that data is shared between the application product families and may have an impact on the overall application architecture. For example, the legal entity that you create as part of your financial architecture may also be a government reporting entity within HR. Make sure that the same organization is correctly defined for the integrated application architecture. If you are sharing other classifications of organizations in the HR organization hierarchy with financials and manufacturing, it is again important to provide consistency of operation.

HR employee data is also referenced within the financial and manufacturing products, such as Oracle Purchasing and Engineering. If you plan to create an HR application architecture with multiple business groups, this may have implications for the ability to view employee data within the financial products.

Reference: *Oracle Human Resources Implementation Manual* (Release Specific).

Financial Application Implementations

If you are only implementing financial applications, you do not need to concern yourself with the detailed definition of a manufacturing or distribution application architecture. However, because you may still need to share some of the setup parameters across business functions (for example, the parts master), you may not be able to totally disregard the manufacturing and distribution sections of the deliverable. You may also need to consider the human resources business groups that you create to store the employee data, which is shared with other applications.

Manufacturing and Distribution Application Implementations

Although you may only be implementing particular manufacturing or distribution application products, you may still need to define the accounting structure that defines the financial operating environment for a particular inventory center or warehouse.

Linkage to Other Tasks

The Application Architecture is an input to the following tasks:

- BR.030 - Map Business Requirements
- BR.100 - Define Application Setups
- TA.090 - Define Application and Database Server Architecture

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	60
Business Analyst	40

Table 4-11 Role Contribution for Define Application Architecture

Deliverable Guidelines

Use the Application Architecture deliverable to define the cross-application and cross-organization setup parameters.

This deliverable should address the following:

- multi-organization architecture including entities, sets of books, operating units and inventory and supply-chain organizations

Deliverable Components

The Application Architecture consists of the following components:

- Introduction
- Integrated Application Business Architecture
- Integrated Application Functional Architecture

Introduction

This component describes the purpose of the deliverable and list the key application setup parameters that affect the architecture of the application deployment and the partitioning of data in the application database.

Integrated Application Business Architecture

This component details the relationship between the key setup application setup parameters, and relate them to the financial and operating structure of the business.

Integrated Application Functional Architecture

This component describes the relationship between the key setup application setup parameters, and relate them to the main business functions.

Tools

Deliverable Template

Use the Application Architecture template to create the deliverable for this task.

Template Examples

The template is based directly on functional architecture of Oracle Applications and uses Oracle's terminology. The example in the Integrated Application Business Architecture component shows an Oracle multi-organization business structure with sets of books, legal entities, operating units, and inventory organizations. If you are not implementing the multi-organization architecture, you may not need to show the operating units on your architecture. However, you may wish to retain the legal entities that you would implement as balancing entities within the chart of accounts, rather than as Oracle organizations with a classification of legal entity. If you are implementing human resources as well, you need to find a way to represent the HR organizations in the manufacturing and financial business architecture. This may require a separate diagram.

The example in the Integrated Application Functional Architecture shows just the set of books and inventory organizations across application products (business functions). If you are implementing a multi-organization architecture, you need to show the operating units and legal entities also in the same kind of diagram.

TA.050 - Define System Availability Strategy (Core)

In this task, you develop a strategy to provide the minimum level of systems availability that the business demands.

This task focuses on the future production environment and not interim environments to support the project activities.

Deliverable

The deliverable for this task is the **System Availability Strategy**. It is the study of the systems and approaches that will be incorporated into the technical architecture of the new system to provide the required level of system availability.

Prerequisites

You need the following input for this task:

❑ **Business Availability Requirements (RD.070)**

The Business Availability Requirements provide details of the agreed upon requirements for the availability of the business systems. It discusses the minimum level of service that the business expects from the information systems that support it, the downtime it will accept, and the contingency measures that will be adopted if unplanned outages of various types occur.

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review task input deliverables and summarize systems availability requirements.	Critical Systems Availability
2.	Summarize procedures or maintenance tasks that require planned downtime.	Planned System Outages
3.	Summarize possible system failure scenarios that could affect systems availability.	Unplanned System Outages
4.	Identify impact of disk and other hardware component failure scenarios and their handling.	Physical (Hardware or Network) Component Failure
5.	Identify impact of database server failure scenarios and their handling.	Physical (Hardware or Network) Component Failure
6.	Identify impact of application file server failure scenarios and their handling.	Physical (Hardware or Network) Component Failure
7.	Identify impact of network failure scenarios and their handling.	Physical (Hardware or Network) Component Failure
8.	Identify impact of client desktop failure scenarios and their handling.	Physical (Hardware or Network) Component Failure

No.	Task Step	Deliverable Component
9.	Identify impact of data center failure scenarios and their handling.	Physical (Hardware or Network) Component Failure
10.	Identify impact of database corruption scenarios and their handling.	Software Component Failure
11.	Identify software code corruption scenarios and their handling.	Software Component Failure
12.	Identify likely causes of Century date failure in software and interfaces.	Century Date Failure
13.	Identify impact of code management and its handling.	Maintenance Outages
14.	Identify impact of database maintenance and its handling.	Maintenance Outages

Table 4-12   Task Steps for Define System Availability Strategy

Approach and Techniques

This task requires the development of a strategy for providing the level of system availability that the IS organization has previously agreed to as part of the business availability requirements. To develop the strategy, you need to review and understand the required system availability metrics, determine all sources of planned and unplanned outages possible, estimate the failure rate and downtime for each of these, and then design solutions for them that will provide the required availability levels. This task requires familiarity with traditional and new fault tolerance mechanisms for the systems being implemented, and a balanced appreciation of the advantages and disadvantages of the different approaches. Keep in mind that this is a strategy study rather than a detailed design for system management procedures and tools.

At this level you determine the approach for handling each type of failure or planned downtime for use later in the detailed design.

Types of System Outages

System outages may be planned or unplanned. Types of planned system outages include:

- downtime for database backups
- downtime for application of code patches or for code upgrade

Unplanned outages generally occur because of an exception condition in the applications, or because a technical component fails. Unplanned system outages include:

- disk or media failure that may affect database files
- database failure caused by hardware or operating system crash or operator error
- hardware or complete data center failure due to disasters such as fire and earthquakes
- network failure that may affect critical interfaces between remote sites or online connections to a server

Thorough Consideration of Failure Points

Consideration of system availability and the degree of system fault tolerance needed is critical to every implementation, and it is one of the areas that the implementation project managers, and the senior management of the business, are keen to discuss. Consider all possible causes of failure. You should adopt the rule that if something in a system can fail, it will, and design accordingly. The fault tolerant components that you design into the system directly affect the technical architecture of the hardware, networks, or databases, and could indirectly affect the application architecture.

Cost, Performance, and Availability Equation

You need to strike a balance between cost, performance, and availability when designing your technical architecture. If having an available system is the most important consideration to your business, design your architecture with that objective overriding any other, and you will incur greater hardware costs and (possibly) sacrifice some performance to achieve the necessary level of confidence in your system and the *mean time to failure* (MTTF) or *mean time to recovery* (MTTR) you require. The Define Architecture Requirements and Strategy task (TA.010) at the beginning of the Application and Technical Architecture process should help you gather high-level requirements for system availability and preferred hardware architectures and vendors.

The Cost of High Availability

Management often asks for 24x7x52 system uptime without a clear idea about what this really means. For an IS organization to guarantee the business 99.99 percent systems availability, they would usually have to make a prohibitively expensive investment in redundant hardware and communications links. Few businesses end up with a fault tolerance strategy that enables this degree of availability once the costs are identified. The system architect should confirm that expectations have been set properly, and that the management of the business has understood the realities of the situation and has not been misled into thinking that they have 24x7x52 operation simply because the database administration staff are performing regular database backups. The definition of the business availability requirements should be where these types of compromises have been discussed, and so at this stage you should have a set of realistically attainable system availability metrics. For more information, see Identify Business Availability Requirements (RD.070).

Safeguards Against System Failures

Some of the more common techniques for providing safeguards against system failures are:

- hot and cold database backups for database recovery
- disk mirroring and RAID arrays for disk failure tolerance
- network redundancy or X.25 packet rerouting for network segment failure
- platform clustering (Oracle® Parallel Server or Dual Host SCSI for standby platform)
- remote hot backup platform for disaster scenarios
- use of built-in hardware capability to reconfigure around failed components such as system bus, CPU boards, and SCSI controllers

Database Backup and Recovery

The strategies for maintaining availability of an Oracle database are well understood and documented. The detailed procedures will be documented later, so at this stage it is only important to understand the mechanisms, advantages, and disadvantages of comparative techniques. If you choose strategies that rely on specific technical infrastructure components such as mirrored disks, it is important to document these within this task.

For more information about backup and database copy procedures, consult the following:

Reference: Velpuri, Rama. *Oracle8 Backup and Recovery Handbook*. Oracle Press. Osborne McGraw-Hill, 1998. ISBN: 0078823897.

Oracle Parallel Server (OPS)

In an OPS configuration, separate Oracle database instances run simultaneously on one or more hardware nodes. The data files of the Oracle database are located on a set of shared disks that all nodes in the cluster can access. Each Oracle instance has its own system global area (SGA) and background processes. A special distributed locking mechanism (the distributed lock manager - DLM) coordinates access to the shared database by the individual nodes in the cluster and manages data contention and concurrency.

OPS prevents single platform failure from affecting database availability. If a node goes down, whether it be network or node failure, a surviving instance takes over and performs instance recovery for the lost node. Users trying to access the database can still connect to one of the surviving nodes (for example, surviving instances) and access the data while recovery is taking place.

OPS provides an extra level availability if a local server fails, but has higher associated hard costs of extra server and software, and higher soft costs for the learning events or expertise that needs to be hired to manage the system. OPS may have other significant disadvantages, such as the possible inability of the cluster minus one hardware node to handle the regular ambient processing load. This illustrates the need for careful consideration of any system solutions that you choose to adopt.

Use of Standby Databases and Servers

Another solution for hardware node failure is to use two identical database servers, one for processing, and the other on standby. System administrators configure the servers as mirrors of each other with the same directory structure, database structure, and software versions. Database administrators configure the production instance to run in archive log mode. The mirror instance has its database mounted but not open, equivalent to the *startup mount* command. When the production instance performs a log switch and the ARCH process creates an archive log, a batch process copies the archive log and the current control file to the standby server. The standby system is maintained in a continuous recovery mode, however, which means that it cannot be used for purposes other than standby.

A cheaper and more cost-effective way to implement the standby server solution is to buy a minimum hardware configuration for the backup server. The system architect can design the backup server to be a temporary fix until the production server is repaired. The minimum configuration allows for smaller CPUs, or a reduced number of parallel CPUs, and a reduced amount of physical memory. The disk configuration can be quite different as well. The tradeoff is that the standby server may not be able to handle the regular processing load of all users and batch processing, and that some mechanism to limit system usage will be needed while the primary server is repaired and brought back online.

It is not possible to estimate accurately the total recovery time for a database running with a standby server configuration. Accurate recovery time estimates are determined by thorough testing. Implementing the standby server allows for a much improved backup methodology. During off peak hours, administrators can perform a cold backup of the mirrored server using multiple backup devices. During this time, the administrator suspends the transfer of archive logs from the production instance to the mirrored instance. After the backup is complete, the transfer routine is re-enabled and the application of the archive logs continues.

Remote Standby Server

When the standby server is maintained in a remote location, such as a backup data center, it can act as a disaster recovery site to preserve system availability while the main data center is repaired and brought back online.

Oracle supports the standby database feature. For more information, see:

Reference: *Oracle Server Reference Manual* (Release Specific).

Reference: *Oracle Server Administrator's Guide* (Release Specific).

Database Symmetric Replication and Fail-Over

Oracle symmetric replication can also be used in a fail-over configuration to protect against site failures. Its throughput capacity is less than the redo log file transfer approach because symmetric replication does not use the Oracle recovery mechanisms to apply replicated data changes. It does, however, offer advantages over the log file transfer approach in that the fail-over system or systems can be used for other purposes in addition to fail-over.

In a symmetric replication fail-over configuration, the fail-over system can be queried at any time (for example, reporting or decision support analysis). The fail-over system can also be updated if the two systems are employing a shared ownership replication model with full update conflict detection and resolution.

Attention: The use of symmetric replication fail-over for ERP or CRM systems as complex as those managed using Oracle Applications is not generally a simple proposition and requires careful analysis, planning, and testing.

Reference: *Oracle Symmetric Replication Guide* (Release Specific).

Reference: *Oracle Server Distributed Systems Guide* (Release Specific).

Disk Mirroring and RAID Technology

Redundant Arrays of Inexpensive Disks (RAID) is a term that applies to a technology that includes various striping and mirroring configurations (or RAID levels) to provide increased reliability in disk systems. Use of RAID requires careful consideration of the tradeoffs in performance, cost, and availability.

One increasingly popular RAID variant is the full disk mirror (RAID level 1). This provides a full online copy of your application data and enables the application system to keep functioning so long as one of the database mirror images is uncorrupted. In systems where the cost is not prohibitive, a triple mirror configuration can enable the database administrator to back up the database to tape without having to keep parts of the database in hot backup mode, while keeping the disk redundancy safety net.

Reference: Millsap, Cary V. *Configuring Oracle Server for VLDB*. OAUG Proceedings, Spring 1996.

Reference: Manning, Paul. *Backing Up Oracle Applications Through Split-Mirror Technology*. OAUG Proceedings, Spring 1999.

Distributed Data Systems

If you are working with a conceptual architecture that has distributed data, you have to consider the effect of various planned and unplanned outages on the overall architecture. While Oracle has very robust distributed failure handling built into the database and it can automate a distributed recovery, the recovery of distributed data systems is not a totally transparent procedure and thought needs to be given to the precise consequences of a failure in a particular component. When complex mission- or business-critical processes span multiple databases across multiple platform servers, the handling of the recovery process is even more complicated and needs careful consideration.

Any interface between two heterogeneous applications or databases is effectively a distributed data system, and the recovery of interfaces after failure requires careful consideration. The simpler interfaces are designed and built as part of the Module Design and Build (MD) process, but any complex or key interfaces that are subprojects within the architecture process need to be addressed here also. The database designers in the Module Design and Build (MD) process should be made aware of the need for consideration of system failures in their design work.

Century Date Compliance

In the past, two-character date coding was an acceptable convention due to perceived costs associated with the additional disk and memory storage requirements of full four-character date encoding. As the year

2000 approached, it became evident that a full four-character coding scheme was more appropriate.

In the context of the Application Implementation Method (AIM), the convention Century Date or C/Date support rather than Year2000 or Y2K support is used. Coding for any future Century Date is now the modern business and technical convention.

Every applications implementation team needs to consider the impact of the century date on their implementation project. As part of the implementation effort, all customizations, legacy data conversions, and custom interfaces need to be reviewed for Century Date compliance.

When designing the application and technical architecture it is essential that all dates be entered, stored, transferred and processed using the full four-digit year for compliance with Century Date standards. This is an important part of the definition of the System Availability Strategy.

Linkage to Other Tasks

The System Availability Strategy is an input to the following tasks:

- TA.070 - Revise Conceptual Architecture
- TA.090 - Define Application and Database Server Architecture
- TA.120 - Define Platform and Network Architecture
- TA.150 - Define System Management Procedures
- PM.020 - Design Production Support Infrastructure

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	70
System Administrator	10
Database Administrator	10

Role	%
Network Administrator	10
Client Staff Member	*
IS Manager	*

Table 4-13 Role Contribution for Define System Availability Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the System Availability Strategy as an input to subsequent tasks that determine the strategy you will use to manage planned and unplanned system outages. This deliverable should address the following:

- Century Date compliance and its impact on systems availability

Deliverable Components

The System Availability Strategy consists of the following components:

- Critical Systems Availability
- Unplanned System Outages
- Planned System Outages
- Physical (Hardware or Network) Component Failure
- Software Component Failure
- Century Date Failure
- Maintenance Outages

Critical Systems Availability

This component describes the major and most critical system availability requirements. The discussion includes the availability needs at different times of the day and different points in the business cycles (month, quarter).

Unplanned System Outages

This component summarizes the main unplanned system outages due to failures in different components of the system and the strategy that will be used to handle each. It identifies each type of failure, its estimated failure rate, and the strategy for handling the failure identified, so as to preserve the agreed upon level of system availability.

Planned System Outages

This component summarizes the planned outages that are predicted to be needed for regular or routine system maintenance. It also identifies each type of planned outage and estimates the frequency and duration of downtime.

Physical (Hardware or Network) Component Failure

This component describes the major failures due to failure of platform hardware or network failure. It describes the approach and steps to recover from such failures.

Software Component Failure

This component describes the most typical types of software component failures and the actions required to recover from these failures.

Century Date Failure

This component describes the most likely causes of century date failures in software and interfaces and the actions required to recover from these failures.

Maintenance Outages

This component describes the planned outage events that are required to perform some form of maintenance on the system. Examples of system maintenance events that this component include are database maintenance such as backups, tuning and space management as well as software maintenance such as software patches and upgrades.

Audience, Distribution, and Usage

Distribute the System Availability Strategy to the following:

- project manager
- information systems manager
- architecture team members
- database, system, and network administrators

Tools

Deliverable Template

Use the System Availability Strategy template to create the deliverable for this task.

TA.060 - Define Reporting and Information Access Strategy (Optional)

In this task, you analyze the reporting and information access requirements for the new business and specify at a high level how you will meet these in the new architecture. This is a systems view of how the reporting and information access requirements will be satisfied in the new architecture, in contrast to the approach taken in Business Requirements Definition (RD) and Business Requirements Mapping (BR).

This task focuses on the future production environment and not interim environments to support the project activities.

Δ If your project includes *complex reporting*, you should perform this task.

Deliverable

The deliverable for this task is the **Reporting and Information Access Strategy**.

Prerequisites

You need the following input for this task:

❑ **Master Report Tracking List (RD.080)**

The Master Report Tracking List provides information about the reporting requirements of the system. It helps specify the reports needed by the various business organizations and sites.

❑ **Information Model (BR.060)**

The Information Model provides information about the information requirements of the system. It helps specify the information to which the various business organizations and sites need read (reporting) access. If Develop Information Model was not performed, this deliverable will not exist. (See the task description for BR.060 for more information on when this task should be performed).

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements, and strategy for the architecture work.

❑ **Preliminary Conceptual Architecture (TA.030)**

The Preliminary Conceptual Architecture document provides the initial framework for the Applications Architecture. If Develop Preliminary Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.030 for more information on when this task should be performed).

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather key reporting and information access requirements collected in Identify Reporting and Information Access Requirements (RD.080).	
2.	Identify the reporting and information access tools acquired to date for the project.	
3.	Review the purpose and refine the scope.	Introduction
4.	Refine the categorization of reporting and information access systems. Create an architecture diagram.	Summary of Architecture
5.	Review and update the report creation/request procedure.	Report Creation Procedure

No.	Task Step	Deliverable Component
6.	Detail the tool use strategy for each tool. Contrast the use of each tool with each other tool.	Reporting and Information Access Strategy
7.	Detail an alternate view of the strategy, by application module.	Application Module Strategy
8.	Detail an alternate view of the strategy, by business process.	Business Process Strategy
9.	Detail an alternate view of the strategy, by business unit.	Business Unit Strategy
10.	Define the Report Distribution and Retention Strategy.	Report Distribution and Retention Strategy
11.	Review the final strategy with the individuals who developed the initial requirements.	
12.	Revise the strategy as required, and seek approval.	Acceptance Certificate (PJM.CR.080)

Table 4-14 Task Steps for Define Reporting and Information Access Strategy

Approach and Techniques

In this task the precise reporting requirements are mapped onto specific systems and applications that will support the different types of reporting.

Different Types of Reporting

The ability to easily obtain meaningful information from a system so that a business can make timely tactical and strategic decisions is a critical success factor for a newly implemented system. There are many different styles of reports and techniques for manipulating data for reporting purposes, but they all fall into one of two major categories:

- operational reporting
- decision support

Decision support includes the following types of reporting systems:

- classic decision support systems (DSS)
- executive information systems (EIS)
- analytical reporting systems including online analytical processing (OLAP)

The dividing line between the two types is not hard and fast, but there are general characteristics that divide the two.

Operational Reporting

Operational Reporting is the routine transaction-based reporting that needs to occur in a business to support general business operations. This type of reporting generally involves relatively small volumes of data, presented in a detail format on a regular reporting schedule.

Decision Support

Decision Support is a type of reporting that is performed for decision-making purposes in a business. The approach and periodicity of this type of reporting are less regular than the operational reporting. Decision support is often interpreted as being synonymous with reporting against a data warehouse, but it is possible to perform decision support type reporting within transaction systems. Data warehousing is the enabling technology for this type of operational reporting, not the reporting category. It is usually more strategic in nature and can include:

- structured reporting across large volumes of historical data
- analytical (OLAP) style reporting to analyze and dissect business data to understand trends and relationships
- data mining to discover hidden trends in large volumes of historical data

Task Complexity and Project Scope

The potential complexity of this task and the reporting strategy you need to put in place depends on the scope and constraints of the project on which you are working.

Single Installation, Fast Track Implementations

If you are working on a fast track implementation with a single installation of the applications and intend to perform all reporting out of the operational (transaction) environment, you may not need to perform the further requirements gathering and analysis in this task. Use the Oracle Application OLTP database as your reporting database, and you should easily be able to consolidate and summarize data across your different business units. You may need to write some custom reports to perform certain types of consolidated reporting, (that may not be provided with the base application) but you will have all the data you need in a single, consolidated operational database. You should *always* consider the effect on performance of your reporting solution, but you may not need any special databases to support your reporting needs.

☞ **Attention:** In the case of Oracle Applications, users typically execute all their batch reports using the Oracle Applications Concurrent Manager and their online inquiries through forms or some other reporting tool.

Complex Enterprise Reporting Needs

Larger businesses typically have more challenging and complex reporting needs, spanning operational, strategic, and analytical reporting. For example, global reporting or online query of operational data for:

- sites, divisions, and business lines within a single application installation and database
- sites, divisions, and business lines across multiple distributed applications and databases
- different statutory and legal reporting requirements to satisfy local legislation and country laws
- integrated reporting across heterogeneous applications and databases, relational and non-relational, legacy, and third-party vendor packages

Senior management may require long-term trend analysis of data for strategic decision support of:

- forecasting
- financial modeling and budgeting
- supply and distribution planning
- sales and marketing
- data mining

You need to categorize the reporting requirements of different groups and business units within the enterprise, before you attempt to propose solutions.

Components of Reporting Systems

Reporting systems have four components:

- a database of data from which to report — usually a single physical database, but may be a logical database consisting of multiple physical databases
- a way to load data into the reporting database

- a reporting engine that accepts input parameters or business rules to restrict and manipulate the data set that will appear in the report
- a way to present the data in report format

In the simplest operational transaction systems that permit reporting, the data is generated directly in the transaction database, and a reporting tool performs the query and manipulation, as well as the presentation of the data in the report. However, you should consider the different components depending on the exact function and source of the reporting system. A separate reporting system is often classified as an architecture subsystem.

Data Warehouses

A simple definition of a data warehouse is a subject-oriented corporate store of information sourced from one or more systems, applications, or databases. A data warehouse can be an enabling technology for both general categories of reporting. Hence you could envision an operational data warehouse, containing consolidated operational data and having a technical design, that supports the routine operational reporting needs of the business. More often though, a data warehouse contains a broad range and large volume of historical corporate data, and is the repository for decision support and analytical applications and systems. Data warehouse-based systems are a good example of an architecture subsystem that would be singled out as a standalone component of the overall information systems architecture. For more information, see Define and Propose Architecture Subsystems (TA.100).

Performance of Reporting Systems

The effect of a reporting load on the performance of any transaction system is as important as the effect of the transaction, and it should not be (but often is) overlooked. In ERP (OLTP) system environments in which significant reporting is permitted, the load can periodically be extremely heavy. At critical times in the business cycle such as financial period end, the operational reporting load itself can consume many of the system resources of a database server and can decimate the transaction performance. Conversely, if the reporting needs to be completed in a certain time window, the performance of the system for the reporting is critical also. For more information about managing the performance quality of reporting systems, see Performance Testing (PT).

Separate Reporting Systems as a System Capacity Management Technique

It is possible to alleviate some or much of the operational reporting load on an OLTP system by offloading the reporting functions onto a separate reporting database on a separate server. An operational reporting database is a database that is a direct point-in-time copy of the transaction system for the purposes of reporting only. Reporting databases are a valuable technique for offloading processing from the transaction system without expending the resources and time of designing and building a specialized data warehouse or reporting system with a reporting-oriented data model. In order to be useful, there must be reporting groups within the enterprise that can perform their reporting on aged data (for example, data that does not contain all transactions up to the immediate point-in-time). Increasing the refresh frequency for data in the reporting database can circumvent some of these problems, but it is usually not feasible or cost-effective to keep the data in a reporting database synchronized with the OLTP system in real-time. For more information, see Conduct Reporting Fit Analysis (BR.070). For general system capacity planning issues, see Define System Capacity Plan (TA.110).

Suggestion: If project timing makes this possible, you should collaborate with the business analysts who are performing the reporting fit analysis to determine whether there are groups that can report using aged data, and keep the number of such users would significantly offload system resources from the OLTP database server. If the architecture tasks occur in advance of the report mapping task, you need to gather this information yourself, and provide it to the report mapping analysts.

Operational Reporting and Information Access

The complexity of the operational reporting strategy depends on the conceptual architecture and business needs for reporting operational data across multiple application installations or different OLTP application products. In a simple, single installation of Oracle Applications you may choose to perform all of your operational reporting out of Oracle Applications, using the batch reporting capability of the concurrent manager and standard package or custom forms to provide inquiry capability. With slightly more complexity, you could add a special ad hoc query environment within the same transaction database.

Cross-Application or Cross-Installation Reporting

Generally the organizations within your business should be able to satisfy their reporting obligations using the operational system in which they record their particular transactions, but some organizations may need access to data in multiple, physically separate operational systems.

If you need to provide consolidated operational reporting across multiple installations of the same applications suite or across heterogeneous application products, you need to consider some special mechanism to enable such organizations to consolidate data from multiple operational environments. The data can be consolidated in one of the applications or installations through the use of custom interfaces. You then perform consolidated reporting out of the consolidating application or instance.

Another approach is to design an operational data store, which is an independent standalone database, but which may reside in one of the database instances corresponding to the separate application installations. The data store can be as comprehensive or complex as you wish.

Operational Reporting Database

If you have special security concerns for reporting users, or wish to offload report processing load to another server, you can use an operational reporting database. This type of reporting database is different from a data warehouse, because it is a direct copy of the transaction database, without reorganization of the data. When it is copied onto a separate database server, it also offloads processing capacity.

Suggestion: If the business can tolerate daily downtime for operational reporting database refreshes, a UNIX cold backup file copy mechanism is the simplest way to refresh the database.

Operational Reporting Systems — Ad Hoc Queries

Providing users with a means to easily locate, retrieve, format, and display online the information they need can empower users and alleviate some development overhead needed to migrate existing reports from legacy systems to a newly implemented system. However, creating an ad hoc query environment in which users can be productive requires investment of information systems resources. Do not assume that because you make a report or inquiry development tool accessible to the users that this automatically constitutes an ad hoc query capability for users. The main problem is that a relational data model optimized for OLTP performance is not easily understood or decipherable by users. In order to facilitate the use of an ad hoc query tool you may have to create an environment that translates the OLTP data model into the form of business objects that a user understands.

Key-User Layer (Meta Layer)

This technique adds a layer on top of the relational OLTP data model to present the data in the operational transaction system in the form of easily understood and familiar business objects, such as sales orders or purchase orders. You can build this layer on top of the OLTP database directly and allow ad hoc queries in the transaction environment, or you can add the layer to a direct copy of the transaction database, that is mounted as a reporting database.

Operational Data Warehouse

Another type of system that you can use for ad hoc queries on operational data is an operational data warehouse. This type of data warehouse is different from the type that supports strategic reporting or analysis because it supports operational reporting needs, it may only contain the most recent operational data, and it may not have the summarized views of data that provide a long-term strategic business view. The data in the warehouse is incremented on a regular basis and may be completely refreshed after a certain time period. The warehouse will usually have a report- or query-friendly data model, so programs need to be developed both to extract operational data from OLTP systems and to load data into the data warehouse. An example of where this might be useful is in the consolidation of order entry and point-of-sale information, where you might totally refresh the data every quarter or annually.

☞ **Attention:** A data warehouse where operational reporting is performed and data is refreshed from time-to-time is sometimes referred to as an *operational data store* to distinguish it from a “true” data warehouse, where data agglomeration is cumulative for historical analysis.

Decision Support Strategy

Decision support systems are specialized applications that enable users to summarize and organize transaction data into formats that are oriented towards reporting and analysis. The data model for a decision support database may be very different than the data model for the transaction databases that provide the source data. Often the database for a decision support system is a data warehouse that stores transaction data from multiple, different transaction systems over many financial periods. Raw operational data may have been summarized or aggregated to some degree within the warehouse. A data warehouse or decision support system is a common standalone architecture subsystem that might need custom development or custom integration work.

☞ **Attention:** One benefit of a custom data warehouse system that is often overlooked is the fact that the data model is owned by the business. By performing as much of the reporting from the data warehouse as possible, the business insulates itself from the effects of data model changes in new releases of packaged applications.

Online Analytical Processing (OLAP)

This technique is the manipulation and analysis of data in multiple dimensions to support business decisions. The manipulations performed include *slicing and dicing* data across different dimensions, pivoting, selective data summary, consolidation, and drill-down.

OLAP systems are usually specialized in their internal architecture and are subsystems within their own right. OLAP system functions may be supported directly from a transaction database; they may or may not have built-in persistent data caches (effectively a local database). A very powerful solution is to combine a backend data warehouse that contains structures appropriate for OLAP with an intuitive, user-friendly OLAP tool to perform the data manipulation. If an OLAP system is included, you need to define exactly how it should interact with other reporting databases or warehouses and design the architecture accordingly. Of particular importance are the requirements surrounding the aging and refresh of the data. These requirements are driven by how close to a real-time view of the business analysts need.

Data Marts

Data Marts are data warehouses that are smaller in scale than an enterprise data warehouse and are oriented more towards department-level decision support. They may be organizational or functional slices of the complete data set in the enterprise data warehouse. Often they are installed on cheaper workgroup or functional departmental server platforms, which provides a distributed, scaleable, secure environment for either decision support or analytical (OLAP) reporting.

Impact of Data Warehouses

It is relatively straightforward to specify the systems with which a data warehouse will interact and how the data warehouse will be used in the overall reporting strategy, but integrating a package corporate data warehouse solution or custom developing a corporate data warehouse is a major undertaking. An analysis of the requirements and the very detailed technical and database design is beyond the scope of this process in AIM. An architecture subproject should be initiated to perform the design of this part of the architecture solution. If it is apparent that a data warehouse system is required, generate the Architecture Subsystem Proposal (TA.100).

Report Storage and Report Distribution

Report storage and report distribution are related issues that affect how reports should be handled and are concepts that are often discussed in the context of workflow. If many users need to view a particular report, it is a waste of resources to have individual users creating their own copy of a report by executing the report program multiple times. By providing systems to support sensible business practices in this area, the information systems organization can reduce waste of paper and precious platform system resources. Include any special systems needed to support this in the architecture.

Types of systems you might consider include:

- storage of reports or report images in a database or in a computer output to laser disk (COLD) system
- automated or manual routing of electronic reports to users after execution, by email or fax
- intranet web-based report publishing, where reports are converted into HTML format, and can be browsed by users across the internal corporate network

Suggestion: The OAUG conferences that take place twice a year usually have a number of presentations devoted to reporting techniques and tools to use with Oracle Applications. The proceedings that OAUG publishes from the conference papers are a valuable reference source of information.

Linkage to Other Tasks

The Reporting and Information Access Strategy is an input to the following tasks:

- BR.070 - Conduct Reporting Fit Analysis
- TA.070 - Revise Conceptual Architecture
- PT.020 - Identify Performance Test Scenarios

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	80
Business Analyst	20
Business Line Manager	*
IS Manager	*

Table 4-15 Role Contribution for Define Reporting and Information Access Strategy

* Participation level not estimated.

Deliverable Guidelines

Use the Reporting and Information Access Strategy deliverable in subsequent tasks to define the reporting systems that support the reporting needs of system users within your new architecture.

Deliverable Components

The Report and Information Access Strategy consists of the following components:

- Introduction
- Summary of Architecture
- Report Creation Procedure
- Reporting and Information Access Strategy
- Application Module Strategy
- Business Process Strategy
- Business Unit Strategy
- Report Distribution and Retention Strategy

Introduction

This component describes the strategy for reporting and information access and the purpose for it. It also describes the downstream utility of this deliverable for other project work.

Summary of Architecture

This component summarizes the reporting and information systems in the future architecture and the functions and organizations they support.

It also may include a diagram showing the relationship between the reporting and transaction systems.

Report Creation Procedure

This component describes the procedures for requesting the development of production reports.

Reporting and Information Access Strategy

This component describes the reporting and information access strategy including a detailed discussion of the specific tools that may be used for ad-hoc query and production reporting. This view of the strategy is broad, from the tool perspective.

Application Module Strategy

This component describes in detail the reporting and information access strategy from the specific application module perspective. Some reporting and information access tools are specific to particular modules, some work across many application modules.

Business Process Strategy

This component describes the reporting and information access strategy from the business process strategy.

Business Unit Strategy

This component describes the reporting and information access strategy as it applies to each business unit in the enterprise. This is an alternative view of the strategy.

Report Distribution and Retention Strategy

This component describes how the requirements in this area will be met by the information systems architecture. It indicates how the architecture solution will satisfy the original goals for the distribution or storage solution, such as obviating the need to rerun reports multiple times, securing the business-critical report execution system from uncontrolled report requests, report self-service, and so on.

Tools

Deliverable Template

Use the Reporting and Information Access template to create the deliverable for this task.

Oracle's Query Tools

Oracle has products that can help you build the end-user layer (EUL) business views and ad hoc query environment. The Oracle Discoverer™ and Business Intelligence (BI) tools are geared towards reporting and query environments. You can use these tools to build a view-based EUL on top of the Oracle Applications database, and then write reports or allow browsing tools to access the Oracle Applications database through these layers.

Web Site: For more information about Oracle's products and services, see Oracle's world wide web home page at: http://www.oracle.com

Many third-party vendors also have reporting and query products that you can use for this task.

Oracle's OLAP Products

Oracle Express Products

Oracle has a number of products in the Oracle® Express® OLAP suite. The Oracle Express product set enables businesses to develop their own OLAP systems. A small sampling of these products include:

- Oracle® Express® Server
- Oracle Personal Express
- Oracle® Express® Analyzer
- Oracle® Express® Objects

Reference: *Oracle Express Analyzer Reference Guide* (Release Specific).

Also see other Oracle Express product guides and manuals.

Oracle® Financial Analyzer

Oracle Financial Analyzer is an enterprise solution for financial reporting, multi-dimensional analysis, budgeting, and planning. It is integrated with the operational Oracle® General Ledger application and includes built-in capability for extracting data out of Oracle General Ledger and loading into the OLAP analyzer engine.

Reference: *Oracle Financial Analyzer User's Guide* (Release Specific).

Reference: *Integrating Oracle Financial Analyzer with Oracle General Ledger* (Release Specific).

Oracle® Sales Analyzer

Oracle Sales Analyzer is an OLAP-based application that enables a business to perform sales and marketing analysis and reporting. It uses the Oracle Application Data Warehouse as the enabling backend technology.

Reference: *Oracle Applications Installation Manual for Unix, Linix, NT* (or Other Operating System). (Operating System and Release Specific).

Oracle Discoverer™ Tools

The Oracle Discoverer products can also support some OLAP data manipulation functions, including summaries, data pivoting, and drill-down.

Oracle Applications Data Warehouse

Oracle Applications Data Warehouse is a packaged pre-built data warehouse that integrates with various Oracle Application modules. It provides a relational, enterprise-wide data warehouse from which the Express and Discoverer products can extract and present application and legacy data using a common meta-data management layer.

TA.070 - Revise Conceptual Architecture (Optional)

In this task, you review information that you have gathered and analyzed since you created the initial draft of the Conceptual Architecture deliverable and incorporate any relevant new findings and decisions into the deliverable.

After incorporation of changes suggested during the final review procedure, the revised deliverable becomes the final, agreed upon conceptual architecture model for the new business system. This deliverable is important for downstream activities and marks a key milestone in the project.

This task focuses on the future production environment and not interim environments to support the project activities.

Δ If your project is of *medium or high complexity* or includes *complex architecture changes* and Develop Preliminary Conceptual Architecture (TA.030) is performed, you should perform this task.

Deliverable

The deliverable for this task is the revised **Conceptual Architecture**.

Prerequisites

You need the following input for this task:

- ❑ **Process and Mapping Summary (RD.030)**

If a Process and Mapping Summary is developed in the project, it documents key process and mapping decisions. This provides details of the future business model, including business processes and mapping decisions made to date. If Establish Process and Mapping Summary was not performed, this deliverable will not exist. (See the task description for RD.030 for more information on when this task should be performed.)

❑ Business Volumes and Metrics (RD.040)

The Business Volumes and Metrics provide the business volumes information. Use it to identify specific volumes you may need for estimating the I/O throughout and database performance and completing the conceptual architecture.

❑ Information Model (BR.060)

The Information Model provides details of the key information flows corresponding to the future business model and business processes. Understanding these information flows is an important part of the revision of the conceptual architecture, since they may not have been available when the initial conceptual architecture model was created and are the information flows that support the key business processes. The information flows may have implications for the candidate architectures or provide information about the complexity of key interfaces needed.

❑ Current Technical Architecture Baseline (TA.020)

The Current Technical Architecture Baseline contains details of the current architecture in place, and may offer solutions for re-deployment or reuse of existing technical infrastructure. If Identify Current Architecture was not performed, this deliverable will not exist. (See the task description for TA.020 for more information on when this task should be performed.)

❑ Preliminary Conceptual Architecture (TA.030)

The Preliminary Conceptual Architecture provides a high-level view and initial statement of the favored future application and technical architecture and will indicate the approximate deployment of applications in this architecture model. This is the deliverable undergoing revision in this task. If Develop Preliminary Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.030 for more information on when this task should be performed.)

❑ System Availability Strategy (TA.050)

The System Availability Strategy provides information about the strategies used for providing the level of system availability the business needs. If there are special systems or architecturally significant components that will help to maintain the desired availability, this deliverable will contain information about them.

❑ **Reporting and Information Access Strategy (TA.060)**

The Reporting and Information Access Strategy provides information about the systems strategies used for meeting the different reporting needs in the business. If there are special systems, applications, or architecturally significant components that provide the reporting capability, this deliverable will contain information about them. If Define Reporting and Information Access Strategy was not performed, this deliverable will not exist. (See the task description for TA.060 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review information gathered since creation of the Preliminary Conceptual Architecture (TA.030).	
2.	Rework the Preliminary Conceptual Architecture (TA.030), incorporating relevant new project information and decisions and update all affected components.	
3.	Review revised Conceptual Architecture with project management (if applicable) or with select project team leaders.	
4.	Secure acceptance for final Conceptual Architecture.	Acceptance Certificate (PJM.CR.080)

No.	Task Step	Deliverable Component
5.	Publish the Conceptual Architecture deliverable to a wider project team audience.	

Table 4-16 Task Steps for Revise Conceptual Architecture

Approach and Techniques

This task requires a revalidation of the Preliminary Conceptual Architecture (TA.030), taking into account any new requirements that have emerged from the architecture process or other processes in the project, and architecture analysis work performed since the initial version. Often there are few changes to make, or the changes are superficial and do not affect the fundamental strategy. In this situation, the acceptance of the revised Conceptual Architecture should be a formality. If, however, the new information to incorporate precipitates a rethinking of the architecture approach and strategy, you need to go through the same detailed review procedures that you went through for the Preliminary Conceptual Architecture.

Sources of Revision Changes

After the preliminary version of the Conceptual Architecture was created by the architecture team, new requirements and implementation decisions become available, and you should incorporate these into the revised deliverable. The information comes from two main sources:

- tasks completed and deliverables created in Business Requirements Definition (RD) and Business Requirements Mapping (BR)
- tasks completed and deliverables created in Application and Technical Architecture (TA)

Suggestion: If you need to make drastic changes to the original conceptual architecture model, because of subsequent findings and decisions, present it again to project management to obtain management acceptance. If the changes are relatively minor in impact, you can present it to select members of the project team to solicit feedback and acceptance.

Business Requirements Definition and Business Requirements Mapping Tasks

Business Requirements Definition (RD) and Business Requirements Mapping (BR) provide the business requirements input into the architecture and generally proceed concurrently with the Application and Technical Architecture process. The BR process is the repository of all mapping requirements and the business solutions to meet the business scenario requirements. In cases where the package application does not meet the business scenario requirements, the mapping team proposes a solution to fit the need. The preliminary version of the Conceptual Architecture should be an input to the BR process, to help make sure that the mapping team is aware of the proposed conceptual architecture model. If there are mapping decisions affected by the Conceptual Architecture, the mapping team will be aware of the issues.

Generally the Conceptual Architecture has a minor impact on the mapping exercise for implementations where the operational data is centralized in a single database and there are no special reporting systems. The converse is not necessarily true, though, and some proposed solutions may be significant to the architecture of the system, and may even be solutions that the architecture team takes ownership of, because of their complexity and impact on the overall system architecture. The system architect on the team should be aware of key mapping decisions and be able to translate them into requirements or constraints on the architecture design process.

Examples of mapping requirements that would affect architecture include:

- changes in how business processes map onto application functions or onto different applications
- changes in key business processes based on better understanding by the mapping teams of the way the package applications support specific business functions

- definition of the Information Model (BR.060) for the movement of information and data within the business
- decisions about key application setup parameters such as sets of books or inventory organizations
- extension to the package application security capability to support special application-level security requirements
- identification of a critical and complex interface that affects the functioning of a major part of the new system

For more information about the mapping process and tasks, see Business Requirements Mapping (BR).

Linkage to Other Tasks

The Conceptual Architecture is an input to the following tasks:

- TA.080 - Define Application Security Architecture
- TA.090 - Define Application and Database Server Architecture
- TA.100 - Define and Propose Architecture Subsystems
- TA.110 - Define System Capacity Plan
- TA.120 - Define Platform and Network Architecture
- TA.130 - Define Application Deployment Plan
- PT.110 - Prepare Performance Test Environment
- PM.010 - Define Transition Strategy
- PM.030 - Develop Transition and Contingency Plan

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	70
Business Analyst	20

Role	%
Project Manager	10
Client Staff Member	*
IS Manager	*
Project Sponsor	*

Table 4-17 Role Contribution for Revise Conceptual Architecture

* Participation level not estimated.

Deliverable Guidelines

Use the Preliminary Conceptual Architecture deliverable as input to subsequent tasks.

Detail the changes between the preliminary architecture and the revised architecture and document the reasons for the changes. Your documentation of the changes and reasons should be in proportion to the number and extent of changes you need to make. If you need to completely rework your architecture model and revert back to one of the originally unfavored candidate conceptual architectures, your documentation of the changes and the reasons will need to be correspondingly more thorough.

Tools

Deliverable Template

Use the Preliminary Conceptual Architecture (TA.030) to record the changes to the Conceptual Architecture. To secure final acceptance of the Conceptual Architecture, use the Acceptance Certificate (PJM.CR.080) component of the original template.

TA.080 - Define Application Security Architecture (Optional)

In this task, you design the application and database security architecture, incorporating the requirements generated by the business mapping process and the security requirements identified during the design of the conceptual architecture.

This task focuses on the future production environment and <u>not</u> interim environments to support the project activities.

Δ If your project is of *medium or high complexity*, you should perform this task.

Deliverable

The deliverable for this task is the **Application Security Architecture**.

Prerequisites

You need the following input for this task:

- ❑ **Process and Mapping Summary (RD.030)**

If a Process and Mapping Summary document is in use in the project to document key process and mapping decisions, it provides details of the future business model, including business processes and mapping decisions made to date. The process and mapping decisions may include discussion of security requirements. If Establish Process and Mapping Summary was not performed, this deliverable will not exist. (See the task description for RD.30 for more information on when this task should be performed.)

- ❑ **Information Model (BR.060)**

The Information Model provides details of the key information access requirements for the business organizations in the future business model. This document represents the organization-level analysis of the security requirements of the business and is a direct source of input for the design of the security architecture. If Create Information Model was not performed, this deliverable will not exist. (See the task description for BR.060 for more information on when this task should be performed.)

- ❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements, and strategy for the architecture work. The security requirements will be identified in detail in this document.

- ❑ **Conceptual Architecture (TA.070)**

The Conceptual Architecture provides a high-level view of the favored future application and technical architecture and will indicate the approximate deployment of applications in this architecture model. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.070 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review security requirements derived from other tasks and deliverables.	
2.	Review security-related requirements and decisions uncovered in the mapping process.	
3.	Map security requirements onto standard application functionality and identify requirements not met.	Application Security Mapping
4.	Create detailed database security design.	Database Security Design

No.	Task Step	Deliverable Component
5.	Design high-level database security configuration and implementation.	Database Security Configuration
6.	Create detailed application security design.	Application Security Design
7.	Design high-level application security configuration and implementation.	Application Security Configuration
8.	Review design approach with application mapping and module design and build teams.	Acceptance Certificate (PJM.CR.080)

Table 4-18 Task Steps for Define Application Security Architecture

Approach and Techniques

This task focuses on the design of the high-level mechanisms that will meet the application security requirements. The Architecture Requirements and Strategy (TA.010) should contain discussions of any high-level, corporate-wide, or site security requirements. The Information Model (BR.060) describes the more specific aspects of the information security model for the business. The Process and Mapping Summary (RD.030) contains any process or mapping decisions that affect this area. Synthesizing these sources of information should give a complete security requirements analysis, which can then be used for the design work. The design work itself requires a knowledge of the different security mechanisms that are available within the applications and database for implementing security, with the possibility of having to design feasible custom solutions for specific requirements not met by the base applications and which affect the fundamental architecture or functioning of the applications.

☞ **Attention:** The person that assumes the role of system architect and designs the security architecture should have a good knowledge of the security functionality in the applications being implemented, including capabilities and limitations. The skills and expertise needed are similar to those required for Define Application Deployment Plan (TA.130).

User Function or Role-Based Security

This type of security is oriented towards the functions, transactions, forms, or reports to which a user can gain access. This approach to security secures data by preventing users from gaining access to any data in a particular table or from performing a particular business function. The security analysis work done in Business Requirements Mapping (BR) typically takes this approach and is concerned with identifying the different types of roles for the new business system and the menus, forms, reports, and batch programs to which each role or user can gain access. It usually has less impact on the basic architecture of the applications than organization-based security. For more information, see Design Security Profiles (BR.110).

Business Organization-Based Security

This type of security analysis is oriented towards securing the data for a particular business organization or user group. Unlike the function- or role-based security, this area typically deals with securing data for organizations as a whole and not just for individuals or roles within an organization. Often, business organizations may be creating rows in the same tables in a database, but do not wish other organizations to be able to view or transact their data in the same table. The division between this approach to security and the function/role-based approach is not hard and fast, and it may be possible to satisfy the business organization-based security requirements by judicious use of function/role-based security and the standard application features that support it.

The design work in this task may include both function/role-based security and business organization-based security. Generally the details of the function/role-based security are not architecturally significant and are best handled in Business Requirements Mapping (BR). The business organization-based security can have architectural significance and should be handled here. If requirements are met using function/role-based security, mention them in the application security architecture design for a complete view of the solution, but refer to the mapping process for details.

Business Organization-Based Security Design

The design of solutions for business organization-based security is sometimes more difficult than the function / role-based security, because it affects the storage and partitioning of data in the database. The application functional architecture and key setup parameters may have an effect on the security and partitioning of data also, so this task links closely with the task Define Application Architecture (TA.040). The approach you should take to satisfy a particular organizational requirement is detailed below. Whether one solution is more complex than another will depend on the exact circumstances for the implementation project and the business itself:

- Use standard application features that provide organization-based security.
- Use standard application function / role-based security to provide *the effect of* business organization-based security.
- Use procedures and policies to provide the security.
- Use limited custom extensions to application modules.
- Use general database-level custom extensions.
- Use separate application installations.

Many security requirements may be met using standard application features. Before exploring more complex options, be sure to identify the security requirements that can be met by standard application security features for both function / role-based security and organization-based security. The cost and extra risk of custom security mechanisms may force a rethink of whether a particular security requirement is really needed.

Attention: In Oracle Applications you can use various features to meet function / role-based security requirements, such as custom menus, responsibilities, and profile options, or report security sets. For more information about the security options in Oracle Applications, consult the system administrator manuals.

Mechanisms for handling the business organization-based security requirements are most likely be related to application-specific setup parameters, such as *GL Sets of Books, Subledger Operating Units, Inventory, or HR Organizations*. If you can map the secure business organizations to these types of setup parameters, you may be able to use the built-in security mechanisms.

Reference: *Oracle Applications System Administrator Reference Manual* (Release Specific).

Reference: *Oracle Applications System Administrator Guide* (Release Specific).

Use of Policies and Procedures

In many cases, business procedures and policies are an effective and much simpler approach to securing data than designing system-level controls, especially when the required controls are not readily available through functionality built into the applications suite you are implementing. System auditing also falls into this category of approach. Determine whether security across organizations must be enforced by system controls, before embarking on design activities.

Custom Security Extensions Design

This section assumes that you are implementing Oracle Applications, although the general principles are appropriate for other types of applications also.

The most general approach to securing data across organizations is to use database-level views. The security rules are defined in the database, and the security applies for online or batch access, regardless of the application tool used to access the data. This approach is used in the Oracle Applications Multi-Organization application architecture.

Custom Database Extensions for Security

If system controls are required, one technique for partitioning data is to use a combination of custom ORACLE schemas (usernames), views, responsibilities, and descriptive flexfield columns to partition different rows in a table. The technique entails creating a separate ORACLE username for each organization. Reserve a mandatory descriptive flexfield column on the table that maintains the secure data entity, and create a separate descriptive flexfield value for each organization. Create views in each organization's ORACLE schema that filter rows not accessible for that organization. Finally, create a responsibility for each organization that points to the appropriate ORACLE username. If the views do not include table joins, forms and batch programs will be able to insert and update table data through the views. There are several limitations and restrictions to this approach:

- The custom database objects and schemas may need special handling for patches and upgrades.
- Some system and parameter settings cannot be partitioned across organizations.
- As the number of ORACLE usernames and views increases, system administrators have more components to monitor.
- If the number of data entities that require security is large, there may be many potential permutations of access requirements for each organization, depending on the function being performed. Make sure you limit the actual number of secure ORACLE usernames to a reasonable number.
- Referential integrity issues may exist between secured and non-secured entities with foreign keys that require special handling. For business objects with multiple physical tables (for example, order headers and order lines), you should carefully analyze business processing requirements to determine under what circumstances secure data may be accessed, and make sure that any conflicts between secure and non-secure entities are properly handled.

Generally, creating database extensions incurs more risk, and this is particularly true when the custom extensions create a number of custom-secured tables in the database. Careful design is necessary to mitigate this risk, followed by extensive testing. Work closely with the business and technical analysts to draw up comprehensive test plans. Also feed this design work back into Business Requirements Mapping (BR).

Separate Application Installations

The most complete means for securing the data for a business organization is simply to create a separate application installation for each secure organization. This mechanism provides absolute security for the data of each organization and usually does not permit any cross-installation viewing, consolidation, and transacting of data. The disadvantage of this approach is that if there are cross-installation communication requirements, it may be necessary to build custom extensions or interfaces to support the transactions or data viewing required. The following are examples of cross-installation functions:

- synchronized master data, such as parts and customer master (data registries)
- ability to view consolidated and summarized financial data across the organizations
- transacting from one business organization to another, such as intercompany financial or buy-sell relationships

Carefully consider the tradeoffs before proposing this solution, because security requirements for an organization are rarely absolute and some cross-organization sharing of data is often needed.

Linkage to Other Tasks

The Application Security Architecture is an input to the following tasks:

- BR.110 - Design Security Profiles
- TA.090 - Define Application and Database Server Architecture

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Application Specialist	60
Business Analyst	30
Technical Analyst	10

Table 4-19 Role Contribution for Define Application Security Architecture

Deliverable Guidelines

Use the Application Security Architecture as an input to subsequent tasks that describes the database and application-level configuration required to implement your security requirements.

This deliverable should address the following:

- mapping of security requirements to standard features, including gap definitions and solutions
- the application security configuration, describing approaches that will be used to support the security requirements
- the database security configuration and use of standard application database security features or modifications to the standard database organization to support the security requirements

Deliverable Components

The Application Security Architecture consists of the following components:

- Introduction
- Application and Database Security Architecture

- Application Security Mapping
- Application Security Design
- Application Security Configuration
- Database Security Design
- Database Security Configuration

Application and Database Security Architecture

This component describes the approaches used for architecting data-level security for the applications are being implemented.

Application Security Mapping

This component discusses the mapping of the security requirements to the standard functionality and features of the applications. When the Business Requirements Mapping team has undertaken this already, this component summarizes their findings and identify gaps that may require a more complex architecture solution.

Application Security Design

This component discusses the security requirements are met by the design of custom extensions to the applications.

Application Security Configuration

This component describes the security requirements that are met by standard functions and setups of the applications and the effect that the requirements have on the application mapping and setup.

Database Security Design

This component discusses the security requirements that are met by design of custom extensions to the database schema.

Database Security Configuration

This component discusses the security requirements that are met by the standard application database partitioning and logical structure, and relates this to the application security configuration.

Tools

Deliverable Template

Use the Application Security Architecture template to create the deliverable for this task.

TA.090 - Define Application and Database Server Architecture (Optional)

In this task, you design the new logical and physical architectures for the database tier and the application server tier. This task also defines the configuration for both the database and application tiers of the architecture.

This task focuses on the future production environment and not interim environments to support the project activities. This includes on going production supporting testing and learning environments.

Δ If your project is of *medium or high complexity*, you should perform this task. (The *Low Complexity Projects* section in the approach and techniques section below describes how you may want to address this activity on a low complexity project.)

Deliverable

The deliverable for this task is the **Application and Database Server Architecture**. It is the blueprint for the logical and physical architecture of the application and database servers. These servers participate in the three tier architecture. This deliverable also details the configuration of these servers.

Prerequisites

You need the following input for this task:

- ❑ **Business Volumes and Metrics (RD.040)**

The Business Volumes and Metrics provide the business volumes information. Use it to identify specific volumes you may need for estimating the I/O throughout and database performance.

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work.

❑ **Application Architecture (TA.040)**

The Application Architecture provides information about the key critical setup parameters and system functions that affect the precise architecture of the application installations. If Define Applications Architecture was not performed, this deliverable will not exist. (See the TA.040 task description for more information on when this task should be performed.)

❑ **System Availability Strategy (TA.050)**

The System Availability Strategy provides information about the strategies used for providing the level of system availability the business needs. If there are special database architectures and systems needed to provide the desired level of system availability, this deliverable will contain information about them.

❑ **Conceptual Architecture (TA.070)**

The Conceptual Architecture provides a high-level view of the favored future application and technical architecture, and indicates any architecture subsystems that might be required for the new architecture. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.070 for more information on when this task should be performed.)

❑ **Application Security Architecture (TA.080)**

The Application Security Architecture provides details of the approach taken to meet the security requirements for the architecture. This deliverable includes the definition of key setup parameters necessary for application security and any custom extensions needed to meet extra security requirements beyond what is provided by the base applications.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review the design standards for creation of custom code and database objects.	
2.	Review the standard installation architecture for the package applications you are implementing. This would include the installation and architecture of all of the server categories.	
3.	Review the existing Current Technical Architecture Baseline (TA.020) (if available) to establish an understand of the available platforms for repurposing.	
4.	Review the Architecture Requirements and Strategy (TA.010).	
5.	Review and identify the scope for the Applications and Database Server Architecture	Introduction
6.	Revise the diagrams provided for your key environments to reflect your high-level plan.	Database and Applications Server Architecture Summary

No.	Task Step	Deliverable Component
7.	Define the server architecture for your principal (corporate) data center for each of the following: *Database Tier, and Application Tier.*	Data Center / Hosting Facility - Corporate
8.	Define the server architecture for all of your other data centers by repeating this subsection for each of the following: *Database Tier, and Application Tier.*	Data Center / Hosting Facility - Others
9.	Define the detailed configuration for the desktop client tier at each deployment site.	Desktop Client Tier
10.	Update the diagrams in the introduction component to reflect your completed design	Introduction
11.	Distribute this document for comments and update as required.	

Table 4-20 Task Steps for Define Application and Database Server Architecture

Approach and Techniques

Low Complexity Projects

If your project is of a *low complexity*, you may be able to consolidate a subset of this task into your Architecture Requirements and Strategy (TA.010). In the case of a single site implementation with a limited number of users, you may be able to run all of the servers for both the database and application tier on the same platform. As this would reduce the overall complexity of the study, you may be able to state the architecture and configuration for these servers in elsewhere.

In some cases, a site may acquire a single or multiple pre-configured servers. These servers would have all of the required servers and database files pre-staged, ready to run. In such cases, the statement of the architecture is assumed to be provided by the vendor or the pre-configured environment.

Resources

The application and technical system architects performing this task should have a good knowledge of the application and technical architecture of the applications and the database being implemented. The knowledge needed includes application technical configuration, logical database architecture, supported installation structures, and high-level application functional architecture.

Task Complexity and Project Scope

The complexity of this task varies with the scope of the implementation project:

- single *vanilla* installation, no custom extensions
- single installation, custom extensions including new code or database objects
- multiple installations including new code or database objects

Single *Vanilla* Installation, No Custom Extensions

This installation will typically apply to smaller, less complex, fast track implementations where you are not customizing or extending the applications. In this case, generally all you need to do for this task is to document the vanilla installation structure of the file code trees on client and server platforms, as well as the database logical architecture, for future reference by the technical architect for physical database design and capacity planning and for reference by the information systems organization and staff.

Installations with Custom Extensions

This is the more usual situation in package application implementations, where new extension components have been designed and built to be part of the overall solution. The application file structure contains custom modules and the database will have custom objects, both of which necessitate extra design work to isolate the custom file modules or database objects from the base installed database files and objects. The Customization Strategy may already have the strategy outlined for this; if not, you should include the design of any extra schemas you might need. Other factors that may contribute to the complexity of the logical design are:

- multiple distributed applications databases
- specific application's functional architecture (for example, implementation of Oracle Applications multi-organization)
- custom reporting systems (for example, end-user layers (EUL) that reschematize the OLTP database and present the data to users in an intuitive business object format for ad hoc reporting)
- implementation of multilingual requirements

Application Installations

Database Tier

The database tier holds all data and data-intensive programs, and processes all SQL requests for data. The database tier includes the Oracle Database Server, the administration server, and the concurrent processing server. The concurrent processing server executes traditional batch processing tasks such as transactional interfaces and reports.

By definition, platforms in this tier do not communicate directly with applications users, but rather with platforms on the application tier that mediate these communications, or with other servers on the database tier.

This tier includes the following types of servers:

Database Servers

The database server contains the transactional and schema data associated with the Oracle Applications.

Administrative Servers

The administration server is the platform from which you maintain the data in your Oracle Applications database.

Concurrent Processing Servers

The concurrent processing server runs background or batch jobs like reports.

Application Tier

The application servers form the middle tier between the desktop clients and database servers. They provide load balancing, business logic, and other functionality.

This tier includes the following types of servers:

Forms Servers

The forms server is a specific type of application server that hosts the Oracle Forms Server engine. The Oracle Forms Server is an Oracle Developer component that mediates between the desktop client and the Oracle Database Server, displaying client screens and causing changes in the database records based on user actions.

Web Servers

The web server is another type of application server, which runs an HTTP listener. The HTTP listener (also called a web listener) is a component of an HTTP server, such as Microsoft Internet Information Server, or Netscape Enterprise Server. This listener accepts incoming HTTP requests (or URLs) from desktop clients, via the web browser or appletviewer. These requests are either immediately processed — for example, by returning an HTML document — or are passed on to the Forms server.

Desktop Client Tier

The desktop client runs a Java applet using a Java-enabled web browser or appletviewer. The applet sends user requests to the forms server and handles such responses as screen updates, pop-up lists, graphical widgets, and cursor movement. It can display any Oracle Applications screen and supports field-level validation, multiple coordinated windows, and data entry aids like lists of values. A web browser or appletviewer manages the downloading and storage of the Forms client applet on each user’s desktop. They also supply the Java Virtual Machine (JVM) that runs the Forms client applet.

Design Considerations

Although the design of a database architecture is a complex undertaking, fortunately there are methods and standards that experienced Oracle database practitioners have developed and documented. Although these methods have not entirely automated the design process, their use does help mitigate many of the risks.

Optimal Flexible Architecture (OFA)

The most fundamental Oracle database design method is the Optimal Flexible Architecture method, which has become a de facto standard for the physical design of Oracle databases. By adopting OFA principles, you help make sure that the database not only performs well, but that it can be managed by a database administrator without reduced risk of interruption of service or data corruption.

Reference: Millsap, Cary V. *An Optimal Flexible Architecture for a Growing Oracle Database*. White Paper, Revised 1993.

Reference: Millsap, Cary V. *"Optimal Flexible Architecture," Oracle Magazine*. Vol. IX, Nos. 5 and 6, 1995.

Fault Tolerant and Load Balancing Design

Oracle provides automatic load balancing among multiple forms servers. In a load-balancing configuration, a single coordinator called the Metrics Server is on one application server. Metrics Clients located on the other application servers periodically send load information to the Metrics Server so it can determine which has the lightest load. When a client issues a request to download the Forms client applet, the Metrics Server provides the name of the least-loaded host for the applet to connect to. This process is made possible through mediation by the web server and Oracle® Web Application Server.

Automatic failover capabilities are inherent in this load-balancing system. If an application server becomes unavailable for any reason, the Metrics Server ceases to route requests to the server until it comes back online. While the application server is offline, requests are routed to one of the other application servers.

Reference: *Setting up Load Balancing, Oracle® Developer: Deploying Applications on the Web, Oracle Applications Installation* (Release Specific).

Very Large Database (VLDB) Design

In highly centralized processing environments, the potential size of the databases in an architecture increases, and as they gradually transition into the Very Large category, other considerations complicate the more straightforward analysis that is applicable in environments with smaller data volumes.

There is no unambiguous definition of the minimum size for a database before it may be considered very large. The relative size depends on the nature and rate of the transactions executed, the relative mix of online and batch transactions, whether there are significant volumes of reporting, availability requirements, and the age of business data retained in the database. As a very rough rule, an applications database that is 10's of GBs in size may be considered small-to-normal, whereas a database that is 100 GB or more is moving towards being considered very large. However, this can easily vary with the industry sector of the business and the practices and operational model. The *method* used for the design of very large databases does not differ fundamentally from that of more conventional-size databases, but the supporting physical hardware needed will be more complex, and the maintenance techniques and procedures that are feasible for smaller data volumes become untenable for very large databases.

Reference: Millsap, Cary V. *Configuring Oracle Server for VLDB*. OAUG Proceedings, Spring 96.

Impact of Multilingual Requirements

In larger multinational implementations, there may be specific requirements relating to the appearance of the text on application forms and reports in multiple languages and for the languages in which the system data are entered and displayed. These requirements will affect the way the applications are configured at the database level and may have an impact on the number of separate installations (database instances) required.

National Language Support

Oracle Corporation supplies translated versions of the application forms and reports in many foreign languages as standard, but the functionality in the products may not support the entry and display of the same data element in multiple languages for the release you are implementing.

Suggestion: For the latter requirement, Oracle Consulting has partnered with Oracle Applications development to offer a standard consulting solution. This solution is called **Multi-Lingual Support (MLS)**. You can obtain more information on the MLS solution from your Oracle project or account manager.

Multiple Language Support

There may be issues with the specific support for data entry in applications in different languages. Lack of support in base application products for the multiple language entry can create extra complexity in the logical architecture. Note that this is not always the case.

When applicable, the limitations here result from the fact that currently there is no universally supported codeset covers all languages. The codeset is the hardware encoding scheme that relates a set of characters in a language to their binary representation. Some codesets such as US7ASCII are fairly common across hardware vendors and operating systems. Other codesets have been devised by hardware vendors individually and may not be standard. When an Oracle database is created on a server, a codeset has to be selected for the data in the database. Therefore, if a codeset does not exist to support all required languages, you may not be able to run the translated application code properly for all the languages you require. This also applies to the application code. The current rules for code and databases are as follows:

- Mixing European languages on a single platform / Oracle database is possible because a codeset exists that covers European languages.
- Mixing English and one other language on a single platform / Oracle database should also work (all codesets contain English as a subset).
- Generally you cannot mix an Asian multibyte language, English, and another language on the same platform.
- Other language permutations may not work.

These are only general rules to show the extent of the problem and are not guaranteed to give the affirmative or negative answer for the particular hardware platform on which you want to install your application database.

☞ **Attention:** Always check with Oracle for the availability of a codeset that supports all languages you want to support in your applications database in the hardware platform you are running. You can do this through your Oracle project manager or through Oracle Support.

Advanced Database Features and Options

If you are implementing advanced database features or options such as:

- Oracle Parallel Query
- Oracle Parallel Server
- Multi-Threaded Server
- Distributed Data Option
- Internet Commerce Server

You need to modify the architecture design to take into account the special requirements and behavior of these features. You should review any special Oracle documentation accompanying these features or options and design accordingly.

The Cost, Performance, and Availability Equation

When designing your database architecture, you need to strike a balance among cost, performance, and availability. If having an available system is the most important consideration to your business, you will design your database and I/O subsystem with that objective overriding any other. In so doing, you incur greater hardware costs and possibly sacrifice some performance to give you the level of confidence in your system and the *mean time to failure* (MTTF) or *mean time to recovery* (MTTR) you require.

Generally, if the overriding consideration is availability, you look for a disk mirror I/O subsystem solution. If it is performance, you look for a disk striping solution, and if it is cost, then possibly a higher RAID level solution. In practice the equation is never totally dominated by one factor, so be flexible in your approach and compromise if necessary. Oracle Applications are usually implemented into business or mission critical environments, and larger implementations often have availability and performance as their main concerns. This is even more likely at present, given the trend toward decreasing real cost of disk capacity.

Data Storage and I/O Throughput Estimation

When sizing the physical data storage requirements, try to estimate the initial and maximum blocks needed for key tables and indexes and the blocks needed for temporary segments. An important parameter in the calculation of storage requirements and I/O throughout is the size of a row in the table or an index entry. You should be able to derive this information from the overall disk capacity planning work performed as part of the capacity planning effort. For more information, see Define System Capacity Plan (TA.110).

Tablespace Partitioning

Partitioning Segments

The OFA standards define that database segments fall into one of several types:

- system
- temporary
- data
- index
- rollback
- tools
- miscellaneous user

Categorize your segments according to the constraints above and then partition them across different tablespaces to facilitate database administration, optimize system availability, and efficiently use the disk space available to the database.

Constraints Affecting Tablespace Partitioning

A number of constraints affect how you partition your segments into tablespaces:

- I/O performance
- resilience to system outage
- space management
- quota management

Keep these constraints in mind as you design your tablespace structure and map to physical data files.

Mapping Tablespaces to Physical Data Files

Having partitioned the segments into tablespaces, you then need to map the tablespaces to operating system files. You will need to factor in key database architecture strategies that were identified to satisfy the competing system availability, the performance requirements, and the cost requirements, and architect the different categories of tablespaces accordingly. Examples of these strategies include:

- use of RAID technology
- use of disk mirroring
- striping

For more information, see Define System Availability Strategy (TA.050).

Linkage to Other Tasks

The Application and Database Server Architecture is an input to the following tasks:

- TA.110 - Define System Capacity Plan
- TA.120 - Define Platform and Network Architecture
- TA.130 - Define Application Deployment Plan
- TA.140 - Assess Performance Risks
- TA.150 - Define System Management Procedures
- PM.020 - Design Production Support Infrastructure
- PM.030 - Develop Transition and Contingency Plan
- PM.040 - Prepare Production Environment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
Database Designer	40
System Architect	30
Technical Analyst	20
Database Administrator	10
Client Staff Member	*

Table 4-21 Role Contribution for Define Application and Database Server Architecture

* Participation level not estimated.

Deliverable Guidelines

Use the Application and Database Server Architecture deliverable as an input in subsequent tasks.

Detail the installations of applications and database servers in all affected data centers, and the logical structure of the servers you will install. Remember that this document is important for all the technical aspects of the project, so accuracy and appropriate level of detail are important.

Complete Architecture Blueprint Document

You can incorporate all your applications and database servers (including non-Oracle ones) architectures, so that the deliverable becomes the detailed blueprint and information source for all applications that you have or will install in your data centers for the new architecture.

Consider the following types of applications:

- basic application suite being implemented (for example, Oracle Applications)
- custom developed applications built using Oracle tools and database
- custom developed architecture subsystems that are part of your new architecture
- custom developed applications built using another vendor's tools and database
- legacy applications that you will retain in the new architecture
- packaged applications purchased from other vendors

Deliverable Components

The Application and Database Server Architecture consists of the following components:

- Introduction
- Database and Applications Server Architecture Summary
- Data Center / Hosting Facility - Corporate (Repeating)
- Desktop Client Tier

Introduction

This component describes the Applications and Database Server Architecture and the purpose for it. It also describes the downstream utility of this deliverable for other project work.

Database and Applications Server Architecture Summary

This component describes the Applications and Database Server Architecture at a high level. It includes a pictorial representation of each environment.

Data Center/Hosting Facility - Corporate

This component is repeated for each Data Center / Hosting Facility in the architecture. It describes the Applications and Database Server Architecture in detail. This detail is broken down by:

- **Database Tier** — Database Servers, Administrative Servers, Concurrent Processing Servers
- **Applications Tier** — Forms Servers, Web Servers

It also describes the planned configuration for each of the servers in all of the sever level tiers of the architecture broken down by Database Tier and Applications Tier.

Desktop Client Tier

This component describes the detailed configuration for the desktop client tier at each of the planned deployment sites. This tier includes: Web Browsers, Web Appliances, Applet Viewers, and other applications.

Audience, Distribution, and Usage

Distribute the Applications and Database Server Architecture to the following:

- IS manager
- architecture team members
- system and database administrators
- performance testing team
- technical analysts and database designers with an interest in the physical database architecture

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Is the strategy used for the database architecture clearly defined?
- Was a method used for the design?
- Are the key database parameters and their values documented?
- Is there evidence of the consideration of cost, system availability requirements, performance, and database maintenance?

Complete Architecture Blueprint Document

You can incorporate all your applications and databases (including non-Oracle ones) into the Application and Database Server Architecture, so that the deliverable becomes the detailed blueprint and information source for all applications that you have or will install in your data centers for the new architecture.

Consider the following types of applications:

- basic application suite being implemented (for example, Oracle Applications)
- custom developed applications built using Oracle tools and database
- custom developed architecture subsystems that are part of your new architecture
- custom developed applications built using another vendor’s tools and database
- legacy applications that you will retain in the new architecture
- packaged applications purchased from other vendors

Tools

Deliverable Template

Use the Application and Database Server Architecture template to create the deliverable for this task.

Sources of Oracle Applications Architecture Information

The server installation, administration, architecture, and reference manuals that accompany the software also contain much useful information on the subject of the physical design and layout of the required servers. They describe the basic concepts and detailed technical architecture of Oracle servers.

Reference: *ORACLE Server for Unix, Linix, NT Administrator's Reference Guide*. (Operating System and Release Specific).

Suggestion: The International Oracle User Group conference every year has many papers devoted to various aspects of Oracle database management and configuration. You can order a copy of the proceedings of each conference from the International Oracle User Group.

Reference: *Oracle Applications Installation Manual for Unix, Linix, NT* (or other operating system). (Operating System and Release Specific).

Reference: Kodjou, Paule. *Architecting Your Applications Environment for the Development and Preservation of Customizations*. OAUG Proceedings, Spring 96.

TA.100 - Define and Propose Architecture Subsystems (Optional)

In this task, you identify requirements for components or systems to support the architecture that were initially not identified within the project scope. Examples of architecture subsystems include enterprise-wide data distribution (interface) systems, (global) data registries, data warehouses, business intelligence, and operational reporting systems. The subsystems may either be purchased from Oracle or from another vendor, or you may design and built them in-house as custom development projects, but even systems that are purchased often require at least some degree of custom integration work. Custom development projects use the requirements from this task as a starting point in their Definition, Analysis and Design phases.

Δ If your project is of *medium or high complexity* or includes *complex architecture changes,* you should perform this task. (Gaps in the fulfillment of requirements determine the actual need to perform this task.)

Deliverable

The deliverable for this task is the **Architecture Subsystems Proposal**. It is a definition and proposal for a set of architecture subsystems that need to be incorporated into the architecture.

Prerequisites

You need the following input for this task:

❑ **High-Level Gap Analysis (BR.010)**

The High-Level Gap Analysis provides information about gaps that may not be accommodated by the acquired applications or within the initial scope of the project.

❑ **Mapped Business Requirements (BR.030)**

The Mapped Business Requirements provides information about gaps that may not be accommodated by the acquired applications or within the initial scope of the project.

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements, and strategy for the architecture work.

❑ **Conceptual Architecture (TA.070)**

The Conceptual Architecture provides a high-level view of the favored future application and technical architecture, and indicates any architecture subsystems that might be required for the new architecture. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the task description for TA.070 for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Identify components of the Conceptual Architecture that are architecture subsystems. Create one document for each.	
2.	Gather any further requirements or details you need to complete the analysis of the subsystem.	
3.	Complete the description and specification of each subsystem.	Executive Summary
4.	Identify possible approaches for developing or acquiring the subsystem.	Executive Summary

No.	Task Step	Deliverable Component
5.	Identify both the hard and soft benefits and the approach for implementation of the subsystem.	Executive Summary
6.	Identify the assumptions taken into account during the development of the proposal, as well as the assessed risks.	Executive Summary
7.	Describe in detail the subsystem	Background
8.	Describe the integration between the planned subsystem of the rest of the architecture.	Background
9.	Review the completed deliverable with the project manager and secure acceptance.	Acceptance Certificate (PJM.CR.080)
10.	Review the proposal with the steering committee and the project sponsor. Seek approvals.	

Table 4-22 Task Steps for Define and Propose Architecture Subsystems

Approach and Techniques

Definition of an Architecture Subsystem

Architecture subsystems are architecture components that are standalone subsystems within their own right and provide enterprise-level services to the information systems architecture. These types of architecture components generally affect multiple different applications or installations. Examples of such subsystems might include:

- data distribution system (interface infrastructure) that links and distributes data between multiple application databases
- global data registry subsystems that distribute key reference business objects across an enterprise (and would need a data distribution subsystem)
- operational data repository that is synchronized to provide real-time information about the state of business (and would need a data distribution subsystem)
- a critical third-party application package that needs to integrate with multiple transaction systems at the enterprise level
- data warehouse receiving input from multiple transaction systems
- electronic commerce system that processes multiple types of inbound and outbound business documents and transactions
- intranet interface to application systems for use by business users across the enterprise

Typically, these types of subsystems are only important in larger scale projects, but the concept of a non-localized, enterprise-level component to an architecture, affecting multiple applications, interfaces, or databases is entirely general.

The example of the synchronized operational data repository above illustrates the lack of a hard and fast definition of architecture subsystems. In some projects, an interface to a critical third-party application product may be specified and managed within a general interfaces development team. In reality, some interfaces are more complex and stringent in their requirements than others, and the third-party application which they link may be tightly integrated into many of the functions in the core application suite. Development of infrastructure to support such interfaces is often better managed in a separate development subproject, where there is visibility to the impact on the overall systems architecture.

Because of the possible impact of these subsystems on the overall systems architecture and on multiple individual technical groups working in the project, and because of the risk associated with such complex custom development, it is often prudent to separate the specification, design, and build of these systems into separate subprojects linked to the core application and technical architecture process. Therefore a subsystem may be synonymous with a development subproject, but not necessarily.

Gap Identification

The requirements for architecture subsystems are normally identified as gaps during the execution of the Map Business Requirements (BR.030), Define Reporting and Information Access Strategy (TA.060) or Analyze High-Level Gaps (BR.010) tasks. These requirements are often not identified during the application acquisition process or the early strategy and definition activities within the project.

Buy or Build Determination

The applications or infrastructure components that are classified as architecture subsystems may be sold by package application or system vendors. For example, it is possible to purchase a predefined and built application data warehouse from Oracle that will integrate with Oracle Applications transaction systems. It is also possible to purchase messaging systems that can provide the data transport infrastructure to intelligently link applications together, thereby removing some of the development overhead in creating complex interfacing mechanisms to transfer data between multiple databases. Once the need for an architecture subsystem has been established, the information systems department of the business is then faced with the inevitable buy-or-build decision.

Even if the subsystem components are purchased as prebuilt packaged applications, there may still be significant integration work needed to integrate them.

Global Data Registries

A data registry is a set of tables that provide a common definition for a business object that is used by multiple application installations (for example, the customer master list). When a data registry is the source for all information in the global enterprise, the data registry is often termed a global data registry. Depending on the final application architecture and configuration, this may be automatically handled by using a single database and application installation for all organizations. However, if the final configuration requires multiple installations of the applications on one or more databases, or if several applications from different vendors require access to the same reference data, it becomes necessary to design a global registry mechanism for that business object. The information modeling tasks in the Business Requirements Mapping (BR) process help identify requirements for data sharing across organizations — see Create Information Model (BR.060).

Even if you only have a single installation of the applications, there may still be a need for a data registry.

☞ **Attention:** The older, non-multi-organization application architecture for release 10 of Oracle Applications required multiple installations of the subledger applications such as AP and AR to provide multiple sets of books capability. The multiple installations could not share tables, however, so an extension to the applications was needed to share reference data such as vendors, customers, and chart of accounts.

Warning: Make sure that you are familiar with the precise functionality and capability of the application release you are implementing, before assuming the existence or non-existence of a critical piece of functionality.

Reference: *Oracle Applications Installation Manual for Unix, Linix, NT or other operating system* (Operating System and Release Specific).

A data registry can be implemented as a set of database synonyms to one physical table. However, performance requirements may impose restrictions upon cross-instance transactions, especially where the individual instances of an application are physically separated in different databases, linked by a wide area network. This implies that some form of data copying mechanism or replication should be used so that applications can access a local copy of the global data registry. In addition, large companies frequently have many custom and third-party applications that also need access to the global data registry. In this case, it may be more appropriate to specify a custom data model for the business object in the data registry, so that it can record the values of the superset of attributes needed for all Oracle and non-Oracle Application data models.

Dynamic Operational Data Repositories

A dynamic operational data repository is a dynamically changing repository of transaction data spanning multiple business functions, installations and/or sites. An example of an enterprise-level dynamic operational data repository is a global Order Entry ATP (Available-to-Promise) engine, which determines the available shipping date of an item by netting supply and demand at multiple inventory sites. In its simplest form, the ATP engine would provide a real-time result to a query from orders placed in a remote order entry database, but the supply and demand information would still need to be updated constantly with new data as the amounts of inventory vary over time. A subsystem of this type is much more complex than an operational reporting data warehouse that is refreshed asynchronously, and maybe only once a day.

Distributed Data Interfaces

A distributed data interface transports data between Oracle systems or between an Oracle system and a third-party system. Any interface between two separate applications with their own data stores constitutes a distributed data interface. In order to interface applications together you always need two basic components:

- data distribution system (the infrastructure that moves data around an enterprise)
- Application Programming Interfaces (APIs) to get data into or out of application databases

When you create interfaces between applications in a piecemeal fashion, the source and destination of data is typically hard-coded into the interface code, and there is no shared enterprise data distribution system. Each interface has its own data transport mechanism. In more complex situations, the data distribution system may need to link multiple financial or manufacturing sites worldwide and route business objects or transactions between the sites. This more elegant and flexible latter system constitutes an architecture subsystem that can be developed to support enterprise application interfaces.

Complexity of Distributed Data Interface Development

The degree of risk in the development of interfaces varies depending on the exact specifications. Interfaces that are higher risk to design and develop typically involve linking applications that need to be tightly integrated, and that have bi-directional communication. Often both applications are mission-critical, and by implication, so is the interface between them. Generally interfaces are more risk prone and difficult to develop if:

- data needs to be transferred in real time (synchronous data flow from a business event in one of the applications)
- data flow is bi-directional
- the business functions or processes that rely on the interface span the two systems
- the interface synchronizes data across more than two separate databases

Oracle Distributed Data Interfaces

If your implementation spans separate Oracle Applications databases, developing the external interfaces needed to transfer data between the databases is a major task. This is always the case for multi-site installations with multiple data centers that have at least some interaction between their databases. Whenever a business is run with processes that require the manipulation of data in multiple application databases, both data and data ownership potentially need to be transferred between the applications. The physical transfer of data occurs through applications interfaces and is easier to analyze than the transfer of data ownership, but an understanding of both is needed if you are going to develop a robust solution that meets *all* the requirements.

Transaction interfaces that link two distributed Oracle Applications databases are likely to fall to the more complex end of the interface development spectrum for several reasons:

- there are likely bi-directional communication in distributed Oracle Applications interfaces

- the hand-off of business activities is less clear than in heterogeneous application interfaces
- transaction interfaces are more likely to need real-time or near real-time synchronization between the core Oracle Applications data in different sites
- transaction interfaces require control of data integrity across highly normalized and tightly integrated tables

Given that you will probably need to develop a data distribution system, the whole data interfacing and data distribution work should be treated as an architecture subsystem.

Decision Support Systems and Data Warehouses

Decision support systems are specialized applications that summarize and organize transaction data into formats that are oriented towards reporting and analysis. The data model for a decision support database may be very different from the data model for the transaction databases that provide the source data. Often the database is a data warehouse that stores transaction data from multiple different transaction systems over many financial periods. A data warehouse, or decision support system, is a common standalone architecture subsystem that you might custom develop or perform custom integration on.

Enterprise Electronic Commerce

Enterprise electronic commerce involves the transaction of data between the enterprise and the outside world using Electronic Data Interchange (EDI) or the internet. An integrated EDI or web-based system is a good candidate for an architecture subsystem that functions as a separate component of the overall architecture.

Subsystem Proposals

Undertake the task of creating proposals for architecture subsystems after consultation with the project manager. Prior discussion with the project manager enables you to elaborate in detail on the request for the creation of the subsystem, and the reason why a subsystem project may be appropriate for managing the development of a critical and separate piece of the systems architecture infrastructure.

Depending on the project circumstances and the nature of the business environment, you may not need to create a formal proposal or justification for the subsystem; a simple statement of work may be sufficient. However, the mechanism for initiating the subsystem project should communicate the need for a separate project structure that recognizes the critical nature of the architecture component the team will develop, and the need for close collaboration with business analysts. The proposed subsystem may be mission-critical in nature and essential for the overall project to cut over into live production. It may also require especially stringent standards for development and testing procedures.

Once a subsystem proposal is accepted, it should be planned and managed according to the standards and guidelines in Oracle's Project Management Method (PJM).

Linkage to Other Tasks

The Architecture Subsystems Proposal is not an input to any subsequent tasks.

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	80
Project Manager	20
IS Manager	*

Table 4-23 Role Contribution for Define and Propose Architecture Subsystems

* Participation level not estimated.

Deliverable Guidelines

Use the Architecture Subsystems Proposal to identify and propose any required architecture subsystems you may need as part of the information systems architecture. Document each subsystem as thoroughly as possible, but keep the discussion to a high level, describing how the subsystem fits into the overall system architecture, how it satisfies any particular architecture requirements or policies, and what the possible approaches are for developing or acquiring the subsystem, assuming it does not currently exist. If a candidate system does already exist in the application and technical infrastructure, you can discuss how the system may support the identified needs of the new architecture and what extension work might be necessary for the existing system.

Deliverable Components

The Architecture Subsystems Proposal deliverable consists of the following components:

- Executive Summary
- Background

Executive Summary

This component summarizes the proposal for executive audiences. This summary includes a brief introduction, background and problem statement, scope of the proposed subsystem, financial costs and returns, and risks.

Background

This component describes the subsystem requirements, the high-level integration of the subsystem into the architecture (including a diagram), and suggested approaches for custom development or purchase of the subsystem from an application vendor.

Quality Criteria

Use the following criteria to check the quality of this deliverable:

- Are the project scope and objectives clearly identified?
- Are specific critical success factors and risks documented?
- Does this document take into account the impact of dependent projects?
- Are the architecture subsystem requirements clearly defined?
- Is the executive summary clear and concise?

- Are the financial and soft benefits realistic and achievable?
- Can the executive reader just read the executive summary and understand the proposal?

Tools

Deliverable Template

Use the Architecture Subsystems Proposal template to create the deliverable for this task.

☞ **Attention:** The extension of the architecture process to include separate subsystems is subject to change management control, and you should follow the standards in the project management plan in proposing what might be a change to the project scope. The proposal template is only intended to be a sample and does not conform to the rigorous standards that often exist in businesses for the proposal and acceptance of new subsystem project, where there may need to be separate project management, work management, and possibly separate finance arrangements from the main implementation project. If your organization or business has specific, agreed upon practices for this situation, you should follow those practices and use the approved document standards.

TA.110 - Define System Capacity Plan (Core)

In this task, you plan the capacity for the new information systems at production cutover and for some prescribed period of future operations, allowing for business growth and any other known factors that could affect the system capacity requirements. The general term *system capacity* refers to the ability of a system to support numbers of user processing sessions, transaction rates, and data volumes without unacceptable detriment to the performance of the system.

This task focuses on the future production environment and <u>not</u> interim environments to support the project activities. This includes on going production supporting testing and learning environments.

Deliverable

The deliverable for this task is the **System Capacity Plan**. It is the statement of the system capacity requirements for database servers, networks, and client desktop platforms, and application client file servers.

Prerequisites

You need the following input for this task:

- ❑ **Business Volumes and Metrics (RD.040)**

The Business Volumes and Metrics provide the business volumes information on user counts, transaction rates, and volumes of data needed.

- ❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work.

- ❑ **Current Technical Architecture Baseline (TA.020)**

The Current Technical Architecture Baseline provides a consolidated inventory of the applications, databases, interfaces, hardware platforms, and networks in the existing information systems infrastructure and may offer solutions for re-deployment or reuse of existing technical infrastructure. If Identify Current Technical Architecture Baseline was not performed, this deliverable will not exist. (See the task description for TA.020 for more information on when this task should be performed.)

❑ **Conceptual Architecture (TA.070)**

The Conceptual Architecture document provides a high-level view of the favored future application and technical architecture. The conceptual architecture document will contain any initial system capacity estimates performed. These initial estimates can be the basis for a more detailed and accurate capacity plan. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the TA.070 task description for more information on when this task should be performed.)

❑ **Application and Database Server Architecture (TA.090)**

The Application and Database Server Architecture provides the logical and physical definition of various installed application databases, including their schemas and objects. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the TA.090 task description for more information on when this task should be performed.)

❑ **Testing Environments (TE.060)**

The Testing Environments deliverable identifies those testing environments that will remain following production cut-over for production support.

❑ **Project Team Learning Environment (AP.040)**

The Project Team Learning Environment deliverable identifies the learning environment that will remain following production cutover for production support.

Optional

You may need the following input for this task:

- ❑ **Existing System Capacity Strategy or Analysis Documentation**

The information system organization may already have a system capacity strategy in place and should be able to provide copies of documents that discuss the strategy.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review any existing strategy for system capacity planning, taking into account architecture design work to date. Update the introduction as required.	Introduction
2.	Document the capacity planning strategy.	Capacity Planning Strategy
3.	Confirm existing business volumes information and the predictions for future user headcount and business volumes.	
4.	Summarize the business volume requirements.	Business Volume Requirements
5.	Establish minimum lifetime of new technical architecture and hence identify capacity planning horizon.	Capacity Planning Strategy
6.	Research capabilities and requirements for the production server platforms.	Production Environment Servers

No.	Task Step	Deliverable Component
7.	Perform sizing exercise for production servers.	Production Environment Servers
8.	Document recommendations for current or future capacity shortfall.	Production Environment Servers
9.	Research capabilities and requirements for the user learning environment server platforms.	User Learning Environment Servers
10.	Perform sizing exercise for the user learning environment servers.	User Learning Environment Servers
11.	Document recommendations for current or future capacity shortfall.	User Learning Environment Servers
12.	Research capabilities and requirements for the user learning environment server platforms.	Testing Environment Servers
13.	Perform sizing exercise for the user learning environment servers.	Testing Environment Servers
14.	Document recommendations for current or future capacity shortfall.	Testing Environment Servers
15.	Research application requirements for desktop client platforms.	Desktop Client Machines
16.	Document recommendations for desktop client platforms.	Desktop Client Machines

No.	Task Step	Deliverable Component
17.	Research capabilities of network segments.	Network Capacity
18.	Perform sizing exercise for network segments.	Network Capacity
19.	Document recommendations for current or future network capacity shortfall.	Network Capacity
20.	Document assumptions in capacity plan.	Assumptions
21.	Document risks in capacity plan.	Risks
22.	Summarize capacity planning results and recommendations.	Introduction
23.	Review deliverable with project and business management and seek acceptance.	Acceptance Certificate (PJM.CR.080)

Table 4-24 Task Steps for Define System Capacity Plan

Approach and Techniques

The strategy for capacity planning may have been decided when the Architecture Requirements and Strategy (TA.010) was created. If this is the case, then the existing strategy should be reassessed, taking into account subsequent architecture design work, before being executed in this task. If the subsequent design work has revealed issues and results that force a change of capacity planning strategy, then an initial step of agreeing on a new strategy may be necessary. For more information, see Define Architecture Requirements and Strategy (TA.010).

You may have performed an initial system capacity analysis as part of the overall design of the conceptual architecture or during the applications acquisition phase prior to the commencement of the implementation project. These initial capacity studies should be the basis for the work performed in this task, although here it is undertaken in more detail and to completion. Combine all the sizing results, predictions for future growth, benchmark results, and the strategy plan to meet future shortfalls to fully create the System Capacity Plan.

Tight Linkage to Other Tasks

The task of system capacity planning, designing the physical server architecture, and designing the platform and network architecture are tightly linked, and you often work on these tasks as a group within a project. You may need to perform iterations of each task before you arrive at a self-consistent solution to the detailed technical configuration of the system and its expected capacity at production cut-over and for some period into the future. For more information, see Define Application and Database Server Architecture (TA.090) and Define Platform and Network Architecture (TA.120).

Warning: Many performance problems that occur after a system has gone into production can be traced back to poor planning of processing capacity (CPU, memory) in critical servers. At least part of the reason for this are the inherent obstacles to accurate system capacity planning in modern open systems (see below). A detailed and thorough system capacity plan is a critical success factor for the architecture of any implementation, especially one that involves the replacement of business- or mission-critical system applications. When a comprehensive and detailed system capacity plan is performed, it is a major factor in mitigating a major source of risk to projects.

System capacity planning for system implementation requires technical expertise and familiarity with the problems of capacity planning for large, complex, tightly integrated, critical systems. If the implementation does not have the required expertise for planning Oracle systems capacity, contact your Oracle project manager or account representative for assistance.

Definition of a System Capacity Plan

A system capacity plan is a study of the system and network resources needed to support the new information systems at production cut-over and for some prescribed period of future operations, allowing for business growth and any other known factors that could impact the system capacity requirements.

What Is Included in System Capacity Planning?

The following are aspects of the system capacity planning task for the implementation of a new application product or suite:

Database Tier

- database servers
- administration servers
- concurrent processing servers

Application Tier

- forms servers
- web servers

Client Tier

- desktop client machines
  - number of CPUs needed
  - amount of RAM needed
  - amount of disk space needed for installed code, data files, and swap files
  - I/O subsystem capacity needed
- browser or web capabilities
- local printing

Infrastructure

- wide and local area network routing, capacity, and topology
  - latency (round trip time)
  - maximum rated bandwidth
  - actual network throughput
- print servers
- backup services (servers, storage media)

Relationship Between System Capacity and Performance

The relationship between system capacity and performance is an indirect and approximate one. In very simple terms, as you increase the online or batch transactions per second executed against a database server, or increase the number of transactions per second through a particular network segment, the processing performance of the transactions stays approximately constant or slightly decreases incrementally. At a certain level of transactions per second, the performance begins to degrade dramatically because some limiting system resource in CPU cycles, memory, or network bandwidth has been exhausted. It is this "knee" in the performance curve that represents the system capacity for that resource.

It is important to understand the limitations of the capacity studies of CPU and memory processing for servers. The System Capacity Plan is an attempt to understand the limits of the *relative performance* of the server (for example, the performance with a certain load of users and transactions relative to an alternative load). The capacity study of the CPU and memory of a server makes no prediction about the *absolute performance* of a particular transaction or software program in the system. The absolute performance of the server is affected by the intrinsic speed and frequency of the processors, the scalability of the processors against the system bus in a Symmetric Multiprocessor (SMP) platform, specific processor cache sizes, and so on. The absolute performance of a transaction cannot be estimated without a complex study of these variables. In order to predict the absolute performance of the new system, it must be measured empirically in a simulation of the future system, or inferred from analogous implementations.

Relationship Between Network Architecture/Capacity and Performance

Traditionally, capacity planning and infrastructure design have focused on the hardware platform servers that support the servers and executing applications. However, with the nature of the internet computing model, there is a definite need to focus on network design and capacity.

As web-client/server applications such as Oracle Applications present more that just textual materials to the client device, there is a need to architect and size with these other media in mind. For example, the attachment of images, word processing, audio, and video documents illustrate this challenge.

Determining where a bottleneck is or will appear is a considerable challenge in this somewhat distributed environment. Over engineering everything is one approach, however, it is often not cost effective. What may appear to be a performance problem in the eyes of the end-user, may in fact be related to a network latency issue imputed by a single network transacting device (for example, Router) in the path between the end-user client device and the Applications Tier, Forms Server. Or perhaps it is problem with shared traffic on a link between the Forms Server and the Database Server on the Database Tier.

Capacity Plan and Architecture Scope and Complexity

The complexity and detail in the system capacity plan you produce will depend on the exact scope of the implementation and the complexity of the technical architecture.

If you are conducting an enterprise-level architecture process in which you are architecting an entire enterprise of multiple applications installations or subsystems, you may not need to perform a detailed system capacity plan for each installation. The architecture process may only be a high-level feasibility study for the technical configuration aspects of the conceptual architecture. Conversely, if you are designing a detailed architecture for a single data center or system installation, you need to perform a detailed capacity study.

Obstacles to Accurate System Capacity Planning

In contrast to older mainframe environments where the technical configuration was limited to a small number of possibilities, the modern open systems web-client/server environment will generally be a different permutation of application, platform, and network vendors;

versions; and products for each new business system. The widespread disparity of technical configurations and the ever-changing technology stack means that there are not the same detailed metrics for response time and transactions/sec for the new applications infrastructure, as compared with older mainframe environments. Coupling this variability of technology with the different industry-specific business processes and functions for different companies usually means it is impossible to use a purely theoretical analysis for processing capacity *and* have confidence in the results.

General Approaches to System Capacity Planning

You can take various approaches to system capacity planning, each with its own advantages, disadvantages, and level of effort. This section describes a few of the possible different approaches.

Reference: Millsap, Cary V. *Designing Your System to Meet Your Requirements*. OAUG Proceedings, Fall 95.

Reference: Shallahamer, Craig A. *Methods of Planning Capacity*. OAUG Proceedings, Fall 95.

Analogous Implementations

A very sure way of planning your capacity is to find another implementation project that is implementing the same type of business, in an identical technical environment and of roughly the same business volumes and user counts. This is a rare occurrence, so you may be forced to look for other implementations that are analogous in some way, but do not directly match your implementation. If you can find another implementation that does not have precisely the same technical environment or the same business needs, but is implementing on the same database server platform (including operating system version), you may be able to make useful comparisons and get direct empirical capacity planning information. If the operating system version, or the CPU ratings in two platforms from the same vendor are different, this reduces the usefulness of the empirical metrics you might obtain, and you will need to conduct your metrics gathering exercise for your specific configuration.

Rule-of-Thumb Method

The simplest way of planning capacity for platforms is to use *rule-of-thumb* metrics for memory per user and batch process, TPS or TPC-C per user and batch process, network bandwidth and latency minimums. While this technique is better than nothing, it is extremely unsophisticated and can only give order of magnitude estimates of CPU capacity and network requirements with a high degree of risk.

Suggestion: Most hardware platform vendors provided metrics and sizing tools for Oracle Applications. Review the hardware vendor's web site for information on sizing, or contact their representatives. High-level sizing information is provided in the Oracle Applications documentation as a supplement to vendor supplied information.

For memory and network bandwidth estimation, this type of approach is somewhat better, but should still be treated with caution.

Warning: While this technique is better than doing no capacity planning at all, *it is not much better*, and it is an extremely unsophisticated tool. At best, it may give some rough indication about the class of database server you may need to purchase from your hardware platform vendor, but it cannot give accurate predictions of numbers of CPU boards needed, or how much spare online or batch capacity you will have on a platform, before performance starts to suffer.

Statistical Summation Method

This method attempts to discover meaningful linear mathematical relationships between a few measurements of key system resource metrics and individual system or user processes (transaction flows). It is more complex than the estimation model and requires experience and technical expertise to perform.

System Modeling

A more detailed analysis would look at the major transaction flows to be executed in the system architecture and determine the system and network resource usage by transaction flow. Multiplication of these resource usages per flow by the number of sessions executing the flow will give a better system capacity estimate, especially when a CPU queuing model is built into the study. This is also a complex mathematical technique that requires experience and technical expertise to perform.

Capacity Planning by Simulation

This is the ultimate capacity planning technique: you build a simulation of the system architecture to the level of complexity required for a certain degree of confidence in the results and measure the system resource usage directly. It is the technique that is most able to generate accurate, reliable results, but it can be costly and time-consuming to obtain a high degree of accuracy. This technique is discussed in detail in Performance Testing (PT), and capacity planning studies is one reason you might seek to instigate a system performance test.

System Resources Planning

In spite of the complexity and variation in possible technical configurations, some system capacity measures can be reduced to simple metrics that are relatively independent of precise business processes or system component vendors. Using a simple *rule-of-thumb* model, the following system resource measurements can be predicted to some degree of precision:

- memory consumption
- network bandwidth requirements between servers and tiers
- network latency tolerances between servers and tiers
- disk capacity requirements for stored data

Specific Capacity Planning Techniques

Even if a more detailed analysis is done, there is still risk that the analysis is incomplete. For example, the capacity study may have overlooked other significant applications that will execute on the same servers as Oracle Applications or may have overlooked other network traffic that will compete for the same bandwidth as the Oracle components of the system.

Partner with the Hardware Vendor

Hardware vendors have up-to-date metrics for the capacity and performance of their servers. The vendor may also offer technical consulting services with consultants that are familiar with the applications you are attempting to implement. You should discuss your capacity planning requirements with your hardware platform vendor representative, and establish what levels of assistance they can provide for this task.

Capacity Planning Scenarios

In order to relate the capacity planning to real processing situations in your new system, you need to define system scenarios that you will plan capacity against. These system capacity planning scenarios are analogous to the Performance Test Scenarios (PT.020) in Performance Testing (PT). They enable you to relate the results of your capacity planning (or performance testing) in a top-down method to system processing situations that concern the business, either because of extreme system resource load (as might be the case at quarter end, for example), or because of concern about the ability of the system to provide the requisite service level to external customers or internal users.

When mapping the capacity planning scenarios to real point-in-time snapshots of system processing, you need to consider:

- Is there a clear definition of the future period for which the capacity plan is assumed appropriate (the capacity planning horizon); for example, are there projected changes to headcount or business volumes that would render it obsolete?
- Are there plans to extend the number of transactions or end user sessions perhaps as a result of extension of the system through the supply or selling chain? Are there plans to allow customers to transact with the system via in internet in the future?
- Are there fault tolerance requirements in the network, providing alternate connection paths between tiers?
- How many numbers of simultaneous application sessions will each user group will require to perform their jobs?
- Are you planning for peak processing periods; for example, quarter end or heavy nighttime batch processing?
- Are there other applications that are currently processing on the same servers as Oracle Applications or will move there at some point in the future?

For more information about identifying scenarios for performance testing/capacity planning, see Identify Performance Test Scenarios (PT.020).

System Capacity Shortfall

If a System Capacity Plan predicts a shortfall in capacity after a certain period, the technical architecture of the system needs to be upgraded or altered to provide extra capacity. The shortfall usually arises because a business is growing in user headcount or transaction rates and the system resources that were adequate at initial production cutover are insufficient after a period of growth. A thorough capacity plan should address the growth in future operations and foresee a potential need for hardware or network upgrade or re-architecture.

If the shortfall is in disk capacity, it is usually a relatively simple matter to add more disk space to a disk farm. If the disk farm has maximum disk usage, it may be necessary to migrate to an alternative disk configuration. The disk capacity planning should predict this eventuality and suggest a course of action.

If the shortfall is in the capacity of the Forms or Web servers at the application tier, then perhaps the addition of additional servers tied to the metrics server will resolve the issue.

If the shortfall is in network bandwidth on a particular connection path, then adding additional circuits or upgrading existing circuits may resolve the requirement.

If the problem is related to latency, then a segment by segment (point to point) analysis of network traffic and latency may drive the upgrade of a particular router or network bridge device.

Platform Server Hardware or Operating System Upgrade

If the shortfall is in a platform server CPU and/or memory, it may be possible to purchase extra CPU boards or memory for your server to relieve the shortfall. A careful analysis of the capacity requirements versus the maximal capability of the platform server should indicate whether the increase in capacity requirements can be met by server upgrades over the lifetime of the plan.

If the business capacity requirements exceed the maximum capability of a platform server in CPU and/or memory, the capacity need will necessitate some other (potentially more costly) solution. If the hardware vendor has scheduled dates for the release of a new server (with faster CPUs and/or more memory available), or is releasing a new version of the operating system which promises higher SMP scalability, you may be able to continue with the existing server and then migrate to the new server when it becomes available. Depending on the application of the platform server (Database, Forms, and Web Server), you may be able to add additional services to service the demand in parallel.

This demands that you will have sufficient system capacity for production cutover, followed by the ability to migrate your applications to the new processing platform at some point in the future, before the capacity shortfall becomes chronic. This solution introduces a dependency of the business on the hardware platform vendor release schedules and is not often acceptable, unless there is a high degree of confidence that the new hardware platforms will be available when needed.

Use of Concurrent Processing Servers

A concurrent processing server configuration attempts to create extra processing capacity on the database tier by offloading the reporting and batch load to other platform servers connected to the application database server by a high-speed local network connection.

Multi-Threaded Server

This is a way to relieve memory shortages on a memory-bound database server (and can also help with the resource management of system processes in a system with unduly large numbers of process connections to the server). It needs special database tuning skills and monitoring to mitigate the risk of negative performance effects from an extra database resource bottleneck.

Oracle Parallel Server (OPS)

This is a way to distribute the processing across database servers that combine to provide total system resources to process a database. Users or batch programs can be partitioned across the individual platforms in an OPS cluster with inter-node locking and memory management handled transparently. This approach can provide greater scalability and can distribute the total load across multiple database servers, but does have the drawback of having higher hard costs for extra hardware, and higher soft costs for learning events or hiring of expertise in tuning and management.

Network Capacity Planning

Networks are constrained by two factors – bandwidth and latency. Bandwidth is the actual capacity of the physical and data link (physical) or protocol (logical organization of data) layers. For example, Ethernet is available at 10Mbps, 100Mbps, s and 1Gbps. A T1 link runs at 1.536Mbps. Asynchronous Transfer Mode (ATM) is available at up to 155Mbs. Latency can be described as delays in routing of network traffic causing under-utilization of bandwidth. A network transaction often has to travel through many devices from the point of origin to the point of destination.

If a particular device is undersized in terms of performance, it may impose a delay in the forwarding of the transaction on to the next segment in the path. This often results in high bandwidth links being under-utilized as network traffic is delayed somewhere else in the path, however appearing as delays nevertheless. Often bandwidth is increased with little or no overall improvement in network traffic management and performance due to unresolved latency issues. Therefore the overall design of the network and the sizing of the capacity of instream network devices is critical.

As bandwidth and latency combined determine the overall performance of a network path or channel, careful sizing and designing of both links and routing is critical to achieving the goal of high performance for reasonable cost.

Effects of Latency

Oracle Applications and other Web-Client/Server applications are sensitive to the effects of latency. Not only for the Wide-Area-Network (WAN), but also for Local-Area-Networks (LANs). Consider a situation where the Web-Client (Client Tier) is separated from the Applications Form Server (Application Tier) by two bridges and one router hop with a total one way delay of only 30ms (return) across a 1.536Mbs T1 link. For example, if the transport window is 1.5kb the actual data rate would be 0.317Mbps and not 1.536Mps. This model does not take into account queuing that may occur if simultaneous requests are queuing for the same data path. The actual data rate here would be factored as a consumer of bandwidth, but only at .317Mbps out of the capacity of 1.536Mbs of the path. Optimization of this path would involve reducing the latency imposed by the instream devices as well as adjustments to the transport window (frame) size to be optimal for the nature of the traffic. The faster the switching and routing, the more effective the utilization of the available bandwidth.

When there is a need to optimize routing and switching, there are many options. One approach uses very high speed devices that cache routing tables in very high-speed memory called switching fabric. These devices run one or more very-high performance processors and architectures. It is simply a factor of how much one wants to spend.

Disk Sizing

Planning the capacity of disk space is usually one of the less riskier aspects of capacity planning, unless the implementation hardware platform budget has slim margins for error, and the disk space needs to be calculated very accurately. The trend towards decreasing real cost of disk capacity means that the amount of disk purchased from disk vendors for new implementation projects has grown to provide enough safety margin in the estimates. If disk estimates are performed by the system architect as part of the conceptual architecture design, they are likely to be based on initial business volume estimates with only an approximate idea of archive or purge cycles. At the conceptual architecture stage of the process, the benefit derived from the disk sizing exercise may be to identify what the likely disk requirements are to a margin of, for example, 50 percent. Therefore, it may identify an approximate figure for the required disk space, but not much better than this.

As the project proceeds, the required business volume information, archive, and purge processes will be better characterized and the disk estimates correspondingly better. As with the processing capacity, the disk capacity requirements should be estimated for production cutover, and then for a prescribed period of production operations with business growth of transactions included, if possible. The volumes of data converted in the conversion process are important for estimating the initial disk capacity required, but may be irrelevant for the disk estimates after the prescribed period of operations. One or more archive or purge cycles for transaction data may remove the initial extra volumes from converted data, so that the disk space required for a particular data entity is a function of the maximum volume prior to archive or purge. This needs to be addressed on an entity by entity basis. It may also be possible to omit some data entities that contribute minimal disk space to the overall total, and only focus on the main entities for each application product that contribute 90 percent of the total disk space.

Production Monitoring of Disk Space

The client staff can closely monitor disk space during production operations and set a minimum free space volume that triggers purchase of new disk space. For example, in a small-to-medium implementation, if the free disk space falls below 1GB, this could trigger a new purchase of extra disk space. The exact free space trigger value will be determined by the disk purchase lead time and the rate at which the business uses disk space.

Linkage to Other Tasks

The System Capacity Plan is an input to the following tasks:

- TA.120 - Define Platform and Network Architecture
- TA.140 - Assess Performance Risks
- TA.150 - Define System Management Procedures
- PM.140 - Propose Future Technical Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	50
Database Administrator	20
System Administrator	20
Network Administrator	10
Client Staff Member	*
IS Manager	*

Table 4-25 Role Contribution for Define System Capacity Plan

* Participation level not estimated.

Deliverable Guidelines

Use the System Capacity Plan deliverable to understand the capacity requirements for the new architecture.

This deliverable should address the following:

- overall capacity planning strategy for production and production support environments
- database server disk and processing requirements
- concurrent processing server disk and processing requirements
- forms / web sever disk and processing requirements
- network capacity requirements in terms of bandwidth and latency

Deliverable Components

The System Capacity Plan consists of the following components:

- Introduction
- Capacity Planning Strategy
- Business Volume Requirements
- Production Environment Servers
- User Learning Environment Servers
- Testing Environment Servers
- Desktop Client Machines
- Network Capacity
- Assumptions
- Risks

Introduction

This component describes the purpose of the deliverable and the scope of the system capacity planning work. The scope should describe the technical infrastructure components included in the study.

Capacity Planning Strategy

This component describes the strategy that is to be used for the system capacity planning work. The architecture strategy deliverable also contains a discussion of this strategy, in which case this component can either reference the component in the architecture strategy deliverable, or for ease of use, reproduce that component here. In particular, this component should identify the capacity planning horizon.

Business Volume Requirements

This component summarizes the key business volume information used for the capacity planning work. It should include any projections for future headcount or transaction rates. The information recorded here may have been captured by Business Requirements Definition (RD) team. If this is the case, you should refer to the Business Volumes and Metrics (RD.040) deliverable, and summarize the main metrics you will use for the capacity planning.

Production Environment Servers

This component describes the capacity planning work performed on the production servers. Depending on the scope of the capacity planning task, it may include capacity planning sections for the following:

- Processing Capacity — Server Class and CPUs
- Memory Requirements
- Disk Capacity Requirements

User Learning Environment Servers

This component describes the capacity planning work performed on the user learning environment servers. Depending on the scope of the capacity planning task, it may include capacity planning sections for the following:

- Processing Capacity — Server Class and CPUs
- Memory Requirements
- Disk Capacity Requirements

Testing Environment Servers

This component describes the capacity planning work performed on the Testing environment servers. Depending on the scope of the capacity planning task, it may include capacity planning sections for the following:

- Processing Capacity — Server Class and CPUs
- Memory Requirements
- Disk Capacity Requirements

Desktop Client Machines

The desktop client platforms component summarizes the capacity requirements for the desktop client platforms in the new architecture. It includes an assessment of the specifications for these platforms depending on the type of user and the processing capacity needed.

Network Capacity

This component summarizes the capacity planning work undertaken on the local and wide area networks that are part of the network infrastructure for the future system, including estimates of the bandwidth needed and assessment of latency requirements.

Assumptions

This component describes the assumptions that were made in the course of capacity planning (for example, assumptions made because of the timing of the architecture project or missing volumetric information).

Risks

This component describes the risks inherent in the system capacity plan analysis and in the volume metrics used for the work.

Audience, Distribution, and Usage

Distribute the System Capacity Plan to the following:

- project managers of the main project
- information system manager
- architecture team members
- other process or subproject leaders who have dependent tasks with the performance testing process/subproject

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Does the content in the deliverable match the scope and description of the document in the introduction?
- Is there a clear strategy for the system capacity planning?
- Does the system capacity plan cover database server processor, memory and disk requirements?
- Does the system capacity plan address capacity issues after live production cutover?
- Has the system capacity plan taken account of future business growth and other applications or systems that will compete for the same system resources?
- Have the client platforms or networks been considered?

Tools

Deliverable Template

Use the System Capacity Plan template to create the deliverable for this task.

TA.120 - Define Platform and Network Architecture (Core)

In this task, you define the physical platform and network configuration to support your final platform and network architecture. You map the physical application and database server architecture onto specific computing platforms.

This task focuses on the future production environment and not interim environments to support the project activities. This includes on going production supporting testing and learning environments.

Deliverable

The deliverable for this task is the **Platform and Network Architecture**. It describes the deployment of the key platform (hardware) and network components of the new system and their relationship to the application and database architecture.

Prerequisites

You need the following input for this task:

- ❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work.

- ❑ **Current Technical Architecture Baseline (TA.020)**

The Current Technical Architecture Baseline provides a consolidated inventory of the applications, databases, interfaces, hardware platforms, and networks in the existing information systems infrastructure and may offer solutions for re-deployment or reuse of existing technical infrastructure. If Identify Current Technical Architecture was not performed, this deliverable will not exist. (See the TA.020 task description for more information on when this task should be performed.)

❑ **System Availability Strategy (TA.050)**

The System Availability Strategy provides information about the strategies used for providing the level of system availability the business needs. If special hardware platforms and network architectures or components are intended to provide the desired level of system availability, this deliverable will contain information about them.

❑ **Conceptual Architecture (TA.070)**

The Conceptual Architecture document provides a high-level view of the favored future application and technical architecture. It contains a conceptual view of the hardware and network configuration for the new system. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the TA.070 task description for more information on when this task should be performed.)

❑ **Application and Database Server Architecture (TA.090)**

The Application and Database Server Architecture provides the logical and physical definition of various installed application databases, including their schemas and objects. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the TA.090 task description for more information on when this task should be performed.)

❑ **System Capacity Plan (TA.110)**

The System Capacity Plan provides the database sizing metrics needed for estimating the I/O system metrics.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Gather technical requirements not addressed by prior tasks. Update introduction as required.	Introduction

No.	Task Step	Deliverable Component
2.	Document hardware and network standards.	Platform and Network Standards
3.	Define the enterprise-level aspects of the hardware platform and network architecture.	Enterprise Platform and Network Architecture
4.	Create overview of the technical architecture for each data center / hosting facility.	Platform and Network Architecture for Data Center / Hosting Facility
5.	Define the detailed hardware platform architecture for each data center / hosting facility and map the servers to the actual hardware platforms on which they will run.	Platform and Network Architecture for Data Center / Hosting Facility
6.	Define the detailed network architecture for each data center.	Platform and Network Architecture for Data Center / Hosting Facility
7.	Define the detailed configuration of the client desktop tier for each deployment site.	Client Desktop Tier for Deployment Site
8.	Review the deliverable with the project manager and seek approval.	Acceptance Certificate (PJM.CR.080)

Table 4-26 Task Steps for Define Platform and Network Architecture

Approach and Techniques

In this task, you design the platform and network architecture for the new system. The Conceptual Architecture (TA.070) provides the high-level view of the technical infrastructure of the new system. Using this as the basis for the design, you can expand the details of the architecture using architecture deliverables created after the Conceptual Architecture. If gaps in the detailed hardware and network design are not covered by requirements already captured in the architecture process, you need to discuss these areas with the client staff members and information systems manager. Once you create the architecture design, have the project manager and IS manager review it.

Consider the following architecture components when you design the hardware architecture:

- data servers for OLTP, reporting, and data warehouse systems
- file servers
- web servers and internet commerce/EDI entry points
- failover servers and standby servers/data centers
- gateways connecting applications and databases
- disk farm arrays used by key servers
- client desktop platforms

and for network architecture:

- local area networks and subnets
- wide area network connections
- routers and bridges
- network protocol and topologies

Tight Linkage with System Capacity Planning

The tasks of designing the platform and network architecture, designing the physical database architecture, and developing a System Capacity Plan (TA.110) are tightly linked. You often work on these tasks together as a group within a project. The exact platform and network architecture you choose depends on whether they can support the system capacity plan requirements, and how the servers are physically laid out across disks and I/O subsystems. You may need to perform iterations of each of these tasks before you arrive at a self-consistent solution to the detailed technical configuration of the system.

Communication Between Groups Interested in the Architecture

Several teams may be involved with different components of the hardware and network architecture. Typically, large businesses have separate groups that manage the network architecture and the application and computer hardware architecture. It is important for the application system architects to help the technical system architects and corporate information systems administrators to understand the requirements based on business usage patterns and the application and database deployment, and to communicate those requirements to the appropriate information systems group. System architects need to remember that there can be long lead times in getting changes made to a network or the shipping of new servers (or server components) from a hardware vendor. Therefore, they may need to forewarn the technical information systems groups early on in the process about possible new infrastructure for the architecture.

Linkage to Other Tasks

The Platform and Network Architecture is an input to the following tasks:

- TA.130 - Define Application Deployment Plan
- TA.140 - Assess Performance Risks
- TA.150 - Define System Management Procedures
- PM.020 - Design Production Support Infrastructure
- PM.030 - Develop Transition and Contingency Plan
- PM.040 - Prepare Production Environment
- PM.130 - Propose Future Business Direction

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	45
System Administrator	35
Network Administrator	10
Database Administrator	10
Client Staff Member	*
IS Manager	*

Table 4-27 Role Contribution for Define Platform and Network Architecture

* Participation level not estimated.

Deliverable Guidelines

Use the Platform and Network Architecture deliverable to plan the platform (hardware) and network layout in relation to your existing technical infrastructure and the logical application and database architecture.

The deliverable should contain the necessary details to facilitate implementation. It can either include just the changes to the infrastructure for the new systems, or, if the deliverable is intended to be a complete architecture, it can include both the new systems and the existing systems.

Suggestion: Diagrams are an excellent way to communicate the new hardware and network architecture. You should use diagrams in the deliverable wherever possible.

Deliverable Components

The Platform and Network Architecture consists of the following components:

- Introduction
- Platform and Network Standards
- Enterprise Platform and Network Architecture
- Platform and Network Architecture for Data Center / Hosting Facility (Repeating)
- Client Desktop Tier for Deployment Site (Repeating)

Introduction

This component describes the purpose of the Platform and Network Architecture deliverable and the process used to complete it.

Platform and Network Standards

This component describes the enterprise or data center / hosting facility standards to which the architecture needs to conform. The discussion makes clear who owns the standards and their scope. For example, if data centers / hosting facilities are autonomous in their choice of standards, the component indicates their standards ownership.

The Architecture Requirements and Strategy (TA.010) deliverable may already contain discussion of these standards, in which case this component can either reference the component in the requirements and strategy deliverable or, for ease of use, can reproduce that text here. Similarly, if the standards are detailed in another document, you should reference that other documentation.

Enterprise Platform and Network Architecture

This component describes the high-level enterprise architecture, covering one or more individual data centers, sites or installations. For example, it could show the geographical data center deployment, the business organizations they service, and the network links between data centers. Inclusion of this component is entirely dependent on the scope of the architecture process. If the architecture process only covers a single site or data center, you may not wish to include this component in the deliverable.

Platform and Network Architecture for Data Center/Hosting Facility

This repeating component describes the detailed hardware platform, server mapping and network architecture for each data center / hosting facility and the enterprise as a whole. There is one component for each data center / hosting facility in the scope of the design. The areas that are included in the discussion include:

- technical configuration in the data center
- database and other servers
- data center network configuration

Suggestion: If you are only creating an architecture for a single installation of applications or for a single data center or hosting facility, you should elevate the data center subsections to be the components of the deliverable.

Client Desktop Tier for Deployment Site

This repeating component describes the client desktop tier, printing and peripheral device support for each deployment site.

Audience, Distribution, and Usage

Distribute the Platform and Network Architecture to the following:

- project manager of the main implementation project
- information systems manager
- architecture team members
- production system, network and database administrators
- client staff member providing input to the project

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Does the content in the deliverable match the scope and description of the document in the introduction?
- Is there a clear distinction between the enterprise-level aspects of the hardware and network architecture and the more detailed data center architecture?
- Does the deliverable define the technical configuration in the data centers, the architecture for the data servers, and the desktop client environment?

Tools

Deliverable Template

Use the Platform and Network Architecture template to create the deliverable for this task.

Technical Configurations for Oracle Applications Servers

For information and suggestions about Oracle Applications data server configurations, see the following:

Reference: *Oracle Applications Installation Manual for Unix, Linix, NT or other operating system.* (Operating System and Release Specific).

TA.130 - Define Application Deployment Plan (Optional)

In this task, you determine the details of the deployment of applications across data centers within the scope of the new architecture.

This task focuses on the future production environment and not interim environments to support the project activities. This includes on going production supporting testing and learning environments.

Δ If your project is of *medium or high complexity,* you should perform this task.

Deliverable

The deliverable for this task is the **Application Deployment Plan**. It is the map for the future deployment of the applications.

Prerequisites

You need the following input for this task:

❑ **Process and Mapping Summary (RD.030)**

The Process and Mapping Summary provides information about the future business model, including business processes and mapping decisions made to date. If Establish Process and Mapping Summary was not performed, this deliverable will not exist. (See the RD.030 task description for more information on when this task should be performed.)

❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements and strategy for the architecture work.

❑ Current Technical Architecture Baseline (TA.020)

The Current Technical Architecture Baseline provides a consolidated inventory of the applications, databases, interfaces, hardware platforms, and networks in the existing information systems infrastructure and may offer solutions for re-deployment or reuse of existing technical infrastructure. If Identify Current Technical Architecture Baseline was not performed, this deliverable will not exist. (See the TA.020 task description for more information on when this task should be performed.)

❑ Conceptual Architecture (TA.070)

The Conceptual Architecture provides a high-level view of the favored future application and technical architecture and will indicate the approximate deployment of applications in this architecture model. If Revise Conceptual Architecture was not performed, this deliverable will not exist. (See the TA.070 task description for more information on when this task should be performed.)

❑ Application and Database Server Architecture (TA.090)

The Application and Database Server Architecture provides the logical and physical definition of various installed application databases, including their schemas and objects. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the TA.090 task description for more information on when this task should be performed.)

❑ Platform and Network Architecture (TA.120)

The Platform and Network Architecture provides information about the hardware platforms and network deployment that will support the application deployment and the processing of the databases. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the TA.120 task description for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review deliverables from the architecture process and other supporting documentation. Update the introduction as appropriate.	Introduction
2.	Create high-level application rollout map, if appropriate.	Introduction
3.	Define phase rollout of data center / hosting facilities hosting of database and application tier servers. Repeat section if multiple facilities involved.	Application Rollout
4.	Define deployment of desktop client tier to sites. Repeat if multiple sites involved.	Phase - One
5.	Identify any assumptions related to the deployment plan.	Assumptions
6.	Review plan with project manager and obtain acceptance.	Acceptance Certificate (PJM.CR.080)

Table 4-28 Task Steps for Define Application Deployment Plan

Approach and Techniques

The Preliminary Conceptual Architecture (TA.030) provides information about the basic framework and direction for the new application architecture; this task fills in the details of the deployment. The Process and Mapping Summary (RD.030) provides information on the use of Oracle Applications and applications from other vendors that might be mapped to a particular business process. This should provide you with details you need to specify the exact application modules you will need in each site or data center. If the architecture project spans an enterprise multi-phase application rollout, you need to create application deployments for the intermediate phase milestones on the way to the full rollout target end state. The Architecture Requirements and Strategy (TA.010) should describe the rollout strategy as it pertains to the architecture project.

Task Complexity and Project Scope

The complexity of the application deployment map you need to produce will depend on the scope and complexity of the implementation and the conceptual architecture. If you have designed a centralized architecture in a single site, then the deployment map only includes a single data center and the deployment map reduces to a definitive listing of the applications in the data center. If, however, the architecture scope includes applications in multiple, separate data centers, you need to produce a deployment list for each separate data center, and determine the evolution from the existing application and database deployment architecture to the new architecture for each. The evolution may need to take into account rollout strategy across an enterprise, with intermediate stages where one or more data centers have converted to the new architecture, whereas others have not.

Specific Deployment Considerations

The details of the deployment depend on the rollout strategy and the characteristics of the included applications. Deciding which applications will be deployed in a data center requires some care in a complex architecture. Even if a data center is not explicitly supporting a particular business function, you may still need to partially or fully install the application corresponding to that business function to support other business functions or operations taking place within the data center. This depends on precise business processes and can also depend on the solutions adopted for application gaps.

Full/Partial Installation of Dependent Oracle Application Modules

When you have determined precisely which business organizations and functions each data center will support, you map these onto application modules that provide the required functionality. In drawing up the lists of modules needed you should remember that a particular Oracle Application module may require a full or partial install of other application modules. For example, in release 10.6 a full installation of Oracle Order Entry in a data center requires a full installation of Oracle Receivables also. Use the appropriate installation manual to determine any dependencies that may exist.

Reference: *Oracle Applications Installation Manual for Unix, Linix, NT or other platform* (Operating System and Release Specific).

Specific Application Versions

If there are specific requirements for application versions and the rollout of the applications is dependent on these versions, you should include them and indicate their effect on the deployment.

Linkage to Other Tasks

The Application Deployment Plan is an input to the following tasks:

- PM.020 - Design Production Support Infrastructure
- PM.030 - Develop Transition and Contingency Plan

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	85
Business Analyst	15
IS Manager	*

Table 4-29 Role Contribution for Define Application Deployment Plan

* Participation level not estimated.

Deliverable Guidelines

Use the Application Deployment Plan to define the rollout plan for the new architecture. This deliverable is the roadmap to single or multi-phased rollouts of the applications.

Deliverable Components

The Application Deployment Plan consists of the following components:

- Introduction
- Phase - One (repeated for all phases)
- Assumptions

Introduction

This component describes the purpose for the deployment plan as well as the structure of the document.

Phase - One

This component is repeated for each phase in the application deployment. A phase may include one or more data centers or hosting facilities for the database and application tier as well as one or more desktop client tier deployment sites. Subsequent phases may just include additional desktop client tier deployment sites, or they may also include additional data centers or hosting facilities for the application tier.

This component describes the following for each phase:

- Technical Infrastructure Rollout
- Oracle Applications
- Other Applications
- Criteria for Phase Completion
- Database Tier
- Application Tier (repeated for all data centers/hosting facilities)
- Desktop Client Tier Site (repeated for all deployment sites)

Assumptions

This component documents assumptions made in determining the application deployment. Examples of assumptions that may influence the deliverable include:

- application releases or versions
- impact/completion of ongoing information systems projects
- mapping work that is in process

Audience, Distribution, and Usage

Distribute the Application Deployment Plan to the following:

- Architecture team members use the information to construct more detailed architecture designs for using the application deployment as a framework.
- Business Requirements Mapping (BR) team members use the information to help identify application integration points and custom interfaces.
- Module Design and Build (MD) team members *may* need the information to help specify and design customizations.

- Production Migration (PM) team need to understand the application deployment when planning the migration process.
- Client staff members may use this as a reference to understand which applications will be installed where.

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Does the content in the deliverable match the scope and description of the document in the introduction?
- Does the deliverable cover all data centers/hosting facilities within scope?
- Does the deliverable cover all applications within scope?
- Are there diagrams summarizing the deployment?

Tools

Deliverable Template

Use the Application Deployment Plan template to create the deliverable for this task.

TA.140 - Assess Performance Risks (Core)

In this task, you identify any performance risks that are apparent based on the proposed architecture and suggest techniques to mitigate the risks.

This task focuses on the future production environment and not interim environments to support the project activities. This includes on going production supporting testing and learning environments.

Deliverable

The deliverable for this task is the **Performance Risk Assessment**. It is the assessment of the performance risks associated with the application and technical architecture.

Prerequisites

You need the following input for this task:

- ❑ **Business Volumes and Metrics (RD.040)**

The Business Volumes and Metrics provide information about business volumes. Use this deliverable to identify specific processing risk areas based on the volumes.

- ❑ **Architecture Requirements and Strategy (TA.010)**

The Architecture Requirements and Strategy provides information on the scope, requirements, and strategy for the architecture work.

- ❑ **Application and Database Server Architecture (TA.090)**

The Application and Database Server Architecture provides the logical and physical definition of various installed application databases, including their schemas and objects. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the TA.090 task description for more information on when this task should be performed.)

❑ System Capacity Plan (TA.110)

The System Capacity Plan provides the database sizing metrics needed for estimating the I/O system metrics.

❑ Platform and Network Architecture (TA.120)

The Platform and Network Architecture provides information about the hardware platforms and network deployment that will support the application deployment and the processing of the databases. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the TA.120 task description for more information on when this task should be performed.)

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review deliverables from the architecture process and other supporting documentation. Update the introduction as appropriate.	Introduction
2.	Identify risks for the network.	Risk Area: Network
3.	Identify risks for the database tier.	Risk Area: Database Tier
4.	Identify risks for the applications tier.	Risk Area: Application Tier
5.	Identify risks for the desktop client tier.	Risk Area: Client Tier
6.	Identify risks for the infrastructure.	Risk Area: Infrastructure

No.	Task Step	Deliverable Component
7.	Identify risks for other areas of the architecture.	Risk Area: Others
8.	Update the Key Recommendations section with those items on which action is recommended.	Introduction
9.	Review the deliverable with the project manager and seek acceptance.	Acceptance Certificate (PJM.CR.080)
10.	Review the key risks and proposed mitigation with the project sponsor or steering committee.	

Table 4-30 Task Steps for Assess Performance Risks

Approach and Techniques

In this task, you review the strategies and approaches used for the various parts of the architecture design and identify those that are potential performance risk areas.

Sources of Performance Risk in Architecture Projects

The following sections discuss some typical performance risk areas:

Custom Integration or Subsystem Components

Whenever custom extensions are designed and built in projects, a performance risk is associated with that development.

Inadequate or Inaccurate Business Volume Metrics

The business volume metrics may be incomplete or unavailable. The business metrics are important for the assessment of the performance of any components in a system, so if these are inaccurate or incomplete, this causes general risk in all aspects of system performance assessment.

Capacity Planning

There is always risk associated with the capacity planning exercise in an architecture project and in particular, planning the capacity of the database server and the number of CPUs required. The reasons why this might lead to risk include:

- only a high-level capacity planning exercise was performed
- capacity planning is unable to determine the precise dynamic effects that occur in a system, especially in a high-volume batch parallel processing environment
- the database server is identified as having insufficient capacity at some future time, with uncertainty about the precise timing of the future capacity shortfall
- inaccurate business volume metrics

High-Volume Processing Environments

In addition to the risk of inadequate or unknown capacity requirements, high-volume processing environments are intrinsically more prone to performance risks. The *critical processing periods* can be a useful tool for identifying potential bottlenecks during critical processing time frames such as the end of a quarter. This worksheet indicates the day that key processes are run during a critical time span. Projects where these details can be determined in advance can reduce the risk of unanticipated performance problems. By identifying potential bottlenecks early in the implementation cycle, you have more time to identify potential solutions.

Wide Area Network (WAN) Configurations

As the technical architecture includes web-client/server configurations separated by a wide area network connection, there can be performance issues if the capability of the WAN segment is not sufficient for the application characteristics and user load from remote business sites.

☞ **Attention:** This is especially relevant WAN implementations. If custom extensions will be deployed over a WAN connection, include these within the risk discussion.

Advanced Database and Hardware Features

If the technical architecture includes Oracle Parallel Server or multi-threaded server configurations, there is additional technical performance risk because of tuning and configuration requirements that should be mentioned.

Risk Mitigation

The following sections list a few typical performance risk mitigation approaches. This list is incomplete, and the approaches adopted depend on the precise nature of the risk and the project circumstances.

Detailed Capacity Planning

A more detailed capacity study may be necessary to mitigate capacity planning risks. The capacity planning efforts can continue after production cutover using metrics from production system by way of a continuous monitoring of capacity requirements.

High-Volume Processing Environments

Review the Business Volumes and Metrics (RD.040) document that was produced in the Gather Business Volumes and Metrics task. The Oracle project manager can help gather information about other similar implementations to determine whether the transaction volumes are within the bounds of other known similar production sites. Oracle user groups can also be a source of this information.

Prototype Architecture Components of Concern

If there are particular architecture configurations or components that cause performance concerns, build a prototyping exercise and formal performance test into development or deployment schedules.

Performance Testing

In some cases, reliable predictive information may not be available for large processing requirements. This can be due to the availability of a new application release, new hardware platform, or other factors. If project resources are available, a performance test is a very good way to provide performance data based on the planned production configuration and usage patterns. Conducting a performance test also provides a hands-on learning opportunity for production system and database administrators before the production cutover date. For more information, see Performance Testing (PT).

Production System Readiness

The risks and mitigation that are identified in this analysis should be communicated to the production migration team. During the assessment of the readiness for production operations, the areas of risk should be revisited to make sure that they have been addressed or that the risk is within an acceptable limit of exposure. For more information, see Verify Production Readiness (PM.070).

Linkage to Other Tasks

The Performance Risk Assessment is an input to the following tasks:

- PM.030 - Develop Transition and Contingency Plan
- PM.090 - Measure System Performance

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Architect	100
IS Manager	*
Project Sponsor	*

Table 4-31 Role Contribution for Assess Performance Risks

* Participation level not estimated.

Deliverable Guidelines

Use the Performance Risk Assessment to evaluate all of the risks to the performance of the new production environment including production support environments such as learning and testing.

This deliverable should address the following:

- risk assessments for all of the tier of the architecture
- risk assessment for the network
- risk assessment for the infrastructure
- risk assessment for other general aspects of the architecture

Deliverable Components

The Performance Risk Assessment deliverable consists of the following components:

- Introduction
- Risk Area: Network
- Risk Area: Database Tier
- Risk Area: Application Tier
- Risk Area: Client Tier
- Risk Area: Infrastructure
- Risk Area: Others

Introduction

This component summarizes the analysis of performance risk areas in the architecture, the results of the analysis, and the recommended risk mitigation approaches for the benefit of executive management or project sponsors.

Risk Area: Network

This component describes the performance risk areas associated with the network aspects of the architecture design. It specifically identifies the risk areas and the reasons why there are concerns. The consequences of these risks are identified, as well as pre-planned contingencies to be taken if these risks occur. Early mitigation that may be taken to prevent these risks from occurring is also identified in this component.

Risk Area: Database Tier

This component describes the performance risk areas associated with the database tier of the architecture design. It specifically identifies the risk areas and the reasons why there are concerns. The consequences of these risks are identified, as well as pre-planned contingencies to be taken if these risks occur. Early mitigation that may be taken to prevent these risks from occurring is also identified in this component.

Risk Area: Application Tier

This component describes the performance risk areas associated with the application tier of the architecture design. It specifically identifies the risk areas and the reasons why there are concerns. The consequences of these risks are identified, as well as pre-planned contingencies to be taken if these risks occur. Early mitigation that may be taken to prevent these risks from occurring is also identified in this component.

Risk Area: Client Tier

This component describes the performance risk areas associated with the client tier of the architecture design. It specifically identifies the risk areas and the reasons why there are concerns. The consequences of these risks are identified, as well as pre-planned contingencies to be taken if these risks occur. Early mitigation that may be taken to prevent these risks from occurring is also identified in this component.

Risk Area: Infrastructure

This component describes the performance risk areas associated with other aspects of the architecture design. It specifically identifies the risk areas and the reasons why there are concerns. The consequences of these risks are identified, as well as pre-planned contingencies to be taken if these risks occur. Early mitigation that may be taken to prevent these risks from occurring is also identified in this component.

Risk Area: Others

This component describes the performance risk areas associated with the network aspects of the architecture design. It specifically identifies the risk areas and the reasons why there are concerns. The consequences of these risks are identified, as well as pre-planned contingencies to be taken if these risks occur. Early mitigation that may be taken to prevent these risks from occurring is also identified in this component.

Audience, Distribution, and Usage

Distribute the Performance Risk Assessment to the following:

- project managers of the main project
- information systems manager
- architecture team members
- performance testing process team leader
- business requirements definition process team leader (if there are issues with business volume metrics)

Quality Criteria

Use the following criteria to check the quality of this deliverable:

- Does the content in the deliverable match the scope and description of the document in the introduction?
- Does the deliverable cover the performance risk areas across the entire architecture process?
- Does the deliverable suggest risk mitigation approaches?

Tools

Deliverable Template

Use the Performance Risk Assessment template to create the deliverable for this task.

TA.150 - Define System Management Procedures (Core)

In this task, you design the procedures and specify the tools that the client staff will need to manage the new system. After you design the procedures in this task, you need to test and refine them later in the project, prior to incorporating them into the System Management Guide (DO.090) and conducting learning events for the system support staff.

This task focuses on the future production environment and <u>not</u> interim environments to support the project activities. This includes on going production supporting testing and learning environments.

Deliverable

The deliverable for this task is the **System Management Procedures**. It is a list of system management procedures and tools that the operations personnel will use to manage and monitor the new system.

Prerequisites

You need the following input for this task:

- ❑ **System Availability Strategy (TA.050)**

The System Availability Strategy provides information about the strategies for achieving the level of system availability the business needs. If special database architectures and systems are needed to provide the desired level of system availability, this deliverable contains information about them.

- ❑ **Application and Database Server Architecture (TA.090)**

The Application and Database Server Architecture provides the logical and physical definition of various installed application databases, including their schemas and objects. If Define Application and Database Server Architecture was not performed, this deliverable will not exist. (See the TA.090 task description for more information on when this task should be performed.)

- ❑ **System Capacity Plan (TA.110)**

The System Capacity Plan provides the database sizing metrics needed for estimating the I/O system metrics. The plan also addresses the horizon for the capacity established, and steps to be taken to measure consumption leading to capacity forecasts as well as how to extend capacity when indicated.

- ❑ **Platform and Network Architecture (TA.120)**

The Platform and Network Architecture provides information about the hardware and network deployment that supports the application deployment and the processing of the databases. If Define Platform and Network Architecture was not performed, this deliverable will not exist. (See the TA.120 task description for more information on when this task should be performed.)

Optional

You may need the following input for this task:

- ❑ **Current Technical Architecture Baseline (TA.020)**

The Current Technical Architecture Baseline provides a consolidated inventory of the applications, databases, interfaces, hardware and networks in the existing information systems infrastructure. If Identify Current Technical Architecture was not performed, this deliverable will not exist. (See the TA.020 task description for more information on when this task should be performed.)

- ❑ **System Management Documents Describing Current Tools and Procedures**

The client staff may have documentation, manuals, or guides on the current tools and procedures in place to manage the existing systems. You can use these documents to understand the existing tools and procedures.

Task Steps

The steps for this task are as follows:

No.	Task Step	Deliverable Component
1.	Review existing system management procedures, tools, and strategy.	
2.	Identify missing requirements and gather appropriate information. Update the introduction as appropriate.	Introduction
3.	Review the text and the suggestion for establishment of a Network and Server Operations Center. Revise as appropriate.	Overview of the Network and Server Operations Center
4.	Summarize system management policies and preferences.	Enterprise Management Standards and Policies
5.	Review and update the information related to planned maintenance for the enterprise systems in the architecture.	Planned Maintenance Schedule
6.	Identify all tools that will be used to manage the systems and servers within the architecture.	Tools Summary
7.	Identify all procedures for managing the servers within the database tier of the new architecture	Database Tier

No.	Task Step	Deliverable Component
8.	Identify all possible failure scenarios for the servers within the database tier of the new architecture.	Database Tier
9.	Identify all procedures for managing the servers within the application tier of the new architecture	Applications Tier
10.	Identify all possible failure scenarios for the servers within the application tier of the new architecture.	Applications Tier
11.	Identify all procedures for managing the software and systems within the desktop client tier of the new architecture	Desktop Client Tier
12.	Identify all possible failure scenarios for the software and systems within the desktop client tier of the new architecture.	Desktop Client Tier
13.	Design login security mechanisms for database servers.	Security and Accounts Management
14.	Design application login security mechanisms.	Security and Accounts Management
15.	Design procedures for password validation and management.	Security and Accounts Management
16.	Identify possible security breaches.	Security and Accounts Management

No.	Task Step	Deliverable Component
17.	Design security monitoring and protection procedures.	Security and Accounts Management
18.	Design procedures for adding a new user to the system.	Security and Accounts Management
19.	Identify hardware and networks management requirements and techniques.	Hardware and Network Management
20.	Identify all possible hardware and networks failure scenarios.	Hardware and Network Management
21.	Design procedures to recover from hardware and networks failure scenarios.	Hardware and Network Management
22.	Identify hardware and networks management and monitoring tools needed.	Hardware and Network Management
23.	Design change control procedures for hardware and networks.	Hardware and Network Management
24.	Identify software management requirements and techniques.	Software Management
25.	Identify tools needed to manage software.	Software Management
26.	Identify capacity planning and performance management monitoring necessary in production system.	Capacity Planning and Performance Management

No.	Task Step	Deliverable Component
27.	Identify capacity planning and performance monitoring tools needed.	Capacity Planning and Performance Management
28.	Summarize planned outage schedule for the system, taking into account all sources of outages.	Planned Maintenance Schedule
29.	Summarize the system management tools that will be used for the management of the system.	System Management Tools Summary
30.	Seek acceptance for the deliverable from the project manager and the system administrator.	Acceptance Certificate (PJM.CR.080)

Table 4-32 Task Steps for Define System Management Procedures

Approach and Techniques

Task Complexity and Project Scope

If you are working on an enterprise-level architecture that spans multiple sites or databases, you may not need to perform this task in detail. Even at the enterprise level there may be a need to specify the standard tools and common procedures that are used across all data centers and installations. This would be especially true when the enterprise architecture is part of a wider program management structure that is helping to define standards and strategies across all the key areas in an enterprise implementation project.

When you are designing an architecture for a more limited scope project (for example, a single site project or a smaller business) you need to perform this task to a detailed completion level. The deliverable from the task should be a detailed design document that discusses the tools and procedures for the management of the local system.

Early Consideration of Systems Management

When you implement a new production system of any type, you need procedures and tools to manage, maintain, and troubleshoot the new system. When you replace major portions of your system, the amount of work to put in place the new procedures increases dramatically, especially when the systems replacement is accompanied by a change in technology.

The tools and procedures required to maintain a new system are often not considered until the end of implementation projects, leaving little time for testing, refinement, documentation, and new procedure learning events. The result is that the client staff are not fully prepared for the new system cutover, and procedures that may look viable on paper are unfeasible in practice, since they have not been tested and optimized.

By considering the design of the system management procedures and tools as part of architecture, you can design the procedures as you design the technical architecture of the system and can assess the system management impact during this process. The other benefit is that the architecture process naturally occurs early in projects, so you have designed procedures in place earlier in a project, giving you time to refine, test, and document later.

Classification of System Management

System management is a major area for consideration, especially when the new systems being implemented support critical business processes. The systems usually require new tools and techniques for management, and dedication of project time and resources to this task helps make sure that the new system is managed smoothly, with minimal interference to users and normal business operations.

Although it is impossible to predict every conceivable event or situation in a complex system, the goal of the work performed in this task is to predict as many of them as possible. A well-designed system management strategy can preempt the great majority of system outages and prevent the necessity of designing procedures as situations unfold in production operations.

One of the difficulties in discussing system management is that the area covers a diverse variety of system events and procedures. The confusion is compounded by a plethora of system management tools available from different vendors. Because the subject of system management is so large, it is useful to subdivide it into separate topical areas that can be worked on independently by different administrators with special relevant expertise.

A suggested subdivision of the different system management areas is:

- Database Management
- Security and Accounts Management
- Scheduling Management
- Hardware and Network Management
- Software Management
- Capacity Planning and Performance Management

Other groupings are possible as well. The administrator responsible for defining the procedures and tools for an area should consider the following subtopics and identify those that are relevant:

- Proactive and Reactive Monitoring
- Normal Management and Maintenance Procedures
- System Failure Analysis and Recovery
- Long Term Resource Planning
- Tools and Utilities

Database Management

The following list provides some brief suggestions to focus on within the overall database management area:

Proactive and Reactive Monitoring

- space utilization and management
- database fragmentation
- communication of exception conditions or system failures to the appropriate operations administrator

Normal Management and Maintenance Procedures

- backup procedures
- archive and purge procedures
- Oracle cost-based optimizer analysis of tables

System Failure Analysis and Recovery

- diagnosing different database errors
- recovery after loss of an Oracle data file
- recovery after loss of a redo log file
- recovery after loss of a control file
- recovery after operator or administrator error
- recovery after disk failure
- recovery after server failure

Tools and Utilities

- backup tools
- database monitoring tools
- data storage media (for example, magnetic tapes, computer output to laser disk, and optical jukeboxes)

Specific procedures for managing the database components of a system are documented in:

Reference: Millsap, Cary V. *An Optimal Flexible Architecture for a Growing Oracle Database*. White Paper, Revised 1993.

Reference: Velpuri, Rama. *Oracle8 Backup and Recovery Handbook*. Oracle Press. Osborne McGraw-Hill, 1998. ISBN: 0078823897.

Security and Accounts Management

The following list provides some brief suggestions for things to focus on within the overall security and accounts management area. This is only meant to be illustrative, and is not necessarily complete.

Proactive and Reactive Monitoring

- security breaches from external sources, including internet servers
- user login auditing
- data and transaction auditing
- denial of service
- communication of exception conditions or system failures to the appropriate operations administrator

Normal Management and Maintenance Procedures

- authorization of accounts for a new hire
- cancellation of system access for a terminated employee
- automated password expire
- single network login

Tools and Utilities

- firewalls to protect corporate systems
- security systems (for example, kerberos and network encryption services)
- remote access authentication

Management Scheduling

The following list provides some brief suggestions for things to focus on within the overall scheduling management area:

Proactive and Reactive Monitoring

- monitoring of batch queues, backlog, and timing
- communication of exception conditions or system failures to the appropriate operations administrator

Normal Management and Maintenance Procedures

- batch processing and queue stop and restart for scheduled outage events

System Failure Analysis and Recovery

- system cleanup after batch job failure
- stop and restart of batch processing after server, database, disk, or network failure

Tools and Utilities

- batch queue monitoring
- automated stop and restart of batch queue processing

☞ **Attention:** In Oracle Applications, the batch processing is managed within the concurrent manager, and this automates much of the batch queue management and job submission and resubmission. The concurrent manager still needs monitoring and is often overlooked in implementations.

Hardware and Network Management

The following list provides some brief suggestions for things to focus on within the overall hardware and network management area:

Proactive and Reactive Monitoring

- system availability metrics and statistics
- database server failure
- network segment failure
- client file server failure
- communication of exception conditions or system failures to the appropriate operations administrator

System Failure Analysis and Recovery

- diagnosis of hardware or network failure point
- database server failure recovery
- client code file server failure recovery
- user client desktop failure recovery
- migration from standby server back to primary server after system repair

Tools and Utilities

- hardware and network monitoring tools
- central system console for an integrated complete view of all systems

Software Management

The following list provides some brief suggestions for things to focus on within the overall software management area:

Proactive and Reactive Monitoring

- system infection by viruses
- communication of exception conditions or system failures to the appropriate operations administrator

Normal Management and Maintenance Procedures

- testing of software patches prior to migration into production
- migration of software patches to production database servers
- migration and distribution of software patches to production user client desktop platforms and file servers
- software upgrade procedures
- backup of production system software
- source control for system software

Tools and Utilities

- client software distribution tools
- source control systems
- configuration management systems

Capacity Planning and Performance Management

The following list gives some brief suggestions for things to focus on within the overall capacity planning and performance management area:

Proactive and Reactive Monitoring

- usage of system resources during different system processing scenarios
- identification of inefficient software programs and individual database queries
- database tuning metrics, such as cache hit ratios

Normal Management and Maintenance Procedures

- analysis of database statistics for cost-based optimizer
- analysis of batch queue processing metrics
- analysis of online usage metrics and software program efficiency

Long-Term Resource Planning

- projected future usage of system resources
- identification of future system capacity constraints
- strategies to manage future system capacity demands

Tools and Utilities

- system resource usage
- restriction of the maximum number of online sessions per user
- the governors for system resource usage by programs and ad hoc queries

Production Monitoring of Disk Space

It can be effective for the client staff to closely monitor disk space during production operations and set a minimum free space volume that triggers purchase of new disk space. For example, in a small-to-medium implementation, if the free disk space falls below 1 GB, this could trigger a new purchase of extra disk space. The exact free space trigger value is determined by the disk purchase lead time and the rate at which the business uses disk space.

Cost-Based Optimizer and ANALYZE

If you are using the Oracle cost-based optimizer for the SQL statements in your applications, you should schedule reanalysis of the affected tables on a periodic basis to maintain continuing good performance.

☞ **Attention:** In the current release of Oracle Applications, the rule-based optimizer (RBO) is the default optimization method used, but a small percentage of SQL queries use cost-based optimizer (CBO) hints. To maintain continuing optimal performance for these queries, you should schedule to run the ANALYZE facility against the affected tables on a regular basis (for example, once per month), but the precise frequency depends on how the business populates the affected tables.

This may change in the future. Please review the Oracle Applications Installation Guide for your operating system for the latest information pertaining to this issue.

Linkage to Other Tasks

The System Management Procedures are an input to the following tasks:

- DO.090 - Publish System Management Guide
- PM.020 - Design Production Support Infrastructure
- PM.040 - Prepare Production Environment

Role Contribution

The percentage of total task time required for each role follows:

Role	%
System Administrator	40
Network Administrator	20
Database Administrator	20
System Architect	20
IS Manager	*
Client Staff Member	*

Table 4-33 Role Contributions for Define System Management Procedures

* Participation level not estimated.

Deliverable Guidelines

Using the System Management Procedures as an input, describe of all of the designs for the procedures and tools needed to manage the new system environment.

It is important to distinguish this deliverable from the System Management Guide (DO.090) that is created within the documentation process or from any documentation that help desk client staff may use to diagnose an exception condition or issue that a user reports. This deliverable is a design document and does not discuss the procedures in file detail or in a reference guide format. It is produced primarily by the technical systems management experts in the project, who are responsible for creating workable procedures and selecting appropriate tools for the management of the new systems. The resulting deliverable is a primary input to the System Management Guide, which is a separate manual produced by a technical writer. For more information, see Publish System Management Guide (DO.090).

Deliverable Components

The System Management Procedures consist of the following components:

- Introduction
- Overview of the Network and Server Operations Center
- Enterprise Management Standards and Policies
- Planned Maintenance Schedule
- Tools Summary
- Database Tier
- Applications Tier
- Desktop Client Tier
- Security and Accounts Management
- Hardware and Network Management
- Software Management
- Capacity Planning and Performance Management

Introduction

This component describes the need for systems management procedures.

Overview of the Network and Server Operations Center

This component describes the role and service level commitments of the Network and Server Operations Center.

Enterprise Management Standards and Policies

This component describes the standards and policies to which the system management aspects of the architecture must conform. The discussion makes clear who owns the standards and policies and their scope.

The Architecture Requirements and Strategy (TA.010) deliverable may already contain discussion of these standards, in which case this component should either reference the applicable component in the Architecture Requirements and Strategy deliverable or, for ease of use, or reproduce that text here. Similarly, if the standards are detailed in another document, it should reference that other documentation.

Planned Maintenance Schedule

This component summarizes all planned maintenance that affects the system across system management areas.

Tools Summary

This component summarizes the tools that will be used for managing the production systems across all system management areas.

Database Tier

This component discusses the procedures and tools needed for managing the servers in the database tier of the new system. It covers both normal maintenance of the databases, as well as possible failure scenarios. It also describes the tools that will be needed, both those developed in-house and those that must be purchased.

Applications Tier

This component discussed the procedures and tools needed for managing the servers in the application tier of the new systems. It covers both normal maintenance of the servers as well as possible failure scenarios. It also describes the tools that will be needed, both those developed in-house and those that much be purchased.

Desktop Client Tier

This component discussed the procedures and tools needed for managing the software in the desktop client tier of the new systems. It covers both normal maintenance of the software environment as well as possible failure scenarios. It also describes the tools that will be needed, both those developed in-house and those that much be purchased.

Security and Accounts Management

This component discusses the procedures and tools needed for managing access to systems and accounts and for preventing and responding to security breaches.

Hardware and Network Management

This component discusses the procedures and tools needed for management of the hardware and network configuration of the new system. It also includes discussion of procedures for the change management and testing of technical configurations.

Software Management

This component discusses the procedures and tools needed for management of all the software components of the new system. It includes configuration management of clients and servers, source control, and patch and upgrade procedures.

Capacity Planning and Performance Management

This component discusses the procedures and tools needed for managing the performance of the new systems and the proactive future planning of system capacity.

Audience, Distribution, and Usage

Distribute the System Management Procedures to the following:

- information systems manager
- project manager
- client staff member
- members of the architecture process team

Quality Criteria

Use the following criteria to help check the quality of this deliverable:

- Is the deliverable comprehensive in its coverage of different systems management areas?
- Does the deliverable address the various types of procedures and tools needed?
- Are the procedures documented to a degree of detail consistent with the scope of the architecture process?

Tools

Deliverable Template

Use the System Management Procedures template to create the deliverable for this task.

Oracle Alert®

Oracle Alert® is a tool for monitoring aspects of the Oracle Applications environment. It is an alert mechanism that that can track applications database events as they occur, or on a periodic condition. It can track database-level exception conditions and can alert an administrator by generating a report, an email, or an email to a paging device.

Auditing Oracle Applications

Oracle Applications has built-in auditing mechanisms for both user login auditing and data auditing. While the use of the built-in tools for auditing is very convenient, like other auditing mechanisms it does create a system overhead on performance. You should use the data auditing features judiciously, rather than enabling audit for many transactions and tables where there is not a strict business need. For more information, see:

Reference: *Oracle Applications System Administrator Reference Manual.* (Release Specific).

System Management Tools

There are many vendors in the system management arena that offer a plethora of different tools and applications for monitoring, tuning, and managing relational databases. These tools vary in their capabilities and the degree of integration with networked Oracle environments.

Oracle's System Management Technology Initiative (SMTI)

The Oracle SMTI program helps vendors of system management tools and applications integrate their products with the Oracle's servers. Vendors that are part of the SMTI program are more likely to provide tools that are integrated with Oracle servers.

For further information about the SMTI program, see:

Reference: *Oracle's SMTI Partners*. Contact the local Oracle Office for the current list of partners.

Reference: *Oracle SMTI Fact Sheet*. Contact the local Oracle office for the latest version of this fact sheet.

Oracle's System Management Tools

Oracle offers a number of system management tools that are intended for the administration and management of Oracle servers including concurrent (batch) job management in a networked environment. These tools include the modules:

- Enterprise Manager
- Enterprise Manager Performance Pack (for performance tuning)
- Replication Manager
- Network Manager
- Concurrent (batch) Job Manager
- Oracle Client Software Manager
- Oracle Enterprise Backup

The Oracle system management platform, which these tools are built on, supports the Simple Network Management Protocol (SNMP) and the integration of different agents. Oracle databases can be monitored through a central console alongside other distributed network components, such as hosts, routers, and bridges. The adoption of the protocol standard also means that Oracle's products can be launched through the products of other vendors.

For information about Oracle's System Management products, you should consult the following reference:

Web Site: Oracle's World Wide Web Home page at: http://www.oracle.com

For information about Oracle Enterprise Manager, see:

Reference: *Oracle Enterprise Manager Concepts Guide* (Release Specific).

There are many sources for discussions of systems management tools and techniques. The International Oracle User Group holds conferences every year (IOUW) at which papers are presented and many of these papers deal with the issues and techniques for managing and monitoring Oracle-based systems. For a sample of papers, you can refer to the following:

Reference: International Oracle User Group *International Oracle User Week Conference Proceedings* (Current Year).

Reference: *Oracle Magazine* (Current Year).

Configuration of Database Objects and Application Files

Keeping track of the database objects and application files in a production environment is a complex undertaking. It is made difficult where you have:

- many application tier servers to synchronize
- multiple production databases (distributed data) or separate applications installations
- bug fixes or patches to modules in the production environments
- extensions or modifications to vendor software package code

The management of software environments is also sometimes called *configuration management*. You will need to develop or purchase some system to track your software and hardware configuration and record changes to environments in the system. A sophisticated configuration tracking system would have:

- the ability to track and cross-validate client and server software code and objects
- integration of the configuration system with a bug or issue tracking system
- integration of the configuration system with a source control system for both client and server code

Oracle® Designer

If you are using Designer to develop your custom code you may be able to use this to track your configuration.

Intersolv™ PVCS

Intersolv's PVCS software configuration management products are integrated with the Oracle Developer product and can be used to manage the code developed using the Oracle products.

Web Site: For more information see Intersolv's World Wide Web home page at: http://www.intersolv.com